

A LOCAL PROOF OF LANGLANDS'S SECOND MAIN LEMMA FOR LOCAL EPSILON FACTORS OVER NONARCHIMEDEAN LOCAL FIELDS

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ABSTRACT. We give a local proof of Langlands's Second Main Lemma for local epsilon factors over nonarchimedean local fields, in mixed and equal characteristic. The lemma compares local constants of characters of two intermediate fields in a bicyclic extension. It is one of the identities used in Langlands's construction of epsilon factors of local Weil representations.

The proof builds on the work of Dwork, Langlands, and Lakkis. Langlands attributes the Second Main Lemma to Dwork, but Dwork did not publish a complete proof. Lakkis gave a detailed local treatment of the Second Main Lemma. For odd prime degree, his argument proves the required equality. In degree two, however, it proves only that the equality holds up to sign. Both Dwork and Lakkis worked in the mixed-characteristic setting. The surviving part of Langlands's manuscript does not contain a complete proof of the Second Main Lemma.

The new point in the present paper is the wild dyadic case, i.e. biquadratic extensions over a nonarchimedean local field of residue characteristic two. For a biquadratic extension the First Main Lemma determines the square of the required equality, leaving a possible sign. We determine this sign by comparing the finite sums occurring in Lamprecht's formula. This completes the proof of the Second Main Lemma, including the equal-characteristic case.

OpenAI ChatGPT was used extensively in the development of this proof. A substantial part of the argument consists of long local calculations, which are in principle accessible by standard methods, but would have required a very large amount of time. ChatGPT was therefore used to accelerate this technical work: to check calculations and compare them with the arguments of Dwork, Langlands, and Lakkis, and detect inconsistencies in intermediate versions of the proof. The author checked the final mathematical arguments and assumes responsibility for the results.

Date: September 2026.

2020 Mathematics Subject Classification. 11S37, 11S40, 11R42.

Key words and phrases. local epsilon factors, Artin L -functions, local constants, Gauss sums, ramification, Lamprecht formula, biquadratic extensions, equal characteristic.

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1. INTRODUCTION

1.1. The Second Main Lemma. Langlands's Main Lemmas are identities between local constants associated to characters over different nonarchimedean local fields. They are used to define epsilon factors of local Weil representations by induction from one dimensional representations [3]. The Second Main Lemma compares characters of distinct intermediate fields that have the same norm pullback.

Let F be a nonarchimedean local field with residue characteristic p . Let K/F be a Galois extension with bicyclic Galois group $C_\ell \times C_\ell$, where ℓ is prime. For a degree- ℓ intermediate field L , put

$$S(L/F) = \{\nu \in \text{Hom}_{\text{cont}}(F^\times, \mathbf{C}^\times) : \nu \circ N_{L/F} = 1\}.$$

Here Hom_{cont} denotes continuous group homomorphisms. The elements of $S(L/F)$ are called norm characters. Fix a nontrivial continuous additive character ψ_F and set $\psi_L = \psi_F \circ \text{Tr}_{L/F}$.

Let $\Theta : K^\times \rightarrow \mathbf{C}^\times$ be a continuous quasi-character invariant under $\text{Gal}(K/F)$. For every degree- ℓ intermediate field L , choose a continuous quasi-character θ_L of $L^\times$ such that

$$\theta_L \circ N_{K/L} = \Theta.$$

We call (θ_L) a *compatible family*. It is called *primitive* if there is no continuous quasi-character θ_F of $F^\times$ with $\Theta = \theta_F \circ N_{K/F}$.

Theorem 1.1 (Second Main Lemma). *For every primitive compatible family (θ_L) , the quantity*

$$\mathcal{A}_L(\theta_L) = \Delta_L(\theta_L, \psi_L) \prod_{\nu \in S(L/F)} \Delta_F(\nu, \psi_F)$$

is independent of the degree- ℓ intermediate field L . This holds in mixed and equal characteristic, for every residue characteristic p and every prime ℓ .

Here $\Delta_E(\theta, \psi)$ is the local constant defined in (1); see [3] and [7].

The characters θ_L exist for every invariant Θ . Indeed, Lemma C.1 describes the kernel of $N_{K/L}$, and then by Galois-invariance Θ gives a character of $N_{K/L}(K^\times) \subset L^\times$. Norm filtration shows that this character is trivial on a sufficiently deep unit group of L , and Lemma C.2 extends it to $L^\times$. Conversely, if characters of two distinct degree- ℓ intermediate fields L_i have the same norm pullback, that pullback is invariant under both subgroups $\text{Gal}(K/L_i)$, which generate $\text{Gal}(K/F)$. Thus a prescribed compatible pair extends to a compatible family in the sense above, without changing either character.

The primitivity assumption implies that $\theta_L^\sigma / \theta_L$ is nontrivial for every $1 \neq \sigma \in \text{Gal}(L/F)$, hence generates $S(K/L)$. Indeed, if θ_L were $\text{Gal}(L/F)$ -invariant, Hilbert 90 would give $\theta_L = \lambda \circ N_{L/F}$ for some quasi-character λ of $F^\times$, and hence $\Theta = \lambda \circ N_{K/F}$. Conceptually, primitivity means that the corresponding 1-dimensional representation $\hat{\Theta}$ of W_K does not extend to a character of W_F ; consequently, for every degree- ℓ intermediate field L , any extension $\hat{\theta}_L$ of $\hat{\Theta}$ to W_L induces an irreducible representation of W_F .

1.2. Historical background and contribution of the present paper. Langlands attributed the First and Second Main Lemmas to Dwork in [3]. He later stated that Dwork had proved both lemmas, but that Dwork never published the proofs. Thus the existence of a complete proof by Dwork is known from Langlands's account, not from a surviving published proof of Dwork. We therefore do not claim that any particular calculation below was absent from Dwork's unpublished proof.

Lakkis gave a detailed local treatment of Dwork's calculations. For odd prime degree, his argument proves the equality required in the Second Main Lemma. In degree two, however, his calculation proves the desired equality only up to sign. Both Dwork and Lakkis worked in the mixed-characteristic setting.

Langlands formulated the Second Main Lemma in both mixed and equal characteristic. The surviving Chapter 12 of [3] states the lemma and proves preliminary conductor reductions. When one degree- ℓ intermediate field is unramified, the surviving Chapter 12 proves the conductor-one case and the case in which the character conductor is at least twice the conductor attached to

the ramified degree- ℓ extension. It then begins the remaining intermediate-conductor case, but the manuscript ends while constructing the stationary coefficient needed for that comparison. No proof of the totally ramified case survives. Thus the surviving manuscript does not contain the completion of the wild unramified–ramified case, the totally ramified odd-prime case, or the general quadratic sign calculation. Langlands later explained that, after Deligne found a shorter proof using global functional equations, he stopped preparing a complete version of the local argument.

The new point in the present paper is the exact treatment of the bi quadratic dyadic case. For a biquadratic extension, the First Main Lemma gives

$$\left(\frac{\mathcal{A}_{L_1}(\theta_{L_1})}{\mathcal{A}_{L_2}(\theta_{L_2})}\right)^2 = 1,$$

so it leaves a possible sign. Lakkis’s dyadic calculation leaves the same sign undetermined. We determine it by comparing the actual finite sums in Lamprecht’s formula. This proves

$$\mathcal{A}_{L_1}(\theta_{L_1}) = \mathcal{A}_{L_2}(\theta_{L_2})$$

in residue characteristic two for every ramification type and every character conductor. Together with the odd-prime and tame calculations, this gives the Second Main Lemma for all prime ℓ .

Finally, let us mention that the current paper also incorporates several minor simplifications suggested by the Lean formalization; see Remarks 4.9, 7.3, 7.13, 8.10, 11.3, and 12.14.

1.3. Outline of the proof. Local constants of conductor zero or one are evaluated directly. If the conductor is greater than one, Lamprecht’s formula expresses the local constant as the product of a character value, an additive character value, and a finite-sum factor; see Lemma 4.1. The proof compares these factors for two degree- ℓ intermediate fields.

Let I be the inertia subgroup of $\text{Gal}(K/F)$.

If $\ell \neq p$, the extension is tame. There is a unique unramified degree- ℓ intermediate field. Theorem 5.3 compares its normalized induction expression with that of every ramified degree- ℓ intermediate field. Hence $\mathcal{A}_L(\theta_L)$ is independent of L . This includes $\ell = 2$ when p is odd. The proof uses finite-field Gauss sums at conductor one and stable twisting at higher conductor.

Now suppose that $\ell = p > 2$. Corollary 4.3 gives

$$\left(\frac{\mathcal{A}_{L_1}(\theta_{L_1})}{\mathcal{A}_{L_2}(\theta_{L_2})}\right)^p = 1.$$

If $|I| = p$, the comparison is carried out in Section 6. If $|I| = p^2$, it is carried out in Sections 7–9. In both cases the calculations show that the same quotient is a fourth root of unity. Since p is odd, the quotient is 1. When $p = 3$, the additional identity required in the calculation is Lemma 6.13. The norm, trace, and stationary-phase identities used in these sections are stated in a form valid in both mixed and equal characteristic. In equal characteristic the terms containing the integer p are zero; in mixed characteristic their valuations are estimated.

It remains to consider $\ell = p = 2$. Proposition 4.4 gives

$$\frac{\mathcal{A}_{L_1}(\theta_{L_1})}{\mathcal{A}_{L_2}(\theta_{L_2})} \in \{1, -1\},$$

so equality of the squares does not determine the sign. The proof therefore compares the critical functions in Lamprecht’s formula. Lemmas 4.5–4.7 compare these functions when their stationary representatives are norms of the same element of K . If the induced norm map on the required unit quotient has image of index two, Lemma 4.8 compares the two finite sums; its hypotheses are proved using the third quadratic intermediate field.

If $|I| = 2$, this argument is carried out in Section 10. Suppose instead that $|I| = 4$, so that K/F is totally ramified. In characteristic two, Section 11 uses Artin–Schreier equations and local residues. In mixed characteristic put $e = v_F(2)$. The three lower breaks are either

$$(2a - 1, 2e, 2e), \quad 1 \leq a \leq e,$$

or

$$(2a - 1, 2r - 1, 2r - 1), \quad 1 \leq a \leq r \leq e.$$

These two cases are treated in Sections 12 and 13, respectively.

Thus the proof is divided as follows:

hypotheses	section
$\ell \neq p$	5
$\ell = p > 2, I = p$	6
$\ell = p > 2, I = p^2$	7–9
$p = \ell = 2, I = 2$	10
$p = \ell = 2, I = 4, \text{char } F = 2$	11
$p = \ell = 2, I = 4, (2a - 1, 2e, 2e)$	12
$p = \ell = 2, I = 4, (2a - 1, 2r - 1, 2r - 1)$	13.

Section 14 proves that these cases exhaust all $C_\ell \times C_\ell$ extensions and all conductors occurring in Theorem 1.1, and then proves the theorem. The appendices prove the additional finite-field, residue, descent, and ramification results used in the preceding sections.

1.4. Organization of the paper. Sections 2–4 collect the results used throughout the proof. Section 2 defines the local constants and recalls the First Main Lemma. Section 3 records the trace, norm, conductor, and ramification formulas used later. Section 4 gives the form of Lamprecht’s formula used in the proof and establishes the comparison lemmas for the resulting finite sums.

Section 5 proves the Second Main Lemma when $\ell \neq p$. The wild odd-prime case is treated in Sections 6–9. Section 6 treats the case in which the inertia subgroup has order p . For the totally ramified case, Section 7 proves the required Artin–Schreier, trace, and norm estimates, Section 8 constructs the compatible characters and their stationary coefficients, and Section 9 proves the resulting finite-sum identity.

The case $\ell = p = 2$ is treated in Sections 10–13. Section 10 treats extensions having an unramified quadratic intermediate field. Section 11 treats totally ramified extensions in characteristic two. Sections 12 and 13 treat the two possible ramification patterns for totally ramified biquadratic extensions of mixed characteristic.

Section 14 proves that the preceding cases exhaust all possibilities and deduces the Second Main Lemma. Appendix A proves the leading Gauss congruence used in the tame calculation, Appendix B proves the trace compatibility of local residues used in characteristic two, Appendix C proves the descent and elementary ramification results used in the main text, and Appendix D proves the norm-subgroup and ramification statements for $C_\ell \times C_\ell$ extensions.

2. LOCAL CONSTANTS AND THE FIRST MAIN LEMMA

2.1. Valuations, conductors, and local constants. For a local field E , write $\mathcal{O}_E$ for its ring of integers, $\mathfrak{p}_E$ for its maximal ideal, k_E for its residue field, and $v_E(\mathfrak{p}_E) = 1$. Put $U_E^0 = \mathcal{O}_E^\times$ and $U_E^j = 1 + \mathfrak{p}_E^j$ for $j \geq 1$. The multiplicative conductor $m_E(\theta)$, also denoted $a_E(\theta)$ in the later case calculations, is the least $m \geq 0$ for which $\theta|_{U_E^m} = 1$. The additive conductor $n_E(\psi)$ is specified by $\mathfrak{p}_E^{-n_E(\psi)}$, the largest ideal on which ψ is trivial. It is not generally the full kernel of ψ .

We use the local constant of [7]. For a nonzero complex number z put $\text{ph}(z) = z/|z|$. If $m = m_E(\theta) > 0$, an element $\Gamma \in E^\times$ is called *admissible* if $v_E \Gamma = m + n_E(\psi)$. Choose such a Γ and set

$$(1) \quad \Delta_E(\theta, \psi) = \theta(\Gamma) \text{ph} \left(\int_{\mathcal{O}_E^\times} \psi(u/\Gamma) \theta(u)^{-1} du \right).$$

Here du is a positive Haar measure on $\mathcal{O}_E^\times$. The integral is nonzero and is a positive multiple of a finite Gauss sum. The value of (1) is independent of Γ ; see [7, Proposition 2.4]. For $m = 0$ this convention gives $\Delta_E(\theta, \psi) = \theta(\pi_E)^{n_E(\psi)}$. Multiplying the Haar measure by a positive constant does not change the phase. No unitarity assumption is imposed on θ . In particular, the factor $\theta(\Gamma)$ in (1) is not replaced by its phase.

For a totally ramified extension E/F of degree ℓ , our valuation conventions give

$$v_E|_F = \ell v_F, \quad v_F(N_{E/F}x) = v_E(x).$$

The different exponent $D_{E/F}$ is measured in E . Conjugation of a character means $\theta^\sigma(x) = \theta(\sigma^{-1}x)$. In a tower $A \subset B_i \subset B$, we write $N_i = N_{B/B_i}$ and $n_i = N_{B_i/A}$. Trace notation will be specified in each section.

2.2. The First Main Lemma and elementary identities. For a cyclic extension E/F of prime degree and a quasi-character χ of $F^\times$, the First Main Lemma is

$$(2) \quad \Delta_E(\chi \circ N_{E/F}, \psi_F \circ \text{Tr}_{E/F}) \prod_{\nu \in S(E/F)} \Delta_F(\nu, \psi_F) = \prod_{\nu \in S(E/F)} \Delta_F(\chi \nu, \psi_F).$$

We also use

$$(3) \quad \Delta_E(1, \psi) = 1,$$

$$(4) \quad \Delta_E(\theta, \psi(c \cdot)) = \theta(c) \Delta_E(\theta, \psi) \quad (c \in E^\times),$$

and, for a finite-order character μ ,

$$(5) \quad \Delta_E(\mu, \psi) \Delta_E(\mu^{-1}, \psi) = \mu(-1).$$

These are [7, Theorem 1.1 and Lemmas 2.8, 2.10–2.11]. A change of variables in (1) gives

$$(6) \quad \Delta_E(\theta^\sigma, \psi_E) = \Delta_E(\theta, \psi_E) \quad \text{if } \psi_E \circ \sigma = \psi_E.$$

3. RAMIFICATION AND NORM CHARACTERS

3.1. Trace ideals, norm filtration, and conductors. For a finite separable extension E/F of ramification index $e_{E/F}$,

$$(7) \quad \text{Tr}_{E/F}(\mathfrak{p}_E^j) = \mathfrak{p}_F^{\lfloor (j+D_{E/F})/e_{E/F} \rfloor}, \quad n_E(\psi_F \text{Tr}) = e_{E/F} n_F(\psi_F) + D_{E/F}.$$

Here $j \in \mathbf{Z}$. The definition of the inverse different gives

$$(8) \quad \{x \in E : \text{Tr}_{E/F}(x \mathfrak{p}_E^r) \subset \mathcal{O}_F\} = \mathfrak{p}_E^{-r-D_{E/F}} \quad (r \in \mathbf{Z}).$$

To prove the first equality in (7), write $\text{Tr}_{E/F}(\mathfrak{p}_E^j) = \mathfrak{p}_F^c$. The inverse different is the trace-dual of $\mathcal{O}_E$. For $y \in F$, the condition $y \text{Tr}(\mathfrak{p}_E^j) \subset \mathcal{O}_F$ is therefore equivalent to $y \mathfrak{p}_E^j \subset \mathfrak{p}_E^{-D_{E/F}}$, or $e_{E/F} v_F(y) + j \geq -D_{E/F}$. Comparing the two principal dual ideals proves the first equality in (7). The conductor formula follows by applying it to the largest trivial ideal of ψ_F . The cyclic-prime instances used most often below are [7, Theorem 3.7 and Corollary 3.8]. For a totally ramified cyclic extension E/F of prime degree ℓ and ramification break t , $D_{E/F} = (\ell - 1)(t + 1)$. Write $\psi_{E/F}$ for the inverse Herbrand function; on nonnegative real arguments it is s up to t , and $t + \ell(s - t)$ above t . By [7, Theorem 3.7], the induced graded norm is bijective except at depth t , where its image has index ℓ . For integers $r > t$,

$$(9) \quad N_{E/F}(U_E^{\psi_{E/F}(r)}) = U_F^r.$$

For an unramified extension the same equality holds for $r \geq 1$, with $\psi_{E/F}(r) = r$. Every nontrivial norm character of the totally ramified extension has conductor $t + 1$.

For a character χ of $F^\times$ of conductor m , write $m' = m_E(\chi \circ N_{E/F})$. Then

$$(10) \quad m' = \begin{cases} 1 + \psi_{E/F}(m - 1), & m > t + 1, \\ m, & m \leq t. \end{cases}$$

At $m = t + 1$ it is at most m , with a drop precisely when twisting by a norm character lowers the conductor of χ . These are [7, Corollary 3.9 and Proposition 3.10]. Unramified norm pullback preserves conductor.

We also use the following norm approximation: for $A \in F^\times$ and $0 \leq j \leq t$ there is $x \in E^\times$ with $v_E x = v_F A$ and $N_{E/F} x / A \in U_F^j$. This is [7, Lemma 6.6]. The restriction $j \leq t$ is essential

in its applications below. For a wild cyclic extension E/F of degree p , let $E_i(x)$ denote the i th elementary symmetric function of the conjugates of x . For $1 \leq i < p$, one has

$$(11) \quad v_F(E_i(x)) \geq \left\lfloor \frac{iv_E(x) + (p-1)(t+1)}{p} \right\rfloor.$$

This holds also for $v_E(x) < 0$; see [7, Lemma 3.4]. The stationary-parameter results of [7, Proposition 6.8] are used in Sections 6–9. The Hasse–Davenport identities are recalled in Section 5.

3.2. Additional classical local results. We record the additional local results and the locations of their proofs.

Hilbert 90 and extension of characters. For a finite cyclic extension with generator σ , the multiplicative norm kernel consists of $y/\sigma(y)$ (equivalently $\sigma(y)/y$), and the additive trace kernel consists of $\sigma(y) - y$. Both assertions are proved in Lemma C.1. Lemma C.2 extends a character from a subgroup of $E^\times$ containing a unit group on which it is trivial. Its applications require the prescribed character formulas to agree on the intersections of their domains.

Norm characters in $C_\ell \times C_\ell$ extensions. The order and conductor of a cyclic-prime norm quotient are supplied by [7, Corollary 3.9]. Proposition D.4 proves that the subgroups $N_{L_i/F}(L_i^\times)$ are distinct for the pairs of intermediate fields used below. Lemma D.5 then identifies $S(K/L_i)$ with the pullback of $S(L_j/F)$ by $N_{L_i/F}$, for $i \neq j$. In the biquadratic case, Lemma D.6 applies to all three pairs and proves $\omega_1\omega_2 = \omega_3$. These statements also follow from local reciprocity; see [4, Chapters XIII–XIV]. Our proofs use the cyclic-prime norm results of [7].

Ramification and differentials. The cyclic-prime trace, different, norm, and conductor formulas are quoted from [7]. Lemma D.2 proves the cyclic unramified quotient, the tame inertia and Frobenius action, and the subgroup rule for lower numbering. Lemma D.3 derives the ramification breaks of the intermediate extensions from the different formula. These results are special cases of the theory in [4, Chapters III–V]. Lemma C.4 proves transitivity of the different, Lemma C.5 gives the order-discriminant bound, and Lemma C.3 gives the first wild ramification term, including at negative valuations. Lemma D.1 gives the Newton approximation used to choose roots. We also use the monogenic different formula, elementary Galois theory, and finite-field theory. Elementary cyclic coordinates are recalled at the start of Appendix D; the equal-characteristic coefficient field is constructed in Theorem B.4.

Local residues in equal characteristic. For a finite separable extension of Laurent-series fields, we use

$$(12) \quad \mathrm{Tr}_{k_E/k_F} \mathrm{Res}_E(\eta) = \mathrm{Res}_F(\mathrm{Tr}_{E/F} \eta).$$

Here the trace of $x d\varpi_F$ is $\mathrm{Tr}_{E/F}(x) d\varpi_F$. Equation (12) is proved in Theorem B.4; see also [6]. Lemma B.5 proves $d \log N(u) = \mathrm{Tr}(d \log u)$. These identities and the Cartier operator give the Artin–Schreier norm-character formula in Lemma 11.2.

Finite sums. Ordinary finite-field Hasse–Davenport is taken from [7]. The Gauss congruence used in Section 5 is Theorem A.3, proved by Jacobi sums and induction on the sum of the base- p digits. We also use character orthogonality for finite abelian groups.

4. STATIONARY REPRESENTATIVES AND LAMPRECHT’S FORMULA

4.1. Stationary representatives and a change of additive character. Let E be a local field, let θ be a continuous quasi-character of $E^\times$, and let ψ_E be a nontrivial continuous additive character of E . Suppose $m = m_E(\theta) > 1$, let Γ be admissible for (θ, ψ_E) , and let $b \in \mathcal{O}_E^\times$ satisfy $\theta(1+z) = \psi_E(bz/\Gamma)$ for $z \in \mathfrak{p}_E^{[m/2]}$. We call b a *stationary representative*. It represents a class modulo $\mathfrak{p}_E^{[m/2]}$. The coefficient b/Γ of z will also be used.

If $\Psi_E = \psi_E(\alpha \cdot)$, the formula $\theta(1-z) = \Psi_E(Cz)$ corresponds to $b/\Gamma = -\alpha C$. The character and additive factors in Lamprecht’s formula are then

$$\theta((-\alpha C)^{-1}) \Psi_E(-C).$$

Thus the sign and the factor α are determined by the choice of $1 - z$ and Ψ_E . We use this convention in Lemma 4.1.

4.2. Lamprecht's formula and stable twists. The following form of Lamprecht's formula is [7, Theorem 4.5 and Lemma 4.6], with the coefficient written as $-\alpha Z$. For the stationary representative b and admissible Γ above, [7, Lemma 4.7] also gives

$$(13) \quad \Delta_E(\theta\nu, \psi_E) = \nu(\Gamma/b)\Delta_E(\theta, \psi_E) \quad (m_E(\nu) \leq \lfloor m_E(\theta)/2 \rfloor).$$

Lemma 4.1 (Lamprecht's formula for $\Psi_E(x) = \psi_E(\alpha x)$). *Let E be a local field, $\alpha \in E^\times$, and $\Psi_E(x) = \psi_E(\alpha x)$ with largest trivial ideal $\mathfrak{p}_E^J$. Write the conductor of θ as $m = 2d + \varepsilon > 1$, with $\varepsilon \in \{0, 1\}$. Suppose $\theta(1 - z) = \Psi_E(Zz)$ for $z \in \mathfrak{p}_E^{d+\varepsilon}$, with $v_E Z = J - m$. Then*

$$(14) \quad \Delta_E(\theta, \psi_E) = \theta((- \alpha Z)^{-1})\Psi_E(-Z)g_E.$$

For even m , $g_E = 1$. For odd m , choose $v_E \delta = d$ and set

$$(15) \quad H_E(\bar{x}) = \Psi_E(-Z\delta x)\theta(1 + \delta x)^{-1}, \quad g_E = |k_E|^{-1/2} \sum_{x \in k_E} H_E(x).$$

The function H_E is well defined, $H_E(0) = 1$, and $|H_E(x)| = 1$. Its polar pairing is

$$(16) \quad \frac{H_E(x + y)}{H_E(x)H_E(y)} = \Psi_E(-Z\delta^2 \widetilde{xy}) \quad (x, y \in k_E).$$

This pairing is nondegenerate, and

$$(17) \quad \left| \sum_{x \in k_E} H_E(x) \right| = |k_E|^{1/2}, \quad |g_E| = 1.$$

More generally, $\Psi_E(-Zv)\theta(1 + v)^{-1}$ is constant as v varies modulo $\mathfrak{p}_E^{d+1}$ in $\mathfrak{p}_E^d$ when m is odd; it is identically one on $\mathfrak{p}_E^d$ when m is even.

Proof. Apply [7, Theorem 4.5 and Lemma 4.6] with $b/\Gamma = -\alpha Z$. In odd conductor, $v_E(Z\delta^2) = J - 1$, so the character induced by $z \mapsto \Psi_E(-Z\delta^2 z)$ on k_E is nontrivial. Thus (16) is nondegenerate, and character orthogonality gives (17). For the last assertion, suppose $v' - v \in \mathfrak{p}_E^{d+1}$ and write $1 + v' = (1 + v)(1 + w)$ with $w \in \mathfrak{p}_E^{d+1}$. The quotient of the two expressions is $\Psi_E(-Zvw) = 1$, since $v_E(Zvw) \geq J$; the linear part cancels by stationarity. In even conductor stationarity holds already on $\mathfrak{p}_E^d$. $\square$

4.3. Character identities and comparison of critical sums. The following results will be used in several cases of the proof. The pointwise identity in Lemma 4.7 uses only compatibility and triviality of norm characters on norms. Its final interpretation as an identity of critical functions, and Lemma 4.6, use stationary coefficients $N_{K/L_i}C$ and $N_{L_i/F}(\text{Tr}_{K/L_i} C)$. The required elements C are constructed in Sections 10–13.

Lemma 4.2 (Conjugate characters and restriction to $F^\times$). *Let ℓ be prime, let K/F be finite Galois with group C_ℓ^2 , and let L_1, L_2 be distinct degree- ℓ intermediate fields with distinct subgroups $N_{L_i/F}(L_i^\times)$ of $F^\times$. Put $N_i = N_{K/L_i}$, $n_i = N_{L_i/F}$ and $\epsilon_i = \prod_{\omega \in S(L_i/F)} \omega$. Let $\theta_i N_i = \Theta$ be primitive compatible characters. For a generator σ_i of $\text{Gal}(L_i/F)$, the quotient $\mu_i = \theta_i^{\sigma_i}/\theta_i$ generates $S(K/L_i)$. Every character in that group is fixed by $\text{Gal}(L_i/F)$, and the conjugates of θ_i are precisely its twists by $S(K/L_i)$. Moreover*

$$(18) \quad D = D_\theta = \theta_i|_{F^\times \epsilon_i}$$

is independent of i . For odd ℓ , $\epsilon_i = 1$; for $\ell = 2$, $\epsilon_i = \omega_i$ is the nontrivial lower norm character. The required distinctness follows from Proposition D.4 for the pairs specified there, and from Lemma D.6 for all biquadratic pairs.

Proof. The character Θ is invariant under both $\text{Gal}(K/L_i)$, which generate $\text{Gal}(K/F)$. Thus μ_i is trivial on $N_i(K^\times)$. If $\mu_i = 1$, Lemmas C.1 and C.2 would give a character λ of $F^\times$ with $\theta_i = \lambda \circ n_i$. This would imply $\Theta = \lambda \circ N_{K/F}$, contrary to primitivity. Since $L_i^\times/N_i(K^\times)$ has prime order, μ_i generates $S(K/L_i)$. Lemma D.5 identifies that group with $S(L_j/F) \circ n_i$; it is therefore invariant under σ_i . Iterated conjugation gives all ℓ twists.

For $x \in L_j^\times$, compatibility gives $\theta_i(n_j x) = \theta_j(x)^\ell$, whereas multiplication over the conjugates gives

$$\theta_j(n_j x) = \theta_j(x)^\ell \mu_j(x)^{-\ell(\ell-1)/2}.$$

For odd ℓ the last factor is one, and the characters in each $S(L_i/F)$ pair with their inverses, so $\epsilon_i = 1$. For $\ell = 2$, that factor is $\mu_j(x) = \omega_i(n_j x)$ and $\omega_j(n_j x) = 1$. In either case $\theta_i|_{F^\times \epsilon_i}$ and $\theta_j|_{F^\times \epsilon_j}$ agree on $n_j L_j^\times$, and by symmetry on $n_i L_i^\times$. These distinct index- ℓ subgroups generate $F^\times$, proving the result. $\square$

Corollary 4.3 (Odd-degree powers). *In Lemma 4.2, suppose ℓ is odd and use trace-compatible additive characters. Then*

$$\Delta_{L_i}(\theta_i, \psi_{L_i})^\ell = \Delta_K(\Theta, \psi_K), \quad \prod_{\omega \in S(L_i/F)} \Delta_F(\omega, \psi_F) = 1.$$

In particular, $(\mathcal{A}_{L_1}/\mathcal{A}_{L_2})^\ell = 1$.

Proof. For an odd-prime cyclic extension, (5) gives $\Delta(\omega)\Delta(\omega^{-1}) = \omega(-1) = 1$; use (3) for the trivial character. Apply (2) to K/L_i . By Lemma 4.2, its twists are conjugate, and their local constants agree by (6). This proves the power relation as well as both product claims. $\square$

Proposition 4.4 (Squares in the biquadratic case). *For a primitive compatible biquadratic family with trace-compatible additive characters, let $\mathcal{A}_i = \Delta_{L_i}(\theta_i)\Delta_F(\omega_i)$. Then*

$$\mathcal{A}_i^2 = \Delta_K(\Theta) \prod_{j=1}^3 \Delta_F(\omega_j).$$

Thus the ratio of two such expressions belongs to $\{1, -1\}$.

Proof. By Lemma 4.2 and (2) for K/L_i , $\Delta_{L_i}(\theta_i)^2 = \Delta_K(\Theta)\Delta_{L_i}(\omega_j n_i)$ for $j \neq i$. Equation (2) for L_i/F , applied to ω_j , gives $\Delta_{L_i}(\omega_j n_i)\Delta_F(\omega_i) = \Delta_F(\omega_j)\Delta_F(\omega_k)$, where $\{i, j, k\} = \{1, 2, 3\}$. Use $\omega_i \omega_j = \omega_k$ from Lemma D.6 and multiply the two equalities. $\square$

Lemma 4.5 (The second elementary symmetric function). *For a biquadratic extension K/F , $C \in K$, and any quadratic intermediate field L_i ,*

$$E_2(C) = \text{Tr}_{L_i/F}(N_i C) + n_i(S_i C), \quad S_i = \text{Tr}_{K/L_i},$$

where $E_2(C)$ is the second elementary symmetric function of the four conjugates of C .

Proof. Write the four conjugates, paired by the involution fixing L_i , as c_1, c_2 and c_3, c_4 . The two terms on the right are $c_1 c_2 + c_3 c_4$ and $(c_1 + c_2)(c_3 + c_4)$. Adding gives the six terms defining $E_2(C)$. $\square$

Lemma 4.6 (Norms as stationary coefficients). *Use the biquadratic setting of Lemma 4.2. Fix $\alpha \in F^\times$ and put $\Psi_E = \psi_E(\alpha \cdot)$. Put $m_i = m_{L_i}(\theta_i)$ and $T_i = m_F(\omega_i)$, and assume $m_i, T_i > 1$. Let $C \in K^\times$ satisfy $S_i C \neq 0$, and suppose that*

$$Z_i = N_i C, \quad W_i = n_i(S_i C)$$

satisfy $\theta_i(1 - z) = \Psi_{L_i}(Z_i z)$ for $z \in \mathfrak{p}_{L_i}^{[m_i/2]}$ and $\omega_i(1 - z) = \Psi_F(W_i z)$ for $z \in \mathfrak{p}_F^{[T_i/2]}$. Let g_i and h_i be the corresponding factors defined in (15), with value one in even conductor. Then

$$\mathcal{A}_i(\theta_i) = D(-\alpha^{-1})\Theta(C)^{-1}\Psi_F(-E_2(C))g_i h_i.$$

Consequently $\mathcal{A}_i(\theta_i)$ is independent of i if $g_i h_i$ is independent of i .

Proof. By (14), the two local constants are

$$\theta_i(-\alpha^{-1})\theta_i(Z_i)^{-1}\Psi_{L_i}(-Z_i)g_i, \quad \omega_i(-\alpha^{-1})\omega_i(W_i)^{-1}\Psi_F(-W_i)h_i.$$

Compatibility gives $\theta_i(Z_i) = \Theta(C)$. Since $W_i = n_i(S_iC)$, one has $\omega_i(W_i) = 1$. Multiply, apply Lemma 4.2, and use Lemma 4.5. $\square$

Lemma 4.7 (Products of critical functions). *Let K/F be biquadratic, with quadratic intermediate fields L_i , and put*

$$N_i = N_{K/L_i}, \quad S_i = \text{Tr}_{K/L_i}, \quad n_i = N_{L_i/F}.$$

Let $\theta_i N_i = \Theta$ be compatible characters, and let ω_i be the nontrivial norm character of L_i/F . Fix a nontrivial additive character ψ_F , put $\psi_{L_i} = \psi_F \text{Tr}_{L_i/F}$, fix $\alpha \in F^\times$, and write $\Psi_E(x) = \psi_E(\alpha x)$. Let $C, C' \in K^\times$ satisfy $S_i C, S_i C' \neq 0$ for every i , and put

$$\begin{aligned} Z_i &= N_i C, & W_i &= n_i(S_i C), \\ 1 + z_i &= N_i C' / N_i C, & 1 + w_i &= n_i(S_i C') / n_i(S_i C). \end{aligned}$$

Then the products

$$\Psi_{L_i}(-Z_i z_i) \theta_i(1 + z_i)^{-1} \Psi_F(-W_i w_i) \omega_i(1 + w_i)^{-1}$$

are independent of i . Their common value is

$$\Theta(C'/C)^{-1} \Psi_F(-E_2(C') + E_2(C)).$$

If, in addition, $m_i = m_{L_i}(\theta_i) > 1$, $T_i = m_F(\omega_i) > 1$, Z_i, W_i are stationary coefficients for θ_i, ω_i , and

$$z_i \in \mathfrak{p}_{L_i}^{\lfloor m_i/2 \rfloor}, \quad w_i \in \mathfrak{p}_F^{\lfloor T_i/2 \rfloor},$$

then Lemma 4.1 identifies these products with the products of the corresponding critical functions.

Proof. Compatibility gives $\theta_i(1 + z_i) = \Theta(C'/C)$. Since $1 + w_i = n_i(S_i C' / S_i C)$, one has $\omega_i(1 + w_i) = 1$. The additive exponent is

$$-\text{Tr}_{L_i/F}(N_i C' - N_i C) - n_i(S_i C') + n_i(S_i C),$$

which is $-E_2(C') + E_2(C)$ by Lemma 4.5. This proves the first assertion without any stationary or primitivity hypothesis. The last assertion follows from Lemma 4.1. $\square$

Lemma 4.8 (Cancellation from a common pullback). *Let I be an index set with at least two elements. For $i \in I$, let V_i be finite abelian groups of the same order, and let $H_i : V_i \rightarrow \mathbf{C}^\times$ satisfy $H_i(0) = 1$. Suppose*

$$B_i(x, y) = \frac{H_i(x + y)}{H_i(x)H_i(y)}$$

is a bicharacter and that $B_i(w, \cdot)$ is nontrivial for every $0 \neq w \in V_i$. Let Z be a finite abelian group and let $p_i : Z \rightarrow V_i$ be group homomorphisms whose kernels have order two and whose images I_i have index two. Suppose

$$H_i \circ p_i = Q$$

for the same function Q on Z , and suppose the kernels of the p_i are pairwise distinct. Then, for every i ,

$$\sum_{x \in V_i} H_i(x) = \frac{1}{2} \sum_{z \in Z} Q(z).$$

In particular the sums divided by $|V_i|^{1/2}$ are equal.

Proof. Fix i and choose $j \neq i$. Let a be the nonzero element of $\ker p_j$. Since the kernels are distinct and both have order two, $a \notin \ker p_i$. Put $w_i = p_i(a)$. Then $w_i \neq 0$ and

$$H_i(w_i) = Q(a) = H_j(0) = 1.$$

If $x = p_i(z) \in I_i$, then

$$H_i(w_i + x) = Q(a + z) = H_j(p_j(a + z)) = H_j(p_j z) = Q(z) = H_i(x).$$

Hence $B_i(w_i, x) = 1$ for every $x \in I_i$. Since $w_i \neq 0$ and the pairing is nondegenerate, $B_i(w_i, \cdot)$ is nontrivial on V_i . It is therefore the nontrivial character of V_i/I_i .

Translation by w_i preserves $V_i \setminus I_i$ and changes the sign of every summand there, so the sum over that coset is zero. On I_i , every element has exactly two preimages under p_i . Summing $H_i p_i = Q$ gives the stated identity. $\square$

Remark 4.9. The Lean formalization suggested this shorter argument. The original proofs of Propositions 11.10 and 13.12 identify the character on the missing coset by explicit coefficient calculations. The common pullback, the distinct kernels, and nondegeneracy of the polar pairing already imply the required cancellation, so those coefficient calculations are not needed for either proposition.

Lemma 4.10 (A norm in a prescribed multiplicative coset). *Let L/F be cyclic of prime degree and let ω be a nontrivial character with kernel $N_{L/F}(L^\times)$. Let $H \subset F^\times$ be a subgroup such that $\omega(H) = \omega(F^\times)$. For $A_* \in F^\times$, there exists $u \in H$ with $A_*/u \in N_{L/F}(L^\times)$. If $\alpha \in F^\times$, then $(\alpha u)(A_*/u) = \alpha A_*$.*

Proof. Choose $u \in H$ with $\omega(u) = \omega(A_*)$. Then $\omega(A_*/u) = 1$, so A_*/u is a norm. The final identity is immediate. $\square$

The applications of Lemma 4.10 replace α by αu and A_* by A_*/u . The effect on other character formulas must be checked separately; see Lemmas 8.18, 12.8, and 13.8.

5. FINITE-FIELD IDENTITIES AND THE TAME COMPARISON

5.1. Gauss sums and the leading congruence. For a finite field k , a nontrivial additive character ψ_k , and a multiplicative character χ , put

$$\tau_k(\chi) = - \sum_{x \in k^\times} \chi(x)^{-1} \psi_k(x).$$

Thus $\tau_k(1) = 1$. For a finite extension κ/k all additive characters are composed with trace. The Hasse–Davenport identities proved in [7] are

$$(19) \quad \begin{aligned} \tau_\kappa(\chi \circ N) &= \tau_k(\chi)^{[\kappa:k]}, \\ \chi(s^s) \tau_k(\chi^s) \prod_{j=1}^{s-1} \tau_k(\eta^j) &= \prod_{j=0}^{s-1} \tau_k(\chi \eta^j), \end{aligned}$$

where η has order s . The second formula is used only when s is prime to the characteristic, as required for such a character.

We use Theorem A.3 in the following notation. Fix an embedding of the cyclotomic values into $\mathbf{Q}_p(\mu_{p^f-1}, \zeta_p)$, put $\varpi = \zeta_p - 1$, and let ω be the Teichmüller character of $\mathbf{F}_{p^f}^\times$. For the additive character $x \mapsto \zeta_p^{\text{Tr}(x)}$ and $0 < a < p^f - 1$, write $a = \sum a_i p^i$. Then

$$(20) \quad \tau_{\mathbf{F}_{p^f}}(\omega^a) \equiv_{\times} \frac{\varpi^{\sum a_i}}{\prod a_i!}.$$

Here $X \equiv_{\times} Y$ means $X/Y \in 1 + \mathfrak{m}$ in this local cyclotomic field. The sign in (20) is positive because our τ is minus the Gauss sum. Equivalently, in the usual plus-sum convention $G(\omega^{-a}) = \sum_{x \neq 0} \omega^{-a}(x) \zeta_p^{\text{Tr}(x)}$, the leading term is $-\varpi^{\sum a_i} / \prod a_i!$; our definition is $\tau(\omega^a) = -G(\omega^{-a})$. The formula also applies at $p = 2$, with $\varpi = \zeta_2 - 1 = -2$.

For later use, Wilson’s theorem gives, for every nonnegative integer n with base- p digits n_i ,

$$n! \equiv_{\times} (-p)^{(n - \sum n_i)/(p-1)} \prod n_i!.$$

Indeed separate the multiples of p , reduce each full nonzero residue block to $(p-1)! \equiv -1$, and repeat for $\lfloor n/p \rfloor!$. Since $\varpi^{p-1} \equiv_{\times} -p$, we may group (20) into base- q digits for any power q of p :

$$(21) \quad a_* = \sum_{i=0}^{r-1} \gamma_i q^i \implies \tau_{\mathbf{F}_{q^r}}(\omega^{a_*}) \equiv_{\times} \frac{\varpi^{\sum \gamma_i}}{\prod \gamma_i!}.$$

5.2. The primitive finite-field identity. Let ℓ be prime, $\ell \mid q - 1$, and let κ/k have degree ℓ , where $|k| = q$. Suppose

$$(22) \quad \chi_\kappa^\ell = \chi_0 \circ N_{\kappa/k}, \quad \chi_0|_{\mu_\ell} \neq 1.$$

Let μ be a character of $k^\times$ of order ℓ .

Theorem 5.1. *With the trace-compatible additive characters,*

$$(23) \quad \tau_\kappa(\chi_\kappa) = \chi_0(\ell) \tau_k(\chi_0) \prod_{j=1}^{\ell-1} \tau_k(\mu^j).$$

Proof. First take the canonical additive characters. Put $S = (q^\ell - 1)/(q - 1)$ and $h = (q - 1)/\ell$. Write $\chi_0 = \omega_k^a$, $1 \leq a \leq q - 2$, $\ell \nmid a$. The exponent of χ_κ is aS/ℓ modulo $(q^\ell - 1)/\ell$. Multiplication by q changes it by $a(q^\ell - 1)/\ell$; hence all ℓ possibilities are Frobenius conjugates. Their Gauss sums are equal, so we may take the exponent exactly $a_* = aS/\ell$.

Set

$$L = \frac{\tau_\kappa(\chi_\kappa)}{\chi_0(\ell) \tau_k(\chi_0) \prod_{j=1}^{\ell-1} \tau_k(\mu^j)}.$$

The order- ℓ characters over κ are the norm lifts of μ^j . Their twists of χ_κ are the Frobenius conjugates just considered. The product and lifting formulas (19), and $\chi_\kappa(\ell^\ell) = \chi_0(\ell)^\ell$, give

$$(24) \quad L^\ell = 1.$$

To prove $L = 1$, we use (21). Write $a = \ell m + k_0$, $1 \leq k_0 < \ell$. The unordered base- q digits of aS/ℓ are

$$(25) \quad \gamma_j = m + jh + \mathbf{1}_{j \geq \ell - k_0}, \quad 0 \leq j \leq \ell - 1.$$

Here “unordered” is sufficient for their sum and factorial product. For completeness, multiplication of $k_0 S/\ell$ by ℓ , using $q = \ell h + 1$, gives successive residues $i_j = ((\ell - 1 - j)k_0 \bmod \ell)$ and carries $c_j = \lfloor (\ell - 1 - j)k_0/\ell \rfloor$. The zeroth digit has the additional $k_0 - c_0 = 1$; at a later digit the additional $c_{j-1} - c_j$ is one exactly when $i_j \geq \ell - k_0$. The i_j permute $0, \dots, \ell - 1$, proving (25). All digits lie between zero and $q - 1$, since $m \leq h - 1$. In particular

$$(26) \quad \sum \gamma_j = a + \sum_{j=1}^{\ell-1} jh.$$

We claim the multiplicative factorial congruence

$$(27) \quad \ell^a \prod_{j=0}^{\ell-1} \gamma_j! \equiv_\times a! \prod_{j=1}^{\ell-1} (jh)!.$$

In fact

$$\frac{\prod \gamma_j!}{\prod (jh)!} = \prod_{j=0}^{\ell-1} \prod_{n=1}^m (n + jh) \prod_{j=\ell-k_0}^{\ell-1} (m + 1 + jh).$$

For the first product, $\ell(n + jh) = \ell n - j + jq \equiv_\times \ell n - j$: the positive integer $\ell n - j$ is less than $q = p^f$, so its p -valuation is less than f . The numbers $\ell n - j$ exhaust $1, \dots, \ell m$. For the second product the same argument applies to $\ell(m + 1) - j$, which exhausts $\ell m + 1, \dots, a$, again all less than q . Multiplication proves (27), even when some of its factorials are divisible by p .

Apply (21) to the numerator and each factor of the denominator of L . The Teichmüller value $\chi_0(\ell)$ reduces to ℓ^a . Equations (26) and (27) give $L \equiv_\times 1$. Reduction is injective on the group of ℓ th roots of unity because $\ell \neq p$. Together with (24), this proves $L = 1$.

For an arbitrary additive character of k , change the canonical one by multiplication by $c \in k^\times$. The two sides acquire factors $\chi_\kappa(c)$ and $\chi_0(c) \prod_{j=1}^{\ell-1} \mu^j(c)$. These agree. Indeed the restriction exponent of χ_κ is aS/ℓ modulo $q - 1$: it is a for odd ℓ , and $a + (q - 1)/2$ for $\ell = 2$ because a is odd. This is exactly the exponent of $\chi_0 \prod \mu^j$. This proves (23) for every nontrivial additive character. $\square$

5.3. The tamely ramified extension. Let F have residue characteristic p , let K/F have Galois group C_ℓ^2 with $\ell \neq p$, and let U/F be its unique unramified degree- ℓ subextension. Let E be any other degree- ℓ intermediate field. Its tame cyclic inertia gives $\ell \mid |k_F| - 1$; K/E is unramified and K/U is totally ramified. Suppose $\theta_U N_{K/U} = \theta_E N_{K/E}$ is invariant and does not descend from F . With $S(L/F)$ as defined in Section 1, put

$$\epsilon_L = \prod_{\nu \in S(L/F)} \nu, \quad \Lambda_L = \prod_{\nu \in S(L/F)} \Delta_F(\nu, \psi_F).$$

For odd ℓ , $\epsilon_L = 1$; for $\ell = 2$ it is the nontrivial quadratic norm character. All additive characters are trace compatible.

5.4. Conductors greater than one. Put $m = m_U(\theta_U)$. Its conjugate quotient has conductor one, so $m \geq 1$. If $m > 1$, unramified Hilbert 90 on U_U^1 , followed by norm surjectivity, gives a character λ of $F^\times$ such that

$$\theta_U = \chi_U(\lambda n_U), \quad \theta_E = \chi_E(\lambda n_E), \quad m(\chi_U) = m(\chi_E) = 1, \quad m(\lambda) = m.$$

Here $n_L = N_{L/F}$. To justify the conductor-one assertion, remove the descended restriction on U_U^1 . The nontrivial conjugate quotient forces χ_U to have conductor exactly one; its twists by $S(K/U)$ are its conjugates. The conductor formula of [7, Proposition 3.10] gives conductor one for its pullback to K ; unramified pullback from E preserves conductor.

Choose a stationary representative β for λ and put $g = \beta/\Gamma$, where $v_F(\Gamma) = m + n_F(\psi_F)$. The following local calculation shows that both norm pullbacks have stationary coefficient g . Their conductors are m over U and $M = 1 + \ell(m - 1)$ over E , and $n_E(\psi_E) = \ell n_F(\psi_F) + \ell - 1$. Hence the same $\Gamma \in F$ is admissible in both.

Lemma 5.2 (A norm-pullback calculation). *Let L/F be cyclic of prime degree ℓ , let $m > 1$, and let λ have conductor m . Suppose $g \in F^\times$ satisfies*

$$\lambda(1 + z) = \psi_F(gz) \quad (z \in \mathfrak{p}_F^{\lceil m/2 \rceil}),$$

and put $\psi_L = \psi_F \text{Tr}_{L/F}$. If L/F is unramified, then

$$(\lambda \circ N_{L/F})(1 + x) = \psi_L(gx) \quad (x \in \mathfrak{p}_L^{\lceil m/2 \rceil}).$$

If L/F is totally ramified, then the same identity holds for

$$x \in \mathfrak{p}_L^s, \quad s = \left\lceil \frac{1 + \ell(m - 1)}{2} \right\rceil.$$

Proof. First suppose L/F is totally ramified and take $x \in \mathfrak{p}_L^s$. For the j th elementary symmetric function of the conjugates of x , the symmetric estimate gives

$$v_F(E_j(x)) \geq \left\lceil \frac{js}{\ell} \right\rceil.$$

For $j \geq 2$ this is at least m , since $2s \geq 1 + \ell(m - 1)$. Hence

$$N_{L/F}(1 + x) \equiv 1 + \text{Tr}_{L/F} x \pmod{\mathfrak{p}_F^m}.$$

Moreover

$$v_F(\text{Tr}_{L/F} x) \geq \left\lceil \frac{s}{\ell} \right\rceil \geq \left\lceil \frac{m}{2} \right\rceil.$$

The last inequality follows separately for $m = 2b$ and $m = 2b + 1$ from the definition of s . The quotient of the two principal units in the congruence belongs to U_F^m , on which λ is trivial. Evaluating λ gives the required formula.

If L/F is unramified, put $r = \lceil m/2 \rceil$ and take $x \in \mathfrak{p}_L^r$. The j th symmetric term has depth at least jr , so every term of degree at least two lies in $\mathfrak{p}_F^m$, while $\text{Tr}_{L/F} x \in \mathfrak{p}_F^r$. The same norm expansion proves the assertion. $\square$

Applying Lemma 5.2 to U/F and E/F gives the stationary coefficient g for both norm pullbacks.

The conductor-one characters are in the stable-twist range on both sides, including $\ell = 2$ and $m = 2$. Thus

$$(28) \quad \Delta_L(\theta_L) = \chi_L(g^{-1})\Delta_L(\lambda n_L).$$

Equations (2) and (13) give

$$(29) \quad \Delta_L(\lambda n_L)\Lambda_L = \prod_{\nu \in S(L/F)} \Delta_F(\nu\lambda) = \epsilon_L(g^{-1})\Delta_F(\lambda)^\ell.$$

All lower norm characters have conductor zero or one, hence at most $\lfloor m/2 \rfloor$. Combining (28)–(29) with Lemma 4.2 for the compatible pair (χ_U, χ_E) proves

$$(30) \quad \Delta_U(\theta_U)\Lambda_U = \Delta_E(\theta_E)\Lambda_E$$

for every $m > 1$.

5.5. Conductor one and the sign. Now $m_U = m_E = 1$. Let $k = k_F = k_E$, $\kappa = k_U$, and denote the two residue characters by χ_κ, χ_0 . Norm compatibility gives $\chi_\kappa^\ell = \chi_0 N_{\kappa/k}$. The Frobenius conjugate quotient is a nontrivial character of order ℓ . Writing the exponents as in the proof of Theorem 5.1 shows that this is exactly $\ell \nmid a$, or $\chi_0|_{\mu_\ell} \neq 1$. In particular both residue Gauss sums are nontrivial-character sums.

Write $n = n_F(\psi_F)$ and choose $\Gamma = \pi_F^{n+1} \in F$. It is admissible in both fields: $n_E = \ell n + \ell - 1$, while $n_U = n$. The conductor-one formulas for the local constants give

$$\begin{aligned} \Delta_U(\theta_U) &= -\theta_U(\Gamma) \text{ph } \tau_\kappa(\chi_\kappa), \\ \Delta_E(\theta_E) &= -\theta_E(\Gamma) \chi_0(\ell) \text{ph } \tau_k(\chi_0). \end{aligned}$$

The factor $\chi_0(\ell)$ is present because the residual additive character over E is $x \mapsto \psi_k(\ell x)$, not $\psi_k(x)$. For the lower norm-character products,

$$\Lambda_U = (-1)^{(\ell-1)n}, \quad \Lambda_E = (-1)^{\ell-1} \epsilon_E(\Gamma) \prod_{j=1}^{\ell-1} \text{ph } \tau_k(\mu^j).$$

Indeed $\epsilon_U(\pi_F) = (-1)^{\ell-1}$ by the product of the ℓ unramified characters, and every nontrivial member of $S(E/F)$ has conductor one. Lemma 4.2 gives $\theta_U(\Gamma)/\theta_E(\Gamma) = \epsilon_E(\Gamma)/\epsilon_U(\Gamma)$. Since $\epsilon_U(\Gamma) = (-1)^{(\ell-1)(n+1)}$, all the displayed signs cancel in the quotient of (30). What remains is precisely the phase of the quotient in Theorem 5.1, namely one. This proves (30) at conductor one as well.

Theorem 5.3 (The tame case). *Theorem 1.1 holds when $\ell \neq p$.*

Proof. The unramified degree- ℓ intermediate field is unique. The preceding arguments compare it with every other degree- ℓ intermediate field, so transitivity compares any two. The higher-conductor case is (28)–(29); the conductor-one case is the final calculation. $\square$

6. WILD ODD DEGREE: THE UNRAMIFIED–RAMIFIED CASE

6.1. Statement and notation. Let F be a nonarchimedean local field of residue characteristic $p > 2$. Let K/F be Galois with group $C_p \times C_p$, and suppose its inertia subgroup has order p . Let U be the unramified degree- p intermediate field and E any ramified degree- p intermediate field. Then $K = EU$, K/E is unramified, and E/F and K/U are totally ramified cyclic of degree p , with the same break $t \geq 1$. Set

$$T = t + 1, \quad D_0 = (p - 1)T, \quad q_0 = \lceil T/p \rceil, \quad e_0 = \lfloor D_0/p \rfloor = T - q_0.$$

No hypothesis $p \nmid t$ is imposed. All valuations are normalized on the field displayed. In particular $v_K|_E = v_E$, $v_K|_U = pv_U$, and $v_E|_F = pv_F$. Put

$$n_E = N_{E/F}, \quad n_U = N_{U/F}, \quad N_E = N_{K/E}, \quad N_U = N_{K/U}.$$

Fix a nontrivial additive character e_F and set $e_L = e_F \circ \text{Tr}_{L/F}$ for all four fields. Let $\Delta_L(\theta, e_L)$ denote the local constant (1).

Theorem 6.1. *Suppose θ_U and θ_E have the same norm pullback*

$$\theta_K = \theta_U \circ N_U = \theta_E \circ N_E,$$

and this character does not descend through $N_{K/F}$. Then

$$\Delta_U(\theta_U, e_U) = \Delta_E(\theta_E, e_E).$$

By Corollary 4.3, the products over $S(U/F)$ and $S(E/F)$ are both one, so this proves $\mathcal{A}_U(\theta_U) = \mathcal{A}_E(\theta_E)$. The assertion holds in mixed and equal characteristic.

We use the one-dimensional results recalled in Sections 2–4, together with the subcritical norm representatives of [7, Lemma 6.6] and [7, Proposition 6.8] on stationary representatives of norm pullbacks. For a totally ramified degree- p extension L/M with break t , these results give

$$(31) \quad \text{Tr}_{L/M}(\mathfrak{p}_L^a) = \mathfrak{p}_M^{\lfloor (a+D_0)/p \rfloor},$$

$$(32) \quad v_M(e_i(y)) \geq \lfloor (iv_L(y) + D_0)/p \rfloor \quad (1 \leq i < p),$$

$$(33) \quad v_M(Ny) = v_L(y).$$

Here $e_i(y)$ is the i th elementary symmetric function of the p conjugates, even when y lies in a proper subfield. The inequalities hold also for negative valuations. For unramified extensions, the norm is onto every positive unit group and norm pullback preserves conductors. If $c \in M^\times$ has valuation a and $0 \leq r \leq t$, there is $x \in L^\times$ with

$$(34) \quad v_L(x) = a, \quad c - N(x) \in \mathfrak{p}_M^{a+r}.$$

We also use Lemmas C.1 and C.2.

6.2. Truncated logarithms and norm characters. Put

$$P(Z) = \sum_{j=1}^{p-1} \frac{Z^j}{j}.$$

Its coefficients lie in $\mathbf{Z}_{(p)}$. Our sign convention is that $P(z)$ is the truncation of $-\log(1-z)$.

Lemma 6.2 (Truncated logarithms on unit groups). *The polynomial $P(z+w-zw) - P(z) - P(w)$ has no monomial of total degree less than p . Consequently, for integers $1 \leq r \leq M$ with $pr \geq M$, the map*

$$1 - z \mapsto P(z)$$

is a group isomorphism $U_L^r/U_L^M \rightarrow \mathfrak{p}_L^r/\mathfrak{p}_L^M$. The map $P : \mathfrak{p}_L^r \rightarrow \mathfrak{p}_L^r$ is bijective and preserves valuations.

Proof. Compare degrees less than p in the formal logarithm identity over $\mathbf{Q}$. The truncated expression has coefficients in $\mathbf{Z}_{(p)}$, so the assertion is valid in both characteristics. The congruence is additive modulo $\mathfrak{p}_L^M$ under the stated inequality. Finally $P(z) = z + z^2 R(z)$ with R integral and $P'(z) \equiv 1 \pmod{\mathfrak{p}_L}$. Hensel's lemma solves $P(z) = w$ uniquely in $\mathfrak{p}_L^r$, or equivalently one can invert successively on its successive quotients. This also proves the valuation assertion and the group isomorphism. $\square$

Lemma 6.3 (Truncated logarithms of norms). *Let L/M be a totally ramified cyclic extension of degree p and break t . For $h \geq 0$ and $z \in \mathfrak{p}_L^{h+q_0}$,*

$$(35) \quad P(1 - N(1 - z)) \equiv \text{Tr } P(z) + N(P(z)) \pmod{\mathfrak{p}_M^{T+h}}.$$

For an unramified degree- p extension L/M and $z \in \mathfrak{p}_L^{q_0}$, one has

$$(36) \quad P(1 - N(1 - z)) \equiv \text{Tr } P(z) \pmod{\mathfrak{p}_M^T}.$$

Proof. Let $z_1, \dots, z_p$ be indeterminates and e_i their elementary symmetric functions. Give e_i weight i and put

$$u = 1 - \prod_{a=1}^p (1 - z_a) = e_1 - e_2 + \dots + e_p, \quad \mathcal{D} = P(u) - \sum_a P(z_a) - \prod_a P(z_a).$$

The polynomial $\mathcal{D}$ belongs to $\mathbf{Z}_{(p)}[e_1, \dots, e_p]$. Every monomial has weight at least p and at least two elementary-symmetric factors. Indeed the formal logarithm identity cancels all terms of weight less than p , and the weight- p term before the final product is $(\sum_a z_a^p - e_1^p)/p$. Newton's identity gives coefficient 1 for e_p in this polynomial; the final product cancels it. There is no possible one-factor monomial of larger weight. Moreover every coefficient of a pure power e_p^j is divisible by p : modulo p , specialize $e_1 = \dots = e_{p-1} = 0$, adjoin w with $w^p = e_p$, and note that $\sum_a P(z_a) = 0$ and $\prod_a P(z_a) = P(w)^p = P(e_p)$. The injectivity of $\mathbf{F}_p[e_p] \rightarrow \mathbf{F}_p[w]$ justifies this coefficient test.

Write $q = h + q_0$ and specialize to the conjugates of z . Then $v_M(e_p) \geq q$ and, for $i < p$,

$$v_M(e_i) \geq \lfloor (iq + D_0)/p \rfloor \geq \frac{iq + (p-1)t}{p}, \quad v_M(e_i) \geq e_0.$$

A monomial containing both e_p and a smaller-index factor has valuation at least $q + e_0 = T + h$. A monomial consisting of $r \geq 2$ smaller-index factors and having total weight at least p has valuation at least

$$q + \frac{2(p-1)t}{p} \geq T + h,$$

because $q_0 + (p-2)t/p - 1 \geq 0$. In mixed characteristic $e = v_M(p) \geq e_0$, by $p = \text{Tr}_{L/M}(1)$ and (31). A pure norm-power error has valuation at least $e + 2q \geq T + h$. In equal characteristic it is zero. This proves (35). For an unramified extension, every monomial of weight at least p has valuation at least $pq_0 \geq T$, and the norm of $P(z)$ has that valuation also. This gives (36). $\square$

Choose a nontrivial norm character $\tau_F \in S(E/F)$. It has conductor T . Lemma 6.2 and the additive trace pairing give $\alpha \in F^\times$ such that

$$(37) \quad \tau_F(1 - z) = e_F(\alpha P(z)) \quad (z \in \mathfrak{p}_F^{q_0}), \quad v_F(\alpha) = -n_F(e_F) - T.$$

Set

$$\Psi_L(z) = e_L(\alpha z).$$

The largest ideal on which Ψ_L is trivial is $\mathfrak{p}_L^T$ for each of F, U, E, K . For E/F , this is the different calculation $-pT + D_0 = -T$; the cases of U and K follow by unramified base change. Set $\tau_U = \tau_F \circ n_U$. By Lemma D.5, it is a norm character for K/U . Equation (36) gives $\tau_U(1 - z) = \Psi_U(P(z))$ for $z \in \mathfrak{p}_U^{q_0}$.

Corollary 6.4 (An additive identity from the norm character). *For $L/M = E/F$ or K/U and every $w \in \mathfrak{p}_L^{q_0}$,*

$$(38) \quad \Psi_M(\text{Tr}_{L/M} w + N_{L/M} w) = 1, \quad \Psi_L(w) = \Psi_M(-N_{L/M} w).$$

Proof. Write $w = P(z)$ with $z \in \mathfrak{p}_L^{q_0}$, using Lemma 6.2. The norm of $1 - z$ belongs to $U_M^{q_0}$: the intermediate symmetric terms have valuation at least q_0 by (32), and so does the norm term. The norm character is trivial on $N_{L/M}(1 - z)$. Its formula (37), or its counterpart for K/U , together with Lemma 6.3 at $h = 0$, proves (38). $\square$

6.3. Changing a stationary coefficient.

Lemma 6.5 (Lamprecht's formula for $C = B + \eta$). *Let θ have conductor $m \geq 2$ on a local field L of odd residue characteristic. Let $\Psi_L = e_L(\alpha \cdot)$ have largest trivial ideal $\mathfrak{p}_L^J$, where $J \in \mathbf{Z}$. Put $d = \lfloor m/2 \rfloor$ and $s = \lceil m/2 \rceil$. There is $B \in L$ of valuation $J - m$, determined modulo $\mathfrak{p}_L^{J-d}$, such that*

$$(39) \quad \theta(1 - z) = \Psi_L(BP(z)) \quad (z \in \mathfrak{p}_L^d).$$

If $C = B + \eta$ with $v_L(\eta) \geq J - s$, then $\theta(1 - z) = \Psi_L(Cz)$ for $z \in \mathfrak{p}_L^s$, and

$$(40) \quad \Delta_L(\theta, e_L) = \theta((- \alpha C)^{-1}) \Psi_L(-C) \Psi_L\left(\frac{\eta^2}{2B}\right) g_L, \quad g_L \in \mu_4.$$

For even m , $g_L = 1$ and the correction is 1. If $v_L(\eta) \geq J - d$, the correction is 1 in either parity.

Proof. Since $pd \geq m$, Lemma 6.2 identifies the character on U_L^d with an additive character on $\mathfrak{p}_L^d/\mathfrak{p}_L^m$. The trace pairing gives B and its ambiguity, and the exact conductor gives $v_L(B) = J - m$. On this depth $P(z) \equiv z + z^2/2$ at the character precision, since $3d \geq m$. Thus

$$\theta(1 + z) = \Psi_L(-Bz + Bz^2/2).$$

For an admissible denominator Γ , take the stationary representative $-\alpha C\Gamma$. The character and additive factors in Lamprecht's formula are $\theta((- \alpha C)^{-1}) \Psi_L(-C)$. For odd $m = 2d + 1$, choose $\delta \mathcal{O}_L = \mathfrak{p}_L^d$. The critical function for this representative is

$$\bar{z} \mapsto \Psi_L(-\eta \delta z - B \delta^2 z^2/2).$$

The inequality $v_L(\eta) \geq J - s$ makes $\eta/(B\delta)$ integral. Completing the square produces the factor $\Psi_L(\eta^2/(2B))$. The remaining homogeneous, nondegenerate quadratic Gauss phase is a fourth root of unity. Indeed, writing $G = \sum_{x \neq 0} \chi_2(x) \psi_0(x)$ for a nontrivial residue additive character, orthogonality gives $|G|^2 = |k_L|$, and the substitution $x \mapsto -x$ gives $\bar{G} = \chi_2(-1)G$. Thus $G^2 = \chi_2(-1)|k_L|$; scaling the quadratic coefficient only multiplies G by χ_2 of that coefficient. In even conductor the ordinary linearization already holds on $\mathfrak{p}_L^d$ and Lamprecht has no residual sum. Finally $v_L(\eta^2/B) \geq J$ in the even case, and also whenever the stronger depth $J - d$ holds. This proves every assertion. $\square$

6.4. Characters of conductor T . Let k be the residue field of F, E and κ that of U, K . Write $Q = |k|$ and let ϕ be arithmetic Frobenius in $\text{Gal}(U/F)$. Choose $\bar{d} \in \kappa$ with $\bar{d}^Q - \bar{d} = 1$. Then $\bar{c} = \bar{d}^p - \bar{d} \in k$ has absolute trace 1. Choose any lift $c \in \mathcal{O}_F$ of $\bar{c}$. The polynomial $X^p - X - c$ is irreducible and unramified over F , so we can choose

$$(41) \quad d \in U, \quad d^p - d = c, \quad d \bmod \mathfrak{p}_U = \bar{d}.$$

In mixed characteristic, Hensel's lemma at $d+1$ gives $\phi(d) - (d+1) \in p\mathcal{O}_U$. In equal characteristic this difference is zero. Also

$$(42) \quad e_0 + q_0 = T, \quad \lceil T/2 \rceil \geq q_0, \quad 2e_0 \geq T, \quad e_0 \geq \lceil T/2 \rceil.$$

The elementary ceiling inequalities hold for every $T \geq 2, p \geq 3$.

Proposition 6.6 (Construction of compatible characters). *With τ_F, τ_U, Ψ_L as above, there are characters χ_U^0 and χ_E^0 of conductor T , with a common invariant primitive norm pullback, such that*

$$(43) \quad (\chi_U^0)^\phi / \chi_U^0 = \tau_U,$$

$$(44) \quad \chi_U^0(1 - z) = \Psi_U(d^p P(z)) \quad (z \in \mathfrak{p}_U^{q_0}),$$

$$(45) \quad \chi_E^0(1 - z) = \Psi_E(cP(z)) \quad (z \in \mathfrak{p}_E^{q_0}).$$

Here $\chi^\phi(y) = \chi(\phi^{-1}y)$.

Proof. First define the right side of (44) on $H = U_U^{q_0}$. It is a character by Lemma 6.2, with exact conductor T . Its conjugate quotient is τ_U on H because

$$\phi(d^p) - d^p - 1 \in p\mathcal{O}_U, \quad v_U(p) + q_0 \geq e_0 + q_0 = T.$$

We extend this character so that (43) holds on $U^\times$. Put $V = U_U^1$ and $I = \{\phi^{-1}(v)/v : v \in V\}$. Prescribe on I

$$\kappa(\phi^{-1}(v)/v) = \tau_U(v).$$

This is well defined: the kernel of $v \mapsto \phi^{-1}(v)/v$ is U_F^1 , and $\tau_U(f) = \tau_F(f^p) = 1$ there. If $\phi^{-1}(v)/v \in H$, unramified Teichmüller expansions imply $v = fh$ for $f \in U_F^1$ and $h \in H$; indeed every coefficient of a Frobenius-fixed truncated expansion lies in k . Thus the prescribed character

on $I \cap H$ agrees with (44). The two characters define a character on IH , trivial on U_U^T . Extend it across the finite abelian quotient V/U_U^T to a character of V . Take its value to be 1 on the prime-to- p Teichmüller subgroup and on a fixed uniformizer of F . The character τ_U is trivial on these two factors, so (43) now holds on all of $U^\times$. This gives χ_U^0 of conductor T satisfying (44).

Put $\chi_K^0 = \chi_U^0 \circ N_U$. It is invariant under $\text{Gal}(K/U)$ by definition and under $\text{Gal}(K/E)$ by (43), since τ_U is a norm character. By Lemma C.1, it is trivial on $\ker N_E$. Thus $N_E x \mapsto \chi_K^0(x)$ defines a character of $N_E(K^\times)$. Lemma C.2 extends it to χ_E^0 on $E^\times$. Since K/E is unramified, its norm is surjective on units; only the value at a uniformizer remains to be chosen.

For $z \in \mathfrak{p}_K^{q_0}$ the ramified norm of $1 - z$ is in $U_U^{q_0}$. Equations (44), (35), and (38), using $N_U(d) = d^p$ exactly, give

$$\chi_K^0(1 - z) = \Psi_K(d^p P(z)) \Psi_U(N_U(dP(z))) = \Psi_K((d^p - d)P(z)) = \Psi_K(cP(z)).$$

The unramified congruence (36) shows that this is the pullback of the right side of (45). The norm $U_K^{q_0} \rightarrow U_E^{q_0}$ is onto, so (45) follows on its entire stated subgroup. Since c is a unit and the ideal conductor of Ψ_E is T , χ_E^0 has exact conductor T . If the common pullback descended from F , then χ_U^0 would be a character of the form $\lambda \circ n_U$ times an element of $S(K/U)$. Both are fixed by ϕ , the latter by Lemma D.5. This contradicts its nontrivial conjugate quotient τ_U , proving primitivity. $\square$

Proposition 6.7 (Writing the two characters using one character of $F^\times$). *Choose τ_F using Lemma 4.2, and construct the characters of Proposition 6.6. There is a character λ of $F^\times$ such that, after changing only the unramified value of χ_E^0 if needed,*

$$(46) \quad \theta_U = \chi_U^0(\lambda \circ n_U), \quad \theta_E = \chi_E^0(\lambda \circ n_E).$$

Write $m_U = T + h$. Then $h \geq 0$, $m_E = T + ph$, and for $h > 0$ the character λ has conductor $T + h$. For $h = 0$ its conductor is at most T .

Proof. The quotient θ_U/χ_U^0 is ϕ -invariant. Lemmas C.1 and C.2 therefore give $\theta_U/\chi_U^0 = \lambda \circ n_U$ for a character λ of $F^\times$. Compatibility implies that $\theta_E/(\chi_E^0(\lambda n_E))$ is an unramified norm character for K/E . Absorb it into χ_E^0 ; the formula (45) is unchanged. The conductor of θ_U is at least that of its commutator, namely T . If it exceeds T , then $\lambda \circ n_U$ has the same conductor as θ_U , because χ_U^0 has conductor T . Since U/F is unramified, $a_U(\lambda \circ n_U) = a_F(\lambda)$. All twists of θ_U by $S(K/U)$ are conjugate and have the same conductor. Therefore [7, Proposition 3.10] applies, both at $m_U = T$ and above it; it gives $m_K = T + ph$. Norm pullback from E to the unramified extension K preserves conductors, so $m_E = m_K$. $\square$

6.5. Norm representatives for the coefficient of λ . Use Proposition 6.7 and suppose $0 \leq h \leq t$. Set

$$m_U = T + h, \quad m_E = T + ph, \quad d_U = \lfloor m_U/2 \rfloor, \quad d_E = \lfloor m_E/2 \rfloor.$$

In particular $d_U, d_E \geq q_0$. By Lemma 6.2 and the additive pairing, choose $A_* \in F$ so that

$$(47) \quad \lambda(1 - z) = \Psi_F(A_* P(z)) \quad (z \in \mathfrak{p}_F^{d_U}).$$

For $h > 0$, $v_F(A_*) = -h$ and its ambiguity is $\mathfrak{p}_F^{T-d_U}$. For $h = 0$, its class lies in $\mathcal{O}_F/\mathfrak{p}_F^{T-d_U}$; a zero class may be represented by 0.

Lemma 6.8. *There is $x \in E$ with $A = n_E(x)$ having the following properties. If $0 \leq h < t$, one may take $A_* = A$ in (47). If $h = t$, one has instead*

$$(48) \quad A_* = A + \epsilon, \quad \epsilon \in \mathcal{O}_F.$$

For $h > 0$, $v_E(x) = v_F(A) = -h$. For $h = 0$, x is integral; the zero coefficient is treated by $x = A = 0$.

Proof. For $0 < h < t$, the relative precision of the coefficient class in (47) is $T + h - d_U = \lfloor m_U/2 \rfloor \leq t$. Thus (34) supplies an exact norm in that class. For $h = 0$, a nonzero class has valuation $a \geq 0$; its relative precision is $T - d_U - a \leq \lceil T/2 \rceil \leq t$. Again (34) applies. If the class is zero, use $x = 0$. At $h = t$, $m_U = 2t + 1$ and $d_U = t$. Use (34) at relative precision t , not $t + 1$. Its absolute error is integral, giving (48). The valuation assertions follow from (33). $\square$

Choose x, A, ϵ as in Lemma 6.8, set $\epsilon = 0$ for $h < t$, and put

$$(49) \quad B_U = A + d^p + \epsilon \in U, \quad B_E = A - x + c \in E.$$

Proposition 6.9 (Coefficients for θ_U and θ_E). *For the characters in (46),*

$$(50) \quad \theta_U(1 - z) = \Psi_U(B_U P(z)) \quad (z \in \mathfrak{p}_U^{d_U}), \quad \theta_E(1 - z) = \Psi_E(B_E P(z)) \quad (z \in \mathfrak{p}_E^{d_E}).$$

These formulas allow us to apply Lemma 6.5 with $B = B_U$ and $B = B_E$.

Proof. For the unramified pullback of λ , the logarithm identity through an unramified norm is valid modulo $\mathfrak{p}_F^{m_U}$ at depth d_U , because $pd_U \geq m_U$. It gives coefficient A_* over U . Combine it with (44) to obtain the first formula.

For $z \in \mathfrak{p}_E^{d_E}$ put $u = 1 - n_E(1 - z)$. The symmetric estimates give

$$v_F(u) \geq \min\{d_E, \lfloor (d_E + D_0)/p \rfloor\} \geq d_U.$$

The last inequality follows from $d_E \geq d_U$ and $d_E + D_0 - pd_U = (p - 1)(T + \varepsilon)/2 \geq 0$, where $\varepsilon \in \{0, 1\}$ is the common conductor parity. Also $d_E - h \geq q_0$, since $d_E - h = \lfloor (T + (p - 2)h)/2 \rfloor \geq \lfloor T/2 \rfloor \geq q_0$. For $h = 0$ the same inequalities hold with x integral. Lemma 6.3 therefore gives

$$P(u) \equiv \text{Tr } P(z) + n_E(P(z)) \pmod{\mathfrak{p}_F^{T+h}}.$$

Multiply by A , whose valuation is $-h$ for $h > 0$ and nonnegative for $h = 0$. Equation (38) applies to $xP(z)$, and the *exact* equality $n_E(x) = A$ gives

$$\Psi_F(AP(u)) = \Psi_E(AP(z))\Psi_F(n_E(xP(z))) = \Psi_E((A - x)P(z)).$$

At $h = t$, $d_E = (p + 1)t/2 \geq T$. It follows from the same norm estimate that $u \in \mathfrak{p}_F^T$. Hence the extra coefficient $\epsilon \in \mathcal{O}_F$ in (48) contributes $\Psi_F(\epsilon P(u)) = 1$. Thus the displayed formula for $\lambda \circ n_E$ also holds when $h = t$. Multiply by (45) to obtain the second formula. $\square$

For the coefficients of Proposition 6.9, when $h > 0$, $v_U(B_U) = -h$ and $v_E(B_E) = -ph$, with unique dominant term A . For $h = 0$, if $\xi = \bar{x} \in k$, their residues are $\xi^p + \bar{d}^p$ and $\xi^p - \xi + \bar{c}$. The first is nonzero because $\bar{d} \notin k$; the second has absolute trace 1, so is nonzero also. Thus the coefficients have exactly the valuations required by their stated conductors in every case.

6.6. The norms of $C = x + d$. Put

$$(51) \quad C = x + d \in K^\times, \quad Z_U = N_U(C), \quad Z_E = N_E(C).$$

Because d lies in U , it is fixed under $\text{Gal}(K/U)$. Because x lies in E , it is fixed under $\text{Gal}(K/E)$. Hence, writing $a_i = e_i^{E/F}(x)$,

$$(52) \quad Z_U = d^p + \sum_{i=1}^{p-1} a_i d^{p-i} + A,$$

$$(53) \quad Z_E = x^p - x + c.$$

Equation (53) follows by evaluating $X^p - X - c$ at $-x$. Define

$$\eta_U = Z_U - B_U = \sum_{i=1}^{p-1} a_i d^{p-i} - \epsilon, \quad \eta_E = Z_E - B_E = x^p - A.$$

Lemma 6.10 (Bounds for η_U and η_E). *For $0 \leq h < t$, one has $v_U(\eta_U) \geq T - d_U$ and $v_E(\eta_E) \geq T - d_E$. Hence both correction factors in (40) equal one. For $h = t$, one has $v_U(\eta_U) \geq T - \lceil m_U/2 \rceil$ and $v_E(\eta_E) \geq T - \lceil m_E/2 \rceil$. If also $p > 3$, then $v_E(\eta_E) \geq T - d_E$. For $h = t$ and $b = a_{p-1}$, one further has*

$$(54) \quad \eta_U \equiv bd - \epsilon \pmod{\mathfrak{p}_U}, \quad b \in \mathcal{O}_F.$$

Proof. For integral x , (32) puts all the a_i in $\mathfrak{p}_F^{e_0}$. The characteristic-polynomial identity gives

$$(55) \quad x^p - A = \sum_{i=1}^{p-1} (-1)^{i+1} a_i x^{p-i}, \quad v_E(x^p - A) \geq p v_E(x) + (p-1)t.$$

The bound follows from $p \lfloor (ia + D_0)/p \rfloor \geq ia + (p-1)t$ with $a = v_E(x)$. Thus for $h = 0$ the errors have depths at least e_0 and $(p-1)t$, respectively, both at least $T - \lfloor T/2 \rfloor$. If $x = 0$ the assertions are immediate.

For $h > 0$, (32) and (55) give

$$(56) \quad v_U \left(\sum_{i=1}^{p-1} a_i d^{p-i} \right) \geq \lfloor (p-1)(T-h)/p \rfloor, \quad v_E(\eta_E) \geq (p-1)t - ph.$$

If $h < t$, put $r = T - h \geq 2$. The first bound is $r - \lceil r/p \rceil \geq \lceil r/2 \rceil = T - d_U$. Here $\lceil r/p \rceil \leq \lfloor r/2 \rfloor$ for $r \geq 2$, $p \geq 3$. For the second, $h \leq t - 1$ gives

$$2((p-1)t - ph) - (T - ph) = (2p-3)t - ph - 1 \geq (p-3)t + p - 1 \geq 0;$$

hence $(p-1)t - ph \geq T - d_E$. This proves the bounds for $h < t$.

At $h = t$, $m_U = 2t + 1$, $m_E = (p+1)t + 1$, and the required bounds are 0 over U and $-(p-1)t/2$ over E . The two bounds $v_U(\eta_U) \geq 0$ and $v_E(\eta_E) \geq -t$ suffice. If $p > 3$, also $-t \geq 1 - (p-1)t/2 = T - d_E$, which gives the stronger bound for η_E . Finally, at this boundary a_{p-1} is integral and all a_i with $i \leq p-2$ have valuation at least $\lfloor (t+p-1)/p \rfloor = \lceil t/p \rceil \geq 1$. This proves (54). $\square$

Lemma 6.11 (The quotient of the two local constants). *Let*

$$(57) \quad S = \text{Tr}_{U/F} Z_U - \text{Tr}_{E/F} Z_E.$$

The local constants satisfy

$$(58) \quad \frac{\Delta_U(\theta_U)}{\Delta_E(\theta_E)} \in \mu_4 \Psi_F(-S) \frac{\Psi_U(\eta_U^2/(2B_U))}{\Psi_E(\eta_E^2/(2B_E))}.$$

Furthermore one has the exact identity

$$(59) \quad S = pA - \text{Tr}_{E/F}(x^p) + p \text{Tr}_{E/F}(x).$$

Proof. Apply Lemma 6.5 with $C = Z_U, Z_E$, using Lemma 6.10. The factors at $-\alpha \in F^\times$ cancel by (18), and

$$\theta_U(Z_U) = \theta_K(C) = \theta_E(Z_E).$$

Substitution in (40) gives (58). Newton identities for $X^p - X - c$ give

$$\text{Tr}_{U/F}(d^j) = 0 \quad (1 \leq j < p-1), \quad \text{Tr}_{U/F}(d^{p-1}) = p-1, \quad \text{Tr}_{U/F}(d^p) = pc.$$

Take traces in (52) and (53) and subtract. This gives (59). $\square$

6.7. The cases $h < t$ and $p > 3$.

Lemma 6.12. *One has $S \in \mathfrak{p}_F^T$ if $h = 0$, if $1 \leq h < t$, or if $p > 3$ and $h = t$.*

Proof. For integral x , all smaller elementary coefficients and all power traces $\text{Tr}(x^j)$ have valuation at least e_0 . Newton's identity at degree p puts $\text{Tr}(x^p) - pA$ in $\mathfrak{p}_F^{2e_0}$, and $p \text{Tr} x$ has valuation at least $v_F(p) + e_0 \geq 2e_0$. Use $2e_0 \geq T$.

For $h > 0$ use the following integral universal identity:

$$\text{Tr}(x^p) - pA = a_1^p + pQ(a_1, \dots, a_{p-1}),$$

where every monomial of Q has weight p and at least two factors. Indeed the power sum is congruent to a_1^p modulo p as an integral symmetric polynomial, its a_p coefficient is p , and there is no one-factor monomial of weight p among $a_1, \dots, a_{p-1}$. The estimates

$$v_F(a_i) \geq \frac{-ih + (p-1)t}{p}, \quad v_F(p) \geq \frac{(p-1)t}{p}$$

give the respective lower bounds

$$v_F(pa_1) \geq \frac{2(p-1)t-h}{p}, \quad v_F(a_1^p) \geq (p-1)t-h,$$

and, for every monomial of pQ , a lower bound $3(p-1)t/p-h$. All three bounds are strictly greater than t when $p > 3$ and $h \leq t$, or when $p = 3$ and $h < t$. Since valuations are integral, they are at least T . In equal characteristic the terms divisible by p are zero and the same conclusion follows. This proves the lemma. $\square$

For $h < t$, both corrections in (58) are 1 by Lemma 6.10, so Lemma 6.12 already puts the quotient in μ_4 . At $h = t$ one has $B_U = A + d^p + \epsilon$ with $v_F(A) = -t$ and $\eta_U \equiv bd - \epsilon \pmod{\mathfrak{p}_U}$. Therefore

$$(60) \quad \Psi_U\left(\frac{\eta_U^2}{2B_U}\right) = \Psi_U\left(\frac{(bd - \epsilon)^2}{2A}\right).$$

The error from replacing the numerator has valuation at least $t+1$; the error from replacing the inverse denominator has valuation at least $2t \geq t+1$. Both lie in the trivial ideal of Ψ_U . If $p > 3$, both $\text{Tr}(d)$ and $\text{Tr}(d^2)$ vanish, while the trace of 1 is p . The right side of (60) is $\Psi_F(p\epsilon^2/(2A)) = 1$, since its argument has valuation at least $v_F(p) + t \geq T$ (and is zero in equal characteristic). The correction over E is one by Lemma 6.10. Together with Lemma 6.12, this puts the quotient in (58) in μ_4 also when $p > 3$ and $h = t$.

6.8. The case $p = 3$ and $h = t$. It remains to treat $p = 3$ and $h = t$, without any restriction on divisibility of t . Write

$$a = \text{Tr}_{E/F} x, \quad b = e_2^{E/F}(x), \quad A = n_E(x).$$

The symmetric estimates give

$$(61) \quad v_F(a) \geq \lceil t/3 \rceil \geq 1, \quad v_F(b) \geq 0, \quad v_F(A) = -t, \quad v_E(x) = -t.$$

The coefficients in (50) are $B_U = A + d^3 + \epsilon$ and $B_E = A - x + c$. Because $\text{Tr}_{U/F}(d^2) = 2$ and $\text{Tr}_{U/F}(d) = 0$, (60) gives

$$(62) \quad \Psi_U\left(\frac{\eta_U^2}{2B_U}\right) = \Psi_F(b^2/A).$$

The remaining term $3\epsilon^2/(2A)$ belongs to $\mathfrak{p}_F^T$. Thus ϵ does not affect (62).

The characteristic polynomial of x gives the exact identity

$$(63) \quad \eta_E = x^3 - A = ax^2 - bx.$$

One has $v_E(\eta_E) \geq -t$, and $v_E(1/B_E - 1/A) \geq 5t$. Consequently

$$(64) \quad \Psi_E\left(\frac{\eta_E^2}{2B_E}\right) = \Psi_F\left(\frac{\text{Tr}_{E/F}(\eta_E^2)}{2A}\right),$$

since the omitted E -argument has valuation at least $3t \geq T$. The trace maps $\mathfrak{p}_E^T$ into $\mathfrak{p}_F^T$ by (31).

By (59), $S = 3a - a^3 + 3ab$. The following cancellation uses only the cyclic cubic extension E/F , its break t , the norm-phase identity (38), and the condition $v_E(x) = -t$.

Lemma 6.13 (Cubic cancellation). *Put*

$$\Lambda = 3a - a^3 + 3ab - \frac{b^2}{A} + \frac{\text{Tr}_{E/F}(\eta_E^2)}{2A}.$$

Then

$$\Lambda = 3a + a^3 - \frac{b^2 + b^3}{A} + \frac{a^2(a^2 - 3b)^2}{2A},$$

and $\Psi_F(\Lambda) = 1$.

Proof. Newton's identities give

$$\mathrm{Tr} x^2 = a^2 - 2b, \quad \mathrm{Tr} x^3 = a^3 - 3ab + 3A, \quad \mathrm{Tr} x^4 = a^4 - 4a^2b + 2b^2 + 4aA.$$

Using $\eta_E = ax^2 - bx$, one obtains

$$\mathrm{Tr} \eta_E^2 = a^6 - 6a^4b + 9a^2b^2 - 2b^3 + 4a^3A - 6abA.$$

Substitution gives the displayed identity.

The last summand belongs to $\mathfrak{p}_F^T$: by (61), $v_F(a) \geq 1$, $v_F(a^2 - 3b) \geq 1$, and $v_F(A^{-1}) = t$. For the first pair apply (38) to $a \in F \subset E$. Since $v_E(a) \geq 3\lceil t/3 \rceil \geq t \geq q_0$, it gives

$$\Psi_F(3a + a^3) = 1.$$

For the second pair apply (38) to b/x . It also has E -valuation at least $t \geq q_0$, and

$$\mathrm{Tr}_{E/F}(b/x) = \frac{b^2}{A}, \quad \mathrm{N}_{E/F}(b/x) = \frac{b^3}{A}.$$

Thus

$$\Psi_F((b^2 + b^3)/A) = 1.$$

The three character values prove $\Psi_F(\Lambda) = 1$. □

Equations (58), (62), and (64) now give

$$\frac{\Delta_U(\theta_U)}{\Delta_E(\theta_E)} \in \mu_4 \Psi_F(-\Lambda) = \mu_4.$$

This completes the calculation for $0 \leq h \leq t$.

6.9. The case $h \geq T$ and the proof of Theorem 6.1. Suppose $h \geq T$, so $m = m_F(\lambda) = T + h \geq 2T$. Choose a stationary representative $\beta \in F^\times$ for λ with admissible denominator $\Gamma \in F^\times$. By [7, Proposition 6.8], the same Γ, β apply to its norm pullbacks to U and E . The denominator assertion includes the different:

$$m_E(\lambda n_E) + n_E(e_E) = T + p(m - T) + pn_F(e_F) + (p - 1)T = p(m + n_F(e_F)) = v_E(\Gamma).$$

The characters χ_U^0, χ_E^0 have conductor T , at most half the conductor of each pullback. Equation (13) gives

$$\Delta_L(\theta_L) = \chi_L^0(\Gamma/\beta) \Delta_L(\lambda \circ n_L), \quad L = U, E.$$

The multipliers agree by the common restriction to $F^\times$. By (2) for L/F and (13) over F ,

$$\prod_{\nu \in S(L/F)} \Delta_F(\nu \lambda) = \Delta_F(\lambda)^p \prod_{\nu \in S(L/F)} \nu(\Gamma/\beta) = \Delta_F(\lambda)^p.$$

The character product is trivial in an odd cyclic group, and the product of its individual local constants is 1 by inverse pairing. Therefore $\Delta_U(\lambda n_U) = \Delta_E(\lambda n_E)$ and the stable quotient is exactly 1.

Proof of Theorem 6.1. The preceding argument proves the equality for $h \geq T$. For $0 \leq h \leq t$, (58), Lemmas 6.10–6.12, and Lemma 6.13 show that the quotient belongs to μ_4 . Its p th power is 1 by Corollary 4.3. Since p is odd, $\mu_p \cap \mu_4 = \{1\}$, and the quotient is 1. The lower cyclic products are 1 by the same corollary. These cases exhaust all possible conductors, and every step was valid in either characteristic. □

7. ARTIN–SCHREIER COORDINATES AND TRACE–NORM ESTIMATES

7.1. The ramification data and valuation conventions. Let B/A be totally ramified and Galois with $\text{Gal}(B/A) = C_p^2$, where $p > 2$ is the residue characteristic. Choose distinct degree- p intermediate fields B_1, B_2 with breaks $t = t_1 \leq t_2$, such that B/B_2 and B/B_1 have breaks t and $t' = t + p\delta$, respectively, where $\delta = t_2 - t \geq 0$. Write

$$N_i = N_{B/B_i}, \quad S_i = \text{Tr}_{B/B_i}, \quad n_i = N_{B_i/A}, \quad T_i = \text{Tr}_{B_i/A}.$$

All valuations v_B, v_i, v_A are normalized on their own fields. In particular $v_B|_{B_i} = pv_i$, $v_i|_A = pv_A$, and $v_A(n_i y) = v_i(y)$. The different exponents of B_i/A are $D_i = (p-1)(t_i + 1)$; that of B/B_1 is $D' = (p-1)(t' + 1)$.

For a totally ramified cyclic extension L/M of degree p and break u , (7) and (11) give

$$(65) \quad \text{Tr}_{L/M}(\mathfrak{p}_L^a) = \mathfrak{p}_M^{\lfloor (a+(p-1)(u+1))/p \rfloor},$$

$$(66) \quad v_M(E_i^{L/M}(y)) \geq \lfloor (iv_L(y) + (p-1)(u+1))/p \rfloor, \quad 1 \leq i < p.$$

The exponent a and the valuation of y may be negative. We also use Lemmas C.1 and C.3. In mixed characteristic put $V = v_B(p)$. Applying (65) to 1 on B_2/A gives

$$(67) \quad V \geq p(p-1)t_2.$$

In equal characteristic all expressions divisible by p below are zero.

Lemma 7.1 (Powers and norms). *For every $y \in L$ in the preceding cyclic extension,*

$$v_L(y^p - N_{L/M} y) \geq pv_L(y) + (p-1)u.$$

Proof. For $1 \leq i < p$, (66) implies $pv_M(E_i(y)) \geq iv_L(y) + (p-1)u$. Every intermediate term $E_i(y)y^{p-i}$ of the characteristic polynomial therefore has the displayed valuation. Its constant term is $-Ny$ because p is odd. $\square$

Lemma 7.2 (Best approximation). *Let L/M be a finite totally ramified extension of local fields with ramification index e , and let σ be an M -automorphism of L . If $y \in L$ and $\sigma y \neq y$, there is $a \in M$ such that*

$$q := v_L(y - a) = \max_{b \in M} v_L(y - b), \quad e \nmid q.$$

Moreover

$$(68) \quad q \leq v_L(\sigma y - y).$$

Proof. For every $b \in M$,

$$\sigma(y - b) - (y - b) = \sigma y - y.$$

Since σ preserves the valuation,

$$v_L(y - b) \leq v_L(\sigma y - y).$$

The right side is finite. Thus the possible values $v_L(y - b)$ form a nonempty set of integers bounded above, so they have a largest value q . Choose a attaining it.

Suppose $e \mid q$, say $q = ek$. Choose a uniformizer π_M of M . Then

$$v_L((y - a)\pi_M^{-k}) = 0.$$

Since L and M have the same residue field, choose $u \in \mathcal{O}_M^\times$ with the same residue as $(y - a)\pi_M^{-k}$. Put $c = u\pi_M^k \in M$. Then $v_L((y - a) - c) > q$, so $a + c$ is a better approximation to y , a contradiction. Thus $e \nmid q$. $\square$

Remark 7.3. The formulation of Lemma 7.2 was suggested by the Lean formalization. The explicit bound (68) replaces the closed-subfield argument for the existence of a best approximation. The same lemma applies both in Proposition 7.4 below and in Lemma 9.32.

7.2. An Artin–Schreier equation for B/B_2 .

Proposition 7.4. *There exists $\Delta \in B$ such that*

$$\Delta^p - \Delta = a \in B_2, \quad v_B(\Delta) = -t, \quad v_2(a) = -t.$$

In this situation $p \nmid t$ and $B = B_2(\Delta)$.

Proof. Let σ generate $\text{Gal}(B/B_2)$. First suppose the characteristic is zero and put $\kappa = V - (p-1)t > 0$. Equation (7) for B/B_2 gives

$$S_2(\mathfrak{P}_B^\kappa) = p\mathcal{O}_{B_2}.$$

Choose $\eta \in \mathfrak{P}_B^\kappa$ with $S_2\eta = -p$. Since $S_2(1 + \eta) = 0$, additive Hilbert 90 gives $d_0 \in B$ with

$$\sigma d_0 - d_0 = 1 + \eta.$$

Apply Lemma 7.2 to d_0 over B_2 . Choose $b \in B_2$ and put $d = d_0 - b$, so that

$$q := v_B(d) = \max_{c \in B_2} v_B(d_0 - c), \quad p \nmid q.$$

The displacement is unchanged and has valuation zero. The first ramification term therefore gives

$$0 = v_B(\sigma d - d) = q + t.$$

Thus $v_B(d) = -t$ and $p \nmid t$.

Put $a_0 = d^p - d$. The equation $\sigma d - d = 1 + \eta$, the binomial theorem, and $v_B(d) = -t$ give

$$v_B(\sigma a_0 - a_0) \geq \kappa.$$

If $a_0 \in B_2$, put $a = a_0$ and $\Delta = d$. Suppose instead that $a_0 \notin B_2$. Apply Lemma 7.2 to a_0 and choose $a \in B_2$ with $q_0 = v_B(a_0 - a)$ maximal. Then $p \nmid q_0$, so the first ramification term gives

$$v_B(\sigma a_0 - a_0) = q_0 + t.$$

Hence

$$v_B(a_0 - a) = q_0 \geq \kappa - t = V - pt > 0.$$

For $f(X) = X^p - X - a$, the shifted polynomial $f(d + Z)$ has integral coefficients, unit linear coefficient, and constant coefficient in $\mathfrak{P}_B^{V-pt}$. Indeed $v_B(pd^{p-1}) = \kappa > 0$, and the other intermediate coefficients are integral. Hensel's lemma gives a root $\Delta \equiv d \pmod{\mathfrak{P}_B^{V-pt}}$.

In either case $v_B(\Delta) = -t$, and therefore

$$v_B(\Delta^p - \Delta) = -pt,$$

so $v_2(a) = -t$. Since $p \nmid t$, one has $\Delta \notin B_2$, and hence $B = B_2(\Delta)$.

In characteristic p , additive Hilbert 90 gives d_0 with $\sigma d_0 - d_0 = 1$. Apply Lemma 7.2 and write $d = d_0 - b$ with $b \in B_2$ and $v_B(d)$ maximal. The same first ramification calculation gives $v_B(d) = -t$ and $p \nmid t$. Now $d^p - d$ is fixed by σ , so take

$$\Delta = d, \quad a = d^p - d \in B_2.$$

Then $v_2(a) = -t$ and $B = B_2(\Delta)$ as above. □

Set

$$e_i = E_i^{B/B_1}(\Delta), \quad b = N_1\Delta = e_p, \quad s = -e_{p-1}, \quad K_0 = 1 + pt_2.$$

Then $N_2\Delta = a$ and $v_1(b) = -t$. Because the polynomial over B_2 is exactly $X^p - X - a$,

$$(69) \quad S_2(\Delta^j) = 0 \quad (1 \leq j < p-1), \quad S_2(\Delta^{p-1}) = p-1, \quad S_2(\Delta^p) = pa.$$

7.3. Coefficient extraction and twisted estimates.

Lemma 7.5 (Coefficient extraction). *If $\theta \in B_1$ and $\theta = \sum_{i=0}^{p-1} Y_i \Delta^i$ with $Y_i \in B_2$, then*

$$v_2(Y_i) \geq v_1(\theta) + it \quad (0 \leq i < p).$$

Proof. Since $p \nmid t$, the valuations $pv_2(Y_i) - it$ are distinct modulo p . Each is at least $pv_1(\theta)$, proving the assertion for $i = 0$ and the preliminary bound for every i . For each nonzero $\xi \in \mu_{p-1} \subset A$, a conjugate of Δ is

$$\Delta_\xi = \Delta + \xi + \eta_\xi, \quad v_B(\eta_\xi) \geq V - (p-1)t.$$

This follows by applying Hensel's lemma to the shifted polynomial at $\Delta + \xi$; its constant coefficient has that valuation and its derivative is a unit. In equal characteristic $\eta_\xi = 0$.

Induct simultaneously for all $\theta \in B_1$ on j . The restriction of this conjugating automorphism to B_1 has break t , so $v_1(\theta_\xi - \theta) \geq v_1(\theta) + t$. The error obtained by replacing Δ_ξ by $\Delta + \xi$ in the expansion of θ_ξ has valuation at least

$$pv_1(\theta) + V - (p-2)t.$$

Its Δ^{j-1} coefficient has v_2 at least $v_1(\theta) + jt$, because $V \geq p(p-1)t \geq (p-1)(j+1)t$. The induction hypothesis applied to $\theta_\xi - \theta$ thus yields

$$\sum_{i=j}^{p-1} \binom{i}{j-1} \xi^{i-j+1} Y_i \in \mathfrak{P}_{B_2}^{v_1(\theta)+jt}.$$

Multiply by ξ^{-1} and sum over μ_{p-1} . The power sums cancel every term except $(p-1)jY_j$; this multiplier is a unit. $\square$

Lemma 7.6 (Twisted trace and symmetric functions). *For $Y \in B_2$ and $1 \leq i < p$,*

$$(70) \quad v_1(S_1(Y\Delta^i)) \geq (p-1)t_2 - it + v_2(Y),$$

$$(71) \quad v_1(E_i^{B/B_1}(Y\Delta)) \geq (p-1)t_2 + i(v_2(Y) - t).$$

Also $v_1(S_1Y) \geq (p-1)t_2 + v_2(Y)$. In particular

$$(72) \quad v_1(e_i) \geq (p-1)t_2 - it \quad (i < p), \quad v_1(s) \geq (p-1)\delta.$$

Proof. The assertion for $S_1Y = T_2Y \in A$ follows from (65). For $i \geq 1$ put $c = (p-1)t_2 - it + v_2Y$. By trace duality it suffices to show $T_1(\theta S_1(Y\Delta^i)) \in \mathcal{O}_A$ for every $v_1(\theta) \geq -D_1 - c$. Write $\theta = \sum Y_j \Delta^j$ and use trace transitivity. For $1 \leq k \leq 2p-2$, the only nonzero power traces over B_2 are those of degrees $p-1, p, 2p-2$, with values $p-1, pa, p-1$. Thus

$$S_2(\theta \Delta^i) = (p-1)Y_{p-1-i} + paY_{p-i} + \mathbf{1}_{i=p-1}(p-1)Y_{p-1}.$$

Lemma 7.5 shows that, after multiplication by Y , all three terms lie in $\mathfrak{P}_{B_2}^{-D_2}$. For the first this is exactly

$$v_2Y + (p-1-i)t - D_1 - c = -D_2;$$

for the second use $v_2(pa) \geq -t$, and for the last there is an additional nonnegative $(p-1)t$. This proves the trace estimate. Newton's identity for $E_i(Y\Delta)$ involves the power traces $S_1(Y^j \Delta^j)$, $1 \leq j \leq i < p$. Each product with total weight i has valuation at least $(p-1)t_2 + i(v_2Y - t)$, and i is a unit. $\square$

7.4. The trace-norm congruence.

Theorem 7.7. *One has*

$$T_1(b) - T_2(a) \in \mathfrak{p}_A^{1+t_2}.$$

Proof. Compare the coefficient of Z^p in the exact identity

$$n_1(N_1(1 - Z\Delta)) = n_2(N_2(1 - Z\Delta)).$$

The upper polynomials are

$$N_1(1 - Z\Delta) = 1 + \sum_{i=1}^{p-1} (-1)^i e_i Z^i - bZ^p, \quad N_2(1 - Z\Delta) = 1 - Z^{p-1} - aZ^p.$$

The right coefficient is $-T_2a$. Besides $-T_1b$, the left coefficient contains $-n_1(e_1)$ and traces of monomials in conjugates of the e_i with $i < p$, at least two factors, and total weight p . Indeed the cyclic group permutes the selections from the p factors; every orbit has size p , except the constant selection e_1 from every factor, which gives $n_1(e_1)$. By (72), each trace of a nonconstant selection has valuation at least

$$\left\lfloor \frac{2(p-1)t_2 - pt + D_1}{p} \right\rfloor.$$

This is at least $1 + t_2$ whenever $(p-2)t_2 - t - 1 \geq 0$. The same condition makes $v_A(n_1e_1) = v_1(e_1) \geq (p-1)t_2 - t \geq 1 + t_2$. This covers $p \geq 5$ and $p = 3, t_2 > t$.

For $p = 3, t_2 = t$, first note that $T_1e_1 = T_1S_1\Delta = 0$ by (69). We show $e_1 \in \mathfrak{P}_{B_1}^{t+1}$ directly from the inverse different. Here B_1/A is totally ramified of degree 3 and has different exponent $D_1 = 2(t+1)$. By (8), it is enough to prove

$$T_1(e_1y) \in \mathcal{O}_A \quad \left(y \in \mathfrak{P}_{B_1}^{-D_1-t-1} = \mathfrak{P}_{B_1}^{-3(t+1)} \right).$$

Since the residue degree is one, choose $c \in \mathfrak{p}_A^{-t-1}$ such that

$$y - c \in \mathfrak{P}_{B_1}^{-3(t+1)+1}.$$

Then $T_1(e_1c) = cT_1(e_1) = 0$, while

$$e_1(y - c) \in \mathfrak{P}_{B_1}^{t-3(t+1)+1} = \mathfrak{P}_{B_1}^{-D_1},$$

whose trace is integral by the definition of the inverse different. Thus $e_1 \in \mathfrak{P}_{B_1}^{t+1}$, and hence $v_A(n_1(e_1)) \geq t+1$. Every remaining weight-three selection has one e_1 factor and one e_2 factor. Its trace has valuation at least $\lfloor (t+1+2(t+1))/3 \rfloor = t+1$. $\square$

7.5. Norms of the elements $\Delta + \xi$. Put $H = pt_2 + t$. For $x \in B_2$ and $1 \leq k \leq p$, write

$$X = n_2x, \quad R = v_B(x\Delta^{k-1}) = pv_2x - (k-1)t.$$

Lemma 7.8. *If $R \geq 1$, then*

$$XT_1(b^k) \equiv X \sum_{\xi \in \mathcal{T}} N_1(\Delta + \xi)^k \pmod{\mathfrak{P}_{B_1}^{pt_2+R}}, \quad \mathcal{T} = \{0\} \cup \mu_{p-1}.$$

Proof. It is exact in characteristic p . In mixed characteristic fix $\xi \neq 0$, put $D = \Delta + \xi$ and write the corresponding conjugate as $D(1+u)$. Polynomial division gives the exact polynomial

$$Q_\xi(\Delta) = \frac{(\Delta + \xi)^p - (\Delta + \xi) - (\Delta^p - \Delta)}{\Delta + \xi}.$$

It has degree $p-2$, leading coefficient $p\xi$, and every coefficient is in $p\mathcal{O}_A$. The equation for u is

$$u = Q_\xi + pD^{p-1}u + D^{p-1} \sum_{j=2}^p \binom{p}{j} u^j.$$

Hence

$$v_B u \geq V - (p-2)t, \quad v_B(u - Q_\xi) \geq 2V - (2p-3)t.$$

For $p \geq 5$, the trace of the latter error has valuation at least H : subtracting H from its lower bound gives

$$(p-3)t - \lfloor (p-2)t/p \rfloor + (2p-3)\delta \geq 0.$$

The symmetric functions $E_i(u)$ with $2 \leq i < p$ have at least this precision, and $v_1(N_1u) = v_B u \geq V - (p-2)t \geq H$. Terms of degree at most $p-3$ in Q_ξ have traces of valuation at least $2(p-1)t_2 - (p-3)t \geq H$ by Lemma 7.6.

For $p = 3$ retain one further iterate:

$$Q_\xi = 3\xi\Delta, \quad u = Q_\xi + 3D^2Q_\xi + E, \quad v_BE \geq 3V - 5t.$$

The trace of E has valuation at least $V - t + 2\delta \geq H$. Moreover

$$3D^2Q_\xi = 9\xi\Delta^3 + 18\xi^2\Delta^2 + 9\xi^3\Delta.$$

Using $\Delta^3 = \Delta + a$, each trace on the right has valuation at least $6t_2 - 2t \geq H$. The other norm coefficients satisfy $v_1(E_2(u)) \geq 2V/3 + 2\delta \geq H$ and $v_1(N_1u) \geq V - t \geq H$. Thus in all cases

$$(73) \quad N_1(1+u) \equiv 1 + \xi L \pmod{\mathfrak{P}_{B_1}^H}, \quad L = pS_1(\Delta^{p-2}), \quad v_1L \geq pt + 2(p-1)\delta.$$

Since $2v_1L \geq H$, the same formula raised to the k th power has linear term $k\xi L$.

Now $N_1(\Delta + \xi) - b$ is integral by (72); hence $v_1(N_1(\Delta + \xi)^k - b^k) \geq -kt + t$. After multiplying by X , the error term in (73) is in $\mathfrak{P}_{B_1}^{pt_2+R}$ because $v_1(Xb^k) = R - t$. In the linear term, replacing $N_1(\Delta + \xi)^k$ by b^k gives an error of valuation at least $R + pt + 2(p-1)\delta \geq R + pt_2$. The remaining expression vanishes after summation because $\sum_{\xi \in \mu_{p-1}} \xi = 0$ exactly. Since $\text{Gal}(B/A)$ is abelian, the sum over conjugates is $T_1(b^k)$. $\square$

7.6. Comparison of two power sums. Define

$$P_k = b^k + \sum_{\xi \in \mu_{p-1}} N_1(\Delta + \xi)^k, \quad Q_k = S_1((\Delta^p - \Delta)^k) = T_2(a^k).$$

Lemma 7.9. *With X, R as above and $R \geq 1$,*

$$X(P_k - Q_k) \equiv (p-1)\mathbf{1}_{k=p-1}X(1 + e_{p-1}^p) \pmod{\mathfrak{P}_{B_1}^{pt_2+R}}.$$

Proof. For a monomial $e_\lambda = \prod e_i^{\lambda_i}$ put $r = \sum \lambda_i$, $n = \sum i\lambda_i$, and $\nu = \sum_{i < p} \lambda_i$. The exact Teichmüller power sums show that P_k contains it only if $n \equiv k \pmod{p-1}$, with coefficient

$$(p-1) \binom{k}{r} \frac{r!}{\prod \lambda_i!},$$

plus the separate term b^k . Write $n = (p-1)v + k$. To keep the Newton coefficients integral from the start, introduce indeterminates $Z_1, \dots, Z_p$ and put

$$A = \sum_{i=1}^p Z_i, \quad B = \sum_{i=1}^p iZ_i.$$

For $r > 0$ define

$$D_\lambda = [Z^\lambda] BA^{r-1} \in \mathbf{Z}.$$

This is an integer by definition. Euler's identity for the homogeneous polynomial A^r gives the denominator-free relation

$$rD_\lambda = n \binom{r}{\lambda_1, \dots, \lambda_p}.$$

Equivalently,

$$D_\lambda = \frac{n(r-1)!}{\prod \lambda_i!}.$$

Newton's generating-function formula therefore shows that the coefficient of e_λ in Q_k , for $0 \leq v \leq k$, is the integer

$$(-1)^{r+v} \binom{k}{v} D_\lambda = (-1)^{r+v} \binom{k}{v} \frac{n(r-1)!}{\prod \lambda_i!}.$$

The valuation estimate (72) gives

$$(74) \quad v_1(Xe_\lambda) \geq R - t + (p-1)((r-v)t + \nu\delta).$$

For $v \geq 0$ and $r \geq v + 2$, the inequality $pr - n = v + p(r-v) - k \geq p$ implies $\nu \geq 2$. Then (74) is at least $R - t + 2(p-1)t_2 \geq R + pt_2$. If $r = v$, only $v = r = k$, $e_\lambda = b^k$ is possible; its coefficients are $1 + (p-1) = p$ and p , so they cancel.

Suppose $r = v + 1 \leq k$. Their coefficient difference is exactly

$$\frac{(r-1)!}{\prod \lambda_i!} \left((p-1)r \binom{k}{r} + n \binom{k}{r-1} \right) = \frac{pk \binom{k}{r-1} (r-1)!}{\prod \lambda_i!}.$$

Here $r \binom{k}{r} = (k-r+1) \binom{k}{r-1}$ and $n = (p-1)(r-1) + k$. The factorial denominators are units at p : if $r = p$, then $k = p$ and $n \equiv 1 \pmod{p}$, so no λ_i can equal p . Except for the pure norm monomial, one has $\nu \geq 1$. The added valuation $v_1 p \geq (p-1)t_2$ then makes (74) at least $R + pt_2$. The pure norm possibility is $k = p, r = 1, v = 0, e_\lambda = b$; its coefficient difference is p^2 , which gives valuation at least $R + pt_2$ after applying (67).

If $v = k$ and $r = k + 1$, the first coefficient is absent and the difference is $pk k! / \prod \lambda_i!$. It is divisible by p except when $k = p - 1$ and $e_\lambda = e_{p-1}^p$, in which case it is exactly $p - 1$. Indeed for $k \leq p - 2$ all factorial denominators are units; for $k = p - 1$ the only nonunit denominator comes from that constant selection; for $k = p$ the numerator contains at least three factors of p and the factorial denominator at most one. The remaining monomials with $r \geq v + 2$ were already dealt with.

Finally $v = -1$ contributes only the constant monomial when $k = p - 1$, giving $p - 1$, or e_1 when $k = p$, giving $p(p - 1)e_1$. The latter has valuation at least $R + pt_2$ by (72) and (67). These cases exhaust the possible monomials and leave exactly the two terms displayed in the lemma. $\square$

Theorem 7.10 (A congruence for h). *Define*

$$h(z) = T_1 N_1(\Delta z) - T_2 N_2(\Delta z).$$

For $x \in B_2$, $0 \leq j \leq p - 1$, and $R = v_B(x\Delta^j) \geq 1$,

$$(75) \quad h(x\Delta^j) \equiv (p-1)\mathbf{1}_{j=p-2}n_2(x)(1-s^p) \pmod{\mathfrak{P}_{B_1}^{pt_2+R}}.$$

Proof. Put $k = j + 1$. By multiplicativity of the norm, the difference between $h(x\Delta^{k-1})$ and $n_2(x)(T_1 b^k - T_2 a^k)$ is $T_2((n_2 x - x^p)a^k)$. By Lemma 7.1, the argument has v_2 at least $R - t + (p-1)t_2$. Its trace, viewed in B_1 , has valuation at least $R - t + 2(p-1)t_2 \geq R + pt_2$. Apply Lemmas 7.8 and 7.9. Since p is odd and $s = -e_{p-1}$, $1 + e_{p-1}^p = 1 - s^p$. $\square$

7.7. A congruence for g . Put

$$g(z) = T_1(bS_1 z) - T_2(aS_2 z).$$

Expand $b = \sum_{l=0}^{p-1} Y_l \Delta^l$ over B_2 and $s = \sum s_l \Delta^l$ over B_2 .

Lemma 7.11. *The coefficients satisfy*

$$\begin{aligned} v_2(Y_l) &\geq (p-1)t_2 - (p-l)t & (2 \leq l < p), \\ v_2(Y_0 - a) &\geq (p-1)t_2 - t, & Y_1 = 1 - s_0 + E, \quad v_2 E \geq (p-1)t_2. \end{aligned}$$

Proof. Use $b = a + \Delta + \sum_{i=1}^{p-1} (-1)^i e_i \Delta^{p-i}$. Expand each e_i over B_2 by Lemma 7.5; its Δ^k coefficient has valuation at least $(p-1)t_2 - it + kt$. If $k + p - i \geq p$, reduce once using $\Delta^p = \Delta + a$. The unreduced contribution to Y_l has valuation $(p-1)t_2 - (p-l)t$; each reduced contribution has valuation at least $(p-1)t_2 + (l-1)t$. For $l = 0$ only the terms with $k = i$ contribute, with a factor a . For $l = 1$ the sole low term is the constant coefficient of e_{p-1} , namely $-s_0$; all others have valuation at least $(p-1)t_2$. $\square$

Proposition 7.12. *Let $x \in B_2$ and $0 \leq j < p$. If $j \neq p - 2$ and $x \in \mathfrak{p}_{B_2}$, then*

$$g(x\Delta^j) \in \mathfrak{p}_A^{1+t_2}.$$

If $j = p - 2$ and $R = v_B(x\Delta^{p-2}) \geq 1 + \delta$, there is a fixed $\omega \in B_2$ such that

$$g(x\Delta^{p-2}) = T_2(x(\omega - 1)), \quad x\omega - xs \in \mathfrak{P}_B^{K_0}.$$

Proof. Trace transitivity gives, for every $z \in B$,

$$g(z) = T_2(S_2((b-a)z)).$$

Write

$$b-a = \sum_{l=0}^{p-1} Z_l \Delta^l, \quad Z_0 = Y_0 - a, \quad Z_1 = Y_1, \quad Z_l = Y_l \ (l \geq 2).$$

Lemma 7.11 gives

$$Z_1 \in \mathcal{O}_{B_2}, \quad Z_l \in \mathfrak{p}_{B_2}^{t_2} \ (l \neq 1).$$

The power traces $S_2(\Delta^k)$ are integral for $0 \leq k \leq 2p-2$. Moreover, by (69) and $\Delta^p = \Delta + a$,

$$S_2(\Delta^{j+1}) \in \mathfrak{p}_{B_2}^{t_2} \quad (0 \leq j < p, \ j \neq p-2);$$

for $j = p-1$ this is pa , which has depth at least t_2 by (67), and in the other nonexceptional cases it is zero. Hence, if $x \in \mathfrak{p}_{B_2}$ and $j \neq p-2$, every term in

$$S_2((b-a)x\Delta^j) = \sum_{l=0}^{p-1} xZ_l S_2(\Delta^{l+j})$$

belongs to $\mathfrak{p}_{B_2}^{1+t_2}$. Its trace therefore belongs to $\mathfrak{p}_A^{1+t_2}$, proving the first assertion.

For $j = p-2$, put

$$\omega = 1 + (p-1)Y_1 + paY_2.$$

The exact moment formula gives $g(x\Delta^{p-2}) = T_2(x(\omega-1))$. Lemma 7.11 gives

$$\omega - s = p(1-s_0) + (s_0-s) + (p-1)E + paY_2.$$

After multiplication by x , every term except $s_0 - s$ has valuation at least K_0 , by (67) and Lemma 7.11. For the remaining term, Lemma 7.5 gives

$$v_B(s-s_0) \geq p(p-1)\delta + (p-1)t.$$

Since $v_B x = R + (p-2)t$, the required lower bound differs from this one by $(p-3)t + (p-1)^2\delta \geq 0$. Thus $x\omega - xs \in \mathfrak{P}_B^{K_0}$. $\square$

Remark 7.13. This proof was suggested by the Lean formalization. For $j \neq p-2$ it improves the original argument: the single expansion above only requires $x \in \mathfrak{p}_{B_2}$ and treats all nonexceptional exponents at once. The original proof used the stronger condition $v_B(x\Delta^j) \geq 1 + \delta$ and treated $j = 0, 1 \leq j \leq p-3$, and $j = p-1$ separately.

7.8. Comparison of norms of close elements.

Lemma 7.14. *Let $x \in B_2$, $w \in B_1$, and $y \in B_2$ satisfy*

$$xw - y \in \mathfrak{P}_B^{K_0}, \quad h_0 = v_B(xw) \geq 1 + t_2 + (p-3)t.$$

Then

$$N_1(xw) - n_2(y) \in \mathfrak{P}_{B_1}^{K_0}.$$

Proof. The zero case is immediate, so write $z = xw \neq 0$ and expand $z = \sum_{i=0}^{p-1} y_i \Delta^i$, with $y_i = xc_i \in B_2$. Lemma 7.5 gives

$$v_B(y_i \Delta^i) \geq h_0 + (p-1)it.$$

Thus $v_B(y_0) = h_0$, and every nonconstant term is strictly deeper. The valuations of the nonzero terms in

$$z - y = (y_0 - y) + \sum_{i=1}^{p-1} y_i \Delta^i$$

have distinct residues modulo p , because $p \nmid t$. Since their sum has valuation at least K_0 , each term does. In particular $v_B(y_i \Delta^i) \geq K_0$ for $i \geq 1$ and $v_2(y_0 - y) \geq t_2 + 1$.

First keep the linear term and put $z_\ell = y_0 + y_1 \Delta$. Multiplicativity of the norm gives

$$N_1(z_\ell) = n_2(y_0)N_1(1 + (y_1/y_0)\Delta).$$

We have $v_1(n_2y_0) = h_0$ and $v_2(y_1/y_0) \geq t$. The twisted symmetric estimate therefore bounds the intermediate terms in $N_1(z_\ell) - n_2y_0$ by

$$h_0 + (p-1)t_2 + j(v_2(y_1/y_0) - t) \geq h_0 + (p-1)t_2 \geq K_0.$$

The remaining term is $N_1(y_1\Delta)$ and has v_1 equal to $v_B(y_1\Delta) \geq K_0$. Hence $N_1(z_\ell) - n_2y_0 \in \mathfrak{P}_{B_1}^{K_0}$.

Now put $U = \sum_{i=2}^{p-1} y_i \Delta^i$. Then $v_B(U/z) \geq 2(p-1)t$ and $v_B U \geq K_0$. The exact identity

$$N_1 z - N_1 z_\ell = N_1 z \{1 - N_1(1 - U/z)\}$$

is expanded over B/B_1 , whose different exponent is D' . For each intermediate coefficient the strong symmetric bound gives

$$v_1(N_1 z E_j(U/z)) \geq h_0 + \lfloor [2j(p-1)t + D']/p \rfloor \geq K_0 + \lfloor [(p-3)t + p-1]/p \rfloor \geq K_0.$$

The minimum occurs at $j = 1$, and the second inequality follows by inserting $h_0 \geq 1 + t_2 + (p-3)t$ and $t' = t + p\delta$. The remaining term is $N_1 U$, which has valuation $v_B U \geq K_0$. Thus $N_1 z - N_1 z_\ell \in \mathfrak{P}_{B_1}^{K_0}$.

Finally y_0, y are integral and $v_2(y_0 - y) \geq t_2 + 1$. To check the last norm comparison explicitly, set $a = v_2 y$. If $a \geq t_2 + 1$ (or $y = 0$), both norms have valuation at least $t_2 + 1$. Otherwise factor out y , apply the strong symmetric estimate to $(y_0 - y)/y$, and note that its shallowest intermediate term has valuation at least

$$a + \lfloor [t_2 + 1 - a + (p-1)(t_2 + 1)]/p \rfloor = t_2 + 1 + \lfloor (p-1)a/p \rfloor \geq t_2 + 1.$$

The norm term has valuation $v_2(y_0 - y) \geq t_2 + 1$. Consequently $n_2y_0 - n_2y \in \mathfrak{p}_A^{t_2+1}$, which lies in $\mathfrak{P}_{B_1}^{p(t_2+1)} \subseteq \mathfrak{P}_{B_1}^{K_0}$. Combining the three comparisons proves the assertion. $\square$

8. COMPATIBLE CHARACTERS IN TOTALLY RAMIFIED ODD DEGREE

8.1. Construction of characters of conductors m_1, m_2 .

The local data. Let $A \subset B_1, B_2 \subset B$ be totally ramified, $\text{Gal}(B/A) = C_p^2$, p odd. Write

$$t_1 = t, \quad t_2 = t + \delta, \quad t' = t + p\delta, \quad p \nmid t, \quad \kappa_2 = t_2 + 1, \quad T' = t' + 1.$$

The breaks of B/B_1 and B/B_2 are t' and t , respectively. Put $N_i = N_{B/B_i}$, $n_i = N_{B_i/A}$, $S_i = \text{Tr}_{B/B_i}$, $T_i = \text{Tr}_{B_i/A}$. Normalize each valuation on its own field. Proposition 7.4 constructs $\Delta \in B$ with

$$\Delta^p - \Delta = a \in B_2, \quad v_B \Delta = -t, \quad N_2 \Delta = a, \quad b = N_1 \Delta, \quad s = -E_{p-1}^{B/B_1}(\Delta).$$

The estimates of Section 7 give

$$(76) \quad T_1 b - T_2 a \in \mathfrak{p}_A^{\kappa_2}, \quad v_B(b - a) \geq -t,$$

$$(77) \quad v_1 s \geq (p-1)\delta, \quad v_B(p) \geq p(p-1)t_2.$$

For $w = \sum x_j \Delta^j$, $x_j \in B_2$, the term valuations are distinct modulo p . For a coefficient $x \in B_2$ set

$$g(z) = T_1(bS_1 z) - T_2(aS_2 z), \quad h(z) = T_1 N_1(\Delta z) - T_2 N_2(\Delta z), \quad R = v_B(x \Delta^j).$$

If $R \geq 1 + \delta$, $g(x \Delta^j)$ is in $\mathfrak{p}_A^{\kappa_2}$ except at $j = p-2$. There is $\omega \in B_2$, independent of x , such that in that exception

$$(78) \quad g(x \Delta^{p-2}) = T_2(x(\omega - 1)), \quad x\omega - xs \in \mathfrak{P}_B^{1+pt_2}.$$

For $R \geq 1$, Theorem 7.10 gives

$$(79) \quad h(x \Delta^j) \equiv (p-1) \mathbf{1}_{j=p-2} n_2 x (1 - s^p) \pmod{\mathfrak{P}_{B_1}^{pt_2+R}}.$$

Finally, if $x \in B_2, w \in B_1, y \in B_2$, $v_B(xw) \geq 1 + t_2 + (p-3)t$, and $xw - y \in \mathfrak{P}_B^{1+pt_2}$, then

$$(80) \quad n_2 x w^p \equiv n_2 y \pmod{\mathfrak{P}_{B_1}^{1+pt_2}}.$$

Equations (78) and (80) are Proposition 7.12 and Lemma 7.14. These results remain valid after an unramified extension of A .

Choose a nontrivial norm character $\tau \in S(B_2/A)$. Let $e_L = e_A \text{Tr}_{L/A}$, and choose $\alpha \in A^\times$ so

$$\tau(1-z) = e_A(\alpha P(z)), \quad P(X) = \sum_{j=1}^{p-1} X^j/j, \quad z \in \mathfrak{p}_A^c, \quad c = \lceil \kappa_2/p \rceil.$$

Put $\Psi_L(x) = e_L(\alpha x)$. Their largest trivial ideals are

$$\mathfrak{p}_A^{\kappa_2}, \quad \mathfrak{P}_{B_1}^{T'}, \quad \mathfrak{P}_{B_2}^{\kappa_2}, \quad \mathfrak{P}_B^{T'}.$$

Corollary 6.4 gives

$$(81) \quad \Psi_A(T_2 v + n_2 v) = 1 \quad (v \in \mathfrak{P}_{B_2}^c).$$

In equal characteristic, expressions divisible by p are zero, with valuation $+\infty$.

The groups H_i . Set

$$m_1 = 1 + t + t', \quad m_2 = 1 + t + t_2, \quad q_i = \lceil m_i/p \rceil, \quad q = q_1 = \delta + \lceil (2t+1)/p \rceil, \quad H_i = U_{B_i}^{q_i}.$$

Define characters on these groups by

$$(82) \quad R_1(1-z) = \Psi_1(bPz), \quad R_2(1-z) = \Psi_2(aPz).$$

They are characters because $pq_i \geq m_i$ and the truncated logarithm is additive for the multiplicative group modulo $\mathfrak{P}_{B_i}^{m_i}$. Their conductors are m_i , since $v_1(b) = v_2(a) = -t$.

Lemma 8.1. *One has $q_i \geq 1$, $q \leq t'$, $q_2 \leq t_2$ and*

$$q \geq q_2, \quad \lfloor [q + (p-1)(t+1)]/p \rfloor \geq q_2.$$

Consequently

$$(83) \quad N_1^{-1}(H_1) = U_B^q, \quad N_2(U_B^q) \subset H_2.$$

For any unramified base extension the same assertions hold.

Proof. The inequalities follow by writing $q = \delta + \lceil (2t+1)/p \rceil$ and $q_2 = \lceil (2t+\delta+1)/p \rceil$. For the floor inequality it is enough that $q + (p-1)(t+1) \geq pq_2$; use $pq_2 \leq 2t + \delta + p$ and $(p-3)t + \lceil (2t+1)/p \rceil - 1 \geq 0$. The inequalities $q \leq t'$ and $q_2 \leq t_2$ follow from $2t+1 \leq pt$ for $p \geq 3, t \geq 1$ and $\delta \geq 0$. For B/B_1 and depths less than t' , the norm preserves the first nonzero unit depth; at the critical depth its image is still in that depth. Thus a norm in H_1 has a preimage of depth at least q , including the case $q = t'$. The converse follows from [7, Theorem 3.7]. The second assertion follows coefficientwise from (11), using the two displayed inequalities. Norm valuations first exclude elements outside U_B^1 from the first preimage assertion. $\square$

The value of $\Psi_A(g(x\Delta^j) + h(x\Delta^j))$.

Lemma 8.2. *For $x \in B_2$, $0 \leq j < p$, $v_B(x\Delta^j) \geq q$, one has*

$$\Psi_A(g(x\Delta^j) + h(x\Delta^j)) = 1.$$

Proof. We have $q \geq 1 + \delta$. Outside $j = p-2$ this follows at once from (78)–(79): $h(x\Delta^j)$ belongs to A and to $\mathfrak{P}_{B_1}^{1+pt_2}$, hence to $\mathfrak{p}_A^{\kappa_2}$. For $j = p-2$, $pv_2x \geq q + (p-2)t \geq \kappa_2$, so $v_2x \geq c$. Put $K_0 = 1 + pt_2$. The difference between $p-1$ and -1 in (79) is in $\mathfrak{P}_{B_1}^{K_0}$ because

$$v_1p + pv_2x \geq (p-1)t_2 + q + (p-2)t \geq K_0,$$

and s is integral. Also

$$v_B(xs) \geq q + (p-2)t + p(p-1)\delta \geq 1 + t_2 + (p-3)t.$$

Using (78) and (80) gives $n_2x s^p \equiv n_2(x\omega) \pmod{\mathfrak{P}_{B_1}^{K_0}}$. The approximation $x\omega \equiv xs \pmod{\mathfrak{P}_B^{K_0}}$ also implies $v_2(x\omega) \geq c$. Thus $g(x\Delta^{p-2}) + h(x\Delta^{p-2})$ is congruent in A to

$$-T_2x - n_2x + T_2(x\omega) + n_2(x\omega) \pmod{\mathfrak{p}_A^{\kappa_2}}.$$

Equation (81) gives character value one for each pair. $\square$

A consequence of additivity after unramified extension.

Lemma 8.3. *Let M be a local field of residue characteristic p , and let Ψ_M be a continuous additive character. Write $f(Z) = \sum_{n=1}^D a_n Z^n \in M[Z]$, where $D < p^2$, and suppose $x \mapsto \Psi_M(a_n x)$ has order dividing p on $\mathcal{O}_M$ for each n . Let M'/M be unramified of degree prime to p , with absolute residue degree at least 3, and set $\Psi_{M'} = \Psi_M \circ \text{Tr}_{M'/M}$. For $\lambda \in k_{M'}$, let $[\lambda]$ be its Teichmüller lift (the coefficient-field lift in equal characteristic). Suppose*

$$\lambda \mapsto \Psi_{M'}(f([\lambda]))$$

is additive on $k_{M'}$. Then

$$\Psi_M(f(1)) = \Psi_M(a_1 + a_p),$$

where missing coefficients are zero.

Proof. A Teichmüller carry belongs to $p\mathcal{O}_{M'}$, and hence is killed by each coefficient character. Therefore these characters on Teichmüller lifts are residue additive characters, of the form $\psi_{\mathbf{F}_p} \text{Tr}_{k_{M'}/\mathbf{F}_p}(\beta_n \lambda)$; unramified trace shows that each β_n is the scalar extension of its base coefficient. The displayed function is consequently

$$\psi_{\mathbf{F}_p} \text{Tr}_{k_{M'}/\mathbf{F}_p} \left(\sum_n \beta_n \lambda^n \right).$$

Under the absolute trace, exponents in the same Frobenius orbit are identified. Among positive exponents less than p^2 , with at least three base- p digits available in the residue field, the only identifications are n with pn when both occur. This follows by rotating the base- p digits modulo $|k_{M'}| - 1$: a two-nonzero-digit exponent cannot return below p^2 except at itself. Distinct polynomial monomials of degrees less than $|k_{M'}|$ are linearly independent as functions. An additive function can therefore retain only the orbit of the exponents $1, p$; every other orbit has zero trace contribution. Evaluating at 1 proves the assertion over M' . Both sides are p th roots of unity and the unramified degree is prime to p , so the equality descends to M . $\square$

Compatibility on U_B^q . Define on U_B^q

$$\Xi(u) = R_1(N_1 u) R_2(N_2 u)^{-1}.$$

It is a multiplicative character by Lemma 8.1. For $z \in \mathfrak{P}_B^q$ put

$$(84) \quad F_z(Z) = T_1(bP(1 - N_1(1 - Zz))) - T_2(aP(1 - N_2(1 - Zz))).$$

This is an exact polynomial of degree at most $p(p-1) < p^2$ and $\Xi(1 - Zz) = \Psi_A(F_z(Z))$ for integral base scalars Z .

Lemma 8.4. *Every coefficient of F_z defines an additive character of order dividing p on $\mathcal{O}_A$. Its degree-one coefficient is $g(z)$; its degree- p coefficient is $h(z)$ modulo $\mathfrak{p}_A^{\kappa_2}$.*

Proof. The coefficients of $1 - N_i(1 - Zz)$ belong to $\mathfrak{P}_{B_i}^{q_i}$, by the coefficientwise proof of Lemma 8.1. Products of these coefficients have at least that valuation. Multiplication by b or a loses t . In mixed characteristic $v_i p \geq (p-1)t_2$, and

$$(p-1)t_2 - t + q \geq T', \quad (p-1)t_2 - t + q_2 \geq \kappa_2.$$

For the first inequality the difference is $(p-3)t + \lceil (2t+1)/p \rceil - 1$; the second follows from $(p-3)t + (p-2)\delta + q_2 - 1 \geq 0$. Thus every coefficient defines a character of order dividing p on $\mathcal{O}_A$. In equal characteristic this assertion is automatic.

The linear coefficient follows by differentiation at $Z = 0$. At degree p , the norm terms give $h(z)$. All other terms contain at least two intermediate elementary coefficients of total weight p . If the upper break is u , their valuation after multiplication by b or a is at least

$$q - t + 2(p-1)u/p.$$

For $u = t$ this is strictly greater than $\kappa_2 - 1$; for $u = t'$ it is strictly greater than $T' - 1$, using $q = \delta + \lceil (2t+1)/p \rceil$. Integral valuations therefore put each term in the corresponding trivial ideal. Trace to A puts it in $\mathfrak{p}_A^{\kappa_2}$. $\square$

Theorem 8.5 (Compatibility on U_B^q). *The characters (82) satisfy $R_1 N_1 = R_2 N_2$ on U_B^q . This holds for all odd p , in mixed and equal characteristic.*

Proof. Put

$$M_0 = \max\{q, p \max(m_1, m_2)\}.$$

The norm filtration gives

$$N_i(U_B^{M_0}) \subset U_{B_i}^{m_i} \quad (i = 1, 2),$$

so $R_1 N_1 = R_2 N_2 = 1$ on $U_B^{M_0}$. We descend one step at a time. Suppose $q \leq s < M_0$ and that Ξ is trivial on U_B^{s+1} . Since $2s \geq s+1$, the map

$$z \mapsto \Xi(1-z), \quad z \in \mathfrak{P}_B^s,$$

is additive: the error in replacing multiplication of $1-x, 1-y$ by addition of x, y belongs to U_B^{s+1} .

The same statement holds after an unramified base extension, after pulling Ξ back by the norm; the induction hypothesis remains valid because the norm preserves the unit depth. Choose such an extension of degree prime to p and absolute residue degree at least 3. For $z \in \mathfrak{P}_B^s$, Lemmas 8.3 and 8.4 applied to (84) give

$$\Xi(1-z) = \Psi_A(g(z) + h(z)).$$

Write $z = \sum_{j=0}^{p-1} x_j \Delta^j$. Since $p \nmid t$, the valuations of the terms are distinct modulo p , so every $x_j \Delta^j$ still lies in $\mathfrak{P}_B^s$. Additivity and Lemma 8.2 give

$$\Xi(1-z) = \prod_{j=0}^{p-1} \Xi(1-x_j \Delta^j) = 1.$$

Thus Ξ is trivial on U_B^s . Descending from M_0 to q proves $R_1 N_1 = R_2 N_2$ on U_B^q . $\square$

Extension to $B_i^\times$.

Theorem 8.6. *Every extension θ_2 of R_2 to $B_2^\times$ has a compatible extension θ_1 of R_1 to $B_1^\times$. Both characters have conductors m_i , and their common pullback is primitive.*

Proof. Lemma C.2 gives θ_2 . Put $\Theta = \theta_2 N_2$. Let ρ generate $\text{Gal}(B/B_1)$, whose break is t' . For $z \in B^\times$, the ramification inequality gives $\rho(z)/z \in U_B^{t'} \subset U_B^q$. Its N_1 -norm is 1. By Theorem 8.5 and (83),

$$\Theta(\rho(z)/z) = R_2(N_2(\rho(z)/z)) = R_1(N_1(\rho(z)/z)) = 1.$$

Hence Θ is invariant under $\text{Gal}(B/B_1)$ on all of $B^\times$. Lemmas C.1 and C.2 give a character $\tilde{\theta}_1$ of $B_1^\times$ with $\tilde{\theta}_1 \circ N_1 = \Theta$. On $H_1 \cap N_1 B^\times$ it agrees with R_1 : any norm preimage belongs to U_B^q by (83), where Theorem 8.5 applies. Thus $\tilde{\theta}_1/R_1$ on H_1 is a character of the image of H_1 in $B_1^\times/N_1 B^\times$. Extend it across this finite cyclic quotient and divide $\tilde{\theta}_1$ by the resulting norm character. This gives $\theta_1|_{H_1} = R_1$, without changing the common pullback.

The restrictions R_i are nontrivial at depth $m_i - 1$ and trivial at depth m_i , so the conductors are m_i . For a generator σ of $\text{Gal}(B_2/A)$ and $z \in \mathfrak{P}_{B_2}^t$, $\sigma^{-1}(1-z)/(1-z)$ has leading depth $t + t_2 = m_2 - 1$ and nonzero ramification coefficient t . Because $p \nmid t$, its leading map on the residue line is surjective. Therefore θ_2^σ/θ_2 is nontrivial, of conductor $t+1$. The invariance of Θ under $\text{Gal}(B/A)$ makes this a norm character for B/B_2 . If Θ descended from A , this commutator would be trivial, a contradiction. Here the action of $\text{Gal}(B_2/A)$ on $B_2^\times/N_2(B^\times)$ is trivial, since $|\text{Aut}(C_p)| = p-1$. $\square$

A prescribed primitive pair.

Lemma 8.7 (A lower bound for the conductor). *Let (η_1, η_2) be a primitive compatible pair for B/A . If B_i/A and B/B_i have breaks t_i and u_i , respectively, then*

$$a(\eta_i) \geq 1 + t_i + u_i \quad (i = 1, 2).$$

Proof. Let σ_i generate $\text{Gal}(B_i/A)$. By Lemma 4.2, $\mu_i = \eta_i^{\sigma_i}/\eta_i$ is a nontrivial norm character for B/B_i , of conductor $u_i + 1$. Choose $y = 1 + z \in U_{B_i}^{u_i}$ with $\mu_i(y) \neq 1$. Lemma C.3 gives

$$v_{B_i}(\sigma_i^{-1}z - z) \geq t_i + v_{B_i}z, \quad \sigma_i^{-1}y/y \in U_{B_i}^{t_i+u_i}.$$

The value of η_i on this quotient is $\mu_i(y) \neq 1$. Thus $a(\eta_i) > t_i + u_i$. $\square$

Let (φ_1, φ_2) be a primitive compatible pair. Assume only that φ_2 is trivial on $U_{B_2}^{m_2}$.

Lemma 8.7 gives $a(\varphi_i) \geq m_i$. The upper bound on φ_2 therefore gives $a(\varphi_2) = m_2$. Since $m_1 > t' + 1$ and $m_2 > t + 1$, the common norm pullback and (10) give

$$pa(\varphi_1) - (p-1)(t' + 1) = pm_2 - (p-1)(t + 1).$$

Using $t' = t + p\delta$ and $t_2 = t + \delta$, we obtain

$$(85) \quad a(\varphi_1) = m_1, \quad a(\varphi_2) = m_2.$$

Choose any initial pair from Theorem 8.6. For a generator σ of $\text{Gal}(B_2/A)$, the characters $\varphi_2^\sigma/\varphi_2$ and θ_2^σ/θ_2 generate $S(B/B_2)$ by Lemma 4.2. Choose $j \in \{1, \dots, p-1\}$ so that these characters agree after raising the second to the j th power. Then $\varphi_2\theta_2^{-j}$ is invariant under $\text{Gal}(B_2/A)$. Hilbert 90 and character extension give a character λ of $A^\times$ with

$$(86) \quad \varphi_2 = \theta_2^j(\lambda n_2).$$

Put

$$h_0 = \lfloor t/p \rfloor, \quad c = \lceil (t_2 + 1)/p \rceil, \quad q_0 = h_0 + c.$$

Equation (10) and $a(\lambda n_2) \leq m_2$ imply $a(\lambda) \leq \kappa_2 + h_0$. The following inequalities include the case $p \mid t_2$:

$$(87) \quad q_0 \leq q_2, \quad pq_0 \geq \kappa_2 + h_0, \quad n_2(H_2) \subset U_A^{q_0}, \quad (p-2)t_2 - ph_0 + q_0 - 1 \geq 0.$$

To verify them, write $t = ph_0 + b_0$, $1 \leq b_0 < p$. Then $q_2 = h_0 + \lceil (b_0 + t_2 + 1)/p \rceil \geq q_0$. For the norm inclusion use (7) and $q_2 + (p-1)\kappa_2 \geq pq_0$; the latter follows from $\kappa_2 \geq ph_0 + b_0 + 1$ and the preceding ceiling formula. The last inequality follows by replacing t_2 by t and q_0 by $h_0 + \lceil (t+1)/p \rceil$; the resulting lower bound is nonnegative.

Put $\Psi'_L = \Psi_L(j \cdot)$. Lemma 6.2 gives a coefficient $\gamma \in A$, $v_A\gamma \geq -h_0$, such that

$$\lambda(1-z) = \Psi'_A(\gamma Pz) \quad (z \in \mathfrak{p}_A^{q_0}).$$

If the restriction is trivial take $\gamma = 0$ and $w = 0$. Otherwise [7, Lemma 6.6] gives

$$(88) \quad w \in B_2, \quad k = v_2w = v_A\gamma \geq -h_0, \quad \gamma - n_2w \in \mathfrak{p}_A^{k+t_2}.$$

Lemma 8.8. For $x \in \mathfrak{P}_{B_2}^{q_2}$,

$$(\lambda n_2)(1-x) = \Psi'_2((w^p - w)P(x)).$$

Proof. Set $u = 1 - n_2(1-x)$. It has depth at least q_0 . The error in replacing γ by n_2w contributes valuation at least $k + t_2 + q_0 \geq \kappa_2$, so (88) suffices. The norm-logarithm congruence from Lemma 6.3, now with target $\kappa_2 + h_0$ and input depth $h_0 + c = q_0 \leq q_2$, yields

$$(\lambda n_2)(1-x) = \Psi'_2((n_2w)P(x))\Psi'_A(n_2(wP(x))).$$

Since $k + q_2 \geq c$, (81) changes the second factor to $\Psi'_2(-wP(x))$. Equation (11), applied to the coefficients of the characteristic polynomial of w , gives $v_2(w^p - n_2w) \geq pk + (p-1)t_2$. Multiplication by $P(x)$ puts this error in $\mathfrak{P}_{B_2}^{\kappa_2}$ by (87). This proves the formula. $\square$

Theorem 8.9 (Character formulas for a prescribed pair). *Let (φ_1, φ_2) be a primitive compatible pair and suppose $\varphi_2|_{U_{B_2}^{m_2}} = 1$. Then $a(\varphi_i) = m_i$ for $i = 1, 2$. Moreover there exist $\alpha' = j\alpha$, with $1 \leq j < p$, and $\Delta_\lambda \in B$ such that*

$$\Delta_\lambda^p - \Delta_\lambda = a_\lambda \in B_2, \quad v_B\Delta_\lambda = -t.$$

For some $\rho \in \text{Gal}(B_1/A)$, the character formulas are

$$\begin{aligned} \varphi_2(1-z) &= e_{B_2}(\alpha' a_\lambda P(z)) & (z \in \mathfrak{P}_{B_2}^{q_2}), \\ \varphi_1^\rho(1-z) &= e_{B_1}(\alpha' N_1(\Delta_\lambda)P(z)) & (z \in \mathfrak{P}_{B_1}^{q_1}). \end{aligned}$$

The characters φ_1^ρ and φ_1 have the same local constant.

Proof. The conductor equalities are (85). Set $a_\lambda = a + w^p - w$. Its valuation is $-t$ because $pv_2w \geq -ph_0 = -t + b_0 > -t$. In mixed characteristic the error of $\Delta + w$ in $X^p - X - a_\lambda$ has valuation at least

$$v_B(p) - (p-1)t + pv_2w \geq v_B(p) - pt + b_0 > 0.$$

Its derivative is a unit. Lemma D.1, applied after translating by $\Delta + w$, gives the required exact root, still of valuation $-t$. In equal characteristic $\Delta + w$ is already that root. It generates B/B_2 because $p \nmid t$. Equation (86) and Lemma 8.8 give the stated formula for φ_2 on H_2 . Lemmas 7.5 and 7.6 hold for any coordinate satisfying $\Delta^p - \Delta = a$, $v_B(\Delta) = -t$, $p \nmid t$, and $B = B_2(\Delta)$, with the ramification data fixed in Section 7. They therefore apply to Δ_λ and a_λ , with $b_\lambda = N_1(\Delta_\lambda)$ and $s_\lambda = -E_{p-1}^{B/B_1}(\Delta_\lambda)$. Apply Theorem 8.6 to the new generator and norm character, choosing φ_2 itself as the second extension. It supplies $\tilde{\varphi}_1$ with the stated formula on H_1 and the same norm pullback as φ_2 . Thus $\tilde{\varphi}_1/\varphi_1 \in S(B/B_1)$. For a primitive compatible pair its nontrivial conjugate quotient generates this order- p group, on which $\text{Gal}(B_1/A)$ acts trivially. Hence every such twist is a Galois conjugate. Equation (6) proves equality of their local constants. $\square$

Remark 8.10. The original statement assumed both conductor equalities $a(\varphi_i) = m_i$. The Lean proof suggested the formulation in Theorem 8.9: it is enough to assume that φ_2 is trivial on $U_{B_2}^{m_2}$. Primitivity and compatibility then force both exact conductors.

8.2. The congruence for $n_1(s)$.

Setup. Use $A \subset B_1, B_2 \subset B$, $t_1 = t$, $t_2 = t + \delta$, $t' = t + p\delta$, $p \nmid t$, and N_i, n_i, S_i, T_i as in §7 and §8.1. Put

$$T' = t' + 1, \quad \kappa_2 = t_2 + 1, \quad m_1 = T' + t, \quad q_1 = \lceil m_1/p \rceil, \quad c_t = \lceil (t+1)/p \rceil = \lceil t/p \rceil, \quad q = \delta + c_t.$$

Let $\Delta^p - \Delta = a \in B_2$, $v_B \Delta = -t$, and

$$b = N_1 \Delta, \quad a = N_2 \Delta, \quad s = -E_{p-1}^{B/B_1}(\Delta), \quad H = n_1 s.$$

Choose nontrivial norm characters $\tau_2 \in S(B_2/A)$ and $\tau_1 \in S(B_1/A)$, with coefficients α, β in the truncated-logarithm formulas relative to the same additive character e_A . Write

$$\Psi_L(z) = e_L(\alpha z), \quad \varrho = \beta/\alpha, \quad v_A \varrho = \delta,$$

so $\Psi_A, \Psi_1, \Psi_2, \Psi_B$ have largest trivial ideals $\mathfrak{p}_A^{\kappa_2}, \mathfrak{P}_{B_1}^{T'}, \mathfrak{P}_{B_2}^{\kappa_2}, \mathfrak{P}_B^{T'}$. By Theorem 8.9, we may choose χ in its Galois conjugacy class so that

$$(89) \quad \chi(1-y) = \Psi_1(bP(y)) \quad (y \in \mathfrak{P}_{B_1}^{q_1}), \quad P(X) = \sum_{j=1}^{p-1} X^j/j.$$

Its exact conductor is m_1 . Let σ be the restriction to B_1 of the automorphism of B/B_2 carrying Δ to the root near $\Delta + 1$. Our character convention is $\chi^\sigma(u) = \chi(\sigma^{-1}u)$.

The quotient χ^σ/χ on $U_{B_1}^q$.

Lemma 8.11. For $x \in \mathfrak{P}_{B_1}^q$,

$$\frac{\chi^\sigma(1-x)}{\chi(1-x)} = \Psi_1((\sigma b - b)P(x)).$$

Proof. Put $d = \sigma^{-1}x - x$, so $v_1 d \geq q + t$. The quotient $(1 - \sigma^{-1}x)/(1 - x)$ is $1 - d/(1 - x)$. We have $q + t \geq q_1$, hence (89) applies to this quotient even though it need not apply separately to $1 - x$. As an integral formal power series in x, d ,

$$P(d/(1-x)) - P(x+d) + P(x)$$

is divisible by d and has total degree at least p . Its denominators are products of integers $1, \dots, p-1$ and powers of $1-x$. Its valuation is therefore at least $(p-1)q + (q+t) = pq + t \geq T' + t = m_1$. Multiplication by b puts it in the kernel of Ψ_1 . Finally trace invariance gives $\Psi_1(bP(\sigma^{-1}x)) = \Psi_1((\sigma b)P(x))$. $\square$

Lemma 8.12. *Set*

$$R = T' - q = t + 1 - c_t + (p - 1)\delta.$$

Then $\sigma b - b \equiv 1 - s \pmod{\mathfrak{P}_{B_1}^R}$.

Proof. Norm expansion gives

$$N_1(\Delta + 1) - b = 1 - s + \sum_{i=1}^{p-2} E_i^{B/B_1}(\Delta).$$

Each summand has valuation at least $(p - 1)t_2 - (p - 2)t = t + (p - 1)\delta \geq R$ by Section 7. For the conjugate of Δ , write $\sigma\Delta = (\Delta + 1)(1 + u)$. The elementary Hensel estimate gives $v_B u \geq V - (p - 2)t$, where $V = v_B p$. For intermediate coefficients in $N_1(1 + u) - 1$, the bound in (11) is smallest at the trace coefficient, and after multiplying by $N_1(\Delta + 1)$ it gives at least $-t + \lfloor [V - (p - 2)t + (p - 1)(t' + 1)]/p \rfloor = V/p - t + (p - 1)\delta + c_t \geq (p - 2)t + 2(p - 1)\delta + c_t \geq R$. The norm term has still larger valuation. In equal characteristic $u = 0$. This proves the assertion, including $p = 3$. $\square$

Lemmas 8.11 and 8.12 show that $\mu = \chi^\sigma / \chi$ satisfies

$$(90) \quad \mu(1 - x) = \Psi_1((1 - s)P(x)) \quad (x \in \mathfrak{P}_{B_1}^q).$$

It is a nontrivial norm character for B/B_1 , of conductor T' . The same is true of $\tau_2 n_1$. Proposition 8.15 will prove that these two characters agree.

The character $\tau_2 \circ n_1$. By [7, Lemma 6.6], choose $c \in B_1^\times$ with

$$(91) \quad n_1 c = \varrho \varepsilon, \quad \varepsilon \in U_A^t, \quad v_1 c = \delta.$$

Lemma 8.13. *On $U_{B_1}^q$ one has*

$$(\tau_2 n_1)(1 - x) = \Psi_1((1 - \varrho/c)P(x)).$$

Proof. For $u = 1 - n_1(1 - x)$, the norm filtration gives $v_A u \geq \lceil \kappa_2/p \rceil$, so the truncated-logarithm formula for τ_2 applies. The norm-logarithm congruence of Lemma 6.3 on B_1/A , with target κ_2 and input depth $\delta + c_t = q$, gives

$$(\tau_2 n_1)(1 - x) = \Psi_1(P(x))\Psi_A(n_1 P(x)).$$

The error in replacing the second factor by $e_A(\beta n_1(P(x)/c))$ has valuation, before multiplication by α , at least $q + t \geq \kappa_2$. The error is $n_1 P(x)(\varepsilon^{-1} - 1)$. Since $v_1(P(x)/c) \geq q - \delta = c_t$, the norm-character conversion for τ_1 makes this factor $e_{B_1}(-\beta P(x)/c)$. $\square$

Equality of the two norm characters.

Lemma 8.14 (Reciprocal trace identity). *If $\delta = 0$, then for every $Y \in \mathcal{O}_A$,*

$$\Psi_1(sY/b) = \Psi_2(Y/a) = \Psi_A(-n_1(Y/b)).$$

Proof. Elementary symmetric functions give the *exact* identities

$$s/b = -S_1(1/\Delta), \quad S_2(1/\Delta) = -1/a.$$

The second uses the coefficient -1 of X in $X^p - X - a$. Trace transitivity therefore gives the first equality. As $v_2(Y/a) \geq t \geq \lceil (t+1)/p \rceil$, (81) gives $\Psi_2(Y/a) = \Psi_A(-n_2(Y/a))$. Finally $n_1 b = n_2 a$ by norm transitivity and $Y \in A$, so $n_2(Y/a) = Y^p/n_2 a = n_1(Y/b)$. $\square$

Proposition 8.15. *One has $\mu = \tau_2 \circ n_1$.*

Proof. Both are nontrivial characters of the same cyclic norm quotient of order p . Their restrictions to its last nonzero unit layer distinguish the $p - 1$ possible generators. If $\delta > 0$, $v_1 s \geq (p - 1)\delta$ and $v_1(\varrho/c) = (p - 1)\delta$ are positive. Equations (90) and Lemma 8.13 agree on $U_{B_1}^{t'}$; here $P(x) \equiv x$ at the character modulus. Hence the characters are equal.

If $\delta = 0$, use $x = Y/b$, whose classes as Y varies span $\mathfrak{P}_{B_1}^t/\mathfrak{P}_{B_1}^{t+1}$. On this layer $P(x) \equiv x$. The intermediate coefficients of $n_1(1-x)$ other than trace and norm are in $\mathfrak{p}_A^{t+1}$ by (11). Thus

$$(\tau_2 n_1)(1-x) = \Psi_A(T_1 x + n_1 x) = \Psi_1(x) \Psi_A(n_1 x).$$

Lemma 8.14 identifies this with $\Psi_1((1-s)x)$, which is the value of μ by (90). Their restrictions to $U_{B_1}^t$ agree, so the norm characters are equal. $\square$

The congruence for $H = n_1(s)$.

Theorem 8.16. *For every odd p , in both characteristics and every $\delta = t_2 - t \geq 0$,*

$$(92) \quad H \equiv \varrho^{p-1} \pmod{\mathfrak{p}_A^R}, \quad R = t + 1 - c_t + (p-1)\delta.$$

Consequently, for every integer m with $pm > t$,

$$H \equiv (\beta/\alpha)^{p-1} \pmod{\mathfrak{p}_A^{1+t_2-(p-2)m}}.$$

Proof. Proposition 8.15, (90) and Lemma 8.13 imply

$$\Psi_1((s - \varrho/c)P(x)) = 1 \quad \text{for every } x \in \mathfrak{P}_{B_1}^q.$$

By Lemma 6.2, P is a bijection of $\mathfrak{P}_{B_1}^q$. The additive pairing therefore gives

$$s - \varrho/c \in \mathfrak{P}_{B_1}^{T'-q} = \mathfrak{P}_{B_1}^R.$$

Put $d_s = (p-1)\delta$. Both ϱ/c and s have valuation d_s , since $R - d_s = R_0 := t + 1 - c_t \geq 1$. Their ratio is in $U_{B_1}^{R_0}$. Since $1 \leq R_0 \leq t$, [7, Theorem 3.7] gives $n_1(s)/n_1(\varrho/c) \in U_A^{R_0}$. As $v_A(n_1(\varrho/c)) = d_s$ and $d_s + R_0 = R$, it follows that

$$n_1 s \equiv n_1(\varrho/c) = \varrho^{p-1} \varepsilon^{-1} \pmod{\mathfrak{p}_A^R}.$$

The remaining error has valuation at least $d_s + t \geq d_s + t + 1 - c_t = R$. This proves (92), including $\delta = 0$. Finally $pm > t$ and $p \nmid t$ imply $m \geq c_t$, and

$$R - [1 + t_2 - (p-2)m] = (p-2)(\delta + m) - c_t \geq 0.$$

The final inequality proves the second assertion. $\square$

8.3. Norm representatives and the factors Q_i .

Truncated-logarithm coordinates. Throughout p is odd and $P(X) = \sum_{j=1}^{p-1} X^j/j$. For a field E let e_E have largest trivial ideal $\mathfrak{P}_E^{-d_E}$. A multiplicative character θ of conductor $n \geq 2$ admits a coefficient C with

$$(93) \quad \theta(1-x) = e_E(CP(x)), \quad x \in \mathfrak{P}_E^{\lceil n/p \rceil}, \quad v_E(C) = -d_E - n.$$

By Lemma 6.2, P identifies U_E^q/U_E^n with $\mathfrak{P}_E^q/\mathfrak{P}_E^n$ for $q = \lceil n/p \rceil$. Additive duality gives (93), and the exact conductor gives $v_E(C) = -d_E - n$. The coefficient in (93) is determined modulo $\mathfrak{P}_E^{-d_E-q}$, where $q = \lceil n/p \rceil$.

Lemma 8.17 (A norm representative after changing the coefficient). *Let E/F be ramified cyclic of odd prime degree with break $t \geq 1$. Let ν be a nontrivial norm character and write*

$$\nu(1-z) = e_F(AP(z)), \quad z \in \mathfrak{p}_F^c, \quad c = \lceil (t+1)/p \rceil, \quad v_F(A) = -d_F - t - 1.$$

For any $\gamma \in F^\times$ there is A' with

$$A'/A \in U_F^t, \quad \nu(1-z) = e_F(A'P(z)) \quad (z \in \mathfrak{p}_F^c), \quad \gamma/A' \in N_{E/F}(E^\times).$$

Consequently there is $w \in E^\times$ with $Nw = \gamma/A'$ and $v_E(w) = v_F(\gamma/A')$.

Proof. Apply [7, Lemma 6.6] to γ/A at precision t . It gives $w \in E^\times$ such that

$$h := \frac{\gamma/A}{Nw} \in U_F^t.$$

Put $A' = Ah$. Then $\gamma/A' = Nw$. Since $h - 1 \in \mathfrak{p}_F^t$, the change in the additive coefficient on the prescribed subgroup has valuation at least

$$v_F(A(h-1)P(z)) \geq -d_F - t - 1 + t + c \geq -d_F,$$

so the character formula is unchanged. Total ramification gives $v_F(Nw) = v_E(w)$. $\square$

Lemma 8.18 (Changing α by an element of $U_A^{t_2}$). *Let the totally ramified notation be as in Subsection 8.1, and suppose the characters satisfy (82), with coefficients $\alpha b, \alpha a$ relative to e_{B_i} . For $h \in U_A^{t_2}$, replacing α by αh leaves both formulas in (82) unchanged. It also leaves the truncated-logarithm formula for the norm character of B_2/A unchanged.*

Proof. Write $q_1 = \delta + \lceil (2t+1)/p \rceil$, $q_2 = \lceil (2t+\delta+1)/p \rceil$. The old additive characters $\Psi_i = e_{B_i}(\alpha \cdot)$ have largest trivial ideals with exponents $1 + t'$ and $1 + t_2$. For $z \in \mathfrak{P}_{B_i}^{q_i}$ the change in the argument of Ψ_i is $(h-1)bP(z)$ for $i=1$ and $(h-1)aP(z)$ for $i=2$. Its B_i -valuation is at least $pt_2 - t + q_i$. On B_1 its excess over $1 + t'$ is

$$(p-2)t + \delta + \lceil (2t+1)/p \rceil - 1 \geq 0,$$

and on B_2 its excess over $1 + t_2$ is

$$(p-2)t + (p-1)\delta + q_2 - 1 \geq 0.$$

Thus both formulas in (82) are unchanged. Over A , the change in the argument of Ψ_A is $(h-1)P(z)$, of valuation at least $t_2 + c_2 \geq t_2 + 1$ for $z \in \mathfrak{p}_A^{c_2}$, where $c_2 = \lceil (t_2+1)/p \rceil$. This proves the last assertion. $\square$

Remark 8.19. First apply Theorem 8.9 to fix α and the character formulas. Lemmas 8.17 and 8.18 then allow the norm representatives to be chosen without changing those formulas.

The coefficient of $\lambda \circ N_{E/F}$.

Proposition 8.20. *Let E/F be as in Lemma 8.17. Let λ have conductor $1 + r$, where $r > t$, and let γ be its coefficient in (93) relative to e_F . Choose A, w by that lemma so that $Nw = \gamma/A$. Put $h = r - t$, $n = 1 + t + ph$, and $q = h + \lceil (t+1)/p \rceil = \lceil n/p \rceil$. Then $\lambda \circ N_{E/F}$ has conductor n and*

$$(94) \quad (\lambda N)(1-x) = e_E((\gamma - Aw)P(x)), \quad x \in \mathfrak{P}_E^q, \quad e_E = e_F \text{Tr}_{E/F}.$$

Its coefficient has valuation $p v_F(\gamma)$.

Proof. The conductor follows from (10), with different exponent $(p-1)(t+1)$. One has $v_E(w) = -h$ and

$$v_E(Aw/\gamma) = (p-1)h > 0.$$

This proves the claimed coefficient valuation and matches conductor n , since $d_E = p d_F + (p-1)(t+1)$.

For $x \in \mathfrak{P}_E^q$, the norm $N(1-x)$ lies in $U_F^{\lceil (r+1)/p \rceil}$, where (93) applies to λ . Indeed the highest-degree term in the norm expansion has valuation at least q , and the intermediate terms at least $\lfloor (q + (p-1)(t+1))/p \rfloor$; both are at least the required depth. For the second inequality compare the numerator with $r+p$; its excess is at least $\lceil (t+1)/p \rceil + (p-2)t - 1 \geq 0$. Lemma 6.3, with $h = r - t$ and target modulus $r+1$, gives

$$(\lambda N)(1-x) = e_E(\gamma P(x)) e_F(\gamma N(P(x))).$$

Here its error has depth $r+1$ before multiplication by γ , so is in $\mathfrak{p}_F^{-d_F}$ afterward. Now $wP(x)$ belongs to $\mathfrak{P}_E^{\lceil (t+1)/p \rceil}$. Apply ν to $N(1-wP(x))$ and use Lemma 6.3 modulo $\mathfrak{p}_F^{t+1}$. This gives

$$e_E(AwP(x)) = e_F(-AN(wP(x))) = e_F(-\gamma N(P(x))).$$

Combining the last two equalities proves (94). $\square$

Lamprecht's formula for a truncated-logarithm coefficient.

Lemma 8.21 (Evaluation of Δ_E from (93)). *Under (93), there is an explicitly determined $g_E(\theta, C) \in \mu_4$ with*

$$(95) \quad \Delta_E(\theta, e_E) = \theta(-C^{-1})e_E(-C)g_E(\theta, C).$$

For even n , $g_E = 1$. For odd $n = 2d + 1$, choose any δ_E of valuation d . Then

$$(96) \quad g_E = |k_E|^{-1/2} \sum_{x \in k_E} e_E(-C\delta_E^2 \tilde{x}^2/2).$$

Proof. Apply Lemma 6.5 with $\Psi_E = e_E$, ideal exponent $J = -d_E$, and $B = C$, so $\eta = 0$. Formula (93) is valid on $\mathfrak{p}_E^{\lfloor n/2 \rfloor}$ because $\lfloor n/2 \rfloor \geq \lceil n/p \rceil$ for odd p and $n \geq 2$. Its critical function is therefore $e_E(-C\delta_E^2 \tilde{x}^2/2)$, and the elementary factor is $\theta(-C^{-1})e_E(-C)$. The homogeneous quadratic sum has phase in μ_4 , as in the proof of that lemma. No affine displacement remains. $\square$

Application to $\lambda \circ N_{E/F}$. Let the breaks be $t = t_1 \leq t_2 = t + \delta$, $t' = t + p\delta$, $p \nmid t$. Let the pair have the form

$$\theta_1 = \chi(\lambda n_1), \quad \theta_2 = \varphi(\lambda n_2), \quad \chi N_1 = \varphi N_2,$$

where $a(\chi) = m_1 = 1 + t + t'$, $a(\varphi) = m_2 = 1 + t + t_2$. Suppose $a(\lambda) = 1 + r$, $m = r - t_2$, and $pm > t$. Apply Theorem 8.9 to (χ, φ) . Conjugation on B_1 also conjugates θ_1 , since λn_1 is invariant, and hence does not change its local constant. Denote the resulting coefficients by $\alpha b, \alpha a$, where $b = N_1 \Delta$ and $a = N_2 \Delta$. The scalar α is the coefficient of a nontrivial norm character for B_2/A . Choose γ for λ as in (93), and choose a nontrivial norm character for B_1/A with coefficient β . Apply Lemma 8.17 separately to these two norm characters. Lemma 8.18 shows that its replacement of α does not change the character formulas for χ, φ ; replacing β affects neither formula. Renaming the adjusted coefficients, the exact choices give

$$(97) \quad \begin{aligned} n_2 w_2 &= \gamma/\alpha, & v_2 w_2 &= -m, & n_1 w_1 &= \gamma/\beta, & v_1 w_1 &= -(m + \delta), \\ v_A \alpha &= -d - 1 - t_2, & v_A \beta &= -d - 1 - t, & v_A \gamma &= -d - 1 - r. \end{aligned}$$

The changed coefficients represent the same norm characters on the unit groups specified in Lemma 8.17. The formulas for χ, φ remain

$$\chi(1 - x) = e_{B_1}(\alpha b P(x)), \quad \varphi(1 - x) = e_{B_2}(\alpha a P(x)), \quad x \in \mathfrak{P}_{B_i}^{\lfloor m_i/p \rfloor}$$

for $i = 1, 2$, respectively. Theorem 8.16 therefore applies to these coefficients. Proposition 8.20 gives

$$(98) \quad C_1 = \gamma - \beta w_1 + \alpha N_1 \Delta, \quad C_2 = \gamma - \alpha w_2 + \alpha N_2 \Delta.$$

The conductors are

$$n_1^{\text{char}} = 1 + t' + pm, \quad n_2^{\text{char}} = 1 + t_2 + pm.$$

Write $\alpha_1 = \beta$, $\alpha_2 = \alpha$. The truncated-logarithm formulas with coefficients (98) hold at depths $\lfloor n_i^{\text{char}}/p \rfloor$. Indeed $n_i^{\text{char}} > m_i$, $v_i(\alpha N_i \Delta/\gamma) = pm - t > 0$, and $v_i(\alpha_i w_i/\gamma) = (p - 1)(r - t_i) > 0$. Thus γ has strictly smaller valuation than the other two terms in each coefficient, which proves the stated conductors and domains.

Proposition 8.22 (The factors Q_1, Q_2, Q_3, Q_4). *Define $A_i = \gamma + \alpha N_i \Delta$ and put*

$$\begin{aligned} Q_1 &= \frac{\chi(A_1)}{\varphi(C_2)}, & Q_2 &= \lambda \left(\frac{n_1(A_1)}{n_2(C_2)} \right), \\ Q_3 &= e_{B_1}(-\beta w_1) \theta_1(C_1/A_1), & Q_4 &= e_{B_2}(\alpha w_2). \end{aligned}$$

Then, with the explicit g_i from Lemma 8.21,

$$(99) \quad \frac{\Delta_{B_2}(\theta_2, e_{B_2})}{\Delta_{B_1}(\theta_1, e_{B_1})} = \frac{g_2}{g_1} Q_1 Q_2 Q_3 Q_4 e_A(\alpha[T_1 N_1 \Delta - T_2 N_2 \Delta]).$$

In particular, after the trace-norm estimate of Theorem 7.7 removes the last factor, proving $(Q_1 Q_2 Q_3 Q_4)^{-1} = 1$ implies that the quotient in (99) belongs to μ_4 .

Proof. Compatibility and odd degree give $\theta_1(-1) = \theta_2(-1)$: apply the equal norm pullbacks to $-1 \in B^\times$. Formula (95) therefore gives

$$\frac{\Delta_{B_2}}{\Delta_{B_1}} = \frac{g_2}{g_1} \frac{\theta_1(C_1)}{\theta_2(C_2)} e_{B_1}(C_1) e_{B_2}(-C_2).$$

The two γ traces both equal $p\gamma$ and cancel. Substitute (98); the remaining additive factors are the three shown in (99). Direct multiplication gives

$$Q_1 Q_2 = \frac{\theta_1(A_1)}{\theta_2(C_2)}, \quad Q_1 Q_2 \theta_1(C_1/A_1) = \frac{\theta_1(C_1)}{\theta_2(C_2)}.$$

The remaining additive factors are exactly those of $Q_3 Q_4$. The same trace–norm estimate is $T_1 N_1 \Delta - T_2 N_2 \Delta \in \mathfrak{p}_A^{1+t_2}$; its weight α puts it in $\mathfrak{p}_A^{-d}$, as required. $\square$

9. THE TOTALLY RAMIFIED ODD-PRIME CALCULATION

9.1. The linear trace calculation.

Notation and local estimates. Let $A \subset B_1, B_2 \subset B$ be totally ramified extensions, $\text{Gal}(B/A) = C_p^2$, p odd, with

$$t_1 = t, \quad t_2 = t + \delta, \quad t' = t + p\delta, \quad p \nmid t.$$

Write $N_i = N_{B/B_i}$, $n_i = N_{B_i/A}$, $S_i = \text{Tr}_{B/B_i}$, and put $v = v_B$. Proposition 7.4 and the estimates of Section 7 give

$$\begin{aligned} \Delta^p - \Delta = a \in B_2, \quad v(\Delta) = -t, \quad b = N_1 \Delta, \quad e_i = E_i^{B/B_1}(\Delta), \quad s = -e_{p-1}, \\ (100) \quad v(e_i) \geq p((p-1)t_2 - it), \quad v(S_1(Y\Delta^i)) \geq p(p-1)t_2 - pit + v(Y) \quad (Y \in B_2). \end{aligned}$$

Put $V = v_B(p)$ in mixed characteristic; in equal characteristic every term carrying p is zero. The local trace-ideal bound is $V \geq p(p-1)t_2$. Let $w \in B_2$ have $n_2 w = W$, put

$$u = W^{-1}, \quad v_A(u) = m, \quad M = pm > t, \quad K = 1 + pt_2, \quad S = p(p-1)\delta.$$

Thus $M \geq t + 1$, $v(w) = -M$, $v(W) = -pM$, $v(a) = v(b) = -pt$, and $v(s) \geq S$. All congruences below have modulus $\mathfrak{P}_B^K$ unless a different modulus is displayed. Set

$$Y = W + b, \quad D = Y^p - Y, \quad D_0 = W^p - W + a^p, \quad C = 1 + a/W.$$

Their denominators are nonzero: $v(Y) = -pM$ and $v(D) = v(D_0) = -p^2 M$.

The norms $N_1 \Delta_\xi$. Let $\mathcal{T} = \{0\} \cup \mu_{p-1} \subset A$ and let Δ_ξ be the root congruent to $\Delta + \xi$ at positive relative depth. The construction of Section 7 gives

$$\Delta_\xi = \Delta + \xi + \eta_\xi, \quad v(\eta_\xi) \geq V - (p-1)t.$$

These are all B/B_2 conjugates; in equal characteristic $\eta_\xi = 0$.

Lemma 9.1. *For each $\xi \in \mathcal{T}$,*

$$v_{B_1}(N_1 \Delta_\xi - N_1(\Delta + \xi)) \geq t_2 + 1.$$

Proof. In mixed characteristic put $z = \eta_\xi/(\Delta + \xi)$, so $v(z) \geq V - (p-2)t$. The different exponent for B/B_1 is $D' = (p-1)(t' + 1)$. The strong symmetric estimate gives

$$v_1(N_1(1+z) - 1) \geq \min \left\{ v(z), \left\lfloor \frac{v(z) + D'}{p} \right\rfloor \right\}.$$

Here $v(z) \geq t' + 1$, so the second entry is no larger than the first. Since $v_1 N_1(\Delta + \xi) = -t$, the required difference has valuation at least

$$\frac{V}{p} - t + (p-1)\delta + \left\lceil \frac{t}{p} \right\rceil \geq (p-2)t + 2(p-1)\delta + \left\lceil \frac{t}{p} \right\rceil \geq t_2 + 1.$$

For the last inequality, subtract $t + \delta + 1$ and use $p \geq 3$, $t \geq 1$, $\lceil t/p \rceil \geq 1$. Equal characteristic is exact. $\square$

Define $f(X) = X/(1 + X/W)$. For $v(X) = -pt$ its denominator is a unit. The exact norm polynomial gives

$$N_1(\Delta + \xi) = b + \xi - \xi s + R_\xi, \quad v(R_\xi) \geq p((p-1)t_2 - (p-2)t) \geq pt_2.$$

The exact identity

$$f(X+z) - f(X) = \frac{z}{(1+X/W)(1+(X+z)/W)}$$

and its first-order expansion show

$$(101) \quad f(N_1\Delta_\xi) \equiv \frac{W(b+\xi)}{W+b+\xi} - \frac{\xi s}{(1+b/W)^2} \pmod{\mathfrak{P}_B^{pt_2}}.$$

Indeed, the quadratic s -error has depth at least $pM + 2S \geq pt_2$; replacing $1 + (b+\xi)/W$ by $1 + b/W$ changes the linear term by an element of $\mathfrak{P}_B^{pM+S} \subseteq \mathfrak{P}_B^{pt_2}$. Lemma 9.1 gives an error in $\mathfrak{P}_B^{p(t_2+1)}$.

The power factors, including $p = 3$. Define

$$W_i = S_2 \left((\Delta/w)^i \frac{b}{1+b/W} \right), \quad 1 \leq i \leq p-1.$$

Each summand's power factor has valuation $i(M-t) \geq 1$; therefore (101) may be substituted at modulus K .

For $p \geq 5$, replacing Δ_ξ by $\Delta + \xi$ in that power factor gives an error of valuation at least

$$V + iM - (2p + i - 2)t \geq K.$$

At the weakest values $M = t + 1$, $V = p(p-1)t_2$, the difference from K is $(p^2 - 4p + 2)t + p(p-2)\delta + i - 1 \geq 0$.

Suppose $p = 3$. Hensel's equation at $\Delta + \xi$, $\xi = \pm 1$, gives

$$(102) \quad \Delta_\xi = \Delta + (1 + 3\Delta^2)\xi + \eta'_\xi, \quad v(\eta'_\xi) \geq V - t.$$

To see this, the initial residual is $3\xi\Delta^2 + 3\xi^2\Delta$; substituting the first Hensel correction back into the equation adds terms of valuation at least $2V - 4t$, which is at least $V - t$. Thus the contribution of η'_ξ to W_i has valuation at least $V + iM - (i+3)t \geq K$ for $i = 1, 2$.

Put $\lambda = 1 + 3\Delta^2$ and

$$C_i(\lambda) = w^{-i} \sum_{\xi \in \mathcal{T}} (\Delta + \lambda\xi)^i \frac{W(b+\xi)}{Y+\xi}.$$

Direct summation of the three terms gives the exact identities

$$\begin{aligned} C_1(\lambda) - C_1(1) &= \frac{2(\lambda-1)W^2}{w(Y^2-1)}, \\ C_2(\lambda) - C_2(1) &= \frac{2W(\lambda-1)}{w^2(Y^2-1)} ((\lambda+1)(bY-1) + 2W\Delta). \end{aligned}$$

Their valuations are at least $V - 2t + M$ and $V - 5t + 2M$, respectively, hence at least K . In the signed exceptional sum the exact formulas

$$\sum_{\xi=\pm 1} \xi(\Delta + \lambda\xi) = 2\lambda, \quad \sum_{\xi=\pm 1} \xi(\Delta + \lambda\xi)^2 = 4\lambda\Delta$$

give errors of depths $S + M + V - 2t$ and $S + 2M + V - 3t$, also at least K . Consequently the following congruence holds also when $p = 3$.

Proposition 9.2 (A sum over the conjugates of Δ). For $1 \leq i \leq p-1$,

$$(103) \quad \begin{aligned} W_i &\equiv w^{-i} \sum_{\xi \in \mathcal{T}} (\Delta + \xi)^i \frac{W(b+\xi)}{Y+\xi} \\ &\quad - \frac{s}{w^i(1+b/W)^2} \sum_{\xi \in \mu_{p-1}} \xi(\Delta + \xi)^i. \end{aligned}$$

The second sum is evaluated by the last identity of Lemma 9.3.

Proof. Combine (101) with the power estimates just proved. $\square$

Evaluation of W_i .

Lemma 9.3 (Finite interpolation and power sums). *For $0 \leq i < p$ and $Z \notin -\mathcal{T}$,*

$$\sum_{\xi \in \mathcal{T}} \frac{(\Delta + \xi)^i}{Z + \xi} = \frac{p\Delta^i}{Z} + (p-1) \frac{(\Delta - Z)^i}{Z^p - Z}.$$

For $1 \leq i < p$, the accompanying power sums are

$$\begin{aligned} \sum_{\xi \in \mathcal{T}} (\Delta + \xi)^i &= p\Delta^i + (p-1)\mathbf{1}_{i=p-1}, \\ \sum_{\xi \in \mu_{p-1}} \xi(\Delta + \xi)^i &= (p-1)(\mathbf{1}_{i=p-2} + i\Delta\mathbf{1}_{i=p-1}). \end{aligned}$$

All three identities are exact in mixed and equal characteristic.

Proof. Regard the first identity as an equality of rational functions in Z . Both sides have only simple poles at $-\mathcal{T}$ and vanish at infinity. At zero the residue on each side is Δ^i ; at $-\xi$ with $\xi^{p-1} = 1$, it is $(\Delta + \xi)^i$, since the derivative of $Z^p - Z$ there is $p-1$. Subtraction gives a rational function with no poles and zero value at infinity, hence zero. This works in both characteristics because $p-1$ is nonzero. The last two identities follow by the binomial theorem and $\sum_{\xi \in \mu_{p-1}} \xi^j = 0$ unless $p-1$ divides j , when the sum is $p-1$. $\square$

Apply the lemma with $Z = Y = W + b$. This value has negative valuation, since $v(W) < v(b)$, so all denominators are nonzero. We obtain

$$\begin{aligned} (104) \quad W_i &\equiv \frac{pWb\Delta^i}{w^i Y} + (p-1) \frac{W}{w^{p-1}} \mathbf{1}_{i=p-1} - (p-1) \frac{W^2(\Delta - Y)^i}{w^i D} \\ &\quad - (p-1) \frac{s}{(1+b/W)^2} \left(\frac{\mathbf{1}_{i=p-2}}{w^{p-2}} + \frac{i\Delta\mathbf{1}_{i=p-1}}{w^{p-1}} \right). \end{aligned}$$

The first term has depth at least $V - pt + i(M-t) \geq K$.

The characteristic polynomial of Δ over B_1 gives two useful forms, where $s = -e_{p-1}$:

$$(105) \quad \begin{aligned} b &= \Delta + a + a\omega, & v(\omega) &\geq pt_2 - t, \\ b &= \Delta + a - \Delta s + \rho, & v(\rho) &\geq p(p-1)t_2 - (p^2 - 2p + 2)t. \end{aligned}$$

Each follows by solving the characteristic equation for its constant term and applying (100) to the remaining coefficients.

Lemma 9.4 (Replacing D by D_0). *For $1 \leq i < p$,*

$$\frac{W^2(\Delta - Y)^i}{w^i D} \equiv \frac{W^2(\Delta - Y)^i}{w^i D_0} \pmod{\mathfrak{P}_B^K}.$$

Proof. Write $q = b/W$, $a_1 = a/W$, $d_1 = \Delta/W$, $h_1 = a\omega/W$. Then $q = a_1 + d_1 + h_1$ and

$$D - D_0 = W^p E, \quad E = (1+q)^p - 1 - a_1^p - W^{1-p}q.$$

Since $d_1^p = (\Delta + a)/W^p$, $W^{1-p}q = d_1^p + W^{1-p}h_1$. Every term of E is p -divisible, or contains h_1^p , or contains $W^{1-p}h_1$. Consequently

$$\begin{aligned} v(E) &\geq \min\{V + p(M-t), p(pM + pt_2 - (p+1)t), \\ &\quad p(p-1)M + pM + pt_2 - (p+1)t\} \geq K + M. \end{aligned}$$

The remaining factor in the difference has valuation $(p(p-2) - i(p-1))M \geq -M$. $\square$

Lemma 9.5 (Expansion of $(\Delta - Y)^i$). *Put $A_0 = -a - W$. Modulo $\mathfrak{P}_B^K$,*

$$\frac{W^2(\Delta - Y)^i}{w^i D_0} \equiv \frac{W^2}{w^i D_0} \begin{cases} A_0^i, & i \leq p-2, \\ A_0^{p-1} + (p-1)A_0^{p-2}\Delta s, & i = p-1. \end{cases}$$

Proof. By (105), $\Delta - Y = A_0 + \Delta s - \rho$. Set $Z = S - t$ and $R = p(p-1)t_2 - (p^2 - 2p + 2)t$. For $i \leq p-2$ the first perturbation after the displayed weight has depth at least $2(p-1)M + Z \geq K$. At $i = p-1$, the terms containing ρ have valuation at least $(p-1)M + R \geq K$; the terms of degree at least two in Δs have valuation at least $(2p-1)M + 2Z \geq K$. At $p = 3$, $\delta = 0$, $M = t + 1$ the three differences from K are 3, 1, 4, respectively. $\square$

Lemma 9.6 (The terms containing s). *Modulo $\mathfrak{P}_B^K$,*

$$\frac{s}{w^{p-2}(1+b/W)^2} \equiv \frac{s}{w^{p-2}C^2}.$$

The congruence remains valid after multiplication by Δ/w . For $i = p-1$, the terms linear in Δs in (104) and Lemma 9.5 have sum zero modulo $\mathfrak{P}_B^K$.

Proof. The difference of the inverse squares is

$$\frac{(a-b)/W [2 + (a+b)/W]}{(1+b/W)^2(1+a/W)^2}.$$

Since $v(a-b) \geq -t$, its weighted first error has depth $S + 2(p-1)M - t \geq K$; multiplication by Δ/w only increases it. After extracting the common $(p-1)^2 \Delta s / w^{p-1}$, the two linear terms have bracket

$$-\frac{1}{(1+a/W)^2} + \frac{(1+a/W)^{p-2}}{1-W^{1-p}+(a/W)^p}.$$

The numerator of this difference is $W^{1-p} + \sum_{j=1}^{p-1} \binom{p}{j} (a/W)^j$. The first weighted term has depth $S - t + (p^2 - 1)M \geq K$; the others have depth at least $S - t + (p-1)M + V + p(M-t) \geq K$. $\square$

Theorem 9.7 (Formula for W_i). *For every odd p and $1 \leq i \leq p-1$,*

$$\begin{aligned} W_i \equiv & -(p-1) \frac{s}{w^{p-2}C^2} \mathbf{1}_{i=p-2} + (p-1) \frac{W}{w^{p-1}} \mathbf{1}_{i=p-1} \\ & - (p-1) \frac{W^2(-a-W)^i}{w^i(W^p - W + a^p)} \pmod{\mathfrak{P}_B^{1+pt_2}}. \end{aligned}$$

Proof. Insert Lemmas 9.4–9.6 into (104). $\square$

Remark 9.8. Theorem 9.7 is applied to the logarithm of Q_1 in Subsection 9.3. The term with $i = p-2$ comes from the second sum in (103).

9.2. The quadratic trace calculation.

The character data and local estimates. Retain $A, B_i, B, \Delta, a, b, s, t, t_2, t', \delta$ and the norms and traces of Subsection 9.1. Put $T_i = \text{Tr}_{B_i/A}$. Thus $v_B|_{B_i} = pv_i$ and $v_i|_A = pv_A$. Let $u = \alpha/\gamma \in A$, $m = v_A u$, $M = pm > t$, and choose $w \in B_2^\times$ with $n_2 w = u^{-1}$ and $v_2 w = -m$. These choices are supplied by Lemmas 8.17 and 8.18. Write

$$\begin{aligned} W &= u^{-1}, \quad C_b = 1 + b/W, \quad C = 1 + a/W, \quad D_0 = W^p - W + a^p, \\ q &= p-1, \quad S = p(p-1)\delta, \quad K = 1 + pt_2, \quad F = 1 + t + t' = 1 + 2t + p\delta. \end{aligned}$$

With $v = v_B$, one has

$$v(w) = -M, \quad v(W) = -pM, \quad v(a) = v(b) = -pt, \quad v(C) = v(C_b) = 0, \quad v(D_0) = -p^2M.$$

In mixed characteristic set $e = v_B(p)$; the cyclic ramification bound gives $e \geq p(p-1)t_2$. In equal characteristic all p -divisible terms vanish.

Put $\Psi_E(x) = e_E(\alpha x)$, with trace-compatible e_E . Its largest trivial ideals on A, B_1, B_2 have exponents $1 + t_2, 1 + t', 1 + t_2$, respectively. The character χ of conductor F satisfies

$$(106) \quad \chi(1-x) = \Psi_{B_1}(bP(x)), \quad x \in \mathfrak{P}_1^{[F/p]}, \quad P(X) = \sum_{j=1}^{p-1} X^j/j.$$

This is the formula obtained in Theorem 8.9.

Lemma 7.6 gives

$$(107) \quad v_1(E_j^{B/B_1}(Y\Delta)) \geq (p-1)t_2 + j(v_2Y - t), \quad 1 \leq j < p,$$

$$(108) \quad v_1(S_1(Y\Delta^j)) \geq (p-1)t_2 - jt + v_2Y, \quad 0 \leq j < p.$$

Here $Y \in B_2$. We also use [7, Lemma 3.4]:

$$(109) \quad v_F(E_j^{E/F}z) \geq \left\lfloor \frac{jev_Ez + (p-1)(b_0+1)}{p} \right\rfloor, \quad 1 \leq j < p,$$

for a cyclic extension E/F of break b_0 , together with (7).

Put

$$(110) \quad T = S_1((\Delta/w)^{p-1}), \quad \mathcal{Q} = S_2\left(\frac{b(\Delta/w)^{p-1}T}{C_b^2}\right).$$

Here $T \in B_1$ and $\mathcal{Q} \in B_2$.

Reduction to a finite sum. All congruences in this and the next subsection are in B modulo $\mathfrak{P}_B^K$. Set $f(X) = X/(1+X/W)^2$ and $\mathcal{T} = \{0\} \cup \mu_{p-1} \subset A$. The symmetric-coefficient estimates give

$$(111) \quad v(T) \geq S + qM, \quad v\left(S_1(\Delta^j/w^q)\right) \geq p(p-1)t_2 - pj t + qM \quad (0 \leq j \leq q).$$

The characteristic polynomial of Δ over B_1 also gives

$$(112) \quad b = \Delta + a + a\omega, \quad v(\omega) \geq pt_2 - t.$$

Indeed the error divided by a has terms of valuation at least $(p-1)(pt_2 - jt)$ for $1 \leq j < p$, by (107); their minimum is at least $pt_2 - t$.

Lemma 9.9 (Replacing Δ_ξ by $\Delta + \xi$). *In the sum defining $\mathcal{Q}$ one may replace every B/B_2 conjugate of Δ by its corresponding $\Delta + \xi$, $\xi \in \mathcal{T}$, modulo $\mathfrak{P}_B^K$. This holds for every odd p , including $p = 3$ in mixed characteristic.*

Proof. In equal characteristic the statement is exact. In mixed characteristic, Hensel's lemma applied at $\Delta + \xi$ gives

$$\Delta_\xi = \Delta + \xi + \epsilon_\xi, \quad v(\epsilon_\xi) \geq e - qt.$$

The derivative is a unit since $e - qt > 0$, and the p approximations have distinct residues relative to one another. They are all the conjugates. The outer power difference, after division by w^q , has valuation at least $e - (2p-3)t + qM$. The inner trace difference has this bound plus qt' , by the trace-ideal formula for B/B_1 . Both resulting product errors have valuation at least

$$(113) \quad e + S + 2qM - 3qt.$$

Here $f(N_1\Delta_\xi)$ has valuation $-pt$, and the unchanged outer power and inner trace have bounds $q(M-t)$ and $S + qM$.

For the norm argument itself put $a_0 = e - (p-2)t$. The quotient $\epsilon_\xi/(\Delta + \xi)$ has valuation at least a_0 . By (109),

$$v\{N_1\Delta_\xi - N_1(\Delta + \xi)\} \geq -pt + \min\{pa_0, a_0 + qt'\} = e - qt + S.$$

The equality for this lower bound uses $a_0 \geq t'$, which follows from $e \geq pqt_2$. The exact identity

$$f(X) - f(Y) = (X - Y) \frac{1 - XY/W^2}{(1 + X/W)^2(1 + Y/W)^2}$$

shows that f does not lower this difference valuation. Multiplying by the outer power and the inner trace gives the further bound $e + 2S + 2qM - 2qt$. After $e \geq pq(t + \delta)$ and $M \geq t + 1$, the excesses of the two bounds over K are at least

$$(p^2 - 3p + 1)t + p(2p - 3)\delta + 2p - 3,$$

$$p(p - 2)t + p(3p - 4)\delta + 2p - 3,$$

respectively. Both are positive for $p \geq 3$, so the replacements preserve $\mathcal{Q}$ modulo $\mathfrak{P}_B^K$. $\square$

Lemma 9.10 (A rational sum for $\mathcal{Q}$). *One has*

$$(114) \quad \mathcal{Q} \equiv \frac{T}{w^q} \sum_{\xi \in \mathcal{T}} (\Delta + \xi)^q f(b + \xi).$$

Proof. After Lemma 9.9, expand the translated inner trace as

$$S_1((\Delta + \xi)^q/w^q) = \sum_{j=0}^q \binom{q}{j} \xi^{q-j} \mathcal{U}_j, \quad \mathcal{U}_j = S_1(\Delta^j/w^q).$$

For $j \leq p-3$, the product valuation is at least

$$pqt_2 + 2qM - (2p-1+pj)t \geq K.$$

The weakest case $j = p-3$ has excess at least $(p-1)t + p(p-2)\delta + 2p-3$. For $j = p-2$, its scalar is a multiple of ξ . Subtract the constant product $f(b)\Delta^q/w^q$, whose weighted sum is zero because $\sum_{\xi \neq 0} \xi = 0$. The outer-power difference has valuation at least $qM - (q-1)t$; also $f(N_1(\Delta + \xi)) - f(b)$ has valuation at least zero. Using (111), the two errors are bounded below by

$$pqt_2 + 2qM - (p^2-2)t, \quad pqt_2 + 2qM - (p^2-p-1)t.$$

Their excesses are at least $p(p-2)\delta + 2p-3$ and $(p-1)t + p(p-2)\delta + 2p-3$. Thus only $\mathcal{U}_q = T$ remains.

The exact norm polynomial now reads

$$N_1(\Delta + \xi) = b + \xi - \xi s + r_\xi, \quad v(r_\xi) \geq pt_2.$$

First delete r_ξ using the exact difference formula for f . Next expand in $-\xi s$. Expansion of the inverse square as a geometric power series shows that its Taylor coefficients of degree $j \geq 2$ have valuation at least $(j-1)pM$; no division by a factorial divisible by p is used. The series converges because $v(s/W) > 0$. Finally replace $f'(b + \xi)$ by $f'(b)$; their difference has valuation at least pM . After multiplication by the power and trace factors, the respective error bounds are

$$S + 2qM - qt + pt_2, \quad 3S + (3p-2)M - qt, \quad 2S + (3p-2)M - qt.$$

At $M = t + 1$, their excesses over K are, respectively,

$$qt + S + 2p - 3, \quad qt + p(3p-4)\delta + 3p - 3, \quad qt + p(2p-3)\delta + 3p - 3,$$

all positive. Increasing M only improves them. The only remaining signed term contains

$$\sum_{\xi \neq 0} \xi(\Delta + \xi)^q = q^2 \Delta.$$

Its valuation is at least $2S + 2qM - t$, whose excess is at least $(p-3)t + p(2p-3)\delta + 2p-3 \geq 0$. This proves (114). $\square$

Evaluation of the rational sum. Apply Lemma 9.3 and its derivative with respect to Z .

Theorem 9.11 (Formula for $\mathcal{Q}$). *For $\mathcal{Q}$ defined in (110),*

$$(115) \quad \mathcal{Q} \equiv (p-1) \frac{aT}{w^{p-1}C^2} \pmod{\mathfrak{P}_B^{1+pt_2}}.$$

The right side belongs to B . Lemma 9.25 constructs a congruent element of B_2 .

Proof. Put $L = W + b$, $A_* = \Delta - L$, and $D_L = L^p - L$. Thus $v(L) = v(A_*) = -pM$ and $v(D_L) = -p^2M$. Apply Lemma 9.3 with $i = q$ and differentiate with respect to Z , holding W, b, Δ constant. Writing

$$H(Z) = p\Delta^q/Z + q(\Delta - Z)^q/(Z^p - Z),$$

we obtain the exact equality

$$\sum_{\xi} (\Delta + \xi)^q f(b + \xi) = W^2 \{H(L) + WH'(L)\}.$$

After collecting terms this is

$$(116) \quad \frac{qW^2 A_*^{p-2}(\Delta - b)}{D_L} + \frac{pW^2 \Delta^q b}{L^2} - \frac{pqW^3 A_*^{p-2}}{D_L} - \frac{qW^3 A_*^q(pL^{p-1} - 1)}{D_L^2}.$$

The valuations of the last three terms, splitting the last one at $pL^{p-1} - 1$, are bounded below by

$$e - (2p - 1)t, \quad e - pM, \quad e - pM, \quad p(p - 2)M.$$

The factor T/w^q has valuation at least $S + 2qM$. Consequently their products have the three lower bounds

$$e + S + 2qM - (2p - 1)t, \quad e + S + (p - 2)M, \quad S + (p^2 - 2)M.$$

At $e = pq(t + \delta)$, $M = t + 1$, their excesses over K are, respectively,

$$\begin{aligned} (p^2 - 2p - 1)t + p(2p - 3)\delta + 2p - 3, \\ (p^2 - p - 2)t + p(2p - 3)\delta + p - 3, \\ (p^2 - p - 2)t + p(p - 2)\delta + p^2 - 3. \end{aligned}$$

All are nonnegative for $p \geq 3$; increasing e or M preserves this. In equal characteristic the terms divisible by p are zero.

It remains to treat the first term of (116). By (112), $b - \Delta = a(1 + \omega)$ and $v(a\omega) \geq p\delta - t \geq -t$. Both $A_* / [-(W + a)]$ and $L/(W + a)$ belong to U_B^{pM-t} , and $D_L/L^p = 1 - L^{1-p} \in U_B^{p(p-1)M}$. Since $p - 2$ is odd,

$$\frac{W^2 A_*^{p-2}}{D_L} = -C^{-2}(1 + \eta_0), \quad v(\eta_0) \geq pM - t.$$

The first term is therefore $qaC^{-2}(1 + \omega)(1 + \eta_0)$. After multiplication by T/w^q , the two replacement errors have lower bounds

$$S + 2qM + pt_2 - (p + 1)t, \quad S + (3p - 2)M - (p + 1)t.$$

Their excesses over K , at $M = t + 1$, are $(p - 3)t + pq\delta + 2p - 3$ and $(p - 3)t + p(p - 2)\delta + 3p - 3$. The product error is deeper. Combining this with Lemma 9.10 proves (115). $\square$

9.3. The first factor Q_1 .

A quotient of norms. Retain the notation of Subsection 9.2. Let χ, φ be the compatible characters of Theorem 8.9. Their coefficients in (82) are $\alpha b, \alpha a$, and their conductors are $m_1 = 1 + t + t'$, $m_2 = 1 + t + t_2$. Put $\Psi_L(x) = e_L(\alpha x)$; its largest trivial ideals over A, B_1, B_2 have exponents $1 + t_2, 1 + t', 1 + t_2$, respectively. Let

$$u = \alpha/\gamma, \quad W = 1/u = n_2 w, \quad w = w_2 \in B_2, \quad v_A u = m, \quad pm > t, \quad M = pm.$$

The factor to be evaluated is

$$Q_1 = \frac{\chi(\gamma + \alpha b)}{\varphi(\gamma - \alpha w + \alpha a)}.$$

Lemma 9.12 (A formula for Q_1^{-1}). *For $z = \Delta/w$ and $n = N_1 z = ub$, one has*

$$Q_1^{-1} = \chi\left(\frac{N_1(1 + z)}{1 + n}\right).$$

Proof. The exact Artin-Schreier polynomial gives

$$N_2(1 + z) = 1 - w^{1-p} + a/w^p.$$

The relative difference between this and $1 - uw + ua$ is controlled by

$$(1 - uw + ua) - N_2(1 + z) = (uw^p - 1) \frac{a - w}{w^p}.$$

The cyclic norm–power estimate gives $v_2(uw^p - 1) \geq (p-1)t_2$. Hence the difference has valuation at least

$$(p-1)t_2 + pm - \max(t, m) \geq 1 + t + t_2 = m_2.$$

For $m \leq t$, subtracting m_2 leaves $(p-2)t_2 + pm - 2t - 1 \geq (p-3)t$. For $m > t$ it leaves $(p-2)t_2 + (p-1)m - t - 1 \geq 0$. All compared elements are units, so their φ -values agree. Compatibility gives $\varphi N_2(1+z) = \chi N_1(1+z)$. Finally $\chi(\gamma) = \varphi(\gamma)$ by Lemma 4.2, and $N_1 z = ub$ exactly. This proves the formula. $\square$

Expansion by the truncated logarithm. Put

$$e_i = E_i^{B/B_1}(z), \quad \mathcal{U}_i = S_1(z^i), \quad T = \mathcal{U}_{p-1}, \quad E = \frac{\sum_{i=1}^{p-1} e_i}{1+n},$$

$$W_i = S_2\left(\frac{bz^i}{1+n}\right), \quad \mathcal{Q} = S_2\left(\frac{bz^{p-1}T}{(1+n)^2}\right).$$

Proposition 9.13 (Evaluation of Q_1^{-1} by W_i and $\mathcal{Q}$). *One has*

$$(117) \quad Q_1^{-1} = \Psi_2\left(\sum_{i=1}^{p-1} \frac{(-1)^i}{i} W_i + \frac{\mathcal{Q}}{2(p-1)^2}\right).$$

Proof. By Lemma 9.12, $Q_1^{-1} = \chi(1+E)$. Lemma 7.6 gives

$$v_1(e_i) \geq (p-1)t_2 + i(m-t).$$

Write $m_0 = \lceil t/p \rceil$; then $m \geq m_0$ because $p \nmid t$. All the indicated bounds are increasing in m , and at $m = m_0 \leq t$ their minimum is $\nu_0 = (p-1)(\delta + m_0)$. Writing $t = pa + b_0$, $1 \leq b_0 \leq p-1$, checks directly that

$$\nu_0 \geq \delta + \lceil (2t+1)/p \rceil = \lceil m_1/p \rceil, \quad 3\nu_0 \geq m_1.$$

Thus (106) applies to $1+E$, and

$$\chi(1+E) = \Psi_1(b(-E + E^2/2)).$$

The terms of degree at least three in $P(-E)$ belong to $\mathfrak{p}_{B_1}^{m_1}$ because $3\nu_0 \geq m_1$.

In a nonlinear Newton term, or in a product $e_i e_j$ other than e_{p-1}^2 , the total symmetric index is at most $2p-3$ and there are at least two factors. After multiplication by b , its B -valuation is at least

$$2p(p-1)t_2 + (i+j)(M-pt) - pt.$$

Its worst bound occurs at $M = t+1$ and $i+j = 2p-3$; the excess over $p(1+t')$ is

$$(p-3)t + p(p-2)\delta + p-3 \geq 0.$$

Terms with additional factors have no smaller bound. Consequently only e_{p-1}^2 can contribute among the quadratic terms. Newton's identities, dividing by $1, \dots, p-1$, give

$$\sum_{i=1}^{p-1} e_i \equiv \sum_{i=1}^{p-1} \frac{(-1)^{i-1}}{i} \mathcal{U}_i, \quad e_{p-1}^2 \equiv T^2/(p-1)^2$$

after multiplication by b , modulo $\mathfrak{p}_{B_1}^{1+t'}$. In the second replacement the error has at least three positive-depth symmetric factors and is killed by $3\nu_0 \geq m_1$. The remaining exponent is therefore

$$-\frac{b}{1+n} \sum_i \frac{(-1)^{i-1}}{i} \mathcal{U}_i + \frac{bT^2}{2(p-1)^2(1+n)^2}.$$

Pull one trace S_1 out of each summand, then use $\Psi_B = \Psi_2 S_2$. This is precisely (117). $\square$

Consequences for the final Q_1 expression. By Theorems 9.7 and 9.11, modulo $\mathfrak{P}_B^K$ with $K = 1 + pt_2$, we have

$$W_i \equiv -(p-1) \frac{s \mathbf{1}_{i=p-2}}{w^{p-2}(1+a/W)^2} + (p-1) \frac{W \mathbf{1}_{i=p-1}}{w^{p-1}} - (p-1) \frac{W^2(-a-W)^i}{w^i(W^p - W + a^p)},$$

$$\mathcal{Q} \equiv (p-1) \frac{aT}{w^{p-1}(1+a/W)^2}.$$

Put

$$C = 1 + a/W, \quad F_i = \frac{W}{W^p - W + a^p} \left(\frac{W+a}{w} \right)^i,$$

$$R = W \left(-F_{p-1} + \sum_{i=1}^{p-2} \frac{F_i}{i} + w^{1-p} \right).$$

Let $\mathcal{L}_1 \in B_2$ denote the argument of Ψ_2 in (117). Substitution gives

$$\mathcal{L}_1 \equiv R + \frac{s}{2w^{p-2}C^2} - \frac{aT}{2w^{p-1}C^2} \pmod{\mathfrak{P}_B^K}.$$

Let $y_s, y_T \in B_2$ be the representatives constructed in Lemma 9.25. Since Ψ_2 is trivial on $\mathfrak{p}_{B_2}^{t_2+1}$, the congruence gives

$$Q_1^{-1} = \Psi_2(R + (y_s - y_T)/2).$$

Indeed $(-1)^i$ cancels the $(-1)^i$ in $(-a-W)^i$. For $i < p-1$ one replaces $-(p-1)/i$ by $1/i$ only after its factor p is weighted; $i = p-1$ gives exactly $-F_{p-1}$ and is not changed. The exceptional coefficient $(p-1)/(p-2)$ becomes $1/2$ modulo its weighted kernel, and $1/[2(p-1)]$ becomes $-1/2$ for the quadratic term. For $i \leq p-2$, direct valuation gives

$$v_2(F_i) = (p-1)(p-i)m, \quad v_B(pWF_i) \geq v_B(p) + (p-2)M \geq K.$$

The coefficient of F_{p-1} was never changed. The errors from the two remaining rational-coefficient replacements are weighted by pX_s and pX_T ; their valuations are at least K by the displayed positive source depths in Lemma 9.25. These terms are zero in equal characteristic. Thus the formula is valid for every m with $pm > t$.

9.4. The second factor Q_2 .

An elementary cyclic norm-quotient lemma. Let E/F be totally ramified cyclic of odd prime degree p , with break $t \geq 1$ and different exponent $D = (p-1)(t+1)$. Let λ have conductor $n \geq 2$ and coefficient γ in the following formula for a fixed additive character ψ_F :

$$\lambda(1-z) = \psi_F(\gamma Pz), \quad P(Z) = \sum_{j=1}^{p-1} Z^j/j, \quad z \in \mathfrak{p}_F^{[n/p]}.$$

Thus $v_F\gamma = -n_F(\psi_F) - n$ and $\psi_E = \psi_F \operatorname{Tr}_{E/F}$.

Lemma 9.14. *Let $k \in \mathfrak{P}_E^a$ with $a \geq 1$, and put*

$$b = \left\lfloor \frac{a+D}{p} \right\rfloor.$$

If $2b \geq n$, then

$$(118) \quad \lambda\left(\frac{N_{E/F}(1-k)}{1-N_{E/F}k}\right) = \psi_E\left(\frac{\gamma}{1-N_{E/F}k}P(k)\right).$$

The hypothesis $2b \geq n$ is applied to the quotient, rather than to its two factors.

Proof. Write $e_i = E_i^{E/F}(k)$ and $q_j = \operatorname{Tr}_{E/F}(k^j)$. The exact norm expansion is

$$\frac{N(1-k)}{1-Nk} = 1 + Q, \quad Q = \frac{\sum_{i=1}^{p-1} (-1)^i e_i}{1-Nk}.$$

The denominator is a unit, since $v_F(Nk) = v_E k \geq a > 0$. The strong symmetric estimate gives $v_F(e_i) \geq b$ for every $1 \leq i < p$, and hence $v_F Q \geq b$. As $2b \geq n$, one has $b \geq \lceil n/2 \rceil \geq \lceil n/p \rceil$, and all terms of degree at least two in $P(-Q)$ are killed at conductor n . Consequently

$$\lambda(1+Q) = \psi_F(-\gamma Q).$$

Newton's identities, dividing only by integers $j < p$, give

$$e_j \equiv (-1)^{j-1} q_j/j \pmod{\mathfrak{p}_F^{2b}}, \quad 1 \leq j < p.$$

Indeed q_i also has valuation at least b , by the trace-ideal formula, and every non-leading term in the Newton recursion contains a product of two such positive-depth factors. Therefore

$$-Q \equiv \frac{\sum_{j=1}^{p-1} q_j/j}{1-Nk} = \frac{\operatorname{Tr}_{E/F} P(k)}{1-Nk} \pmod{\mathfrak{p}_F^n}.$$

Multiplication by γ and trace compatibility prove (118). The proof is valid in both characteristics, including $p = 3$. $\square$

Factorization of Q_2 . Use the totally ramified notation $t_1 = t$, $t_2 = t + \delta$, $r = t_2 + m$, $pm > t$, $u = \alpha/\gamma$, $W = 1/u$, $w = w_2$, $n_2 w = 1/u$, $a = N_2 \Delta$, $b_0 = N_1 \Delta$ and $H = n_1(s)$, with $s = -E_{p-1}^{B/B_1}(\Delta)$. Put

$$C = 1 + ua, \quad P_* = n_2 C, \quad q_0 = u^{p-1}, \quad Y = P_* - q_0, \quad Z = \frac{n_1(1 + ub_0)}{Y}, \quad k = \frac{uw}{C}, \quad F_* = \frac{P_*}{Y}.$$

All of C, P_*, Y, Z are units, and

$$n_2 k = q_0/P_*, \quad v_2 k = (p-1)m, \quad F_* = (1 - n_2 k)^{-1}.$$

The factor to be evaluated is

$$Q_2 = \lambda\left(\frac{n_1(\gamma + \alpha b_0)}{n_2(\gamma - \alpha w + \alpha a)}\right).$$

Cancellation of the common γ^p , followed by multiplicativity, gives

$$(119) \quad Q_2^{-1} = \lambda(Z^{-1}) \lambda\left(\frac{n_2(1-k)}{1-n_2 k}\right).$$

The second factor is evaluated by Lemma 9.14.

Weighted trace sums and the congruence for Z . Set $A_0 = n_2 a$, $D_0 = 1 + u^p A_0$. Write $E_i^{(1)}, E_i^{(2)}$ for the lower elementary symmetric functions of b_0, a , and put $F_i = u^i(E_i^{(1)} - E_i^{(2)})$.

Lemma 9.15 (Weighted trace sums). *For $1 \leq j \leq p-1$,*

$$u^j [T_1(b_0^j) - T_2(a^j)] \equiv (p-1) \mathbf{1}_{j=p-1} u^{p-1} (1 - s^p) \pmod{\mathfrak{P}_1^{1+pr}}.$$

Consequently

$$F_i \equiv -\mathbf{1}_{i=p-1} u^{p-1} (1 - s^p) \pmod{\mathfrak{P}_1^{1+pr}}, \quad 1 \leq i < p.$$

Proof. For $j = 1$ Theorem 7.7, multiplied by u , gives the required modulus. For $j \geq 2$, apply Theorem 7.10 to $w^{-j} \Delta^{j-1}$, whose B -valuation is $jpm - (j-1)t > 0$. Its error exponent $pt_2 + jpm - (j-1)t$ is at least $1 + pr$, since their difference is $(j-1)(pm - t) - 1 \geq 0$. The exact discrepancy from multiplicative homogeneity is

$$T_2((u^j - w^{-pj})a^j).$$

The lower cyclic norm-power estimate gives $v_2(u^j - w^{-pj}) \geq jpm + (p-1)t_2$. After adding the different, the trace is in $\mathfrak{p}_A^{r+1}$, because

$$(j-1)pm + (p-2)t_2 - jt - 1 \geq (p-2)t_2 - t + j - 2 \geq 0.$$

This proves the moment congruence, including its exceptional term. Apply Newton's identities to the two integral tuples ub_0 and ua . Induction over $i < p$ introduces only integral factors and division by i , a unit. At $i = p-1$ its leading sign is negative, so division of $(p-1)u^{p-1}(1 - s^p)$ by $-(p-1)$ gives the displayed F_{p-1} . All other F_i vanish at the same modulus. $\square$

Proposition 9.16 (A congruence for Z). *One has*

$$Z \equiv 1 + u^{p-1} \frac{H}{D_0} \pmod{\mathfrak{p}_A^{r+1}}.$$

Proof. The highest-degree terms in the norm expansions of ub_0 and ua agree by norm transitivity. Summing Lemma 9.15 over the remaining symmetric degrees gives

$$n_1(1 + ub_0) - Y \equiv u^{p-1} s^p \pmod{\mathfrak{P}_1^{1+pr}}.$$

The ordinary lower cyclic norm-power estimate gives directly

$$v_1(u^{p-1}(s^p - H)) \geq p(p-1)m + p(p-1)\delta + (p-1)t \geq 1 + pr,$$

because the excess is $p(p-2)(m + \delta) - t - 1 \geq 0$. Both sides after this substitution belong to A , so intersection with A improves the modulus to $\mathfrak{p}_A^{r+1}$.

It remains to prove $u^{p-1}H(Y^{-1} - D_0^{-1}) \in \mathfrak{p}_A^{r+1}$. One has

$$Y - D_0 = \sum_{i=1}^{p-1} E_i^{B_2/A}(ua) - u^{p-1}, \quad v_A E_i(ua) \geq L_0 = \left\lfloor \frac{pm - t + (p-1)(t_2 + 1)}{p} \right\rfloor.$$

The weighted intermediate terms have valuation at least $(p-1)m + (p-1)\delta + L_0 \geq r + 1$. For completeness write $t = pa + b$, $1 \leq b \leq p-1$, and use $m \geq a + 1$. At $\delta = 0$ the excess is at least

$$(p-3)a + p - 2 - b + \left\lfloor \frac{(p-2)b + p - 1}{p} \right\rfloor \geq 0.$$

For $b \leq p-2$ every indicated term is nonnegative; for $b = p-1$ the final floor is $p-2$, giving again a nonnegative result. Increasing δ or m cannot decrease the excess. The weighted final $-u^{p-1}$ term has excess

$$(2p-3)m + (p-2)\delta - t - 1 \geq pm - t - 1 \geq 0.$$

All denominators are units. These estimates prove $u^{p-1}H(Y^{-1} - D_0^{-1}) \in \mathfrak{p}_A^{r+1}$ and hence the proposition. $\square$

Theorem 9.17 (Evaluation of Q_2). *One has*

$$Q_2^{-1} = \Psi_A \left(Wu^{p-1} \frac{H}{D_0} \right) \Psi_2(WF_*P(k)), \quad \Psi_L(x) = e_L(\alpha x).$$

Proof. The conductor of λ is $n = r + 1 = t_2 + m + 1$. For the second factor of (119), Lemma 9.14 has

$$a = (p-1)m, \quad D = (p-1)(t_2 + 1), \quad b = \left\lfloor \frac{(p-1)n}{p} \right\rfloor.$$

For every $n \geq 2$ and odd p , $2\lfloor (p-1)n/p \rfloor \geq n$; this is $n \geq 2\lceil n/p \rceil$, checked at $p = 3$ and then monotone in p . The lemma therefore gives the second displayed phase.

For the first factor, $v_A H \geq (p-1)\delta$. The argument $u^{p-1}H/D_0$ has valuation at least $L = (p-1)(m + \delta)$, and

$$2L - n = (2p-3)m + (2p-3)\delta - t - 1 \geq pm - t - 1 \geq 0.$$

Hence $\lambda((1+x)^{-1}) = e_A(\gamma x)$ for this argument. Proposition 9.16 therefore gives the first factor in the stated formula. $\square$

9.5. An estimate for $u^{p-1}A_0(J + u^{p-1}H)$.

Notation. Let $A \subset B_1, B_2 \subset B$ be totally ramified extensions with $\text{Gal}(B/A) = C_p^2$, p odd. Put

$$t = t_1, \quad \delta = t_2 - t \geq 0, \quad p \nmid t, \quad N_i = N_{B/B_i}, \quad n_i = N_{B_i/A}, \quad S_i = \text{Tr}_{B/B_i}.$$

Let $\Delta^p - \Delta = a \in B_2$, $v_B \Delta = -t$, and set

$$s = -E_{p-1}^{B/B_1}(\Delta), \quad Y = w_2^{-1} \in B_2, \quad v_2(Y) = m, \quad n_2 Y = u, \quad M = pm > t.$$

Thus $v_B(Y) = M$, $v_A(u) = m$. Define

$$T = S_1((Y\Delta)^{p-1}), \quad H = n_1(s), \quad J = n_1(T), \quad A_0 = n_2(a), \quad v_A(A_0) = -t.$$

In (150), the first term is $u^{p-1}A_0(J + u^{p-1}H)/(2D^2)$, where D is a unit. Since Ψ_A is trivial on $\mathfrak{p}_A^{1+t_2}$, we shall prove

$$(120) \quad L_A := u^{p-1}A_0(J + u^{p-1}H) \in \mathfrak{p}_A^{1+t_2}.$$

A valuation bound for $T + Y^{p-1}s$. Write $v = v_B$ in this subsection, and put $z = Y\Delta$. Lemma 7.6 gives

$$(121) \quad v(E_i \Delta) \geq p(p-1)t_2 - pit,$$

$$(122) \quad v(E_i z) \geq p(p-1)t_2 + i(M - pt), \quad 1 \leq i < p.$$

Lemma 7.1 gives

$$(123) \quad v_E(N_{E/F} x - x^p) \geq pv_E(x) + (p-1)\text{break}(E/F).$$

It holds also for negative valuations and in both characteristics. Set

$$R = (p-1)M + p(p-1)\delta, \quad B_0 = R + (p-1)t.$$

Lemma 9.18. *One has*

$$(124) \quad v_B(T + Y^{p-1}s) \geq B_0.$$

Moreover $v_B(T), v_B(Y^{p-1}s) \geq R$.

Proof. The last assertion is the trace estimate in Lemma 7.6 and (121). Subtract Y^p times the characteristic-polynomial identity for Δ from the identity for z . The result is

$$\sum_{i=1}^{p-1} (-1)^i (E_i z - Y^i E_i \Delta) z^{p-i} - N_1 \Delta (n_2 Y - Y^p) = 0.$$

For $i \leq p-2$, (121)–(122) bound the corresponding summand below by

$$pM + p(p-1)t_2 - [p + (p-1)i]t.$$

Equation (123) for B_2/A bounds the last summand below by $pM + p(p-1)t_2 - pt$. Solving for the $i = p-1$ summand and dividing by z , of valuation $M - t > 0$, gives

$$(125) \quad v_B(E_{p-1}z - Y^{p-1}E_{p-1}\Delta) \geq (p-1)M + p(p-1)t_2 - (p-1)^2t = B_0.$$

No division by p occurs.

Newton's identity for degree $p-1$ says

$$S_1(z^{p-1}) = -(p-1)E_{p-1}z + \mathcal{R}(E_1z, \dots, E_{p-2}z),$$

where $\mathcal{R}$ is an integral polynomial whose monomials have weight $p-1$ and at least two elementary-symmetric factors. By (122), every such monomial has valuation at least

$$(p-1)M + p(p-1)(t+2\delta) \geq B_0.$$

Using $E_{p-1}\Delta = -s$ and (125), we obtain $T \equiv (p-1)Y^{p-1}s \pmod{\mathfrak{P}_B^{B_0}}$. In mixed characteristic $v_B(p) \geq p(p-1)t_2$, so

$$v_B(pY^{p-1}s) \geq (p-1)M + p(p-1)(t+2\delta) \geq B_0.$$

In equal characteristic this term is zero. Adding $Y^{p-1}s$ now proves (124). $\square$

Comparison with p th powers.

Lemma 9.19. *In B_1 , put*

$$L_1 = u^{p-1}A_0(T^p + u^{p-1}s^p).$$

Then

$$(126) \quad v_B(L_1) \geq V := 2p(p-1)M - pt + p^2(p-1)\delta.$$

Proof. Let $Z = Y^{p-1}s$ and $E = T + Z$. The identity

$$T^p + Z^p = E^p - \sum_{i=1}^{p-1} \binom{p}{i} T^i Z^{p-i}$$

is exact. Lemma 9.18 gives $v(E) \geq B_0$ and $v(T), v(Z) \geq R$. The weight has valuation

$$v_B(u^{p-1}A_0) = p(p-1)M - p^2t.$$

Its product with E^p therefore has valuation at least V . Every mixed binomial term has valuation at least

$$p(p-1)M - p^2t + v_B(p) + pR \geq V.$$

In equal characteristic these mixed terms vanish.

It remains to replace Z^p by $u^{p-1}s^p$. Equation (123), applied to $Y \in B_2$, gives

$$v_B(u/Y^p - 1) \geq p(p-1)t_2,$$

so

$$v_B(u^{p-1} - Y^{p(p-1)}) \geq p(p-1)M + p(p-1)t_2.$$

After multiplying by $u^{p-1}A_0s^p$, the bound is again at least V by (121). This proves (126). $\square$

Theorem 9.20 (Valuation of L_A). *The element L_A defined in (120) satisfies*

$$(127) \quad v_A(L_A) \geq 2(p-1)m + (p-1)\delta - \lfloor t/p \rfloor \geq 1 + t_2.$$

Therefore $\Psi_A(L_A/(2D^2)) = 1$ for every unit $D \in A$.

Proof. The symmetric-coefficient bounds give $v_1(s) \geq (p-1)\delta$ and $v_1(T) \geq (p-1)(m+\delta)$. Using (123) for B_1/A ,

$$\begin{aligned} v_1(u^{p-1}A_0(T^p - J)) &\geq 2(p-1)M - t + p(p-1)\delta, \\ v_1(u^{2(p-1)}A_0(s^p - H)) &\geq 2(p-1)M - t + p(p-1)\delta. \end{aligned}$$

Both lower bounds are V/p . Lemma 9.19 therefore gives $v_1(L_A) \geq V/p$. Since $L_A \in A$, divide by p again and use integrality of v_A :

$$v_A(L_A) \geq \lceil V/p^2 \rceil = 2(p-1)m + (p-1)\delta - \lfloor t/p \rfloor.$$

Write $t = pa + b$, $a \geq 0$, $1 \leq b \leq p-1$; then $m \geq a+1$. The right side minus $1 + t_2$ is at least

$$(p-3)a + 2p - 3 - b + (p-2)\delta \geq p - 2 \geq 1.$$

This proves (127). Multiplication by a unit denominator D^{-2} and the rational scalar $1/2$ preserves this ideal bound; therefore $\Psi_A(L_A/(2D^2)) = 1$. $\square$

9.6. The product $Q_1Q_2Q_3Q_4$.

The stationary data. Use the notation of Subsections 9.2 and 9.3:

$$\begin{aligned} t = t_1, \quad t_2 = t + \delta, \quad t' = t + p\delta, \quad p \nmid t, \quad \delta \geq 0, \quad p \text{ odd}, \\ N_i = N_{B/B_i}, \quad n_i = N_{B_i/A}, \quad S_i = \text{Tr}_{B/B_i}, \quad T_i = \text{Tr}_{B_i/A}, \quad \Delta^p - \Delta = a \in B_2, \\ b = N_1\Delta, \quad N_2\Delta = a, \quad v_B\Delta = -t, \quad s = -E_{p-1}^{B/B_1}(\Delta). \end{aligned}$$

Let χ, φ be the characters of Theorem 8.9, with equal norm pullbacks and conductors $1 + t + t'$ and $1 + t + t_2$. Put $\theta_1 = \chi(\lambda n_1)$ and $\theta_2 = \varphi(\lambda n_2)$, where $a(\lambda) = 1 + r$, $r = t_2 + m$, $M = pm > t$. The exact choices of Subsection 8.3 give

$$\begin{aligned} u = \alpha/\gamma, \quad W = u^{-1}, \quad v = \beta/\alpha, \quad v_A u = m, \quad v_A v = \delta, \\ n_2 w = \gamma/\alpha = W, \quad v_2 w = -m, \quad n_1 w_1 = \gamma/\beta, \quad v_1 w_1 = -(m + \delta). \end{aligned}$$

Let $\Psi_E(x) = e_E(\alpha x)$, trace compatibly. Its largest trivial ideal on A has exponent $1 + t_2$, and those on B_1, B_2 have exponents $1 + t', 1 + t_2$. The largest trivial ideal for $x \mapsto \Psi_{B_2}(Wx)$ has exponent

$$(128) \quad n_\Gamma = 1 + t_2 + pm.$$

Set

$$\begin{aligned} C = 1 + ua, \quad A_0 = n_2 a, \quad D = 1 + u^p A_0, \quad P_* = n_2 C, \quad H = n_1 s, \\ T = S_1((\Delta/w)^{p-1}), \quad J = n_1 T, \quad S = p(p-1)\delta, \quad K = 1 + pt_2. \end{aligned}$$

Thus C, P_*, D are units and $v_A A_0 = -t$. The symmetric-coefficient bounds give

$$(129) \quad v_1 s \geq (p-1)\delta, \quad v_1 T \geq (p-1)(m + \delta).$$

We use their coefficient version as well: for $z \in B_1$,

$$(130) \quad z = \sum_{i=0}^{p-1} c_i \Delta^i, \quad c_i \in B_2 \implies v_2 c_i \geq v_1 z + it.$$

The cyclic symmetric estimate is

$$(131) \quad v_F(E_j^{E/F} x) \geq \left\lfloor \frac{j v_E x + (p-1)(b_0 + 1)}{p} \right\rfloor, \quad 1 \leq j < p,$$

for a cyclic extension E/F of break b_0 . It implies

$$(132) \quad v_E(Nx - x^p) \geq p v_E x + (p-1)b_0$$

by Lemma 7.1. All these estimates hold in both characteristics.

Theorem 7.7 gives

$$(133) \quad T_1 b - T_2 a \in \mathfrak{p}_A^{1+t_2}.$$

Recall the factors of Proposition 8.22:

$$(134) \quad \begin{aligned} Q_1 &= \chi(A_1)/\varphi(C_2), & Q_2 &= \lambda(n_1 A_1/n_2 C_2), \\ Q_3 &= e_{B_1}(-\beta w_1)\theta_1(C_1/A_1), & Q_4 &= \Psi_{B_2}(w), \end{aligned}$$

where $A_1 = \gamma + \alpha b$, $C_1 = A_1 - \beta w_1$ and $C_2 = \gamma - \alpha w + \alpha a$. By (98), the character θ_1 satisfies $\theta_1(1 - z) = e_{B_1}(C_1 P(z))$ for $z \in \mathfrak{p}_{B_1}^{\lceil (1+t'+pm)/p \rceil}$. We use Proposition 9.13 and Theorems 9.7 and 9.11 to evaluate Q_1 .

Weighted denominator bounds.

Lemma 9.21. *One has*

$$(135) \quad \begin{aligned} u^{p-2}H(P_*^{-2} - D^{-2}) &\in \mathfrak{p}_A^{1+t_2}, \\ u^{p-1}A_0J(P_*^{-2} - D^{-2}) &\in \mathfrak{p}_A^{1+t_2}. \end{aligned}$$

The first assertion also holds with inverse first powers instead of inverse squares.

Proof. The norm expansion and (131) give

$$v_A(P_* - D) \geq \ell := m + \left\lfloor \frac{(p-2)t + (p-1)(\delta+1)}{p} \right\rfloor.$$

Indeed the j th intermediate coefficient has bound $\lfloor [j(pm-t) + (p-1)(t_2+1)]/p \rfloor$, nondecreasing in j . Write $t = pa + b_0$, $1 \leq b_0 \leq p-1$, and $m = a + 1 + c$, $c \geq 0$. Put $f = \lfloor [(p-2)b_0 + (p-1)(\delta+1)]/p \rfloor$. Using (129), the two weighted valuation excesses over $1+t_2$ are bounded below by

$$\begin{aligned} (p-3)a + (p-1)c + p-2 + (p-2)\delta - b_0 + f, \\ (p-3)a + (2p-1)c + 2p-2 + (p-2)\delta - 2b_0 + f. \end{aligned}$$

The first is nonnegative for $b_0 \leq p-2$; if $b_0 = p-1$, use $f \geq p-2$. The second is nonnegative because $b_0 \leq p-1$. All denominators are units; differences of their inverse powers have valuation at least ℓ . This proves every assertion. $\square$

Combining the rational terms in Q_1 and Q_2 . Use the notation of the final formula in Subsections 9.2 and 9.3:

$$F_i = \frac{W}{W^p - W + a^p} \left(\frac{W+a}{w} \right)^i.$$

Put

$$k = \frac{uw}{C}, \quad q_0 = u^{p-1}, \quad F_* = \frac{P_*}{P_* - q_0}, \quad P(X) = \sum_{j=1}^{p-1} \frac{X^j}{j}.$$

These are exactly the variables in Theorem 9.17; in particular $v_2(k) = (p-1)m$. Put $Q = 1 + u^p a^p$ in B_2 , and put

$$\eta = W/w^p, \quad G = \frac{\eta C^p}{Q - q_0}.$$

Both $Q - q_0$ and $P_* - q_0$ are units. Direct multiplication gives

$$(136) \quad F_i = Gk^{p-i}, \quad Gk^p = \frac{q_0}{Q - q_0}.$$

Lemma 9.22 (A congruence for the rational terms). *With n_Γ as in (128),*

$$(137) \quad -F_{p-1} + \sum_{i=1}^{p-2} \frac{F_i}{i} + F_* P(k) \equiv k(1-\eta) \pmod{\mathfrak{P}_2^{n_\Gamma}}.$$

Proof. For $2 \leq j \leq p-1$, $1/(p-j) + 1/j = p/[j(p-j)]$. Thus (136) changes the first two terms on the left into $-GP(k)$, with errors divisible by pk^j for $j \geq 2$. In mixed characteristic their valuation is at least $(p-1)t_2 + 2(p-1)m$, whose excess over n_Γ is $(p-2)(t_2+m) - 1 \geq 0$. In equal characteristic the errors vanish. The coefficient of F_{p-1} has not been changed independently.

The norm–power estimate and the binomial theorem give

$$\begin{aligned} v_2(P_* - C^p) &\geq (p-1)t_2, \\ v_2(C^p - Q) &\geq (p-1)t_2 + pm - t, \\ v_2(1 - \eta) &\geq (p-1)t_2. \end{aligned}$$

In particular $P_* - Q$ has depth at least $(p-1)t_2$. The following identity is exact:

$$(138) \quad F_* - G - (1-\eta) = q_0 \left(\frac{1}{P_* - q_0} - \frac{\eta}{Q - q_0} \right) - \frac{\eta(C^p - Q)}{Q - q_0}.$$

The bracket has depth at least $(p-1)t_2$, as does $F_* - G$. For $j \geq 2$, the valuation of $(F_* - G)k^j$ is at least $(p-1)t_2 + 2(p-1)m \geq n_\Gamma$. For $j = 1$, the first term on the right of (138), multiplied by k , has depth at least

$$(p^2 - 1)m + (p-1)t_2 \geq n_\Gamma.$$

The second has depth at least $(p-1)t_2 + pm - t + (p-1)m$; its excess is

$$(p-2)t_2 + (p-1)m - t - 1 = (p-3)t + (p-2)\delta + (p-1)m - 1 \geq 0.$$

This proves (137). $\square$

Lemma 9.23 (The term containing Q_4). *There is the exact identity in B_2*

$$(139) \quad W[k(1-\eta) + w^{-(p-1)}] - w = \frac{a}{w^{p-1}C}(1 - uw^p) \in \mathfrak{P}_2^{1+t_2}.$$

Proof. Use $Wu = 1$, $\eta = 1/(uw^p)$ and $C-1 = ua$ to obtain the identity by subtraction of the two fractions. The norm-power bound gives $v_2(1 - uw^p) \geq (p-1)t_2$, so the right side has valuation at least $-t + (p-1)m + (p-1)t_2$. Its excess over $1 + t_2$ is $(p-3)t + (p-2)\delta + (p-1)m - 1 \geq 0$. The value of Ψ_{B_2} on this difference is therefore 1. $\square$

The elements $y_s, y_T \in B_2$ and their norms. For $z = \sum_{i=0}^{p-1} z_i \Delta^i$ with $z_i \in B_2$, let $\text{pr}_0 z = z_0$. Define

$$(140) \quad X_s = \frac{s}{w^{p-2}C^2}, \quad X_T = \frac{aT}{w^{p-1}C^2}, \quad y_s = \text{pr}_0 X_s, \quad y_T = \text{pr}_0 X_T.$$

Thus $X_s, X_T \in B$ and $y_s, y_T \in B_2$.

Lemma 9.24 (Constant coefficient and its norm). *Let $z = xw_0$ with $x \in B_2$, $w_0 \in B_1$. Suppose*

$$(141) \quad h := v_B z \geq 1 + t', \quad h \geq 1 + t_2 + (p-3)t.$$

Then $y = \text{pr}_0 z \in B_2$ satisfies

$$(142) \quad z \equiv y \pmod{\mathfrak{P}_B^K}, \quad N_1 z \equiv n_2 y \pmod{\mathfrak{P}_1^K}.$$

If $z \neq 0$, then $v_B y = h$.

Proof. The zero case is immediate. Write $z = \sum y_i \Delta^i$ using (130) for w_0 . It gives

$$v_2 y_i \geq h/p + it, \quad v_B(y_i \Delta^i) \geq h + i(p-1)t.$$

Thus all nonconstant terms are strictly deeper than h and the constant term has valuation h . Since $1 + t' + (p-1)t = K$, all nonconstant terms belong to $\mathfrak{P}_B^K$. This proves the first congruence, with $y = y_0$.

The second bound in (141) and the congruence just proved are precisely the hypotheses of Lemma 7.14, with $K_0 = K$. Applying that lemma to $z = xw_0$ and $y = y_0$ proves the norm congruence in (142). $\square$

Lemma 9.25 (Congruences for y_s, y_T and their norms). *The elements in (140) satisfy*

$$(143) \quad \begin{aligned} X_s &\equiv y_s \pmod{\mathfrak{P}_B^K}, & X_T &\equiv y_T \pmod{\mathfrak{P}_B^K}, \\ n_2 y_s &\equiv u^{p-2} H/D^2 \pmod{\mathfrak{p}_A^{1+t_2}}, & n_2 y_T &\equiv u^{p-1} A_0 J/D^2 \pmod{\mathfrak{p}_A^{1+t_2}}. \end{aligned}$$

Moreover $y_s, y_T \in \mathfrak{P}_2^{c_2}$, where $c_2 = \lceil (1+t_2)/p \rceil$.

Proof. By (129),

$$h_s := v_B X_s \geq (p-2)M + S, \quad h_T := v_B X_T \geq 2(p-1)M - pt + S.$$

At $M \geq t+1$, their excesses over $1+t'$ are at least

$$(p-3)(t+1) + p(p-2)\delta, \quad (p-3)t + 2p-3 + p(p-2)\delta,$$

and their excesses over $1+t_2 + (p-3)t$ are at least

$$p-3 + [p(p-1)-1]\delta, \quad 2p-3 + [p(p-1)-1]\delta.$$

All are nonnegative, including $p = 3, \delta = 0$. Lemma 9.24 therefore applies to both products and proves the first two assertions of (143). It also gives $v_B y_s, v_B y_T \geq 1 + t_2$, hence the stated depth c_2 in B_2 . The norm part of that lemma gives

$$n_2 y_s \equiv \frac{u^{p-2} s^p}{P_*^2}, \quad n_2 y_T \equiv \frac{u^{p-1} A_0 T^p}{P_*^2} \pmod{\mathfrak{P}_1^K}.$$

Here N_1 fixes B_1 with norm the p th power, and its restriction to B_2 is n_2 . By (132) on B_1/A ,

$$\begin{aligned} v_1(u^{p-2}(s^p - H)) &\geq (p-2)M + p(p-1)\delta + (p-1)t, \\ v_1(u^{p-1}A_0(T^p - J)) &\geq 2(p-1)M + p(p-1)\delta - t. \end{aligned}$$

The first excess over K is at least $(p-3)(t+1) + p(p-2)\delta$, and the second at least $(p-3)t + 2p-3 + p(p-2)\delta$. Thus the two powers may be replaced by H, J . Now both sides lie in A , so $A \cap \mathfrak{P}_1^K = \mathfrak{p}_A^{1+t_2}$ applies. Lemma 9.21 replaces P_*^{-2} by D^{-2} at precisely that weighted modulus, proving the remaining assertions. $\square$

Lemma 9.26 (The values at $y_s/2$ and $-y_T/2$). *One has*

$$\begin{aligned} \Psi_{B_2}(y_s/2) &= \Psi_A(-u^{p-2}H/(2D^2)), \\ \Psi_{B_2}(-y_T/2) &= \Psi_A(u^{p-1}A_0J/(2D^2)). \end{aligned} \tag{144}$$

Proof. The norm character of B_2/A has coefficient α in (93). For any $y \in \mathfrak{P}_2^{c_2}$ choose x with $P(x) = y$, using bijectivity of P on that positive ideal. The cyclic coefficient estimate gives $n_2(1-x) \in U_A^{c_2}$, where that formula applies. Its value on this norm is 1. The norm-logarithm identity of Lemma 6.3, with break t_2 and $h = 0$, gives at modulus $\mathfrak{p}_A^{1+t_2}$

$$P(1 - n_2(1 - x)) \equiv T_2 P(x) + n_2(P(x)).$$

Consequently $\Psi_{B_2}(y)\Psi_A(n_2y) = 1$, as in Corollary 6.4. For the half, the product $\Psi_{B_2}(y/2)\Psi_A(n_2y/2)$ has square 1 and has p -primary order. Indeed the image of a continuous additive character of a local field of residue characteristic p is p -primary: sufficiently large p -powers of each argument lie in an open trivial ideal. As p is odd, this product is 1. Apply it to y_s, y_T and use Lemma 9.25. The errors after multiplication by $1/2$ still lie in the whole trivial ideal $\mathfrak{p}_A^{1+t_2}$. $\square$

Proposition 9.27 (Evaluation of $Q_1 Q_2 Q_4$). *One has*

$$(Q_1 Q_2 Q_4)^{-1} = \Psi_A \left(W u^{p-1} \frac{H}{D} - \frac{u^{p-2}H}{2D^2} + \frac{u^{p-1}A_0J}{2D^2} \right). \tag{145}$$

Proof. Proposition 9.13 and Theorems 9.7 and 9.11 express the argument of Ψ_{B_2} in Q_1^{-1} modulo $\mathfrak{P}_B^K$ as

$$-W F_{p-1} + W \sum_{i=1}^{p-2} F_i/i + W/w^{p-1} + \frac{p-1}{p-2} X_s + \frac{1}{2(p-1)} X_T.$$

Replace X_s, X_T by y_s, y_T using Lemma 9.25. The resulting expression belongs to B_2 , so its difference from the argument in (117) lies in $B_2 \cap \mathfrak{P}_B^K = \mathfrak{P}_2^{1+t_2}$ and has trivial Ψ_{B_2} phase.

The two coefficient replacements are

$$\frac{p-1}{p-2} - \frac{1}{2} = \frac{p}{2(p-2)}, \quad \frac{1}{2(p-1)} + \frac{1}{2} = \frac{p}{2(p-1)}.$$

Their errors have B -valuations at least $v_B(p) + h_s$ and $v_B(p) + h_T$, respectively, and hence at least K . They vanish identically in equal characteristic. Thus the surviving exceptional phase is $\Psi_{B_2}((y_s - y_T)/2)$. Theorem 9.17 and Lemma 9.22 combine the two rational sums modulo $\mathfrak{P}_2^{n_\Gamma}$. Including W/w^{p-1} and $Q_4^{-1} = \Psi_{B_2}(-w)$, Lemma 9.23 shows that the remaining additive factor is one. The product is therefore exactly

$$\Psi_A(W u^{p-1}H/D) \Psi_{B_2}((y_s - y_T)/2).$$

Finally (144) gives (145). $\square$

The third factor Q_3 . Put $h_0 = m + \delta$ and $z = \beta w_1/A_1$. Then

$$v_1 z = (p-1)h_0, \quad n_1^{\text{char}} = 1 + t' + pm = 1 + t + ph_0$$

is the conductor of θ_1 .

Lemma 9.28 (Evaluation of Q_3). *One has*

$$(146) \quad Q_3^{-1} = \Psi_A \left(-\frac{W(vu)^{p-1}}{2D} \right).$$

Proof. The inequalities $p(p-1)h_0 \geq n_1^{\text{char}}$ and $3(p-1)h_0 \geq n_1^{\text{char}}$ follow from $pm \geq t+1$: for the latter the excess is $(2p-3)h_0 - t - 1 \geq 0$, and the former is no smaller. Thus the formula for θ_1 in (98), whose coefficient is $C_1 = A_1(1-z)$, applies at z with all terms of degree at least three in the following expression beyond its conductor:

$$Q_3 = e_{B_1}(A_1[(1-z)P(z) - z]) = e_{B_1}(-A_1 z^2/2).$$

The identity used here is exact before deleting the higher terms; for $p=3$ the cubic term is $-z^3/2$ and the same displayed inequality includes it. Consequently

$$Q_3^{-1} = e_{B_1} \left(\frac{\beta^2 w_1^2}{2A_1} \right).$$

Set

$$y = \frac{(\beta/\gamma)w_1^2}{1+ub}.$$

Its valuation is $(p-2)h_0 \geq \rho := \lceil t/p \rceil$. Apply the proof of Lemma 9.26 to B_1/A and its norm character with coefficient β . Here $\rho = \lceil (t+1)/p \rceil$ because $p \nmid t$. It gives

$$e_{B_1}(\beta y/2) = e_A(-\beta n_1 y/2).$$

The exact norm choice $n_1 w_1 = \gamma/\beta$ now implies

$$(147) \quad Q_3^{-1} = e_A \left(-\frac{\gamma}{2} \frac{(\beta/\gamma)^{p-1}}{n_1(1+ub)} \right).$$

The highest-degree term in the norm expansion in the denominator is $n_1(ub) = u^p A_0$, so (131) for B_1/A , whose break is t , gives

$$v_A(n_1(1+ub) - D) \geq m + \left\lfloor \frac{(p-2)t + p - 1}{p} \right\rfloor.$$

The numerator weight $(\beta/\gamma)^{p-1}$ has valuation $(p-1)(m+\delta)$. After subtracting the γ -conductor $r+1$, the weighted denominator error has excess at least

$$(p-1)m + (p-2)\delta + \left\lfloor \frac{(p-2)t + p - 1}{p} \right\rfloor - t - 1 \geq 0.$$

To verify the last inequality write $t = pa + b_0$, $1 \leq b_0 \leq p-1$, and $m = a + 1 + c$. The expression becomes

$$(p-3)a + (p-1)c + p - 2 - b_0 + \left\lfloor \frac{(p-2)b_0 + p - 1}{p} \right\rfloor + (p-2)\delta.$$

It is nonnegative for $b_0 \leq p-2$; for $b_0 = p-1$ the floor is $p-2$. This includes the equality boundary $p=3, b_0=2$. Both denominators are units. Hence (147) permits replacement by D modulo the trivial ideal of e_A , yielding (146) because $\gamma = \alpha W$ and $\beta/\gamma = vu$. $\square$

Evaluation of $Q_1Q_2Q_3Q_4$.

Theorem 9.29 (A formula for $Q_1Q_2Q_3Q_4$). *The factors in (134) satisfy*

$$(148) \quad (Q_1Q_2Q_3Q_4)^{-1} = \Psi_A(\phi),$$

where

$$(149) \quad \phi = -\frac{W}{2} \frac{(vu)^{p-1}}{D} + Wu^{p-1} \frac{H}{D} - \frac{u^{p-2}H}{2D^2} + \frac{u^{p-1}A_0J}{2D^2}.$$

Proof. Combine Proposition 9.27 and Lemma 9.28. $\square$

Corollary 9.30 (The product is 1). *For the characters and coefficients fixed at the start of this subsection, $Q_1Q_2Q_3Q_4 = 1$ in mixed and equal characteristic.*

Proof. Since $Wu = 1$ and $D = 1 + u^pA_0$, direct expansion gives the exact algebraic identity

$$(150) \quad \phi = \frac{L_A}{2D^2} + \frac{u^{p-2}}{2D}(H - v^{p-1}), \quad L_A = u^{p-1}A_0(J + u^{p-1}H).$$

Theorem 9.20 gives

$$v_AL_A \geq 2(p-1)m + (p-1)\delta - \lfloor t/p \rfloor \geq 1 + t_2.$$

Theorem 8.16 gives

$$H - v^{p-1} \in \mathfrak{p}_A^{L+(p-1)\delta}, \quad L = 1 + t - \lfloor t/p \rfloor.$$

The second term of (150) has depth at least $1 + t_2$ because, with $\rho = \lfloor t/p \rfloor$ and $m \geq \rho$,

$$(p-2)m + L + (p-1)\delta - (1 + t_2) = (p-2)(m + \delta) - \rho \geq 0.$$

The unit denominator and rational half preserve these whole ideal bounds. Both terms therefore lie in the largest trivial ideal of Ψ_A , and Theorem 9.29 proves the assertion. $\square$

Corollary 9.31 (Equality when $pm > t$). *For the characters θ_1, θ_2 fixed above, with $pm > t$,*

$$\Delta_{B_1}(\theta_1, e_{B_1}) = \Delta_{B_2}(\theta_2, e_{B_2}).$$

Proof. Proposition 8.22 gives

$$\frac{\Delta_{B_2}(\theta_2, e_{B_2})}{\Delta_{B_1}(\theta_1, e_{B_1})} = \frac{g_2}{g_1} Q_1Q_2Q_3Q_4 \Psi_A(T_1b - T_2a), \quad g_i \in \mu_4.$$

Equation (133) and Corollary 9.30 put this quotient in μ_4 . Corollary 4.3 gives its p th power equal to 1. Since p is odd, $\mu_p \cap \mu_4 = \{1\}$, proving equality. $\square$

9.7. Conductor reduction and completion of the odd-prime case.

Local notation. Let B/A be totally ramified with Galois group C_p^2 , p odd. Choose B_1/A of smallest break and any other degree- p intermediate field B_2 , with breaks

$$t = t_1 \leq t_2 = t + \delta, \quad t' = t + p\delta, \quad p \nmid t.$$

The upper breaks are $u_1 = t'$ and $u_2 = t$. Set $N_i = N_{B/B_i}$, $n_i = N_{B_i/A}$ and $T_i = \text{Tr}_{B_i/A}$ and $e_E = e_A \text{Tr}_{E/A}$. Suppose $\psi_1N_1 = \psi_2N_2 = \psi_B$ is non-descending from A . It is invariant under $\text{Gal}(B/A)$, since $\text{Gal}(B/B_1)$ and $\text{Gal}(B/B_2)$ generate that group.

The exact conductor reduction. We continue to write $a(\theta)$ for the multiplicative conductor exponent.

Lemma 9.32 (Triviality on $U_E^{1+s+u} \cap \ker N_{E/A}$). *Let p be a prime, let M/A be a totally ramified C_p^2 extension, and let E be a degree- p intermediate field. Suppose E/A has break s and M/E has break u . If θ is the character of $E^\times$ in a primitive compatible pair, then*

$$\theta(x) = 1 \quad \text{for } x \in U_E^{1+s+u}, \quad N_{E/A} x = 1.$$

Proof. Choose a generator σ of $\text{Gal}(E/A)$. Lemma C.1 gives $x = \sigma(y)/y$. If $x = 1$ there is nothing to prove. Then $\sigma y \neq y$. Apply Lemma 7.2 to E/A . Choose $a \in A$ such that

$$q := v_E(y - a) = \max_{b \in A} v_E(y - b), \quad p \nmid q.$$

Put $m = v_E(y)$. Since $b = 0$ is allowed, $q \geq m$. The first ramification term is exact at the depth q , so

$$v_E(\sigma y - y) = v_E(\sigma(y - a) - (y - a)) = q + s.$$

On the other hand

$$x - 1 = \frac{\sigma y - y}{y}.$$

Since $x \in U_E^{1+s+u}$, it follows that

$$q - m \geq u + 1.$$

In particular $q > m$, so $a \neq 0$ and $v_E(a) = m$. Replacing y by y/a does not change $\sigma(y)/y$, and

$$y/a \in U_E^{q-m} \subset U_E^{u+1}.$$

By Lemma 4.2, θ^σ/θ is a nontrivial norm character of M/E , of conductor $u + 1$. Hence it is trivial on y/a and its conjugates. Therefore $\theta(\sigma(y)/y) = 1$, as required. $\square$

Proposition 9.33. *Every primitive compatible pair can be written*

$$\psi_1 = \chi(\lambda n_1), \quad \psi_2 = \varphi(\lambda n_2),$$

where $\chi N_1 = \varphi N_2$ is primitive and

$$a(\chi) = m_1 = 1 + t + t', \quad a(\varphi) = m_2 = 1 + t + t_2.$$

If $a(\psi_2) > m_2$, then $a(\lambda) = 1 + r$ with $r > t_2$, and $p(r - t_2) > t$. Thus the condition $p(r - t_2) > t$ of Corollary 9.31 is satisfied.

Proof. By Lemma 9.32, ψ_1 on $U_1^{m_1}$ factors through $n_1(U_1^{m_1})$. The factored character is continuous, because the unit groups are compact and the norm map is a quotient onto its image. Extend it to a continuous character λ of $A^\times$; the image is open, and divisibility of $\mathbf{C}^\times$ extends a character across the resulting discrete quotient. Replace ψ_i by $\psi_i/(\lambda n_i)$ for $i = 1, 2$. The pair remains compatible and primitive, and its first conductor is at most m_1 . Lemma 8.7 makes it exactly m_1 .

By Lemma 8.7, $a(\chi) > t' + 1$ and $a(\varphi) > t + 1$. Equation (10), applied to their common norm pullback, gives

$$p a(\chi) - (p - 1)(t' + 1) = p a(\varphi) - (p - 1)(t + 1).$$

It follows that $a(\varphi) = m_2$. If $a(\psi_2) > m_2$, then $a(\lambda n_2) = a(\psi_2) > m_2$. Equation (10) forces $a(\lambda) = 1 + r > 1 + t_2$ and

$$a(\psi_2) = 1 + t_2 + p(r - t_2) > 1 + t + t_2,$$

which is equivalent to the asserted range. $\square$

The case of conductors m_1, m_2 .

Theorem 9.34. *If $a(\chi) = m_1$ and $a(\varphi) = m_2$, then $\mathcal{A}_{B_1}(\chi) = \mathcal{A}_{B_2}(\varphi)$.*

Proof. Choose Δ as in Theorem 8.9 and write $b = N_1\Delta$, $a = N_2\Delta$. The coefficients in (82) are $C_1 = \alpha b$, $C_2 = \alpha a$, where $e_A(\alpha \cdot)$ has ideal conductor $1 + t_2$. Lemma 8.21 gives

$$\frac{\Delta_{B_2}(\varphi)}{\Delta_{B_1}(\chi)} = \frac{g_2}{g_1} \frac{\chi(\alpha b)}{\varphi(\alpha a)} e_A(\alpha[T_1b - T_2a]), \quad g_i \in \mu_4.$$

Lemma 4.2 cancels the two values at α . Compatibility, evaluated on $\Delta \in B^\times$, cancels $\chi(b)/\varphi(a)$. The trace–norm bound of Theorem 7.7 $T_1b - T_2a \in \mathfrak{p}_A^{1+t_2}$ makes the last factor one. Thus the ratio lies in μ_4 . Corollary 4.3 gives its p th power equal to 1; since p is odd, the ratio is 1. Corollary 4.3 also gives $\prod_{\nu \in S(B_i/A)} \Delta_A(\nu) = 1$ for $i = 1, 2$. The assertion follows. $\square$

Theorem 9.35 (The totally ramified case for odd p). *Theorem 1.1 holds when $\ell = p > 2$ and K/F is totally ramified, in mixed and equal characteristic.*

Proof. First compare the smallest-break field B_1 chosen above with an arbitrary other degree- p intermediate field. If the second conductor is m_2 , the conductor equality in Proposition 9.33 gives a character of $B_1^\times$ of conductor m_1 , and Theorem 9.34 applies. Otherwise Proposition 9.33 writes the pair as $\chi(\lambda n_1), \varphi(\lambda n_2)$ with $p(r - t_2) > t$. Corollary 9.31 gives equality of their local constants. The factors indexed by $S(B_i/A)$ are 1 by Corollary 4.3. For a pair not containing B_1 , extend its invariant common pullback to B_1 by Hilbert 90 and character extension; the two comparisons with B_1 then give the desired equality by transitivity. The preceding proofs include both characteristics and $p = 3$. $\square$

10. DYADIC EXTENSIONS WITH AN UNRAMIFIED INTERMEDIATE FIELD

10.1. Notation. Let F be a nonarchimedean local field of residue characteristic 2, in mixed or equal characteristic. Let K/F be biquadratic. For a quadratic intermediate field L , let ω_L be the nontrivial character of $F^\times / N_{L/F} L^\times$. A primitive compatible pair means characters θ_i of $L_i^\times$ with the same pullback Θ to $K^\times$, where Θ does not descend from F through $N_{K/F}$. Fix ψ_F and use $\psi_L = \psi_F \text{Tr}_{L/F}$. The target is equality of

$$(151) \quad A_L(\theta_L) = \Delta_L(\theta_L, \psi_L) \Delta_F(\omega_L, \psi_F).$$

The omitted trivial-character factor is 1. Write $a_L(\theta)$ for the multiplicative conductor, n_L for the additive conductor in the convention that $\mathfrak{p}_L^{-n_L}$ is the largest trivial ideal, and $v_L(\mathfrak{p}_L) = 1$. For a ramified quadratic extension L/F of break t , put $T = t + 1$. Its different exponent is T , and

$$(152) \quad n_L = 2n_F + T, \quad \text{Tr}_{L/F}(\mathfrak{p}_L^j) = \mathfrak{p}_F^{\lfloor (j+T)/2 \rfloor}, \quad v_F(N_{L/F} x) = v_L(x).$$

For an unramified quadratic extension, additive conductors are unchanged under trace pullback and $v_L|_F = v_F$.

We use the one-dimensional results recalled in Sections 2–4, and the local norm-character identifications of Appendix D.

Put $N_i = N_{K/L_i}$, $n_i = N_{L_i/F}$, and let ν_i denote the nontrivial character of $S(K/L_i)$. Lemma 4.2 gives $\theta_i^{\sigma_i} = \theta_i \nu_i$ and the common restriction $D = \theta_i|_F \omega_i$. Proposition 4.4 leaves a quotient in $\{1, -1\}$. We determine it by evaluating the critical functions in Lemma 4.1.

10.2. A quadratic norm-character identity. Let E/F be ramified quadratic of break $t \geq 1$, put $T = t + 1$ and $q = \lceil T/2 \rceil$, and let τ be the nontrivial character in $S(E/F)$. By additive duality on U_F^q/U_F^T , choose α such that

$$(153) \quad \tau(1 - z) = \Psi_F(z), \quad z \in \mathfrak{p}_F^q, \quad \Psi_L(w) := \psi_L(\alpha w), \quad v_F \alpha = -n_F - T.$$

Both Ψ_F and Ψ_E have largest trivial ideal with exponent T . Formula (153) follows from $2q \geq T$. In mixed characteristic put $e = v_F(2)$; in equal characteristic put $e = +\infty$. The trace of 1 in (152) gives

$$(154) \quad e \geq \lfloor T/2 \rfloor, \quad e + q \geq T.$$

Lemma 10.1 (An identity for Ψ_E and Ψ_F). *For every $z \in \mathfrak{p}_E^q$,*

$$(155) \quad \Psi_E(z) = \Psi_F(N_{E/F} z).$$

Proof. The exact quadratic identity is

$$N(1 - z) = 1 - \text{Tr } z + N z.$$

Both $\text{Tr } z$ and $N z$ have valuation at least q , by (152). Evaluate (153) at $\text{Tr } z - N z$ and use $\tau(N(1 - z)) = 1$. This gives $\Psi_F(\text{Tr } z - N z) = 1$, which is (155). $\square$

Remark 10.2. Equation (155) has a positive norm term, whereas (38) has a negative norm term for odd p .

Lemma 10.3 (The odd quadratic conductor). *If T is odd, then $\text{char } F = 0$ and $T = 2e + 1$. In that case, with k the residue field,*

$$(156) \quad \Psi_F(4z) = (-1)^{\text{Tr}_{k/\mathbb{F}_2}(\bar{z})}, \quad z \in \mathcal{O}_F.$$

Proof. By [7, Lemma 9.23(a)], odd T implies $\text{char } F = 0$ and $v_F(2) = (T - 1)/2$. Thus $v_F 4 = 2e = T - 1$, so $z \mapsto \Psi_F(4z)$ is a nontrivial residue additive character. The equality $(1 + 2z)^2 = 1 + 4(z + z^2)$ and the order of τ show that it kills $\bar{z} + \bar{z}^2$. The image of this Artin–Schreier map is exactly the absolute-trace-zero hyperplane: its kernel has two elements and its image has zero trace. The unique nontrivial character of the quotient is the right side of (156). $\square$

10.3. Characters of conductor T and twists by one character of $F^\times$. Assume from now through Section 10.6 that U/F is unramified quadratic, E/F ramified quadratic of break t , and $K = EU$. Write

$$N_U = N_{K/U}, \quad N_E = N_{K/E}, \quad n_U = N_{U/F}, \quad n_E = N_{E/F}.$$

Let $\eta = \omega_U$ and $\tau = \omega_E$. The norm character on K/U is $\tau_U = \tau n_U$. All four $\Psi_L = \psi_L(\alpha \cdot)$ have largest trivial ideal $\mathfrak{p}_L^T$. Let $k \subset \kappa$ be the residue fields of $F \subset U$, and put $Q = |k|$. Choose $c \in \mathcal{O}_F$ whose residue has absolute trace 1, and choose

$$(157) \quad d \in \mathcal{O}_U, \quad d^2 - d = c.$$

This polynomial gives the unramified quadratic extension. For its nontrivial automorphism σ ,

$$\sigma(d) = 1 - d, \quad \text{Tr}_{U/F} d = 1, \quad n_U(d) = -c.$$

In particular c, d are units and $\bar{d} \notin k$.

Theorem 10.4 (Characters of conductor T). *There exist characters χ_U^0, χ_E^0 , both of conductor T , with a primitive common norm pullback, such that*

$$(158) \quad (\chi_U^0)^\sigma / \chi_U^0 = \tau_U,$$

$$(159) \quad \chi_U^0(1 - z) = \Psi_U(d^2 z) \quad (z \in \mathfrak{p}_U^q),$$

$$(160) \quad \chi_E^0(1 - z) = \Psi_E(cz) \quad (z \in \mathfrak{p}_E^q).$$

Proof. The right side of (159) is a character on $H = U_U^q$, since $2q \geq T$. Its exact conductor is T . Unramified norm expansion shows

$$\tau_U(1 - z) = \Psi_U(z) \quad (z \in \mathfrak{p}_U^q),$$

because $v_F n_U z \geq 2q \geq T$. Further, $\sigma(d^2) - d^2 = 1 - 2d$; the $2d$ error is killed on this whole ideal by (154). Thus the prescribed character on H has the required conjugate quotient on H .

We extend the character of H so that (158) holds on $U^\times$. Put $V = U_U^1$ and $I = \{\sigma(v)/v : v \in V\}$. Prescribe $\kappa_I(\sigma(v)/v) = \tau_U(v)$. This is well defined because $\tau_U(f) = \tau(f^2) = 1$ on U_F^1 . If $\sigma(v)/v \in H$, then v modulo $\mathfrak{p}_U^q$ is fixed, hence is represented by a unit of F ; write $v = fh$ with $f \in U_F^1$, $h \in H$. (For unramified extensions this follows coefficient by coefficient in the uniformizer expansion.) The characters prescribed on I and H consequently agree on the intersection. Extend their product character from IH/U_U^T across the finite abelian group V/U_U^T .

Make it trivial on the odd-order Teichmüller group and on a uniformizer of F . The character τ_U is trivial on both these factors. The resulting χ_U^0 has (158) on all of $U^\times$ and has exact conductor T .

Its pullback to K is invariant under both quadratic automorphisms, since τ_U is trivial on $N_U K^\times$. Hilbert 90 makes this pullback trivial on $\ker N_E$. It therefore defines a character of $N_E(K^\times)$, which extends to $E^\times$ by Lemma C.2. Call the extension χ_E^0 . Since K/E is unramified, $N_E(K^\times)$ contains $\mathcal{O}_E^\times$; it remains only to choose a value on a uniformizer of E . For $z \in \mathfrak{p}_K^q$, the exact ramified quadratic norm gives

$$\chi_U^0(N_U(1-z)) = \Psi_K(d^2 z) \Psi_U(-d^2 N_U z) = \Psi_K((d^2 - d)z) = \Psi_K(cz).$$

The middle equality uses Lemma 10.1 on dz and $N_U(dz) = d^2 N_U z$, an exact equality. The unramified norm onto U_E^q and its expansion modulo $\mathfrak{p}_E^T$ now give (160). Since c is a unit, its conductor is exactly T . Nontriviality of (158) excludes descent of the common pullback from F . $\square$

Proposition 10.5 (Writing the given pair using one character of $F^\times$). *For every primitive compatible pair (θ_U, θ_E) there is a character λ of $F^\times$, after an unramified adjustment of χ_E^0 if necessary, such that*

$$(161) \quad \theta_U = \chi_U^0(\lambda n_U), \quad \theta_E = \chi_E^0(\lambda n_E).$$

There is an integer $h \geq 0$ with

$$(162) \quad a_U(\theta_U) = T + h, \quad a_E(\theta_E) = T + 2h.$$

For $h > 0$, $a_F(\lambda) = T + h$; for $h = 0$, $a_F(\lambda) \leq T$.

Proof. Lemma 4.2 gives $\theta_U^c/\theta_U = \tau_U$. Thus θ_U/χ_U^0 is $\text{Gal}(U/F)$ -invariant. Lemmas C.1 and C.2 give $\theta_U/\chi_U^0 = \lambda n_U$ for a character λ of $F^\times$. The quotient $\theta_E/(\chi_E^0(\lambda n_E))$ belongs to $S(K/E)$. Multiply χ_E^0 by this unramified character to obtain (161). The conductor of θ_U is at least T , the conductor of its commutator. If it is greater than T , then λn_U has the same conductor as θ_U , because χ_U^0 has conductor T ; since U/F is unramified, $a_U(\lambda n_U) = a_F(\lambda)$. If $a_U(\theta_U) = T$, then $a_F(\lambda) \leq T$. Both twists of θ_U by τ_U are conjugate and have the same conductor. The norm-pullback conductor formula [7, Proposition 3.10], including the case of conductor T , gives $a_K(\theta_U N_U) = T + 2h$. Since K/E is unramified, $a_E(\theta_E) = T + 2h$. $\square$

10.4. The element $x + d$ and its norms. For $0 \leq h \leq t$ put

$$s_U = \lceil (T + h)/2 \rceil, \quad s_E = \lceil T/2 \rceil + h, \quad d_U = \lfloor (T + h)/2 \rfloor.$$

Choose $A_* \in F$ such that

$$(163) \quad \lambda(1 - z) = \Psi_F(A_* z) \quad (z \in \mathfrak{p}_F^{s_U}).$$

For $h > 0$, $v_F A_* = -h$; at $h = 0$ the coefficient can be chosen integral. Its ambiguity is $\mathfrak{p}_F^{T-s_U}$.

Lemma 10.6. *There is $x \in E$, with $A = n_E x$, such that A can replace A_* in (163). If $h > 0$, $v_E x = -h$; at $h = 0$, x is integral. The zero coefficient class may be represented by $x = 0$.*

Proof. For $h > 0$ the required relative precision is $T - s_U + h = d_U \leq t$, since $T + h \leq 2T - 1$. Thus the subcritical norm representative theorem applies. For an integral coefficient of valuation $a < T - s_U$ use relative precision $T - s_U - a \leq t$. If the integral class is zero, choose zero. $\square$

Put

$$(164) \quad B_U = A + d^2, \quad B_E = A - x + c, \quad C = x + d, \quad Z_U = N_U C, \quad Z_E = N_E C, \quad b = \text{Tr}_{E/F} x.$$

Lemma 10.7 (Coefficients for θ_U and θ_E). *For the given characters in (161),*

$$(165) \quad \theta_U(1 - z) = \Psi_U(B_U z) \quad (z \in \mathfrak{p}_U^{s_U}), \quad \theta_E(1 - z) = \Psi_E(B_E z) \quad (z \in \mathfrak{p}_E^{s_E}).$$

Their coefficient valuations are $-h$ and $-2h$, respectively.

Proof. For the unramified pullback of λ , the extra norm term has phase valuation at least $-h + 2s_U \geq T$; use (159). For E , $u = \text{Tr } z - n_E z$ belongs to $\mathfrak{p}_F^{s_U}$ for $z \in \mathfrak{p}_E^{s_E}$, because

$$s_E \geq s_U, \quad \lfloor (s_E + T)/2 \rfloor \geq s_U.$$

Then (163) and Lemma 10.1, applied to xz of valuation at least $\lceil T/2 \rceil$, give

$$\lambda(n_E(1 - z)) = \Psi_F(A \text{Tr } z - A n_E z) = \Psi_E((A - x)z).$$

Together with (160) this proves the formulas. For $h > 0$, A is the unique leading term in each coefficient. At $h = 0$, write $\xi = \bar{x} \in k$. The residues are $\xi^2 + \bar{d}^2$ and $\xi^2 - \xi + \bar{c}$. The first is nonzero since $\bar{d} \notin k$, and the second has absolute trace 1. $\square$

The norms of the *same* element C have the exact expressions

$$(166) \quad Z_U = A + bd + d^2,$$

$$(167) \quad Z_E = x^2 + x - c,$$

$$(168) \quad S := \text{Tr}_{U/F} Z_U - \text{Tr}_{E/F} Z_E = 1 + 4c - \mathcal{D}_x, \quad \mathcal{D}_x = b^2 - 4A = (2x - b)^2.$$

Indeed $\text{Tr } d = 1$, $\text{Tr } d^2 = 1 + 2c$, and $\text{Tr } x^2 = b^2 - 2A$.

Write H_L for the critical function and g_L for its normalized sum in Lemma 4.1, with $J = T$.

10.5. The minimal-conductor comparison.

Theorem 10.8. *Every primitive compatible pair with $a_U(\theta_U) = a_E(\theta_E) = T$ satisfies*

$$(169) \quad \Delta_U(\theta_U)\Delta_F(\eta) = \Delta_E(\theta_E)\Delta_F(\tau).$$

This includes the odd-conductor maximal-break case in mixed characteristic and all equal-characteristic minimal pairs.

Even T . Here $T = 2r$ and x in Lemma 10.6 is integral. The differences $Z_U - B_U = bd$ and

$$Z_E - B_E = bx - 2A + 2x - 2c$$

have respective valuations at least r in U and $2r$ in E , by (152) and $v_F 2 \geq r$. Thus Z_U, Z_E are stationary coefficients for θ_U, θ_E in Lemma 4.1. All three conductors T are even, so there are no residual sums. Put $c_0 = -\alpha^{-1}$. Compatibility gives $\theta_U(Z_U) = \theta_E(Z_E)$; Lemma 4.2 gives $\theta_U(c_0)/\theta_E(c_0) = \eta(c_0)\tau(c_0)$. Using (14) and $\Delta_F(\tau) = \tau(c_0)\Psi_F(-1)$ therefore gives

$$(170) \quad \frac{\Delta_U(\theta_U)\Delta_F(\eta)}{\Delta_E(\theta_E)\Delta_F(\tau)} = (-1)^T \Psi_F(1 - S).$$

Here $\Delta_F(\eta) = (-1)^{n_F}$ and $\eta(c_0) = (-1)^{n_F+T}$. Now $2x - b$ is trace zero and has E -valuation at least $2r$. Lemma 10.1 gives $\Psi_F(\mathcal{D}_x) = 1$ since $n_E(2x - b) = -\mathcal{D}_x$. Also $v_F(4c) \geq 2r = T$. Equation (168) proves that (170) is 1.

Odd T : the elementary sign. By Lemma 10.3, $T = 2e + 1$, $\text{char } F = 0$ and $e \geq 1$. We have $x \in \mathcal{O}_E$ and $b \in \mathfrak{p}_F^e$. The differences $Z_U - B_U$ and $Z_E - B_E$ have valuations at least e and $2e$, respectively. Both therefore satisfy the stationary coefficient congruences modulo $\mathfrak{p}_U^e$ and $\mathfrak{p}_E^e$. In addition put

$$(171) \quad P_C = \text{Tr}_{K/E} C = 2x + 1, \quad W_E = n_E(P_C), \quad Q_C = \text{Tr}_{K/U} C = b + 2d, \quad W_U = n_U(Q_C).$$

The element P_C is a unit and $W_E = 1 + 2b + 4A \equiv 1 \pmod{\mathfrak{p}_F^{2e}}$. Thus W_E is a stationary coefficient for τ relative to Ψ_F . Moreover $\tau(W_E) = 1$ because $W_E = n_E(P_C)$.

Lemma 4.5, in the present notation, gives

$$(172) \quad \text{Tr}_{U/F} N_U Y + n_U(\text{Tr}_{K/U} Y) = \text{Tr}_{E/F} N_E Y + n_E(\text{Tr}_{K/E} Y).$$

Use the three stationary coefficients Z_U, Z_E, W_E in (14). Compatibility, (18), and (172) give

$$(173) \quad \frac{\Delta_U(\theta_U)\Delta_F(\eta)}{\Delta_E(\theta_E)\Delta_F(\tau)} = (-1)^T \Psi_F(W_U) \frac{g_U}{g_E g_\tau}.$$

The prefactor on the right is 1. Indeed $a = b/2 \in \mathcal{O}_F$ and

$$W_U = 4n_U(d + a), \quad \overline{n_U(d + a)} = \bar{a}^2 + \bar{a} - \bar{c}.$$

Its absolute trace is 1. Lemma 10.3 gives $\Psi_F(W_U) = -1$, which cancels $(-1)^T = -1$.

Odd T : an identity between three critical functions. Choose a uniformizer Π of E and put $\pi = n_E \Pi$; this is a uniformizer of F and U . In (15) use critical elements π^e for U, F and Π^e for E . For $z \in \mathcal{O}_K$ set

$$Y_z = C(1 + \Pi^e z).$$

The estimates below show that $\text{Tr}_{K/E} Y_z$ is a unit and $v_U(\text{Tr}_{K/U} Y_z) = e$. In particular both traces are nonzero. The three maps on the residue fields are

$$(174) \quad \begin{aligned} \frac{N_U Y_z / Z_U - 1}{\pi^e} &\equiv \bar{z}^2 && \text{in } \kappa, \\ \frac{N_E Y_z / Z_E - 1}{\Pi^e} &\equiv \text{Tr}_{\kappa/k} \bar{z} && \text{in } k, \\ \frac{n_E(\text{Tr}_{K/E} Y_z) / W_E - 1}{\pi^e} &\equiv (\text{Tr}_{\kappa/k}(\bar{C}\bar{z}))^2 && \text{in } k. \end{aligned}$$

To prove the first congruence, note that $\text{Tr}_{K/U}(\Pi^e z)$ has U -valuation at least $\lfloor (e + T)/2 \rfloor \geq e + 1$; its norm has leading term $\pi^e \bar{z}^2$. For the second congruence, the trace term of $N_E(1 + \Pi^e z)$ has leading coefficient $\text{Tr} \bar{z}$, and its quadratic term has depth $2e \geq e + 1$. For the third, the relative trace change has leading E -coefficient $\text{Tr}(\bar{C}\bar{z})$ at depth e , since $\overline{P_C} = 1$; apply the first ramified calculation to its norm. These verifications include the smallest case $e = 1$.

Moreover

$$(175) \quad n_U(\text{Tr}_{K/U} Y_z) - W_U \in \mathfrak{p}_F^T.$$

In fact $v_U Q_C = e$, because $Q_C = 2(d + b/2)$ and $\bar{d} \notin k$. The trace change $\text{Tr}_{K/U}(C\Pi^e z)$ has valuation at least $\lfloor (e + T)/2 \rfloor \geq e + 1$. Expanding its unramified quadratic norm proves (175).

Apply (172) to Y_z and to C , and subtract. The common multiplicative character values cancel by compatibility. The two τ -values of the norms of the E -traces are both 1. Equation (175) removes precisely the remaining U -trace-norm phase. Lemma 4.1 and (174) give the identity of critical functions

$$(176) \quad H_U(z^2) = H_E(\text{Tr}_{\kappa/k} z) H_\tau((\text{Tr}_{\kappa/k}(\bar{C}z))^2), \quad z \in \kappa.$$

The map

$$z \mapsto (\text{Tr} z, (\text{Tr}(\bar{C}z))^2) : \kappa \longrightarrow k \times k$$

is bijective. Write $z = u + v\bar{d}$ and $\bar{x} = \xi \in k$; its coordinates are v and $(u + (\xi + 1)v)^2$. Frobenius on k is bijective. Since Frobenius on κ is bijective too, summing (176) gives

$$\sum_{\kappa} H_U = \left(\sum_k H_E \right) \left(\sum_k H_\tau \right), \quad g_U = g_E g_\tau.$$

Together with (173), this proves Theorem 10.8 when T is odd. The equality (176) compares the critical functions, including their linear terms.

10.6. The cases $1 \leq h < T$. For $1 \leq h < T$, the difference $Z_E - B_E$ need not have the valuation required for a stationary coefficient. We replace Z_E by $(x'/x)Z_E$, where x' is the conjugate of x over F . Lemma 10.10 gives both its coefficient congruence and its character value.

Throughout this subsection let

$$1 \leq h \leq t = T - 1, \quad m_U = T + h, \quad m_E = T + 2h.$$

Write the pair as in Proposition 10.5, and choose $A = n_E x$ in the stationary coefficient class by Lemma 10.6. In particular $v_E x = -h$, so $x \neq 0$. Let x' be its nontrivial E/F conjugate, and put

$$(177) \quad \rho = \frac{x'}{x}, \quad \widehat{Z}_U = A + bd + d^2, \quad \widehat{Z}_E = \rho(x^2 + x - c), \quad b = \text{Tr}_{E/F} x.$$

The trace parity lemma.

Lemma 10.9 (The value of $\Psi_F(-cu)$). *Put $\epsilon = (T - h) \bmod 2$ and $u = b^2/A$. If $\epsilon = 0$, then $v_F u \geq T$, with $v_F 0 = +\infty$. If $\epsilon = 1$, then*

$$v_F u = T - 1, \quad \Psi_F(uz) = (-1)^{\text{Tr}_{k/\mathbb{F}_2}(\bar{z})} \quad (z \in \mathcal{O}_F).$$

Consequently, for the fixed c in (157),

$$(178) \quad (-1)^{T+h} \Psi_F(-cu) = 1.$$

This holds in both mixed and equal characteristic.

Proof. The trace-ideal formula gives

$$v_F b \geq B := \lfloor (T - h)/2 \rfloor.$$

If $T - h$ is even, $2B + h = T$, proving the first assertion. If $T - h$ is odd, the induced map

$$\mathfrak{p}_E^{-h}/\mathfrak{p}_E^{1-h} \longrightarrow \mathfrak{p}_F^B/\mathfrak{p}_F^{B+1}$$

is a nonzero k -linear map between one-dimensional k -spaces. Indeed the trace images of the two consecutive source ideals are exactly $\mathfrak{p}_F^B$ and $\mathfrak{p}_F^{B+1}$. Thus an element of exact valuation $-h$ has trace of exact valuation B . Hence $b \neq 0$ and $v_F u = 2B + h = T - 1$.

For $z \in \mathcal{O}_F$, quadratic norm expansion gives the exact identity

$$(179) \quad n_E \left(1 + \frac{b}{A} xz \right) = 1 + u(z + z^2).$$

Here $v_E((b/A)x) = 2B + h = T - 1 > 0$, so the argument is a unit. Since $T - 1 \geq q$, the norm-character formula (153) applies to the right side. It follows that the residue additive character $z \mapsto \Psi_F(uz)$ annihilates $z + z^2$. It is nontrivial, because Ψ_F has largest trivial ideal $\mathfrak{p}_F^T$ and $v_F u = T - 1$. The image of $z \mapsto z + z^2$ in k is the absolute-trace-zero hyperplane. The character is therefore exactly $(-1)^{\text{Tr}_{k/\mathbb{F}_2} z}$. Since $\bar{c}$ has absolute trace one, $\Psi_F(-cu) = -1$ in this parity and is 1 in the other. Finally $T - h$ and $T + h$ have the same parity, proving (178). $\square$

Stationarity and the exact elementary factor.

Lemma 10.10 (The coefficients $\hat{Z}_U, \hat{Z}_E$). *For $1 \leq h < T$, the elements $\hat{Z}_U, \hat{Z}_E$ in (177) satisfy $\theta_L(1 - z) = \Psi_L(\hat{Z}_L z)$ for $z \in \mathfrak{p}_L^{s_L}$, where $L = U, E$. Moreover*

$$(180) \quad \hat{Z}_U - B_U = bd,$$

$$(181) \quad \hat{Z}_E - B_E = b(1 - c/x),$$

$$(182) \quad \hat{S} := \text{Tr}_{U/F} \hat{Z}_U - \text{Tr}_{E/F} \hat{Z}_E = 1 + cu,$$

$$(183) \quad \frac{\theta_E(\hat{Z}_E)}{\theta_U(\hat{Z}_U)} = (-1)^h.$$

Proof. The first error is immediate. Since $xx' = A$ and $x + x' = b$,

$$\rho(x^2 + x - c) = A + x' - c\rho,$$

whose difference from $A - x + c$ is $b - c(x'/x + 1) = b(1 - c/x)$. As $v_E x = -h < 0$, the latter parenthesis is a unit. With $B = \lfloor (T - h)/2 \rfloor$, the error valuations are at least B over U and $2B$ over E . The required ambiguity depths are

$$T - s_U = B, \quad T - s_E = \lfloor T/2 \rfloor - h.$$

For $\epsilon = (T - h) \bmod 2$,

$$2B - (T - s_E) = \lceil T/2 \rceil - \epsilon \geq 0.$$

Both errors therefore lie in the correct coefficient ambiguity ideals. By Lemma 10.7, $v_U B_U = -h$ and $v_E B_E = -2h$. The two errors have strictly larger valuation, so $v_U \hat{Z}_U = -h$ and $v_E \hat{Z}_E = -2h$.

The traces of ρ and of the corrected coefficients are

$$\text{Tr}_{E/F} \rho = \frac{b^2 - 2A}{A} = u - 2,$$

$$\mathrm{Tr}_{E/F} \widehat{Z}_E = 2A + b + 2c - cu, \quad \mathrm{Tr}_{U/F} \widehat{Z}_U = 2A + b + 1 + 2c.$$

This proves (182). The uncorrected Z_U, Z_E are the two norms of $C = x + d$, so compatibility gives $\theta_U(Z_U) = \theta_E(Z_E)$. Lemma 4.2 on E gives

$$\theta_E(\rho) = \frac{\theta_E(x')}{\theta_E(x)} = \eta(n_E x) = (-1)^{v_F A} = (-1)^h,$$

which proves (183). $\square$

Remark 10.11 (The norms of $x'(x + d)$). The correction can also be recorded using a common element and a base scalar. Put $Y = x'(x + d) \in K$ and $\lambda_0 = A^{-1} \in F$. Then $\lambda_0 N_U Y = \widehat{Z}_U$ and $\lambda_0 N_E Y = \widehat{Z}_E$. The symbol λ_0 is a field element, not the twisting character λ .

Proposition 10.12 (The quotient of local constants). *Apply Lemma 4.1 with coefficients $\widehat{Z}_U, \widehat{Z}_E, 1$ for θ_U, θ_E, τ , respectively. Write g_U, g_E, g_τ for the resulting normalized sums, with $g = 1$ when the conductor is even. Then*

$$(184) \quad \frac{\Delta_U(\theta_U)\Delta_F(\eta)}{\Delta_E(\theta_E)\Delta_F(\tau)} = (-1)^{T+h}\Psi_F(-cu) \frac{g_U}{g_E g_\tau} = \frac{g_U}{g_E g_\tau}.$$

Proof. Let $c_0 = -\alpha^{-1}$. Lemma 4.2 gives $\theta_U(c_0)/\theta_E(c_0) = \eta(c_0)\tau(c_0)$. The product $\eta(c_0)\Delta_F(\eta)$ is $(-1)^T$, since $v_F c_0 = n_F + T$ and $\Delta_F(\eta) = (-1)^{n_F}$. The quotient of the multiplicative stationary values is (183). The additive phase from (14) and the norm-character factor is $\Psi_F(1 - \widehat{S}) = \Psi_F(-cu)$. These exact identities give the first equality in (184). Lemma 10.9 gives the second. $\square$

The unramified residual sum.

Lemma 10.13. *If $T + h$ is odd and $h > 0$, the critical function for θ_U with coefficient $\widehat{Z}_U$ is 1 on the residue subfield $k \subset \kappa$. Its normalized sum is therefore $g_U = 1$.*

Proof. Set $a = (T + h - 1)/2$ and choose the critical element $\delta_U = \pi^a \in F$, for any uniformizer π of F . For $z \in \mathcal{O}_F$ put $v = \delta_U z$. One has

$$2a - s_E = \lfloor T/2 \rfloor - 1 \geq 0.$$

Also $a \geq q$: for even T this follows from odd $h \geq 1$; for odd T the present parity requires even $h \geq 2$. Thus $\theta_E(1 + v)$ can be evaluated by Lemma 10.10, and $\tau(1 + v)$ by (153). The determinant identity and triviality of η on units give

$$\theta_U(1 + v) = \theta_E(1 + v)\tau(1 + v).$$

Substitution into (15) gives

$$H_U(\bar{z}) = \Psi_F((1 - \widehat{S})v) = \Psi_F(-cu \delta_U z) = 1.$$

Indeed $v_F u \geq T - 1$ and $a \geq 1$.

By Lemma 4.1, the polar pairing of H_U on the additive group of κ is nondegenerate. Its restriction to k is 1, so k is isotropic for its polar pairing. Since $|k|^2 = |\kappa|$, nondegeneracy gives $k^\perp = k$. Summation on a coset of k is zero unless the coset is k itself, and the sum on k is $|k|$. Thus $|\kappa|^{-1/2} \sum_\kappa H_U = 1$. $\square$

Reciprocal residual functions at odd T . If T is even, the two remaining factors g_E, g_τ are already 1. For odd T write $T = 2e + 1$, where $e = v_F(2) \geq 1$ by Lemma 10.3. We prove $g_E g_\tau = 1$.

Lemma 10.14 (The identity $\tau(1 + n_E w) = \Psi_E(w)$). *For $w \in \mathfrak{p}_E^e$ one has*

$$(185) \quad \tau(1 + n_E w) = \Psi_E(w).$$

In particular, put

$$\mathcal{H}_\tau(z) = \Psi_F(-z)\tau(1 + z)^{-1}, \quad z \in \mathfrak{p}_F^e,$$

one has

$$(186) \quad \mathcal{H}_\tau(n_E w)^{-1} = \Psi_E(w)\Psi_F(n_E w).$$

Proof. The trace of w has valuation at least $\lfloor (e+T)/2 \rfloor \geq e+1$, and $v_F n_E w \geq e$. The exact identity

$$n_E(1+w) = (1+n_E w) \left(1 + \frac{\text{Tr}_{E/F} w}{1+n_E w} \right)$$

can therefore be evaluated by τ , using its linear formula on the second factor only. Since the left side is a norm,

$$\tau(1+n_E w) = \Psi_F \left(\frac{\text{Tr} w}{1+n_E w} \right) = \Psi_E \left(\frac{w}{1+n_E w} \right).$$

The difference between $w/(1+n_E w)$ and w has E -valuation at least $3e \geq 2e+1 = T$. This proves (185), including $e=1$. The definition of $\mathcal{H}_\tau$ now gives (186). $\square$

Lemma 10.15 (The ramified residual factor is the reciprocal). *For every $1 \leq h < T$ when T is odd,*

$$g_E = g_\tau^{-1}.$$

Proof. The critical E -ideal is $\mathfrak{p}_E^{e+h}$. For v in this ideal, both the formulas (160) and (163) apply. The first is valid since $e+h \geq e+1 = q$. For the second, the two terms in $n_E(1+v)-1 = \text{Tr} v + n_E v$ have F -valuations at least

$$\lfloor (3e+h+1)/2 \rfloor \quad \text{and} \quad e+h,$$

respectively, both at least $s_U = e + \lceil (h+1)/2 \rceil$. It follows that

$$(187) \quad \theta_E(1+v) = \Psi_E(-(A+c)v) \Psi_F(-A n_E v).$$

The difference between $\widehat{Z}_E$ and $A-x+c$ is $b(1-c/x)$. Its product with v has valuation at least

$$2 \lfloor (T-h)/2 \rfloor + e+h = 3e + (h \bmod 2) \geq T.$$

Consequently the critical function at v is

$$\begin{aligned} \mathcal{H}_E(v) &= \Psi_E(-\widehat{Z}_E v) \theta_E(1+v)^{-1} \\ &= \Psi_E(xv) \Psi_F(A n_E v) \\ &= \mathcal{H}_\tau(n_E(xv))^{-1}, \end{aligned}$$

where the last equality is Lemma 10.14, since $v_E(xv) \geq e$ and $n_E(xv) = A n_E v$ exactly.

Choose a uniformizer Π of E and put $\pi = n_E \Pi$. Write $x = \Pi^{-h} u$ with $u \in \mathcal{O}_E^\times$, and use Π^{e+h} and π^e as the two critical elements. On the two residue fields k the map $v \mapsto n_E(xv)$ induces

$$z \mapsto \bar{u}^2 z^2.$$

This is a bijection. Taking normalized sums, as in (15), gives

$$g_E = |k|^{-1/2} \sum_{z \in k} H_\tau(z)^{-1} = \overline{g_\tau} = g_\tau^{-1}.$$

The last equality follows from $|g_\tau| = 1$. $\square$

Theorem 10.16 (Equality for $1 \leq h < T$). *Every primitive compatible pair with $1 \leq h < T$ satisfies (169), in mixed and equal characteristic.*

Proof. Proposition 10.12 leaves $g_U/(g_E g_\tau)$. If $T+h$ is even, $g_U = 1$ by the even-conductor formula; if it is odd, Lemma 10.13 gives $g_U = 1$. If T is even, $g_E = g_\tau = 1$. If T is odd, Lemma 10.15 gives $g_E g_\tau = 1$. Thus the quotient in (184) is 1. $\square$

10.7. The case $h \geq T$.

Lemma 10.17 (Stationary coefficients for a norm pullback of conductor $n > \delta$). *Let L/F be ramified quadratic with different exponent δ , and let λ have conductor $n > \delta$. Let b_F be a stationary coefficient for λ , so $\lambda(1+z) = \psi_F(b_F z)$ for $z \in \mathfrak{p}_F^{\lceil n/2 \rceil}$. A stationary coefficient b_L for $\lambda \circ N_{L/F}$ satisfies*

$$(188) \quad b_L/b_F \in U_L^{n-\delta}.$$

If L/F is unramified and quadratic, b_F is a stationary coefficient for the pullback.

Proof. If L/F is ramified, the pullback conductor is $M = 2n - \delta$, and $n_L(\psi_L) = 2n_F + \delta$, so $v_L b_L = v_L b_F = -2(n + n_F)$. For $z \in \mathfrak{p}_L^n$, its trace belongs to $\mathfrak{p}_F^{\lceil n/2 \rceil}$ because $\lfloor (n + \delta)/2 \rfloor \geq \lceil n/2 \rceil$, and its norm belongs to $\mathfrak{p}_F^n$. The norm expansion and the formula for λ give $(\lambda \circ N_{L/F})(1 + z) = \psi_L(b_F z)$ for $z \in \mathfrak{p}_L^n$. The stationary formula for b_L also applies there. Perfect additive duality gives

$$b_L - b_F \in \mathfrak{p}_L^{-2n_F - \delta - n}.$$

Divide by b_F to obtain (188). For unramified L/F , at depth $s = \lceil n/2 \rceil$, the highest-degree norm term has depth $2s \geq n$ and the trace has depth at least s , so the same expansion gives the final assertion on the stationary ideal. $\square$

Theorem 10.18 (A stable-twist formula). *Let (χ_1, χ_2) be any primitive compatible pair in a biquadratic extension K/F , and put $a_i = a_{L_i}(\chi_i)$ and $\delta_i = a_F(\omega_{L_i})$. Let λ have conductor $n \geq 2$. Assume for every ramified L_i/F that*

$$n > \delta_i, \quad n \geq a_i + \delta_i, \quad n \geq 2\delta_i,$$

and for every unramified L_i/F that $n \geq 2a_i$. For $\theta_i = \chi_i(\lambda n_i)$,

$$(189) \quad A_{L_i}(\theta_i) = D(c_F) \Delta_F(\lambda)^2, \quad c_F = b_F^{-1}, \quad D = \chi_i|_F \omega_{L_i}.$$

In particular $A_{L_i}(\theta_i)$ is independent of i .

Proof. The inequalities imply $a_{L_i}(\chi_i) \leq \lfloor a_{L_i}(\lambda n_i)/2 \rfloor$, so (13) applies. For ramified L_i/F , its stationary depth is $n - \lceil \delta_i/2 \rceil \geq a_i$; the unramified assertion follows from $n \geq 2a_i$. The exact stable-twist formula (13) gives

$$\Delta_{L_i}(\chi_i(\lambda n_i)) = \chi_i(c_i) \Delta_{L_i}(\lambda n_i), \quad c_i = b_{L_i}^{-1}.$$

Lemma 10.17 and $n - \delta_i \geq a_i$ show $\chi_i(c_i) = \chi_i(c_F)$. For unramified L_i/F , take $c_i = c_F$. The First Main Lemma on L_i/F is

$$\Delta_{L_i}(\lambda n_i) \Delta_F(\omega_{L_i}) = \Delta_F(\lambda) \Delta_F(\lambda \omega_{L_i}).$$

The stable-twist formula over F applies since $n \geq 2\delta_i$ (and $\delta_i = 0$ when L_i/F is unramified). Its right side is $\omega_{L_i}(c_F) \Delta_F(\lambda)^2$. Multiply the two formulas and use Lemma 4.2. $\square$

Corollary 10.19. *For $K = UE$, every primitive pair with $h \geq T$ in (162) satisfies $A_U(\theta_U) = A_E(\theta_E)$.*

Proof. Use the characters of conductor T from Theorem 10.4 and Proposition 10.5. Its two conductors are T , the lower different exponents are $0, T$, and $a_F(\lambda) = T + h \geq 2T$. All hypotheses of Theorem 10.18 hold. $\square$

10.8. Completion of the unramified–ramified case.

Theorem 10.20 (The case with an unramified quadratic intermediate field). *Let K/F be biquadratic over a local field of residue characteristic two, and assume it has an unramified quadratic intermediate field U . For every other quadratic intermediate field E and every primitive compatible pair (θ_U, θ_E) , one has*

$$\Delta_U(\theta_U, \psi_U) \Delta_F(\omega_U, \psi_F) = \Delta_E(\theta_E, \psi_E) \Delta_F(\omega_E, \psi_F).$$

Thus $A_L(\theta_L)$ is independent of the quadratic intermediate field L . The assertion holds in mixed and equal characteristic, for every conductor.

Proof. The other quadratic fields are ramified, since an unramified local Galois group is cyclic. For a comparison with either such E , Proposition 10.5 gives conductors $T + h, T + 2h$ with $h \geq 0$. The case $h = 0$ is Theorem 10.8; the cases $1 \leq h < T$ are Theorem 10.16; and every $h \geq T$ is Corollary 10.19. This exhausts the conductors. Comparing each ramified field with U gives independence of all three by transitivity. Two characters of $L^\times$ with the prescribed pullback to K differ by a character in $S(K/L)$ and are conjugate by Lemma 4.2. Their local constants agree by an automorphism change of variables. Thus the conclusion does not depend on those choices either. $\square$

11. TOTALLY RAMIFIED EXTENSIONS IN CHARACTERISTIC TWO

11.1. Statement and setup. Let $F = k((\varpi))$, where k is a finite field of characteristic 2. Let K/F be totally ramified and Galois with group C_2^2 . Write its three nontrivial automorphisms as g_1, g_2, g_3 , with $g_1 g_2 = g_3$, and put

$$L_i = K^{\langle g_i \rangle}, \quad N_i = N_{K/L_i}, \quad S_i = \mathrm{Tr}_{K/L_i}, \quad n_i = N_{L_i/F}, \quad T_i^{\mathrm{tr}} = \mathrm{Tr}_{L_i/F}.$$

We reserve T_i without a superscript for a different/conductor exponent. Let ω_i be the nontrivial character of $F^\times / n_i L_i^\times$.

Theorem 11.1. *Suppose Θ is $\mathrm{Gal}(K/F)$ -invariant and is not a norm pullback from F . Choose characters θ_i of $L_i^\times$ with $\theta_i N_i = \Theta$. For any nontrivial additive character ψ_F , put $\psi_{L_i} = \psi_F T_i^{\mathrm{tr}}$. Then*

$$(190) \quad \mathcal{A}_i(\theta_i) = \Delta_{L_i}(\theta_i, \psi_{L_i}) \Delta_F(\omega_i, \psi_F)$$

is independent of i . Continuous quasi-characters are allowed.

We use the local constants Δ_E defined in (1). In particular, $\Delta_F(1, \psi_F) = 1$, so the trivial-character factor is omitted. Only restrictions to compact unit groups enter the finite calculations.

We use the First Main Lemma, Lamprecht's formula, exact stable twisting, additive scaling and duality, norm filtration, and the trace and conductor formulas from [7], recalled in Sections 2–4. The subcritical norm representatives are those of [7, Lemma 6.6]. For a ramified quadratic extension E/D of break t , these give

$$(191) \quad \mathrm{Tr}_{E/D}(\mathfrak{p}_E^a) = \mathfrak{p}_D^{\lfloor (a+t+1)/2 \rfloor}, \quad v_D(Nx) = v_E(x), \quad a_D(\omega_{E/D}) = t + 1.$$

If $a_D(\lambda) > t + 1$, its norm pullback has conductor $2a_D(\lambda) - (t + 1)$. For $0 \leq j \leq t$, an arbitrary $A \in D^\times$ has a norm approximation with relative error in U_D^j and the correct valuation. We also use Hilbert 90, extension of abelian characters, elementary Artin–Schreier theory, and the local trace–residue identity for separable Laurent-series fields. The residue formula for the norm characters is proved in Lemma 11.2.

Throughout this section K/F is totally ramified. Fix a uniformizer π of K and take

$$(192) \quad \Pi_i = N_i \pi, \quad \varpi = N_{K/F} \pi = n_i \Pi_i.$$

These are compatible uniformizers; the coefficient field k is contained in all four fields. Write $\mathrm{tr} = \mathrm{Tr}_{k/\mathbf{F}_2}$. We first prove the theorem for

$$(193) \quad \psi_F(x) = (-1)^{\mathrm{tr} \mathrm{Res}_F(x d\varpi)}.$$

Its largest trivial ideal is $\mathcal{O}_F$. The passage to an arbitrary additive character is given at the end.

11.2. Local Artin–Schreier characters and a residue identity. For a Laurent differential in a uniformizer v , define

$$\mathcal{C} \left(\sum_n a_n v^n dv \right) = \sum_m a_{2m+1}^{1/2} v^m dv.$$

Coefficient square roots are unique because the residue field is perfect. The identities used below follow from this formula:

$$(194) \quad \mathcal{C}(f^2 \eta) = f \mathcal{C}(\eta), \quad \mathcal{C}(dv/v) = dv/v, \quad \mathrm{tr} \mathrm{Res} \mathcal{C}(\eta) = \mathrm{tr} \mathrm{Res} \eta.$$

For every nonzero Laurent series u ,

$$(195) \quad \mathcal{C}(du/u) = du/u.$$

Write $u = v^n P$, where $P \in k[[v]]^\times$. The term $n dv/v$ is fixed by $\mathcal{C}$, so it remains to treat P . Put

$$q = P'/P.$$

In characteristic two, $P'' = 0$ and differentiation of $P^{-1}P'$ gives

$$q' = q^2.$$

Write $q = \sum_m a_m v^m$. Comparing the coefficient of v^{2m} in this identity gives

$$a_{2m+1} = a_m^2.$$

Hence

$$\mathcal{C}(q dv) = \sum_m a_{2m+1}^{1/2} v^m dv = \sum_m a_m v^m dv = q dv,$$

which proves the assertion.

We use the local trace–residue identity proved in Theorem B.4

$$(196) \quad \mathrm{Tr}_{k_E/k_D} \mathrm{Res}_E(\eta) = \mathrm{Res}_D(\mathrm{Tr}_{E/D} \eta)$$

for a finite separable extension of equal-characteristic local fields. Here the trace of $x d\varpi_D$ is $\mathrm{Tr}_{E/D}(x) d\varpi_D$. We also use $d \log N_{E/D}(u) = \mathrm{Tr}_{E/D}(du/u)$ from Lemma B.5.

Lemma 11.2 (The Artin–Schreier norm character). *Let $z^2 + z = f \in F$ define a nontrivial quadratic extension E/F . Then its nontrivial norm character is*

$$(197) \quad \omega_f(u) = (-1)^{\mathrm{tr} \mathrm{Res}_F(f du/u)}.$$

If $f = f_0 + \sum_{j>0, j \text{ odd}} f_j \varpi^{-j}$ is reduced and its largest pole is $t > 0$, this character has conductor $t + 1$. For every $w \in F$ and $u \in F^\times$, one also has

$$(198) \quad \mathrm{tr} \mathrm{Res}_F((w^2 + w) du/u) = 0.$$

Proof. Equations (194) and (195) give

$$\mathrm{tr} \mathrm{Res}_F(w^2 du/u) = \mathrm{tr} \mathrm{Res}_F \mathcal{C}(w^2 du/u) = \mathrm{tr} \mathrm{Res}_F(w du/u),$$

which proves (198). Thus (197) is unchanged when $w^2 + w$ is added to f . It is multiplicative in u . Every Artin–Schreier class has a reduced representative of the stated form. Remove a negative even power by adding $w^2 + w$ and repeat; remove the positive-depth part using the bijectivity of $w \mapsto w^2 + w$ on the maximal ideal. Constants are taken modulo the same map on k .

For $u = N_{E/F} v$, (196) makes its exponent equal to the absolute trace of $\mathrm{Res}_E((z^2 + z) dv/v)$, which is zero by (198), applied over E . Thus the formula is trivial on norms. For a reduced representative with pole t , it is trivial on U_F^{t+1} , where $f du/u$ has nonnegative valuation. At $u = 1 + c\varpi^t$, the exponent of the induced character on U_F^t/U_F^{t+1} is $\mathrm{tr}(f_t c)$, which is nontrivial as c varies. If instead the class is an unramified nonzero constant, evaluation at a uniformizer is nontrivial. Consequently the formula is nontrivial for every nonzero Artin–Schreier class. The quadratic norm quotient has order two, so this nontrivial character trivial on norms is precisely its norm character. The conductor assertion follows as well. $\square$

Remark 11.3. The Lean proof of Lemma 11.4 suggested that the separate rational Cartier lemma in the original proof is unnecessary. Its identity follows from (198). The deleted proof put $H = 1 + \varpi w^2$, wrote

$$\frac{\varpi f^2}{H} = \varpi(f/H)^2 + \varpi^2(fw/H)^2,$$

and applied Cartier to the two square terms. We use the shorter Artin–Schreier argument directly in Lemma 11.4.

11.3. Artin–Schreier generators and their norms. Choose the labeling so that L_1/F has the smallest break, and take reduced representatives

$$z_i^2 + z_i = f_i \quad (i = 1, 2), \quad z_3 = z_1 + z_2, \quad f_3 = f_1 + f_2.$$

Because K/F is totally ramified, all three nonzero classes have a pole. Write their breaks as $t_1 \leq t_2 = t_3$, all odd, and put

$$(199) \quad T_1 = t_1 + 1 = 2r_1, \quad T_2 = t_2 + 1 = 2r_2, \quad \delta = r_2 - r_1 \geq 0.$$

In the one-break case the two leading pole coefficients are distinct; otherwise their sum would have smaller conductor and would be the first field. In the two-break case $t_2 - t_1 = 2\delta$.

Write $f_i = f_{i,0} + f_i^-$, where f_i^- is the negative-power part. Define

$$(200) \quad \beta_i = \varpi^{t_2} f_i^- = d_i^2 \in \mathcal{O}_F, \quad d_3 = d_1 + d_2.$$

The square roots exist in F : every exponent $t_2 - j$ is even, and k is perfect. Thus $v_F d_1 = \delta$ and d_2, d_3 are units. Set

$$(201) \quad Y = d_1 z_2 + d_2 z_1, \quad B = \beta_1 f_2 + \beta_2 f_1 = \beta_1 f_{2,0} + \beta_2 f_{1,0} \in \mathcal{O}_F, \quad \kappa_i = d_j d_k \quad (\{i, j, k\} = \{1, 2, 3\}).$$

The automorphisms are labeled so that g_1 changes z_2 by 1 and fixes z_1 , while g_2 changes z_1 by 1 and fixes z_2 . Exact identities in the fields are

$$(202) \quad g_i Y - Y = d_i, \quad S_i Y = d_i, \quad a_i := N_i Y = \kappa_i z_i + B.$$

For example $Y^2 = \beta_1 z_2 + \beta_2 z_1 + B$, which proves every formula. In particular,

$$(203) \quad v_K Y = -t_1, \quad v_{L_i} a_i = -t_1, \quad \mathcal{N} := N_{K/F} Y = n_i a_i, \quad v_F \mathcal{N} = -t_1.$$

For $i = 2, 3$, $v_{L_i} \kappa_i = 2\delta = t_2 - t_1$; for $i = 1$ its valuation is zero. These observations prove the second assertion of (203), and the first follows by the norm valuation formula.

The upper breaks are $u_1 = 2t_2 - t_1$ and $u_2 = u_3 = t_1$. Indeed, apply Lemma C.3 to Y in (202). The valuation of d_1 in K is 4δ , and those of d_2, d_3 are zero, proving these upper breaks. Thus the different exponents of K/L_i are

$$(204) \quad D'_1 = 2T_2 - T_1, \quad D'_2 = D'_3 = T_1.$$

Put

$$(205) \quad \alpha = \varpi^{-T_2}, \quad \Psi_F(x) = \psi_F(\alpha x), \quad \Psi_{L_i}(x) = \Psi_F(T_i^{\text{tr}} x).$$

The largest trivial ideals for these characters have exponents

$$(206) \quad J_F = T_2, \quad J_1 = 2T_2 - T_1, \quad J_2 = J_3 = T_2.$$

Since T_2 is even, $\Psi_F(x^2) = 1$ for every $x \in F$. Lemma 11.2, integration by parts for a formal residue, and $df_i = \alpha \beta_i d\varpi$ give

$$(207) \quad \omega_i(1 - z) = \Psi_F(\beta_i z) \quad (z \in \mathfrak{p}_F^{r_i}), \quad r_3 = r_2.$$

Indeed replacing $(1 - z)^{-1}$ by 1 in $f_i dz/(1 - z)$ costs no residue at depth r_i : the error has valuation at least $-t_i + r_i + (r_i - 1) = 0$. Moreover

$$(208) \quad \beta_i = n_i(S_i Y) = n_i(d_i), \quad \omega_i(\beta_i) = 1.$$

The quadratic norm expansion also gives

$$(209) \quad \Psi_{L_i}(\beta_i z) = \Psi_F(\beta_i n_i z) \quad (z \in \mathfrak{p}_{L_i}^{r_i}).$$

Both trace and norm of z lie in $\mathfrak{p}_F^{r_i}$; evaluate (207) on $n_i(1 - z) = 1 + T_i^{\text{tr}} z + n_i z$ and use triviality on norms.

11.4. Construction of compatible characters. Define the minimal conductors and stationary depths

$$(210) \quad \begin{aligned} m_1 &= 2T_2 - 1 = 2t_2 + 1, & s_1 &= T_2, \\ m_2 &= m_3 = T_1 + T_2 - 1 = t_1 + t_2 + 1, & s_2 &= s_3 = r_1 + r_2. \end{aligned}$$

Thus $m_i = J_i + t_1$, $m_i = 2s_i - 1$, and the formulas

$$(211) \quad R_i(1 - z) = \Psi_{L_i}(a_i z), \quad z \in \mathfrak{p}_{L_i}^{s_i}$$

define characters of $U_{L_i}^{s_i}$ with exact conductor m_i . Multiplicativity follows since $2s_i - t_1 \geq J_i$.

Lemma 11.4 (The value of R_1 on $\sigma(v)/v$). *Let σ be the nontrivial automorphism of L_1/F and $\nu_1 = \omega_2 n_1$, the nontrivial character in $S(K/L_1)$. For every $v \in L_1^\times$ with $\sigma(v)/v \in U_{L_1}^{s_1}$,*

$$(212) \quad R_1(\sigma(v)/v) = \nu_1(v).$$

Proof. The description of ν_1 follows from Lemmas D.5 and D.6, whose characteristic-two proof uses only (197) and (196). Multiplication of v by an element of $F^\times$ does not affect either side. Reduce its valuation to 0 or 1. In valuation 1 its conjugate quotient has exact depth $t_1 < s_1$, by Lemma C.3. Thus only units occur. Write $v = a + bz_1$, with $a, b \in F$. The two summands have valuations of opposite parity, since t_1 is odd. Consequently a is a unit and $v_F b \geq r_1$. The conjugate quotient is $1 + b/v$; the assumed depth $s_1 = 2r_2$ forces $v_F b \geq r_2$. Put $w = b/a$, so $v_F w \geq r_2$. The case $b = 0$ is immediate. Exact quadratic trace calculation gives

$$(213) \quad T_1^{\text{tr}} \left(a_1 \frac{b}{v} \right) = \frac{\kappa_1 ab + Bb^2}{n_1 v} = \frac{\kappa_1 w + Bw^2}{1 + w + w^2 f_1}.$$

To evaluate $\nu_1(v)$, put $R = w^2 f_1$, $H = 1 + R$ and $V = H + w$. The elements H and V are units, since $v_F R \geq 2r_2 - t_1 \geq 1$. Factor $V = H(1 + w/H)$. The residue formula evaluates $\omega_2(H)$ exactly, and (207) applies to the second factor. They give

$$\omega_2(V) = \Psi_F \left(\frac{\beta_1 f_2 w^2 + \beta_2 w}{H} \right) = \Psi_F \left(\frac{\beta_2 R + \beta_2 w}{H} \right).$$

In the last equality Bw^2/H was discarded at depth at least T_2 , using $\beta_1 f_2 + \beta_2 f_1 = B \in \mathcal{O}_F$. Write $R_0 = \varpi W^2$, where $W = \varpi^{-r_2} d_1 w \in \mathcal{O}_F$. Then $R - R_0 = f_{1,0} w^2 \in \mathfrak{p}_F^{T_2}$, so it can be discarded in all these unit denominators. Put

$$f = \varpi^{-r_2} d_2 W, \quad H_0 = 1 + R_0, \quad a_0 = \varpi f, \quad b_0 = \varpi H_0.$$

The denominator H_0 is nonzero by parity of valuations. In characteristic two, $db_0 = d\varpi$, and (198) gives

$$0 = \text{tr Res}_F \left((a_0^2 + a_0) \frac{db_0}{b_0} \right) = \text{tr Res}_F \left(\left(\frac{\varpi f^2}{H_0} + \frac{f}{H_0} \right) d\varpi \right).$$

Thus the two residue traces are equal. In the present variables this is exactly

$$\Psi_F \left(\frac{\beta_2 R_0}{1 + R_0} \right) = \Psi_F \left(\frac{d_1 d_2 w}{1 + R_0} \right).$$

It follows that $\omega_2(V) = \Psi_F(\kappa_1 w/H)$ because $\beta_2 + d_1 d_2 = d_2 d_3 = \kappa_1$. Replacing H by V here costs only $\kappa_1 w^2/(HV)$, of depth at least T_2 . The same is true of the Bw^2/V term in (213). Since $n_1 v = a^2 V$ and $\omega_2(a^2) = 1$, (213) proves (212). $\square$

Lemma 11.5 (Equality of $R_i N_i$ on $U_K^{T_2}$). *For $z \in \mathfrak{p}_K^{T_2}$ all $N_i(1 - z)$ lie in the domains of R_i , and $R_i(N_i(1 - z))$ is independent of i .*

Proof. The norm term has L_i -valuation at least T_2 . The upper trace terms have respective depths at least $3r_2 - r_1$ and $r_1 + r_2$; these are at least s_1 and s_2 . The identity of Lemma 4.5 is

$$(214) \quad E_2(C) = T_i^{\text{tr}} N_i C + n_i S_i C.$$

Apply it to $Y(1 - z)$ and Y . Put $h_i = S_i(Yz)/d_i$. The upper trace formula and (203) give

$$v_{L_1} S_1(Yz) \geq r_1 + 3\delta, \quad v_{L_2} S_2(Yz), v_{L_3} S_3(Yz) \geq r_2.$$

After division by d_1 , of L_1 -valuation 2δ , the first bound is also r_2 . Thus $v_{L_i} h_i \geq r_i$ for every i . Since $S_i(Y(1 - z)) = d_i(1 - h_i)$, its lower norm divided by β_i is a norm from L_i/F . Formula (207) applied to that norm quotient shows that the additive phase of $n_i S_i(Y(1 - z)) - \beta_i$ is 1. The remaining term in (214) is $T_i^{\text{tr}}(a_i(S_i z + N_i z))$, exactly the exponent of $R_i(N_i(1 - z))$. Therefore this value is $\Psi_F(E_2(Y(1 - z)) - E_2(Y))$, independent of i . $\square$

Theorem 11.6 (Compatible characters of conductors m_i). *There are characters χ_i on $L_i^\times$, with exact conductors m_i , primitive common norm pullback, and*

$$(215) \quad \chi_i(1 - z) = \Psi_{L_i}(a_i z) \quad (z \in \mathfrak{p}_{L_i}^{s_i}).$$

Proof. Let $I = \{\sigma(v)/v : v \in L_1^\times\}$ and prescribe $\kappa_I(\sigma(v)/v) = \nu_1(v)$. This is well defined because ν_1 is trivial on $F^\times$: it is $\omega_2(n_1x) = \omega_2(x^2)$ there. Lemma 11.4 proves agreement with R_1 on $I \cap U_{L_1}^{s_1}$. Their product character on $IU_{L_1}^{s_1}$ is trivial on $U_{L_1}^{m_1}$. Extend it across the abelian group $L_1^\times/U_{L_1}^{m_1}$. One can first extend on its finite unit subgroup and then choose the uniformizer value. The resulting χ_1 is continuous, has exact conductor m_1 , and satisfies $\chi_1^\sigma/\chi_1 = \nu_1$ globally.

The character $\chi_1 N_1$ is invariant under G : its other conjugate quotient is $\nu_1 N_1 = \omega_2 N_{K/F} = 1$. By Lemma C.1, it is trivial on $\ker N_i$ for $i = 2, 3$. Define $\chi_i^0(N_i x) = \chi_1(N_1 x)$ on $N_i(K^\times)$ and extend χ_i^0 to a character χ_i of $L_i^\times$ by Lemma C.2. Above the upper break t_1 , (9) gives

$$N_i(U_K^{T_2+1}) = U_{L_i}^{s_i} \quad (i = 2, 3),$$

since $2s_i - t_1 = T_2 + 1$. Lemma 11.5 then proves (215) on $U_{L_i}^{s_i}$. Their conductors are at least m_i . The common pullback has conductor $2m_1 - D'_1 = t_1 + 2t_2 + 1$; since $m_i > D'_i$, the high pullback-conductor formula forces their conductors to equal m_i . Finally, descent of the common pullback from F would make χ_1 a product of a norm pullback from F and a character in $S(K/L_1)$. Both are σ -invariant, contradicting $\chi_1^\sigma/\chi_1 = \nu_1 \neq 1$. $\square$

11.5. Writing $\theta_i = \chi_i(\lambda n_i)$ and computing the conductors. Write ν_i for the nontrivial norm character of K/L_i . The common restriction $D_\theta = \theta_i|_{F^\times \omega_i}$ is the one in Lemma 4.2.

Proposition 11.7 (Writing θ_i as $\chi_i(\lambda n_i)$). *For every primitive compatible family (θ_i) there are characters (χ_i) satisfying (215), and a character λ of $F^\times$, such that*

$$(216) \quad \theta_i = \chi_i(\lambda n_i).$$

Put $n = a_F(\lambda)$ and $h = n - T_2$. If $n \leq T_2 + r_1 - 1$, the conductors are exactly m_i . If $n \geq T_2 + r_1$, they are $M_i = 2n - T_i$, all even. These alternatives exhaust all primitive conductors.

Proof. Start with Theorem 11.6. By Lemma 4.2, θ_1/χ_1 is invariant under $\text{Gal}(L_1/F)$. Hilbert 90 and abelian character extension make it λn_1 . For $i = 2, 3$, the remaining quotient is a power of ν_i . Absorb it into χ_i . This does not change its common pullback or (215), because $a_{L_i}(\nu_i) = T_1 \leq s_i$.

Primitivity forces $a_{L_i}(\theta_i) \geq t_i + u_i + 1 = m_i$: on $U_{L_i}^{u_i}/U_{L_i}^{u_i+1}$, the first ramification term in $\sigma(v)/v$ has exact depth $t_i + u_i$. Its coefficient is nonzero since u_i is odd; a character of smaller conductor would make θ_i^σ/θ_i trivial on $U_{L_i}^{u_i}$. For $n > T_i$, the pullback λn_i has conductor $2n - T_i$; for $n \leq T_i$ it has conductor at most T_i . The number $2n - T_i$ is even, whereas m_i is odd, so cancellation at equal conductors is impossible. The same threshold $n = T_2 + r_1$ is obtained for all three fields. This proves the assertions. $\square$

11.6. The norms of $Y + X$, for $X \in L_3$. We compare $\mathcal{A}_1(\theta_1)$ and $\mathcal{A}_2(\theta_2)$. Interchanging L_2, L_3 gives the comparison with $\mathcal{A}_3(\theta_3)$. For $X \in L_3$, put

$$(217) \quad A = n_3 X, \quad a = T_3^{\text{tr}} X, \quad C = Y + X, \quad Q_i = aY + d_i X \in L_i \quad (i = 1, 2).$$

The following are exact algebraic identities:

$$(218) \quad N_i C = a_i + A + Q_i,$$

$$S_i C = d_i + a,$$

$$(219) \quad n_i Q_i = \beta_i A + E,$$

$$E = a(aa_3 + \kappa_3 X) \in F.$$

For example, write $X = p + qz_3$. Then $a = q$, $Q_1 = d_1 p + qd_3 z_1$ and $Q_2 = d_2 p + qd_3 z_2$; taking the two quadratic norms proves (219), with $E = q^2 B + q\kappa_3 p$.

If $v_{L_3} X = -h$, trace ideals and (203) give

$$(220) \quad v_F a \geq r_2 - \lceil h/2 \rceil,$$

$$(221) \quad v_{L_1} Q_1 \geq 2\delta - h,$$

$$v_{L_2} Q_2 \geq -h,$$

$$(222) \quad v_F E \geq r_2 + \delta - h.$$

These statements include negative h and vanishing traces. For the last one, the invariant element $aa_3 + \kappa_3 X$ has L_3 -valuation at least $2\delta - h$, hence F -valuation at least $\delta - \lfloor h/2 \rfloor$; combine this with (220). For the Q_i bounds one can use directly the upper trace of $Y g_i X$, with valuation $-t_1 - 2h$, and the upper differentials (204).

Suppose $\lambda(1-z) = \Psi_F(Az)$ on an ideal of F containing both trace and norm of a variable $z \in L_i$. Whenever $(Q_i/\beta_i)z \in \mathfrak{p}_{L_i}^{r_i}$, equations (209) and (219) give

$$(223) \quad \Psi_{L_i}(Q_i z) = \Psi_F((A + E/\beta_i)n_i z).$$

If also $(E/\beta_i)n_i z \in \mathfrak{p}_F^{T_2}$, the norm expansion gives $(\lambda n_i)(1-z) = \Psi_{L_i}((A + Q_i)z)$.

11.7. Stationary coefficients when $n \leq T_2 + r_1 - 1$. Let the family be realized as in (216), with $n \leq T_2 + r_1 - 1$. Put $q_0 = r_1 + r_2$. For $z \in \mathfrak{p}_{L_i}^{s_i}$, both $T_i^{\text{tr}} z$ and $n_i z$ lie in $\mathfrak{p}_F^{q_0}$. For $i = 1$ the trace bound is exactly q_0 ; for $i = 2, 3$ its bound is at least q_0 . If $n \leq q_0$, take $X = 0$ and $C = Y$ below. Otherwise put $h = n - T_2$, so

$$(224) \quad 1 - \delta \leq h \leq r_1 - 1.$$

Since $2q_0 \geq n$, additive duality supplies

$$\lambda(1-z) = \Psi_F(A_* z) \quad (z \in \mathfrak{p}_F^{q_0}), \quad v_F A_* = -h,$$

with ambiguity $\mathfrak{p}_F^{T_2 - q_0} = \mathfrak{p}_F^\delta$. Choose $X \in L_3$ by norm approximation so that

$$(225) \quad A = n_3 X \equiv A_* \pmod{\mathfrak{p}_F^\delta}, \quad v_{L_3} X = -h.$$

The relative precision requested is $\delta + h \leq r_2 - 1 \leq t_2$, so [7, Lemma 6.6] supplies X . The character formula for λ on $\mathfrak{p}_F^{q_0}$ is unchanged when A_* is replaced by A .

Lemma 11.8 (Stationary coefficients from $Y + X$). *The element C just constructed has valuation $-t_1$ in K . For $i = 1, 2$, one has*

$$(226) \quad \theta_i(1-z) = \Psi_{L_i}(N_i C z) \quad (z \in \mathfrak{p}_{L_i}^{s_i}),$$

$$(227) \quad \omega_i(1-z) = \Psi_F(n_i S_i C z) \quad (z \in \mathfrak{p}_F^{r_i}).$$

If $\delta = 0$, the same single C satisfies these identities for all three indices.

Proof. The case $X = 0$ follows from (215) and (207). Otherwise $-2h > -t_1$ by (224), so C has the valuation of Y . For $i = 1, 2$, (221) gives

$$v_{L_1}((Q_1/\beta_1)z) \geq T_2 - 2\delta - h = 2r_1 - h \geq r_1, \quad v_{L_2}((Q_2/\beta_2)z) \geq r_1 + r_2 - h \geq r_2.$$

The error in (223), at these depths, has valuations at least $T_2 + r_1 - h$ and $3r_2 - h$, respectively; both are at least T_2 . The norm expansion therefore gives $(\lambda n_i)(1-z) = \Psi_{L_i}((A + Q_i)z)$. Multiplication by (215) gives (226), since $a_i + A + Q_i = N_i C$ by (218). Moreover $n_i S_i C = (d_i + a)^2 = \beta_i + a^2$. The bound

$$2v_F a \geq 2r_2 - 2 \lceil h/2 \rceil \geq 2r_2 - r_1 = T_2 - r_1$$

proves (227) for $z \in \mathfrak{p}_F^{r_i}$. It also shows $v_F a > \delta$, so the traces $d_i + a$ are nonzero with the original valuations.

When $\delta = 0$, $r_1 = r_2 = r$ and $h \leq r - 1$. For $i = 3$, (209) gives the coefficient $A + d_3 X$, because $n_3(X/d_3) = A/\beta_3$. Its argument has depth at least $2r - h \geq r$. But

$$N_3 C = a_3 + X^2 + d_3 X = a_3 + A + d_3 X + aX,$$

and $v_{L_3}(aX) \geq 2r - 2h - (h \bmod 2) \geq 0$. Hence $\Psi_{L_3}(aX z) = 1$ for $z \in \mathfrak{p}_{L_3}^{2r}$. Finally $S_3 C = d_3$, so its lower norm coefficient is unchanged. This proves the assertion for all three fields in the one-break case. $\square$

11.8. Comparison at conductors m_i . Use Lemma 4.1 with stationary coefficient A_L and critical element $\Pi_L^{[m/2]}$, writing its critical function and normalized sum as H_L, g_L . In characteristic two the field minus signs in that lemma disappear; complex phases are unchanged.

Use Lemma 11.8 to take $N_i C$ as the coefficient for θ_i and $n_i S_i C$ as the coefficient for ω_i . Since $n_i S_i C$ is a norm from L_i/F and T_i is even, $\omega_i(n_i S_i C) = 1$ and the critical-sum factor for ω_i is 1. Compatibility and (214) therefore give

$$(228) \quad \mathcal{A}_i(\theta_i) = D_\theta(\alpha^{-1})\Theta(C)^{-1}\Psi_F(E_2(C))g_i.$$

For $z \in \mathcal{O}_K$ vary C to $C_z = C(1 + \pi^{t_2}z)$. The relative norms $N_i C_z / N_i C - 1$ have valuations at least

$$\ell_1 = t_2, \quad \ell_2 = \ell_3 = (t_1 + t_2)/2.$$

The changes $S_i(C\pi^{t_2}z)$ have valuations at least $r_1 + 3\delta$ in L_1 and r_2 in L_2 , respectively. After division by $S_i C$, both relative changes have valuation at least $r_2 \geq r_i$. The quotient $n_i S_i C_z / (n_i S_i C)$ can thus be evaluated by (227); its additive phase is 1. Subtracting (214) for C_z, C , and using compatibility and Lemma 4.1, proves

$$(229) \quad H_i(p_i(z)) = Q(z),$$

Here $p_i : k \rightarrow k$ is the map induced by $N_i : U_K^{t_2} \rightarrow U_{L_i}^{\ell_i}$ on the corresponding successive unit quotients, using the uniformizers in (192). The function $Q : k \rightarrow \mathbf{C}^\times$ is independent of i .

Proposition 11.9 (The case $t_1 < t_2$). *If $t_1 < t_2$, every family whose conductors are m_i satisfies $\mathcal{A}_1 = \mathcal{A}_2$.*

Proof. For K/L_1 , the depth t_2 is below the break $2t_2 - t_1$. The trace term is at least one level deeper than the norm term, so $p_1(z) = z^2$, with compatible uniformizers (192). For K/L_2 , the trace induces an isomorphism of the one-dimensional residue lines at source depth t_2 and target depth $(t_1 + t_2)/2$:

$$\lfloor (t_2 + T_1)/2 \rfloor = (t_1 + t_2)/2, \quad \lfloor (t_2 + 1 + T_1)/2 \rfloor = (t_1 + t_2)/2 + 1.$$

The highest-degree norm term is strictly deeper. Hence p_2 is a nonzero k -linear map. Both p_i are bijections of k . Summing (229) gives $g_1 = g_2$, and (228) proves equality. The same comparison with L_3 proves equality for all three intermediate fields when $t_1 < t_2$. $\square$

Proposition 11.10 (The case $t_1 = t_2$). *If $t_1 = t_2 = t$, every family whose conductors are m_i satisfies $\mathcal{A}_1 = \mathcal{A}_2 = \mathcal{A}_3$.*

Proof. Lemma 11.8 supplies C satisfying (226)–(227) for $i = 1, 2, 3$. Put

$$u = \overline{C\pi^t} \in k^\times, \quad c_i = \overline{g_i C - C} \in k^\times.$$

The three c_i are distinct, with $c_1 + c_2 = c_3$. Indeed $g_i Y - Y = d_i$ and the possible adjustment changes the first two by an element of $\mathfrak{p}_F$, leaving their residues unchanged. Expanding the leading term $u\pi^{-t}$ of C , and using oddness of t , gives the critical norm maps

$$(230) \quad p_i(z) = z^2 + (c_i/u)z.$$

Thus

$$\ker p_i = \{0, c_i/u\}.$$

The kernels are pairwise distinct, each has order two, and each image has index two in $(k, +)$.

Equation (16) gives the nondegenerate polar pairing of each critical function H_i . Equation (229) gives the common pullback $H_i \circ p_i = Q$. Lemma 4.8 therefore gives

$$\sum_{x \in k} H_i(x) = \frac{1}{2} \sum_{z \in k} Q(z)$$

for every i . Hence the normalized sums g_i are equal, and (228) proves $\mathcal{A}_1 = \mathcal{A}_2 = \mathcal{A}_3$. $\square$

11.9. The case $T_2 + r_1 \leq n \leq 2T_2 + r_1 - 2$. Suppose now $n \geq T_2 + r_1$, and write $h = n - T_2 \geq r_1$. By Proposition 11.7, the conductors and stationary depths are

$$(231) \quad M_i = 2n - T_i, \quad b_i = M_i/2 = n - r_i; \quad b_1 = T_2 + h - r_1, \quad b_2 = r_2 + h.$$

For this subsection assume

$$(232) \quad r_1 \leq h \leq r_1 + 2r_2 - 2.$$

Put

$$q_F = r_2 + \lfloor (h + r_1)/2 \rfloor.$$

The trace and norm of each variable $z \in \mathfrak{p}_{L_i}^{b_i}$, $i = 1, 2$, belong to $\mathfrak{p}_F^{q_F}$. Moreover $q_F \geq \lceil n/2 \rceil$ and $b_i \geq s_i$. Choose A_* such that $\lambda(1-z) = \Psi_F(A_*z)$ for $z \in \mathfrak{p}_F^{q_F}$. It has valuation $-h$ and ambiguity $\mathfrak{p}_F^{T_2 - q_F}$. A norm from L_3 can represent this class, because its requested relative precision is

$$(233) \quad h + T_2 - q_F = r_2 + \lceil (h - r_1)/2 \rceil \leq 2r_2 - 1 = t_2.$$

Thus choose $X \in L_3$ with $v_{L_3}X = -h$ and $A = n_3X$ in that class, and use (217).

Theorem 11.11 (Equality for $T_2 + r_1 \leq n \leq 2T_2 + r_1 - 2$). *Every family in (232) satisfies $\mathcal{A}_1(\theta_1) = \mathcal{A}_2(\theta_2)$.*

Proof. The elements $(Q_i/\beta_i)z$ in (223) have depths at least r_1 and r_2 , respectively, because

$$-h - 2\delta + b_1 = r_1, \quad -h + b_2 = r_2.$$

The error terms there have depths at least

$$(r_2 + \delta - h) - 2\delta + b_1 = T_2, \quad (r_2 + \delta - h) + b_2 = T_2 + \delta.$$

Thus $(\lambda n_i)(1-z) = \Psi_{L_i}((A + Q_i)z)$ for $z \in \mathfrak{p}_{L_i}^{b_i}$. Combining it with (215) proves that $N_iC = a_i + A + Q_i$ is the stationary coefficient for θ_i . Here $v_KX = -2h < -t_1 = v_KY$, so $C \neq 0$ and $v_{L_i}N_iC = -2h$, exactly the required $J_i - M_i$.

The conductors M_i and T_i are even. Apply Lemma 4.1 with coefficients N_iC for θ_i and β_i for ω_i . This gives

$$\mathcal{A}_i(\theta_i) = D_\theta(\alpha^{-1})\Theta(C)^{-1}\Psi_F(T_i^{\text{tr}}N_iC + \beta_i).$$

By (214) and (218), its last exponent is

$$E_2(C) + (d_i + a)^2 + d_i^2 = E_2(C) + a^2,$$

the same for $i = 1, 2$. Hence $\mathcal{A}_1(\theta_1) = \mathcal{A}_2(\theta_2)$. In fact $\Psi_F(a^2) = 1$, but this extra simplification is not needed. Relabeling the third field proves the other comparison. $\square$

11.10. The case $h \geq r_1 + 2r_2 - 1$. The remaining range is

$$(234) \quad h \geq h_0 := r_1 + 2r_2 - 1, \quad n = T_2 + h.$$

Choose $A \in F^\times$ such that

$$\lambda(1-z) = \Psi_F(Az) \quad (z \in \mathfrak{p}_F^{\lceil n/2 \rceil}).$$

Then $v_FA = -h$; put $c_F = (\alpha A)^{-1}$. For each $i = 1, 2$, choose only an approximate norm

$$(235) \quad n_i x_i = (A/\beta_i)\varepsilon_i, \quad \varepsilon_i \in U_F^{t_i}, \quad v_{L_i}x_i = -h - (T_2 - T_i).$$

Apply [7, Lemma 6.6] to A/β_i at precision t_i to obtain these x_i .

Lemma 11.12 (A coefficient for λn_i and the value of $\chi_i(c_i)$). *A stationary coefficient of λn_i is $A + \beta_i x_i$. If $c_i = (\alpha(A + \beta_i x_i))^{-1}$, then*

$$(236) \quad \chi_i(c_i) = \chi_i(c_F)\Psi_F(\mathcal{N}/A).$$

The correction is identical for both fields.

Proof. For $z \in \mathfrak{p}_{L_i}^{b_i}$, $b_i = n - r_i$, both its trace and norm lie in the base stationary ideal. Also $x_i z$ has valuation at least r_i , so (209) applies. The error from (235), after multiplication by $n_i z$, has F -valuation at least

$$-h + t_i + b_i = T_2 + r_i - 1 \geq T_2.$$

Exact quadratic norm expansion proves the first assertion on the whole stationary ideal. The leading term is A , since

$$v_{L_i}(\beta_i x_i/A) = h + T_2 - T_i > 0.$$

Set $v_i = \beta_i x_i/A$. The lower bound (234) places v_i in $\mathfrak{p}_{L_i}^{s_i}$, where (215) gives

$$\chi_i(c_i/c_F) = \chi_i((1 + v_i)^{-1}) = \Psi_{L_i}(a_i v_i).$$

The value of Ψ_{L_i} has order at most two because $\text{char } F = 2$. To convert it, the variable $a_i x_i/A$ has valuation

$$h - t_1 - (T_2 - T_i) \geq r_i.$$

For $i = 1$ this inequality is precisely $h \geq h_0$; for $i = 2$ it is weaker. Apply (209), using $n_i a_i = \mathcal{N}$, to get

$$\Psi_{L_i}(\beta_i a_i x_i / A) = \Psi_F(\mathcal{N} \varepsilon_i / A).$$

The error on deleting ε_i has valuation at least $h - t_1 + t_i$, which is at least T_2 for $i = 1, 2$ by (234). This proves (236). $\square$

Theorem 11.13 (Equality for $n \geq 2T_2 + r_1 - 1$). *For every family in (234),*

$$(237) \quad \mathcal{A}_i(\theta_i) = D_\chi(c_F) \Psi_F(\mathcal{N}/A) \Delta_F(\lambda, \psi_F)^2 \quad (i = 1, 2).$$

In particular $\mathcal{A}_1(\theta_1) = \mathcal{A}_2(\theta_2)$.

Proof. The conductors m_i satisfy the stable-twist bounds over L_i :

$$b_1 = T_2 + h - r_1 \geq 2T_2 - 1 = m_1, \quad b_2 = r_2 + h \geq T_1 + T_2 - 1 = m_2.$$

The exact stable-twist formula (13) gives $\Delta_{L_i}(\theta_i) = \chi_i(c_i) \Delta_{L_i}(\lambda n_i)$. The First Main Lemma on L_i/F gives

$$\Delta_{L_i}(\lambda n_i) \Delta_F(\omega_i) = \Delta_F(\lambda) \Delta_F(\lambda \omega_i).$$

Since $n \geq 2T_2 \geq 2T_i$, the stable-twist formula over F applies, so the right side is $\omega_i(c_F) \Delta_F(\lambda)^2$. Combine this with Lemma 11.12 and $D_\chi = \chi_i|_{F^\times \omega_i}$ from Lemma 4.2. $\square$

11.11. Completion in characteristic two.

Theorem 11.14. *Theorem 11.1 holds when K/F is totally ramified.*

Proof. Apply Proposition 11.7 to write $\theta_i = \chi_i(\lambda n_i)$. If $n \leq T_2 + r_1 - 1$, use Propositions 11.9 and 11.10. Otherwise $h = n - T_2 \geq r_1$. The interval $r_1 \leq h \leq r_1 + 2r_2 - 2$ is Theorem 11.11, and the remaining integers are Theorem 11.13. Their endpoints are consecutive, including $r_1 = r_2 = 1$. In the two-break case repeat the comparison of L_1 with L_3 . This proves independence of all three. By Lemma 4.2, changing an inducing character by an element of $S(K/L_i)$ gives a conjugate character; (6) preserves its local constant. The passage to an arbitrary additive character is given below. $\square$

Proof of Theorem 11.1. For the canonical additive character this is Theorem 11.14. Every nontrivial additive character of $F = k((\varpi))$ is a scaling $x \mapsto \psi_F(cx)$ of (193), with $c \in F^\times$. By (4), this change multiplies $\mathcal{A}_i(\theta_i)$ by $\theta_i(c) \omega_i(c) = D_\theta(c)$, independent of i by Lemma 4.2. This proves the stated arbitrary-character result. $\square$

12. TOTALLY RAMIFIED MIXED CHARACTERISTIC: THE MAXIMAL BREAK

12.1. Statement and ramification data. Let $F/\mathbf{Q}_2$ be finite, with residue field k , normalized valuation v_F , and $e = v_F(2)$. Suppose K/F is totally ramified, $\text{Gal}(K/F) = C_2^2$. Label its three quadratic fields so that

$$(238) \quad t_1 = 2a - 1, \quad t_2 = t_3 = 2e, \quad 1 \leq a \leq e.$$

Put $T_1 = 2a$, $T_2 = T_3 = T = 2e + 1$. Write g_i for the nontrivial upper automorphism fixing L_i , and

$$N_i = N_{K/L_i}, \quad S_i = \text{Tr}_{K/L_i}, \quad n_i = N_{L_i/F}, \quad \text{tr}_i = \text{Tr}_{L_i/F}.$$

Let ω_i be the nontrivial character in $S(L_i/F)$. Fix a nontrivial additive ψ_F and compose with trace on all fields.

Theorem 12.1. *For every primitive compatible family $\theta_i N_i = \Theta$, where Θ is not a norm pullback from F , the quantities*

$$(239) \quad \mathcal{A}_i(\theta_i) = \Delta_{L_i}(\theta_i, \psi_{L_i}) \Delta_F(\omega_i, \psi_F)$$

are equal. Continuous quasi-characters are allowed. The conclusion holds for every conductor under (238).

We first prove $\mathcal{A}_1(\theta_1) = \mathcal{A}_2(\theta_2)$. Interchanging L_2, L_3 gives $\mathcal{A}_1(\theta_1) = \mathcal{A}_3(\theta_3)$.

We use the results recalled in Sections 2–4. The quadratic-symbol identities needed here are proved below.

For a ramified quadratic extension E/F of different exponent D ,

$$(240) \quad \mathrm{Tr}_{E/F}(\mathfrak{p}_E^j) = \mathfrak{p}_F^{\lfloor (j+D)/2 \rfloor}, \quad v_F(\mathrm{N}x) = v_E x, \quad n_E(\psi_E) = 2n_F(\psi_F) + D.$$

The different exponents of K/L_i are

$$(241) \quad D'_1 = 2T - T_1 = 4e - 2a + 2, \quad D'_2 = D'_3 = 2a.$$

Indeed the upper breaks are $2t_2 - t_1$ and t_1 , by the usual break calculation of Lemma D.3. All valuations below name their field; restriction of v_K to F is $4v_F$.

12.2. Quadratic norm symbols and Kummer generators. For $A, B \in F^\times$ let (A, B) be the quadratic norm symbol: it is 1 exactly when B is a norm from $F(\sqrt{A})$, with the split case interpreted as 1. It is a character in B . Symmetry follows from the symmetric conic equation $z^2 = Ax^2 + By^2$, and hence it is bilinear in both square classes. For $A \neq 1$, one also has $(A, 1 - A) = 1$, since $1 - A$ is the nonzero norm of $1 + \sqrt{A}$. If $A + B \neq 0$, the same conic, under $u = Ax + By$, $v = x - y$, gives

$$(242) \quad (A, B) = (A + B, -AB), \quad (A + B)z^2 = u^2 + ABv^2.$$

The linear change in (x, y) is invertible, proving (242).

Lemma 12.2 (A sufficient unit depth). *If $v_F u = m \geq 1$, $v_F v = n \geq 1$, and $m + n > 2e$, then $(1 + u, 1 + v) = 1$.*

Proof. If $1 + u$ is a square, the assertion is immediate; this includes $m > 2e$ by Hensel's lemma. Otherwise put $r = \lfloor m/2 \rfloor \leq e$. The integral element $(\sqrt{1 + u} - 1)/\varpi^r$ generates an order of discriminant valuation $2e - 2r$. The field discriminant divides the order discriminant, so the quadratic norm character has conductor at most $2e - 2r$; the unramified case also satisfies this bound. If m is even, $n \geq 2e - 2r + 1$; if m is odd, $n \geq 2e - 2r$. In both cases the character kills $1 + v$. $\square$

Put $b = 2e - 2a + 1$, an odd positive integer, and $q = \varpi^b$. There are Kummer generators s, R_0 with

$$s^2 = 1 + qu_0, \quad R_0^2 = qw_0, \quad u_0, w_0 \in \mathcal{O}_F^\times.$$

Here is a verification at the stated breaks. For a uniformizer Π_1 of L_1 , its quadratic minimal polynomial and different imply $v_F \mathrm{Tr} \Pi_1 = a$: the term $2\Pi_1$ has L_1 -valuation $2e + 1 > 2a$. Thus $s = 1 - 2\Pi_1/\mathrm{Tr} \Pi_1$ has the required square defect of valuation b . For a maximal quadratic field, a uniformizer Π has $v_F \mathrm{Tr} \Pi \geq e + 1$, and $\Pi - \mathrm{Tr} \Pi/2$ is a trace-zero uniformizer. Multiplying it by ϖ^{e-a} gives R_0 .

Lemma 12.3 (A choice of S and R). *The generators can be replaced by $S = s(1 + \delta)$ and $R = R_0 v$, with $v \in \mathcal{O}_F^\times$, so that, writing $\epsilon = S^2 - 1$,*

$$(243) \quad v_F \epsilon = v_F R^2 = b, \quad \epsilon/R^2 \in U_F^a.$$

Proof. Take $\delta = \varpi^{e-a+1}w$, $w \in \mathcal{O}_F$. Modulo $\mathfrak{p}_F^a$,

$$\frac{S^2 - 1 - R^2}{q} \equiv u_0 - w_0 v^2 + \varpi(1 + qu_0)w^2.$$

The omitted term $2\delta/q$ has valuation at least a . Because $a \leq e$, $\mathcal{O}_F/\mathfrak{p}_F^a$ has characteristic two and is $k[\varpi]/(\varpi^a)$ after choosing its coefficient field. Such a coefficient field is obtained by the unique lifts of the roots of $X^{|k|} - X$, whose derivative is a unit. In this truncated ring solve

$$w_0 v^2 + \varpi(1 + qu_0)w^2 = u_0.$$

Successively in the degree of ϖ , an even degree determines the next coefficient of v^2 , with unit leading coefficient $\bar{w}_0$; an odd degree determines the next coefficient of w^2 , with leading coefficient 1. Contributions of earlier coefficients are already known. Frobenius on k supplies

the unique required square roots. The constant coefficient of v is nonzero. Lift v, w to F . The displayed congruence proves (243), and the positive-depth correction to u_0 preserves its unit leading coefficient. $\square$

Choose S, R by Lemma 12.3. Set $L_1 = F(S)$, $L_2 = F(R)$, $L_3 = F(SR)$, and define

$$(244) \quad Y = (S + R - 1)/2, \quad a_i = N_i Y, \quad h_i = S_i Y, \quad \beta_i = n_i h_i.$$

For $B_0 = (\epsilon - R^2)/4$ and $C_0 = (\epsilon + R^2)/4$ one has

$$(245) \quad \begin{aligned} a_1 &= (1 - S)/2 + B_0, & h_1 &= S - 1, & \beta_1 &= -\epsilon, \\ a_2 &= -R/2 - B_0, & h_2 &= R - 1, & \beta_2 &= 1 - R^2, \\ a_3 &= -SR/2 - C_0, & h_3 &= -1, & \beta_3 &= 1. \end{aligned}$$

Equation (243) gives $v_F B_0, v_F C_0 \geq 1 - a$; for the plus sign use also $v_F(2R^2) \geq b + a$. Consequently

$$(246) \quad v_K Y = -t_1, \quad v_{L_i} a_i = -t_1, \quad v_{L_1} h_1 = b, \quad v_{L_2} h_2 = v_{L_3} h_3 = 0, \quad \text{Tr}_{K/F} Y = -2.$$

For instance a_2 has the unique leading term $-R/2$ of valuation $1 - 2a$, while B_0 has L_2 -valuation at least $2 - 2a$. The value of Y follows by norm valuation; the other entries follow from (245) and $(S - 1)(S + 1) = \epsilon$.

12.3. Simultaneous coefficients for the norm characters. Choose α using ω_3 :

$$(247) \quad \omega_3(1 - z) = \Psi_F(z) \quad (z \in \mathfrak{p}_F^{e+1}), \quad \Psi_L(x) = \psi_L(\alpha x), \quad v_F \alpha = -n_F(\psi_F) - T.$$

The largest trivial ideals of $\Psi_F, \Psi_{L_1}, \Psi_{L_2}, \Psi_{L_3}$ have respective exponents

$$(248) \quad T, \quad J_1 = 2T - 2a, \quad J_2 = J_3 = T.$$

Additive duality gives (247), since $2(e + 1) \geq T$.

Lemma 12.4 (Coefficients for ω_i). *With $q_1 = a$, $q_2 = q_3 = e + 1$, one has*

$$(249) \quad \omega_i(1 - z) = \Psi_F(\beta_i z) \quad (z \in \mathfrak{p}_F^{q_i}).$$

Each β_i belongs to $n_i(L_i^\times)$, and

$$(250) \quad \Psi_{L_i}(\beta_i x) = \Psi_F(\beta_i n_i x) \quad (x \in \mathfrak{p}_{L_i}^{q_i}).$$

Proof. For $z \in \mathfrak{p}_F^a$ put $C = 1 + \epsilon z$. The Steinberg identity and (242) give

$$(1 + \epsilon, 1 - z) = (1 + \epsilon, -\epsilon(1 - z)) = (C, \epsilon(1 - z)(1 + \epsilon)).$$

Here $v_F(C - 1) \geq b + a$, and $(b + a) + a = 2e + 1$. Lemma 12.2 removes $1 - z$ from the second argument and replaces ϵ by R^2 , since $\epsilon/R^2 \in U_F^a$. By symmetry the result is $(R^2(1 + \epsilon), C) = \omega_3(C)$. Also $b + a = 2e - a + 1 \geq e + 1$, so (247) gives $\omega_1(1 - z) = \Psi_F(-\epsilon z)$ on the *entire* ideal. On U_F^{e+1} the product $\omega_2 = \omega_1 \omega_3$ therefore has coefficient $1 - \epsilon$. Replacing it by $1 - R^2$ costs valuation at least $b + a + e + 1 = 3e - a + 2 \geq T$. This proves (249); the third formula is (247). The norm assertions are (245). Finally for $x \in \mathfrak{p}_{L_i}^{q_i}$ both its trace and norm lie in $\mathfrak{p}_F^{q_i}$ by (240). Evaluate (249) at $n_i(1 - x) = 1 - \text{tr}_i x + n_i x$ and use triviality on norms. This proves (250). $\square$

12.4. Characters of conductors m_i and twists by one character of $F^\times$. Put

$$(251) \quad m_1 = 4e + 1, \quad m_2 = m_3 = 2e + 2a, \quad s_1 = 2e + 1, \quad s_2 = s_3 = e + a.$$

Thus $m_i = J_i + t_1$ and $s_i = \lceil m_i/2 \rceil$. The formulas

$$(252) \quad R_i(1 - z) = \Psi_{L_i}(a_i z), \quad z \in \mathfrak{p}_{L_i}^{s_i},$$

define characters of their indicated unit groups with exact conductor m_i : $2s_i - t_1 \geq J_i$, and the induced character on $U_{L_i}^{m_i-1}/U_{L_i}^{m_i}$ is nontrivial.

Lemma 12.5 (Compatibility before extension). *For $z \in \mathfrak{p}_K^{2e+1}$, all $N_i(1 - z)$ lie in the domains of R_i , and $R_i(N_i(1 - z))$ is independent of i .*

Proof. The highest-degree terms in the norm expansions have valuation at least $2e + 1$ on the target field. The upper traces have depths at least $3e - a + 1$ and $e + a$, hence are in the required ideals. Use the exact identity

$$(253) \quad E_2(W) = \text{tr}_i N_i W + n_i S_i W$$

for the six pair-products of the four conjugates of W . Apply it to $Y(1 - z)$ and Y . The quantities $w_i = S_i(Yz)/h_i$ have valuation at least $e + 1$ on all three fields: the first upper trace bound is $3e - 2a + 2$, from which one subtracts b , and the other two are $e + 1$. Thus $n_i S_i(Y(1 - z))/\beta_i = n_i(1 - w_i)$ is a norm in the domain of (249). Its additive phase implies

$$\Psi_F(n_i S_i(Y(1 - z)) - \beta_i) = 1.$$

The remaining difference in (253) identifies $R_i(N_i(1 - z)) = \Psi_F(-E_2(Y(1 - z)) + E_2(Y))$, independent of i . $\square$

Theorem 12.6 (Compatible characters with restrictions R_i). *There are primitive compatible characters χ_i on $L_i^\times$ of conductors m_i , such that $\chi_i|_{U_{L_i}^{s_i}} = R_i$.*

Proof. First extend R_2 across the abelian group $L_2^\times/U_{L_2}^{m_2}$; extend on the finite unit subgroup and choose a uniformizer value. Call the character χ_2 . Its lower conjugate quotient is determined by R_2 on every element of $L_2^\times$, because every ratio $\sigma(v)/v$ has depth at least $t_2 = 2e \geq s_2$. For $v = N_2 Z$, this ratio is $N_2(g_1 Z/Z)$. The element $g_1 Z/Z$ has depth at least $4e - 2a + 1 \geq 2e + 1$ and its N_1 -norm is 1. Lemma 12.5 therefore proves that the conjugate quotient of χ_2 is trivial on $N_2 K^\times$. It is nontrivial: on $U_{L_2}^{t_1}$ the first nonzero lower ramification term maps onto $U_{L_2}^{t_1+t_2}/U_{L_2}^{t_1+t_2+1}$, since t_1 is odd. The restriction of R_2 to this quotient is nontrivial. Hence $\chi_2^\sigma/\chi_2 = \nu_2$.

Consequently $\chi_2 N_2$ is invariant under all of G ; Hilbert 90 makes it trivial on $\ker N_1$ and $\ker N_3$. The characters defined by $N_i x \mapsto \chi_2(N_2 x)$ on $N_i(K^\times)$ extend to $\tilde{\chi}_1$ on $L_1^\times$ and χ_3 on $L_3^\times$ by Lemma C.2. Norm filtration gives $N_3(U_K^{2e+1}) = U_{L_3}^{e+a}$, since $2(e + a) - t_1 = 2e + 1$. Lemma 12.5 gives $\chi_3|_{U_{L_3}^{s_3}} = R_3$. For N_1 , every preimage of $U_{L_1}^{s_1}$ under N_1 lies in $U_K^{s_1}$: the norm has nonzero graded map at every smaller positive depth, all below the upper break $4e - 2a + 1$, and its residue map is Frobenius. Norm valuation excludes nonunits. Hence $\tilde{\chi}_1$ agrees with R_1 on $U_{L_1}^{s_1} \cap N_1 K^\times$. Their quotient extends across the order-two norm quotient; divide by this norm character to obtain χ_1 with the restriction R_1 and the same pullback. The extensions are trivial on $U_{L_i}^{m_i}$ and nontrivial on $U_{L_i}^{m_i-1}$, by (252) and norm filtration, so their conductors are exactly m_i . If the common pullback were $\lambda N_{K/F}$, then $\chi_2/(\lambda n_2)$ would belong to $S(K/L_2)$. Both factors would be $\text{Gal}(L_2/F)$ -invariant, contrary to $\chi_2^\sigma/\chi_2 = \nu_2 \neq 1$. $\square$

Lemma 12.7 (Writing θ_i as $\chi_i(\lambda n_i)$). *For every primitive compatible family,*

$$\theta_i^\sigma/\theta_i = \nu_i = \omega_j n_i \quad (j \neq i), \quad D_\theta = \theta_i|_{F^\times} \omega_i$$

The character D_θ is independent of i . For comparison of 1 and 2 we may, without changing either local constant or the common pullback, write

$$(254) \quad \theta_i = \chi_i(\lambda n_i) \quad (i = 1, 2)$$

with χ_i as in Theorem 12.6. If $n = a_F(\lambda)$, the conductors are m_i for $n \leq 2e + a$; for $n \geq 2e + a + 1$ they are

$$(255) \quad M_1 = 2n - 2a, \quad M_2 = 2n - T.$$

Proof. Lemma 4.2 gives the stated conjugate quotient and restriction D_θ , both for (θ_i) and for (χ_i) . The quotient θ_2/χ_2 is invariant and is therefore λn_2 . The quotient $\theta_1/(\chi_1(\lambda n_1))$ is either 1 or ν_1 . Conjugate θ_1 in the latter case. Its local constant is unchanged by a field-automorphism change of variables in the defining integral; its norm pullback is unchanged because $\nu_1 N_1 = 1$. This proves (254). Finally [7, Proposition 3.10] gives the alternatives for the norm-pullback conductors: when $n \leq 2e + a$, $a_{L_i}(\lambda n_i) < m_i$; when $n \geq 2e + a + 1$, $a_{L_i}(\lambda n_i) > m_i$. The two conductors cannot be equal: m_1 is odd and $2n - 2a$ is even, whereas m_2 is even and $2n - T$ is odd. $\square$

12.5. Changing α and choosing a norm from L_3 .

Lemma 12.8 (The replacement $\alpha \mapsto \alpha u$). *Replacing α by αu , with $u \in U_F^{2e}$, leaves (249) and (252) unchanged. For any $A_* \in F^\times$, such a replacement can be made so that $A = A_*/u$ is an exact norm from L_3/F . If $\lambda(1-z) = \Psi_F(A_*z)$ on an ideal, the same character formula holds with A_*/u relative to $\psi_F(\alpha u \cdot)$.*

Proof. For the lower formula the change has valuation at least $2e + v_F\beta_i + q_i \geq T$. For (252) the change has L_i -valuation at least $4e - t_1 + s_i \geq J_i$. Thus both formulas are unchanged on their stated ideals. The character ω_3 has conductor $2e + 1$ and is nontrivial on U_F^{2e} . Apply Lemma 4.10 to this subgroup. The norm assertion follows, and $(\alpha u)(A_*/u) = \alpha A_*$. Norm valuation fixes $v_{L_3}X = v_FA$. $\square$

We make this replacement when choosing $A = N_{L_3/F}X$ below.

For comparison of 1, 2 take $X \in L_3$, and put

$$A = n_3X, \quad p_X = \text{tr}_3X, \quad C = Y - X, \quad Q_i = S_i(Yg_iX) \quad (i = 1, 2).$$

Then, exactly,

$$(256) \quad \begin{aligned} N_iC &= a_i + A - Q_i, & S_iC &= h_i - p_X, \\ n_iQ_i &= \beta_iA + \mathcal{E}, & \mathcal{E} &= (X' - X)(a_3X' - a'_3X) \in F, \end{aligned}$$

$$(257) \quad n_iS_iC = \beta_i + \Delta, \quad \Delta = p_X^2 + 2p_X.$$

Primes in the middle line denote the automorphism of L_3/F . These identities follow by multiplying the four conjugates, with X fixed by g_3 ; the last uses $\text{tr}_i h_i = \text{Tr}_K Y = -2$.

Lemma 12.9 (Valuations of p_X , $\mathcal{E}$, and Q_i). *If $v_{L_3}X = -h$, for any integer h , then*

$$(258) \quad \begin{aligned} v_F p_X &\geq e - \lfloor h/2 \rfloor, & v_F \mathcal{E} &\geq T - a - h, \\ v_{L_1} Q_1 &\geq b - h, & v_{L_2} Q_2 &\geq -h. \end{aligned}$$

Proof. Let $r_3 = SR/\varpi^{e-a}$, a trace-zero uniformizer of L_3 . Write $X = P + Qr_3$; parity of the two valuations gives $v_FP \geq -\lfloor h/2 \rfloor$ and $v_FQ \geq -\lceil h/2 \rceil$. The trace is $2P$, proving its bound. Write $a_3 = A_3 + B_3r_3$, with $v_FA_3 \geq 1 - a$ and $v_FB_3 = -a$ by (245). Now $X' - X = -2Qr_3$ and $a_3X' - a'_3X = 2(B_3P - A_3Q)r_3$. Their product has F -valuation at least $2e + 1 - a - h = T - a - h$. The upper trace formula applied to Yg_iX , of K -valuation $-t_1 - 2h$, gives respectively $b - h$ and $-h$ for Q_i . $\square$

The conversion (250) and (256) imply, whenever $(Q_i/\beta_i)z \in \mathfrak{p}_{L_i}^{q_i}$,

$$(259) \quad \Psi_{L_i}(Q_i z) = \Psi_F((A + \mathcal{E}/\beta_i)n_i z).$$

12.6. The case $n \leq 2e + a$. Suppose $n \leq 2e + a$. Both trace and norm of a variable in $\mathfrak{p}_{L_i}^{s_i}$ lie in $\mathfrak{p}_F^{e+a}$, for $i = 1, 2$. If $n \leq e + a$, the twist is trivial there, and use $X = 0$, $C = Y$. Otherwise put $h = n - T$, so

$$(260) \quad a - e \leq h \leq a - 1.$$

Additive duality on U_F^{e+a} supplies a nonzero base coefficient A_* of valuation $-h$, since $2(e+a) \geq n$. Use Lemma 12.8 to make its new value $A = n_3X$ an exact norm.

Lemma 12.10 (The coefficients N_iC and n_iS_iC). *The resulting C has $v_KC = -t_1$. For $i = 1, 2$,*

$$(261) \quad \begin{aligned} \theta_i(1-z) &= \Psi_{L_i}(N_iCz) & (z \in \mathfrak{p}_{L_i}^{s_i}), \\ \omega_i(1-z) &= \Psi_F(n_iS_iCz) & (z \in \mathfrak{p}_F^{q_i}). \end{aligned}$$

Both S_iC are nonzero and have the same valuations as h_i .

Proof. For $X \neq 0$, $-2h > -t_1$, so Y dominates in C . The two arguments of (259) have depths at least $2a - h \geq a + 1$ and $e + a - h \geq e + 1$. The error terms $(\mathcal{E}/\beta_i)n_iz$ have depths at least $T + a - h$ and $T + e - h$, both at least T . The norm expansion therefore gives $(\lambda n_i)(1-z) = \Psi_{L_i}((A - Q_i)z)$

for $z \in \mathfrak{p}_{L_i}^{s_i}$. Multiplying by (252) proves the first formula. For the second, (258) and $h \leq a - 1$ give

$$v_F \Delta \geq \min(2e - 2 \lfloor h/2 \rfloor, 2e - \lfloor h/2 \rfloor).$$

Adding a to either bound gives at least T ; adding $e + 1$ does also. Thus replacing β_i by $n_i S_i C$ preserves (249). Moreover $v_{L_1} p_X \geq 2e - 2 \lfloor h/2 \rfloor > b$, while p_X has positive valuation on L_2 . The traces retain their nonzero leading terms. $\square$

Apply Lemma 4.1, with $c_0 = -\alpha^{-1}$ and residual factors denoted g_θ .

Theorem 12.11 (Equality for $n \leq 2e + a$). *Every family with $n \leq 2e + a$ satisfies $\mathcal{A}_1 = \mathcal{A}_2$.*

Proof. Use the coefficients of Lemma 12.10 in (14). Their norm-character values are 1, and compatibility cancels the two upper multiplicative values. Formula (253) gives

$$(262) \quad \mathcal{A}_i = D_\theta(c_0) \Theta(C)^{-1} \Psi_F(-E_2(C)) g_{\theta_i} g_{\omega_i}.$$

Only θ_1 and ω_2 have odd conductors, with critical depths $2e$ and e respectively. Choose a uniformizer π of K and put $C_z = C(1 + \pi^{2e} z)$ for integral $z \in K$. For $N_1(C_z)/N_1(C)$, the norm term has depth $2e$; its trace has depth at least $3e - a + 1 \geq 2e + 1$. Its residue map is a nonzero scalar times Frobenius, hence bijective. For $N_2(C_z)/N_2(C)$, both terms lie in $\mathfrak{p}_{L_2}^{e+a}$: its trace has depth $e + a$, and its highest-degree norm term has depth $2e \geq e + a$.

The trace change $S_1(C\pi^{2e} z)$ has L_1 -valuation at least $3e - 2a + 1$; after division by $S_1 C$ its depth is at least $e \geq a$. Its lower norm quotient is therefore on the even stationary ideal of ω_1 . For S_2 , the trace of an element of K -valuation $b = 2e - 2a + 1$ induces an isomorphism from its residue line to the L_2 quotient $\mathfrak{p}_{L_2}^e / \mathfrak{p}_{L_2}^{e+1}$: the consecutive trace-ideal exponents are e and $e + 1$. Division by the unit $S_2 C$ preserves this isomorphism. For the lower maximal norm, the highest-degree term has depth e and the trace has depth at least $\lfloor (3e + 1)/2 \rfloor \geq e + 1$. Thus its residue map is again bijective.

Subtract (253) for C_z, C . The multiplicative character values cancel because $\theta_i N_i = \Theta$. The arguments of ω_i are ratios of norms, so their values are 1. The critical functions in (15) therefore satisfy

$$H_{\theta_1}(p(z)) = H_{\omega_2}(q(z)) \quad (z \in k),$$

where both $p, q : k \rightarrow k$ are bijective maps just constructed. All even expressions are 1; all arguments of multiplicative characters are nonzero, since the relative changes have positive depth. Summing gives $g_{\theta_1} = g_{\omega_2}$. Equation (262) proves $\mathcal{A}_1 = \mathcal{A}_2$. $\square$

12.7. The case $n \geq 2e + a + 1$. Now $n \geq 2e + a + 1$, and put $h = n - T \geq a$. The upper stationary depths are

$$(263) \quad b_1 = T + h - a, \quad b_2 = e + 1 + h,$$

with $M_1 = 2b_1$ even and $M_2 = 2b_2 - 1$ odd. Choose A_* such that $\lambda(1 - z) = \Psi_F(A_* z)$ for $z \in \mathfrak{p}_F^{\lfloor n/2 \rfloor}$. Both trace and norm of $z \in \mathfrak{p}_{L_i}^{b_i}$ lie in that base ideal, by (240). Its valuation is $-h$. By Lemma 12.8, we may replace α by αu and A_* by $A = A_*/u$ so that $A = n_3 X$, with $v_{L_3} X = -h$.

Lemma 12.12. *For $i = 1, 2$ and $z \in \mathfrak{p}_{L_i}^{b_i}$, $\theta_i(1 - z) = \Psi_{L_i}(N_i C z)$. Moreover $v_{L_i} N_i C = -2h$.*

Proof. The arguments $(Q_i/\beta_i)z$ of (259) have respective depths at least a and $e + 1$. The weighted errors there have depths

$$(T - a - h) - b + b_1 = T, \quad (T - a - h) + b_2 = 3e + 2 - a \geq T.$$

Consequently exact quadratic norm expansion yields the pulled-back coefficient $A - Q_i$ on the whole indicated ideals. These ideals are contained in $\mathfrak{p}_{L_i}^{s_i}$, so (252) adds the coefficient a_i , giving $N_i C$. The element X dominates Y because $-2h < -t_1$; hence $v_K C = -2h$ and $v_{L_i} N_i C = -2h$, as required by (255). $\square$

Use the coefficients $N_i C$ from Lemma 12.12 for θ_i , and $\beta_i = n_i h_i$ from Lemma 12.4 for ω_i . Equations (253), (257), and (14) give

$$(264) \quad \mathcal{A}_i = D_\theta(c_0) \Theta(C)^{-1} \Psi_F(-E_2(C) + \Delta) g_{\theta_i} g_{\omega_i}.$$

The factor preceding $g_{\theta_i} g_{\omega_i}$ is independent of i . The conductors of θ_1, ω_1 are even, so $g_{\theta_1} = g_{\omega_1} = 1$. The conductors of θ_2, ω_2 are odd.

Theorem 12.13 (Equality for $n \geq 2e + a + 1$). *For every $h \geq a$, one has $\mathcal{A}_1 = \mathcal{A}_2$.*

Proof. Put

$$V = Y \pi^{2e} z, \quad C_z = C + V, \quad Y_z = Y + V \quad (z \in \mathcal{O}_K).$$

We evaluate θ_i at $N_i C_z / N_i C$ and ω_i at $n_i S_i Y_z / \beta_i$. The coefficients for these two characters are $N_i C$ and $\beta_i = n_i h_i$, respectively. The element V has K -valuation at least b and V/C has depth $b + 2h$. For S_1 , the trace has L_1 -valuation at least $3e - 2a + 1 + h \geq b_1$, and its highest-degree norm term has depth $b + 2h \geq b_1$. Thus the upper residual expression is 1 on its even stationary ideal. For S_2 , identify z with the class of V in $\mathfrak{p}_K^b / \mathfrak{p}_K^{b+1}$. Write $\text{gr } N_2$ for the map induced by $x \mapsto N_2(1+x) - 1$, and $\text{gr } n_2$ for the map induced by $x \mapsto n_2(1+x) - 1$. The residue map p is the composition

$$(265) \quad \frac{\mathfrak{p}_K^b}{\mathfrak{p}_K^{b+1}} \xrightarrow{/C} \frac{\mathfrak{p}_K^{b+2h}}{\mathfrak{p}_K^{b+2h+1}} \xrightarrow{\text{gr } N_2} \frac{\mathfrak{p}_{L_2}^{e+h}}{\mathfrak{p}_{L_2}^{e+h+1}}.$$

The first map is an isomorphism because $v_K C = -2h$. For the second, the break of K/L_2 is $2a - 1$ and

$$e + h > 2a - 1, \quad b + 2h = 2(e + h) - (2a - 1).$$

Thus [7, Theorem 3.7] proves that this map, and hence p , is bijective.

For the arguments of ω_i , $S_1 V / h_1$ has depth at least $e \geq a$, so its norm quotient is in the even stationary ideal of ω_1 . For $i = 2$, the residue map q is the composition

$$(266) \quad \frac{\mathfrak{p}_K^b}{\mathfrak{p}_K^{b+1}} \xrightarrow{S_2} \frac{\mathfrak{p}_{L_2}^e}{\mathfrak{p}_{L_2}^{e+1}} \xrightarrow{/h_2} \frac{\mathfrak{p}_{L_2}^e}{\mathfrak{p}_{L_2}^{e+1}} \xrightarrow{\text{gr } n_2} \frac{\mathfrak{p}_F^e}{\mathfrak{p}_F^{e+1}}.$$

By (240), the trace ideals at source depths $b, b + 1$ have depths $e, e + 1$. The induced trace map is therefore a surjection between one-dimensional k -spaces, hence an isomorphism. Division by h_2 is an isomorphism because h_2 is a unit. The last map is bijective by [7, Theorem 3.7], since L_2/F has break $2e$ and $e < 2e$. Thus q is bijective. All traces in these lower character arguments stay nonzero.

Since $S_i C = h_i - p_X$, quadratic norm expansion gives

$$[n_i S_i C_z - n_i S_i C] - [n_i S_i Y_z - \beta_i] = -p_X \text{Tr}_{K/F} V.$$

Using (253), we obtain

$$(267) \quad \text{tr}_i(N_i C_z - N_i C) + (n_i S_i Y_z - \beta_i) = E_2(C_z) - E_2(C) + p_X \text{Tr}_{K/F} V,$$

independent of i . Upper multiplicative values are the common $\Theta(C_z/C)$ and lower ones are 1, because their arguments are norms. Thus (267) and (15) give

$$H_{\theta_2}(p(z)) H_{\omega_2}(q(z)) = 1 \quad (z \in k).$$

Both residue maps are additive bijections. Therefore $\sum H_{\theta_2} = \overline{\sum H_{\omega_2}}$, after reindexing. Each normalized sum has absolute value 1 by (17). Consequently $g_{\theta_2} g_{\omega_2} = 1$. Equation (264) proves the assertion for every $h \geq a$. $\square$

Remark 12.14. The Lean formalization suggested the compositions (265) and (266) in Theorem 12.13. The original proof expanded the trace and quadratic norm terms of both maps and checked directly that their leading terms were nonzero. Factoring the same maps through the graded trace and norm isomorphisms avoids repeating those leading-term calculations.

12.8. Completion of the maximal-break case.

Proof of Theorem 12.1. Construct Y by (244) and apply Lemma 12.4 and Theorem 12.6 to obtain β_i and χ_i . Lemma 12.7 writes the given characters as in (254), after conjugating θ_1 if necessary. This does not change its norm pullback or local constant. Every base conductor is either $n \leq 2e + a$, covered by Theorem 12.11, or $n \geq 2e + a + 1$, covered by Theorem 12.13. Their endpoints are consecutive. Repeat the construction with the two maximal fields interchanged to compare L_1 with L_3 . This proves independence of all three expressions for the original given family. $\square$

13. TOTALLY RAMIFIED MIXED CHARACTERISTIC: NONMAXIMAL BREAKS

13.1. Statement and ramification data. Let $F/\mathbf{Q}_2$ be finite, with residue field k , uniformizer ϖ , and $e = v_F(2)$. Normalize all valuations on their named fields. Let K/F be totally ramified with group C_2^2 . Label its lower quadratic fields $L_i = K^{\langle g_i \rangle}$ so that

$$(268) \quad t_1 = 2a - 1, \quad t_2 = t_3 = 2r - 1, \quad 1 \leq a \leq r \leq e.$$

Put $\delta = r - a$, $T = 2r$, and $T_1 = 2a$, $T_2 = T_3 = T$. These T_i are the lower different exponents and the conductors of the quadratic norm characters ω_i . Write

$$N_i = N_{K/L_i}, \quad S_i = \text{Tr}_{K/L_i}, \quad n_i = N_{L_i/F}, \quad \text{tr}_i = \text{Tr}_{L_i/F}.$$

For an arbitrary nontrivial additive character ψ_F , use $\psi_{L_i} = \psi_F \text{tr}_i$. Set $n_F = n_F(\psi_F)$, so $\mathfrak{p}_F^{-n_F}$ is its largest trivial ideal.

Theorem 13.1. *Suppose $\theta_i N_i = \Theta$ is a primitive compatible family: Θ is invariant under $\text{Gal}(K/F)$ and does not descend by $N_{K/F}$. Then the three expressions*

$$(269) \quad \mathcal{A}_i(\theta_i) = \Delta_{L_i}(\theta_i, \psi_{L_i}) \Delta_F(\omega_i, \psi_F)$$

are equal. This holds for every continuous quasi-character conductor under (268).

A compatible pair extends to a compatible family as explained in Section 1. The factor of the trivial character in $S(L_i/F)$ is 1.

We use the results of Sections 2–4. For a ramified quadratic extension E/D of different exponent d , (7) gives

$$(270) \quad \text{Tr}_{E/D}(\mathfrak{p}_E^j) = \mathfrak{p}_D^{\lfloor (j+d)/2 \rfloor}, \quad v_D(Nx) = v_E x, \quad n_E(\psi_E) = 2n_D(\psi_D) + d.$$

A base character of conductor $n > d$ pulls back with conductor $2n - d$; for $n \leq d$ its pullback has conductor at most d . The descent and order-discriminant facts are proved in Appendix C, and the required crossed norm-character and ramification facts in Appendix D. In particular Lemma D.3 gives the upper quadratic different exponents

$$(271) \quad D'_1 = 4r - 2a, \quad D'_2 = D'_3 = 2a.$$

13.2. A choice of quadratic generators. Choose generators $x \in L_1$, $y \in L_2$ with

$$(272) \quad x^2 + x = f, \quad y^2 + y = g, \quad v_F f = 1 - 2a, \quad v_F g = 1 - 2r.$$

Here is a construction. If a uniformizer of a quadratic field has minimal polynomial $X^2 - cX + b$, its different is the valuation of $2\Pi - c$. The two possible valuations $2e + 1$ and $2v_F c$ have different parities. Different $2a < 2e + 1$ therefore forces $v_F c = a$. Then $x = -\Pi/c$ satisfies (272); use the same argument for y . All conjugations are $x \mapsto -1 - x$, $y \mapsto -1 - y$.

Lemma 13.2. *The generator x may be changed, without changing its field or the valuations in (272), and $d \in F$ may be chosen so that*

$$(273) \quad v_F d = \delta, \quad E := f + d^2 g \in \mathfrak{p}_F^{1-a}.$$

Moreover $k_0 := 1 + d$ is a unit, including when $a = r$.

Proof. Start with x_0, f_0 and put $S_0 = 1 + 2x_0$, $U_0 = S_0^2 = 1 + 4f_0$. Let $b = 2e - 2a + 1$ and $q = \varpi^b$. Replace

$$S_0 \text{ by } S = S_0(1 + \varpi^{e-a+1}w), \quad x = (S - 1)/2, \quad d = \varpi^{r-a}v.$$

The two base variables v, w will be integral, with v a unit. Modulo $\mathfrak{p}_F^a$,

$$\frac{S^2 - 1 + 4d^2g}{q} \equiv u_0 + \varpi U_0 w^2 + v_0 v^2, \quad u_0 = 4f_0/q, \quad v_0 = 4\varpi^{2r-2a}g/q.$$

The omitted linear term has valuation at least a ; u_0, v_0, U_0 are units. Since $a \leq e$, the ring $\mathcal{O}_F/\mathfrak{p}_F^a$ has characteristic two and is $k[\varpi]/(\varpi^a)$. Its coefficient field can be constructed as the unique lifts of the roots of $Z^{|k|} - Z$, using its unit derivative. Solve $v_0 v^2 + \varpi U_0 w^2 = u_0$ successively in powers of ϖ . At an even degree the next coefficient of v^2 is determined, and at an odd degree the next coefficient of w^2 is determined; earlier contributions are known and the leading coefficients are units. Frobenius on k supplies the required residue square roots. The constant coefficient of v is nonzero. Lifting v, w to F proves $v_F E \geq b + a - 2e = 1 - a$. The perturbation of $4f_0$ is strictly deeper than b , so $v_F f = 1 - 2a$ is unchanged.

If $r > a$, then $d \in \mathfrak{p}_F$. Suppose $r = a$ and $\bar{d} = 1$. Then $f + g$ has valuation at least $2 - 2a$, and so does $f + g + 4fg$, since $e \geq a$. The generator $z = x + y + 2xy \in L_3$ satisfies $z^2 + z = f + g + 4fg$. The integral element $\varpi^{a-1}z$ generates an order whose polynomial discriminant has valuation $2a - 2$. The field different is at most this value, contradicting $T_3 = 2a$. Thus $1 + d$ is a unit. $\square$

Put

$$(274) \quad z = x + y + 2xy \in L_3, \quad Y = x + dy, \quad a_i = N_i Y, \quad h_i = S_i Y, \quad \beta_i = n_i h_i.$$

Direct calculation gives

$$(275) \quad z^2 + z = f + g + 4fg,$$

$$(276) \quad h_1 = 2x - d, \quad h_2 = 2dy - 1, \quad h_3 = -k_0,$$

$$(277) \quad \beta_1 = d^2 + 2d - 4f, \quad \beta_2 = 1 + 2d - 4d^2g, \quad \beta_3 = k_0^2,$$

$$(278) \quad a_1 = 2f - E - k_0x, \quad a_2 = E - 2f - dk_0y, \quad a_3 = -E - dz.$$

The right side of (275) has valuation $1 - 2r$. If $r > a$, g is its unique leading term; if $r = a$, Lemma 13.2 excludes cancellation of the leading terms of $f + g$. The term $4fg$ is strictly deeper in both cases. Thus $v_{L_3}z = 1 - 2r$. The constant terms in (278) have base valuation at least $1 - a$, whereas its nonconstant terms have L_i -valuation $1 - 2a$. Hence

$$(279) \quad v_K Y = v_{L_i} a_i = -t_1, \quad v_{L_1} h_1 = 2\delta, \quad v_{L_2} h_2 = v_{L_3} h_3 = 0, \quad \text{Tr}_{K/F} Y = -2k_0.$$

For h_1 , the depth of $2x$ is $2e - 2a + 1 > 2\delta$; the other traces are units. In particular all trace norms in (274) are nonzero, $v_F \beta_1 = 2\delta$, and β_2, β_3 are units.

13.3. The lower stationary coefficients. Use the quadratic norm symbol, the conic identity (242), and the unit-depth estimate of Lemma 12.2. Their hypotheses depend only on F and $e = v_F(2)$, not on maximality of the breaks.

Choose coefficients b_1, b_2 for ω_1, ω_2 such that

$$\omega_1(1 - u) = \psi_F(b_1 u) \quad (u \in \mathfrak{p}_F^a), \quad \omega_2(1 - u) = \psi_F(b_2 u) \quad (u \in \mathfrak{p}_F^r).$$

They exist by additive duality, with valuations $-n_F - 2a$ and $-n_F - 2r$. The product character ω_3 has coefficient $b_1 + b_2$ on $\mathfrak{p}_F^r$. Define

$$(280) \quad \alpha = b_2/\beta_2, \quad \Psi_L(u) = \psi_L(\alpha u), \quad v_F \alpha = -n_F - T.$$

The largest trivial ideals for $\Psi_F, \Psi_{L_1}, \Psi_{L_2}, \Psi_{L_3}$ have exponents

$$(281) \quad T, \quad J_1 = 4r - 2a, \quad J_2 = J_3 = T.$$

Lemma 13.3 (Coefficients for the three norm characters). *With $q_1 = a$, $q_2 = q_3 = r$, one has*

$$(282) \quad \omega_i(1 - u) = \Psi_F(\beta_i u) \quad (u \in \mathfrak{p}_F^{q_i}).$$

Moreover $\omega_i(\beta_i) = 1$, and

$$(283) \quad \Psi_{L_i}(\beta_i v) = \Psi_F(\beta_i n_i v) \quad (v \in \mathfrak{p}_{L_i}^{q_i}).$$

Proof. Write $U = 1 + 4f$, $V = 1 + 4g$. For $u \in \mathfrak{p}_F^a$, put $C = 1 + 4fu$ and $w = (f/g)u$. Then

$$v_F(C - 1) \geq 2e - a + 1, \quad v_F w \geq 2r - a \geq r.$$

Multiplying the second symbol argument by the norm $-4f$ and using (242) gives

$$\omega_1(1 - u) = (C, f(1 - u)U).$$

The ratio $f/(-d^2g)$ lies in U_F^a , by (273). Lemma 12.2 removes this ratio and $1 - u$ from the second argument. The result is $(C, -gU)$. Applying the same transformation with g, w gives $\omega_2(1 - w) = (C, g(1 - w)V) = (C, gV)$. The factor -1 in the product is also killed by Lemma 12.2, since $v_F(-2) = e$. Consequently, for $u \in \mathfrak{p}_F^a$,

$$(284) \quad \omega_1(1 - u) = \omega_2(1 - (f/g)u) \omega_3(1 + 4fu).$$

All uses of the lemma have depth sum greater than $2e$; in particular $(2e - a + 1) + a = 2e + 1$. The last argument has depth at least $2e - a + 1 \geq r$, so the formulas for $b_1, b_2, b_1 + b_2$ apply. Additive duality on $\mathfrak{p}_F^a$ applied to (284) gives

$$(285) \quad b_1 - b_2 R \in \mathfrak{p}_F^{-n_F - a}, \quad R = \frac{f(1 - 4g)}{g(1 + 4f)}.$$

Here multiplication of the variable u by the unit $1 + 4f$ preserves the ideal $\mathfrak{p}_F^a$.

We check that the explicit norm coefficients have exactly this ratio. Direct expansion gives

$$(286) \quad f(1 - 4g)\beta_2 - g(1 + 4f)\beta_1 = E(1 + 2d + 16fg) - 8fgd(d + 2) - 2gd(d^2 + d + 1).$$

Every term has valuation at least $1 - a$. For the middle term, $v_F(d + 2) = \delta < e$, so its valuation is at least $3e + 2 - 4a \geq 1 - a$; for the last it is at least $e - r - a + 1 \geq 1 - a$. The first follows from (273) and $v_F(16fg) \geq 2$. Division by $g(1 + 4f)\beta_2$, of valuation $1 - 2r$, gives $R - \beta_1/\beta_2 \in \mathfrak{p}_F^{2r - a}$. Together with (285) this proves the first lower formula. The second is (280). Finally

$$\beta_3 - \beta_1 - \beta_2 = -2d + 4E \in \mathfrak{p}_F^r,$$

which proves the third formula on $\mathfrak{p}_F^r$. By (274), $\beta_i = n_i h_i$, so their character values are one. For $v \in \mathfrak{p}_{L_i}^{q_i}$ both its trace and norm lie in $\mathfrak{p}_F^{q_i}$. Evaluate (282) at the exact norm $n_i(1 - v) = 1 - \text{tr}_i v + n_i v$. Its character is one, which proves (283). $\square$

13.4. Construction of the minimal characters. Put

$$(287) \quad m_1 = 4r - 1, \quad m_2 = m_3 = 2a + 2r - 1, \quad s_1 = 2r, \quad s_2 = s_3 = a + r.$$

Then $m_i = J_i + t_1 = 2s_i - 1$. The expressions

$$(288) \quad R_i(1 - v) = \Psi_{L_i}(a_i v) \quad (v \in \mathfrak{p}_{L_i}^{s_i})$$

define characters with exact conductor m_i on the indicated groups: $2s_i - t_1 \geq J_i$, and the induced character on $U_{L_i}^{m_i - 1}/U_{L_i}^{m_i}$ is nontrivial.

Lemma 13.4 (The value of R_1 on $\sigma(v)/v$). *Let σ be the nontrivial automorphism of L_1/F and put $\nu_1 = \omega_2 n_1$, the nontrivial character in $S(K/L_1)$. For every $v \in L_1^\times$ with $\sigma(v)/v \in U_{L_1}^{2r}$,*

$$(289) \quad R_1(\sigma(v)/v) = \nu_1(v).$$

Proof. Multiplication of v by an element of $F^\times$ changes neither side. Odd valuation would give conjugate quotient of exact depth $t_1 < 2r$, by the first ramification term, so it is excluded. Otherwise scale to a unit and write $v = P + Qx$. Parity of the two summand valuations makes P a unit and $v_F Q \geq a$. The element $1 + 2x$ is a unit of L_1 . Thus the assumed conjugate-quotient depth forces $w = Q/P \in \mathfrak{p}_F^r$. Put

$$V_0 = 1 - w - fw^2 = n_1(1 + wx), \quad Q_0 = 1 - dw - d^2gw^2 = n_2(1 + dwy), \quad H_0 = 1 + d^2gw^2.$$

All three are units. Write $c = f - d^2g = E - 2d^2g$, so $v_F c \geq 1 - a$. The difference $Q_0 - V_0 = (1 - d)w + cw^2$ lies in $\mathfrak{p}_F^r$. Since Q_0 is a norm from L_2 , (282) gives

$$(290) \quad \nu_1(v) = \omega_2(V_0) = \Psi_F(\beta_2(Q_0 - V_0)/Q_0).$$

On the other hand $\sigma(v)/v = 1 - w(1 + 2x)/(1 + wx)$. Using $a_1 = f - d^2g - k_0x$, an exact quadratic trace calculation yields

$$(291) \quad R_1(\sigma(v)/v) = \Psi_F(B_0/V_0), \quad B_0 = -(1 + 4f)(cw^2 + k_0w).$$

We compare the two arguments of Ψ_F modulo $\mathfrak{p}_F^{2r}$. Both numerators in (290)–(291) lie in $\mathfrak{p}_F^r$. Also

$$V_0 - H_0 = -w - Ew^2 \in \mathfrak{p}_F^r, \quad Q_0 - H_0 = -dw - 2d^2gw^2 \in \mathfrak{p}_F^r.$$

Thus replacing either unit denominator by H_0 costs an element of $\mathfrak{p}_F^{2r}$, exactly the trivial ideal of Ψ_F . With $U = 1 + 4f$, the remaining numerator difference is

$$B_0 - \beta_2(Q_0 - V_0) = -(U + \beta_2)cw^2 - (Uk_0 + \beta_2(1 - d))w.$$

The identities

$$U + \beta_2 = 2k_0 + 4c, \quad Uk_0 + \beta_2(1 - d) = 2(1 + d - d^2) + 4Ek_0 - 8d^2g$$

show that the first coefficient has valuation at least e , and the second at least $e \geq r$. The first summand therefore has valuation at least $2r + e + 1 - a \geq 2r$ and the second at least $2r$. Equations (290)–(291) now prove the assertion. $\square$

For any $Z \in K$ we will repeatedly use the exact identity

$$(292) \quad E_2(Z) = \text{tr}_i N_i Z + n_i S_i Z.$$

Both sides are the sum of the six pair-products of its four conjugates.

Lemma 13.5. *For $v \in \mathfrak{p}_K^{2r}$ all $N_i(1 - v)$ belong to the domains in (288), and $R_i(N_i(1 - v))$ is independent of i .*

Proof. The upper traces have depths at least $3r - a$ and $r + a$, and the highest-degree terms in the norm expansions have depth $2r$. These meet all three domain requirements. Apply (292) to $Y(1 - v)$ and Y . The upper trace changes $S_i(Yv)$ have depths at least $3r - 2a$ and r , respectively. After division by h_i , each relative change has depth at least r , which is at least q_i . Since $n_i(1 + u_i)$ is a norm, Lemma 13.3 makes the phase of $n_i S_i(Y(1 - v)) - \beta_i$ trivial. The other part of (292) is $\text{tr}_i(a_i(-S_i v + N_i v))$. It follows that

$$R_i(N_i(1 - v)) = \Psi_F(-E_2(Y(1 - v)) + E_2(Y)),$$

independent of i . $\square$

Theorem 13.6 (Extension of the characters R_i). *There exist characters χ_i of $L_i^\times$ with primitive common norm pullback and conductors m_i , such that $\chi_i|_{U_{L_i}^{s_i}} = R_i$.*

Proof. Let $I = \{\sigma(v)/v : v \in L_1^\times\}$ and prescribe $\kappa_I(\sigma(v)/v) = \nu_1(v)$. This is well defined since ν_1 is trivial on $F^\times$. Lemma 13.4 proves agreement with R_1 on $I \cap U_{L_1}^{s_1}$. Their product character factors through $IU_{L_1}^{s_1}/U_{L_1}^{m_1}$; extend it across $L_1^\times/U_{L_1}^{m_1}$, first on its finite unit subgroup and then on a uniformizer. The resulting χ_1 has exact conductor m_1 and satisfies $\chi_1^\sigma/\chi_1 = \nu_1$ globally.

Its norm pullback to K is invariant under both involutions, since $\nu_1 N_1 = 1$. By Lemma C.1, it is trivial on $\ker N_i$ for $i = 2, 3$. The character $N_i x \mapsto \chi_1(N_1 x)$ of $N_i(K^\times)$ extends to χ_i on $L_i^\times$ by Lemma C.2. Norm filtration gives

$$N_i(U_K^{2r+1}) = U_{L_i}^{a+r} \quad (i = 2, 3),$$

since K/L_i has break $2a - 1$ for $i = 2, 3$, and $2(a + r) - (2a - 1) = 2r + 1$. Lemma 13.5 gives $\chi_i|_{U_{L_i}^{s_i}} = R_i$. Their conductors are at least $m_i > D'_i$. The conductor of $\chi_1 N_1$ is $2m_1 - D'_1 = 4r + 2a - 2$. Equation (10) then forces the other conductors to be exactly m_i . If $\chi_1 N_1 = \lambda N_{K/F}$, then $\chi_1/(\lambda n_1)$ would belong to $S(K/L_1)$. Both factors would be σ -invariant, contrary to $\chi_1^\sigma/\chi_1 = \nu_1 \neq 1$. Thus the family is primitive. $\square$

13.5. Writing $\theta_i = \chi_i(\lambda n_i)$ and choosing a norm from L_3 .

Lemma 13.7. *For every primitive compatible family,*

$$\theta_i^\sigma/\theta_i = \nu_i = \omega_j n_i \quad (j \neq i), \quad D_\theta = \theta_i|_{F^\times} \omega_i$$

The character D_θ is independent of i . There are characters χ_i as in Theorem 13.6 and a character λ of $F^\times$ such that

$$(293) \quad \theta_i = \chi_i(\lambda n_i).$$

If $n = a_F(\lambda)$, their conductors are m_i for $n \leq 2r + a - 1$; for $n \geq 2r + a$ they are $M_i = 2n - T_i$, all even.

Proof. Lemma 4.2 gives the stated conjugate quotient and restriction D_θ , both for (θ_i) and for (χ_i) . The quotient θ_1/χ_1 is invariant, so Hilbert 90 and character extension give $\theta_1/\chi_1 = \lambda n_1$. For $i = 2, 3$, multiply χ_i by $\theta_i/(\chi_i(\lambda n_i)) \in S(K/L_i)$. These characters have conductor at most $2a \leq s_i = a + r$, so the restrictions R_i are unchanged. This proves (293) for the given family. The pullback of a base character with $n > T_i$ has conductor $2n - T_i$; otherwise it has conductor at most $T_i < m_i$. Since m_i is odd and $2n - T_i$ is even, equal-conductor cancellation does not occur. The common threshold is $n = 2r + a$, giving precisely the alternatives. $\square$

Lemma 13.8 (The replacement $\alpha \mapsto \alpha u$). *For $u \in U_F^{T-1}$, replacing α by αu preserves (282) and (288). For every $A_* \in F^\times$, one may choose such a u so that $A = A_*/u \in n_3(L_3^\times)$. Moreover $(\alpha u)(A_*/u) = \alpha A_*$, so this replacement preserves any character formula with coefficient αA_* .*

Proof. For $z \in \mathfrak{p}_F^{q_i}$ the change in the argument of Ψ_F is $(u - 1)\beta_i z$, with valuation at least $T - 1 + v_F \beta_i + q_i \geq T$. For $z \in \mathfrak{p}_{L_i}^{s_i}$ the change in the argument of Ψ_{L_i} is $(u - 1)a_i z$, with valuation at least $2(T - 1) - t_1 + s_i \geq J_i$. These inequalities include $a = r = 1$. Thus both formulas are preserved. The character ω_3 is onto $\{1, -1\}$ on U_F^{T-1} , so Lemma 4.10 applies and gives the norm assertion. The identity of coefficients is immediate. $\square$

13.6. The norms of $Y - X$, for $X \in L_3$. For the comparison of L_1 and L_2 , take $X \in L_3$ and put

$$(294) \quad A = n_3 X, \quad p_X = \text{tr}_3 X, \quad C = Y - X, \quad Q_i = S_i(Y g_i X) \quad (i = 1, 2).$$

For $X = 0$ all expressions below have their evident zero values. With primes denoting conjugation in L_3/F , direct multiplication gives

$$(295) \quad N_i C = a_i + A - Q_i, \quad S_i C = h_i - p_X,$$

$$(296) \quad n_i Q_i = \beta_i A + \mathcal{E}, \quad \mathcal{E} = (X' - X)(a_3 X' - a'_3 X) \in F,$$

$$(297) \quad n_i S_i C = \beta_i + \Delta, \quad \Delta = p_X^2 + 2k_0 p_X.$$

The last identity uses $\text{tr}_i h_i = -2k_0$, the same for both fields.

Lemma 13.9. *If $v_{L_3} X = -h$ for any integer h , then*

$$(298) \quad \begin{aligned} v_F p_X &\geq r - \lceil h/2 \rceil, & v_F \mathcal{E} &\geq T - a - h, \\ v_{L_1} Q_1 &\geq 2\delta - h, & v_{L_2} Q_2 &\geq -h. \end{aligned}$$

Proof. Write $X = P + Qz$, with z as in (275). Parity of valuations gives $v_F P \geq -\lfloor h/2 \rfloor$ and $v_F Q \geq r - \lfloor h/2 \rfloor$. The trace $2P - Q$ has the asserted bound because $e \geq r$. Using $a_3 = -E - dz$, the exact common error is

$$\mathcal{E} = Q(dP - EQ)(1 + 4(f + g + 4fg)).$$

Its last factor is a unit. The first product term has depth at least $r + \delta - h = T - a - h$; the second at least $1 - a + 2r - 2\lfloor h/2 \rfloor \geq T - a - h$. Finally $Yg_i X$ has K -valuation $-t_1 - 2h$. The upper trace formula (270), with (271), gives the two Q_i bounds exactly as stated. $\square$

For $(Q_i/\beta_i)v \in \mathfrak{p}_{L_i}^{q_i}$, (283) gives

$$(299) \quad \Psi_{L_i}(Q_i v) = \Psi_F((A + \mathcal{E}/\beta_i)n_i v).$$

Suppose $\lambda(1 - v) = \Psi_F(Au)$ on an ideal containing $\text{tr}_i v$ and $n_i v$. If $(\mathcal{E}/\beta_i)n_i v \in \mathfrak{p}_F^T$, then the norm expansion gives $(\lambda n_i)(1 - v) = \Psi_{L_i}((A - Q_i)v)$. If also $v \in \mathfrak{p}_{L_i}^{s_i}$, multiplication by (288) gives the coefficient $a_i + A - Q_i = N_i C$.

13.7. Stationary coefficients when $n \leq 2r + a - 1$. Assume $n \leq 2r + a - 1$ in (293). Both trace and norm of a variable in $\mathfrak{p}_{L_i}^{s_i}$ lie in $\mathfrak{p}_F^{q_F}$, where $q_F = a + r$. If $n \leq q_F$, take $X = 0$. Otherwise $h = n - T$ satisfies

$$(300) \quad 1 - \delta \leq h \leq a - 1.$$

Since $2q_F \geq n$, the base character has a linear formula $\lambda(1 - v) = \Psi_F(A_* v)$ on $\mathfrak{p}_F^{q_F}$, with $v_F A_* = -h$. Apply Lemma 13.8 and choose $A = n_3 X$ exactly, with $v_{L_3} X = -h$. Use the adjusted α henceforth.

Lemma 13.10. *The element C so constructed has $v_K C = -t_1$. For $i = 1, 2$,*

$$(301) \quad \theta_i(1 - v) = \Psi_{L_i}(N_i C v) \quad (v \in \mathfrak{p}_{L_i}^{s_i}),$$

$$(302) \quad \omega_i(1 - v) = \Psi_F(n_i S_i C v) \quad (v \in \mathfrak{p}_F^{q_i}).$$

The traces $S_i C$ are nonzero with the valuations of h_i . If $a = r$, this same C satisfies both formulas for all three indices.

Proof. For $X \neq 0$, the inequality $h \leq a - 1$ gives $-2h > -t_1$, so Y dominates in C . On the two upper ideals the arguments of (299) have respective depths at least $2a - h \geq a$ and $a + r - h \geq r$. Its weighted errors have depths at least $T + a - h$ and $3r - h$, respectively, both at least T . The trace and norm of v lie in $\mathfrak{p}_F^{q_F}$. Thus exact norm expansion and (295) prove (301).

By Lemma 13.9,

$$v_F \Delta \geq \min\{2r - 2\lfloor h/2 \rfloor, e + r - \lfloor h/2 \rfloor\}.$$

Adding a to both bounds gives at least T , since $h \leq a - 1$; adding r does also. Equation (297) therefore proves (302). Moreover $2r - 2\lfloor h/2 \rfloor > 2\delta$ and $r - \lfloor h/2 \rfloor > 0$, so the trace perturbations cannot cancel the leading terms of h_1, h_2 .

When $a = r$, one has $s_3 = T$, $q_3 = r$, and $h \leq r - 1$. For L_3 , equation (283) applied to $(X/k_0)v$ gives the pullback coefficient $A - k_0 X$, since $\beta_3 = k_0^2$. But

$$N_3 C = a_3 + X^2 + k_0 X, \quad (X^2 + k_0 X) - (A - k_0 X) = p_X X - 2A + 2k_0 X.$$

The three terms on the right have L_3 -valuations at least $2r - 2h - (h \bmod 2)$, $2e - 2h$, and $2e - h$, all nonnegative. Their products with $v \in \mathfrak{p}_{L_3}^T$ lie in the trivial ideal of Ψ_{L_3} , proving (301) for $i = 3$. Also $S_3 C = -k_0 - 2X$ is a unit and

$$n_3 S_3 C - \beta_3 = 2k_0 p_X + 4A.$$

Both terms have base valuation at least r , giving the third lower formula. The case $X = 0$ follows from (288) and (282). $\square$

13.8. Comparison at conductors m_i . Put $c_0 = -\alpha^{-1}$ and apply Lemma 4.1. Write H_i for the critical function of θ_i and g_i for its normalized sum.

Take the coefficients $N_i C$ for θ_i and $n_i S_i C$ for ω_i from Lemma 13.10. Since $\omega_i(n_i S_i C) = 1$ and T_i is even, compatibility, Lemma 4.2, and (292) give

$$(303) \quad \mathcal{A}_i = D_\theta(c_0)\Theta(C)^{-1}\Psi_F(-E_2(C))g_i.$$

Choose a uniformizer π of K and compatible lower uniformizers $\Pi_i = N_i \pi$, $\varpi = n_i \Pi_i$. These choices need not be the ones used to construct the generators earlier. Vary C to $C_v = C(1 + \pi^{t_2} v)$ for integral $v \in K$. The critical depths for θ_1 and θ_2, θ_3 are t_2 and $a + r - 1$, respectively. The changes $S_i(C\pi^{t_2} v)$ have depths at least $3r - 2a$ and r . After division by $S_i C$ they have depth at least $r \geq q_i$. Their norm quotients lie in $U_F^{q_i}$ and have ω_i -value 1. Subtracting (292) at C_v, C and using (302) and canceling the common multiplicative value, gives

$$(304) \quad H_i(p_i(\bar{v})) = Q(\bar{v}),$$

Here $p_i : k \rightarrow k$ is induced by N_i on the successive unit quotients of depths t_2 in K and $\lfloor m_i/2 \rfloor$ in L_i , using π, Π_i . The function Q is independent of i . The traces $S_i C_v$ are nonzero because $S_i C_v / S_i C - 1$ has positive valuation.

Proposition 13.11 (The case $t_1 < t_2$). *If $a < r$, every family whose conductors are m_i satisfies $\mathcal{A}_1 = \mathcal{A}_2$.*

Proof. The source depth $t_2 = 2r - 1$ is below the first upper break $4r - 2a - 1$. Thus $p_1(v) = v^2$, with its trace term deeper. For K/L_2 , the consecutive trace ideals at source depths $2r - 1, 2r$ have depths $a + r - 1, a + r$. Hence the trace induces a nonzero k -linear map of residue lines, and the highest-degree norm term is strictly deeper because $r > a$. Thus p_2 is also bijective. Summing (304) proves $g_1 = g_2$. Equation (303) gives the assertion. The relabeled construction compares the first with the third field. $\square$

Proposition 13.12 (The case $t_1 = t_2$). *If $a = r$, every family whose conductors are m_i satisfies $\mathcal{A}_1 = \mathcal{A}_2 = \mathcal{A}_3$.*

Proof. Now $t = t_1 = t_2 = 2r - 1$ and Lemma 13.10 provides the same C for $i = 1, 2, 3$. Put

$$u = \overline{C\pi^t} \in k^\times, \quad c_i = \overline{g_i C - C} \in k^\times.$$

The residues are $\bar{d}, 1, 1 + \bar{d}$, respectively. Equations (274) and (276) give these residues for Y , and the adjustment X changes none: its conjugate differences have positive valuation by $h \leq r - 1$ and the ramification bound. Thus the c_i are distinct.

Expansion of the odd-valuation leading term $u\pi^{-t}$ gives

$$(305) \quad p_i(v) = v^2 + (c_i/u)v.$$

Thus

$$\ker p_i = \{0, c_i/u\}.$$

The kernels are pairwise distinct, each has order two, and each image has index two in $(k, +)$.

Equation (16) gives the nondegenerate polar pairing of each critical function H_i . Equation (304) gives the common pullback $H_i \circ p_i = Q$. Lemma 4.8 therefore gives

$$\sum_{x \in k} H_i(x) = \frac{1}{2} \sum_{v \in k} Q(v)$$

for every i . Hence the normalized sums g_i are equal, and (303) proves $\mathcal{A}_1 = \mathcal{A}_2 = \mathcal{A}_3$. $\square$

13.9. All higher conductors. Now $n = T + h \geq 2r + a$, so $h \geq a$. The upper conductors and stationary depths are

$$(306) \quad M_i = 2n - T_i, \quad b_1 = 2r + h - a, \quad b_2 = r + h, \quad M_i = 2b_i.$$

For a variable in either $\mathfrak{p}_{L_i}^{b_i}$, $i = 1, 2$, its trace and norm belong to the ordinary base stationary ideal $\mathfrak{p}_F^{\lfloor n/2 \rfloor}$. For the first trace this follows from $r + \lfloor (h + a)/2 \rfloor \geq r + \lfloor h/2 \rfloor$; for the second use $r \geq 1$. Choose a coefficient A_* of λ there, of valuation $-h$. Apply Lemma 13.8 to make $A = n_3 X$ exact, and use (294). There is no upper bound on h .

Theorem 13.13 (Equality for $n \geq 2r + a$). *For every $h \geq a$, one has $\mathcal{A}_1 = \mathcal{A}_2$.*

Proof. For $v \in \mathfrak{p}_{L_i}^{b_i}$, the arguments $(Q_i/\beta_i)v$ of (299) have depths at least a and r , respectively:

$$(2\delta - h) - 4\delta + b_1 = a, \quad -h + b_2 = r.$$

The weighted errors have base depths at least

$$(T - a - h) - 2\delta + b_1 = T, \quad (T - a - h) + b_2 = 3r - a \geq T.$$

Thus $(\lambda n_i)(1 - v) = \Psi_{L_i}((A - Q_i)v)$ for $v \in \mathfrak{p}_{L_i}^{b_i}$. Since $b_i \geq s_i$, formula (288) applies and gives $\theta_i(1 - v) = \Psi_{L_i}(N_i C v)$. Moreover X dominates Y , as $-2h < -t_1$. Hence $C \neq 0$ and $v_{L_i} N_i C = -2h = J_i - M_i$.

Use $\beta_i = n_i h_i$ as the coefficient for ω_i . The conductors M_i, T_i are even, so all critical-sum factors are 1. Lemma 4.1 gives

$$\mathcal{A}_i = D_\theta(c_0)\Theta(C)^{-1}\Psi_F(-\mathrm{tr}_i N_i C - \beta_i) = D_\theta(c_0)\Theta(C)^{-1}\Psi_F(-E_2(C) + \Delta).$$

The last identity is (292) and (297). The resulting expression is independent of i . Relabel the third field for the other comparison. $\square$

13.10. Completion of the nonmaximal case.

Proof of Theorem 13.1. Lemmas 13.2 and 13.3 give Y and coefficients β_i for ω_i . Theorem 13.6 extends the characters R_i to compatible characters χ_i . By Lemma 13.7, the given family can be written as $\theta_i = \chi_i(\lambda n_i)$. For $n \leq 2r + a - 1$ use Lemma 13.10 and Propositions 13.11–13.12. For $n \geq 2r + a$ use Theorem 13.13. These two conductor intervals are consecutive. The one-break and two-break ramification possibilities in (268) are both included. $\square$

14. COMPLETION OF THE SECOND MAIN LEMMA

14.1. Ramification possibilities.

Lemma 14.1 (Selection of fields in wild odd degree). *Suppose p is odd and K/F is totally ramified with group C_p^2 . One can choose a degree- p intermediate field B_1 so that, for every other such field B_2 , the breaks of B_i/F are $t \leq t_2 = t + \delta$, and the breaks of $K/B_2, K/B_1$ are $t, t' = t + p\delta$, respectively. Here $\delta \geq 0$ is an integer and $p \nmid t$.*

Proof. Lemma D.3 gives the field B_1 chosen in the statement, the breaks t, t_2 of B_i/F , and the breaks $t' = t + p(t_2 - t), t$ of K/B_i . In particular $\delta = t_2 - t$ is a nonnegative integer. In mixed characteristic, the trace of 1 on B_2/F and (7) imply $v_K(p) \geq p(p-1)t_2$. The proof of Proposition 7.4, before using any assumption $p \nmid t$, constructs an exact Artin–Schreier coordinate by additive Hilbert 90 and best approximation and proves precisely that divisibility assertion from these break and valuation bounds. The equal-characteristic part of that proposition also gives $p \nmid t$. $\square$

Lemma 14.2 (Quadratic conductors in mixed characteristic). *Let $F/\mathbf{Q}_2$ be finite and $e = v_F(2)$. A ramified quadratic norm character has conductor $2a$, $1 \leq a \leq e$, or $2e + 1$. Any two such characters of conductor $2e + 1$ have the same restriction to U_F^{2e} , so their product has conductor at most $2e$.*

Proof. The trace of 1 gives $e \geq \lfloor D/2 \rfloor$ for the quadratic different exponent D , so $2 \leq D \leq 2e + 1$. The even possibilities are as stated. Lemma 10.3 proves that an odd D is $2e + 1$ and proves that the induced character on U_F^{2e}/U_F^{2e+1} , written in coordinates $1 + 4z$, is $(-1)^{\mathrm{Tr}_{k_F/\mathbf{F}_2} \bar{z}}$. This value is independent of the maximal character; hence the product of two is trivial on U_F^{2e} . $\square$

Lemma 14.3 (Ramification cases). *Every Galois extension K/F with group C_ℓ^2 satisfies the hypotheses of one of Theorems 5.3, 6.1, 9.35, 10.20, 11.1, 12.1, or 13.1.*

Proof. By Lemma D.2 the finite residue Galois group is cyclic; a C_ℓ^2 extension cannot be unramified. Its inertia has order ℓ or ℓ^2 . If $\ell \neq p$, wild inertia is trivial and tame inertia is cyclic, so inertia has order ℓ . Abelianness and the residue-Frobenius conjugation action imply

$\ell \mid |k_F| - 1$. There is a unique unramified lower field and every other lower field is tamely ramified. These are exactly the hypotheses of Section 5.

Suppose $\ell = p$. If inertia has order p , there is one unramified degree- p intermediate field and every other degree- p intermediate field is ramified. Theorem 6.1 treats odd p , and Theorem 10.20 treats $p = 2$. If inertia has order p^2 , the extension is totally ramified. For odd p , Lemma 14.1 provides all the hypotheses of the totally ramified proof in Sections 7–9.

It remains to classify totally ramified biquadratic extensions. In characteristic exactly two, the reduced Artin–Schreier representatives of Section 11 have positive odd breaks and the two largest equal, including the one-break case. Theorem 11.1 treats all of them. In mixed characteristic the three nontrivial lower norm characters obey $\omega_3 = \omega_1\omega_2$ by Lemma D.6, and all are ramified. The product of two characters of unequal conductors has the larger conductor, so the maximum of their three conductors occurs at least twice. If it is $2e + 1$, Lemma 14.2 makes the third smaller, and the lower breaks are $(2a - 1, 2e, 2e)$ with $1 \leq a \leq e$, precisely Section 12. Otherwise their conductors are $(2a, 2r, 2r)$ after relabeling, with $1 \leq a \leq r \leq e$, giving the breaks $(2a - 1, 2r - 1, 2r - 1)$ of Section 13. The upper different formulas in those parts follow from Lemma D.3. $\square$

14.2. Proof of the main theorem.

Proof of Theorem 1.1. Apply the theorem specified by Lemma 14.3. It compares $\mathcal{A}_L(\theta_L)$ for a chosen degree- ℓ intermediate field with the corresponding expression for every other such field. Equality for any pair follows by transitivity. The choice of a character θ_L with the prescribed norm pullback does not affect its local constant: two choices differ by a character in $S(K/L)$ and are conjugate by Lemma 4.2, so (6) gives equal local constants. The proof of Theorem 11.1 includes the passage from (193) to an arbitrary nontrivial additive character; all other cases use ψ_F as given. This proves the assertion for the stated family. $\square$

APPENDIX A. THE LEADING GAUSS CONGRUENCE

We prove the Gauss congruence used in Section 5; see also [5]. The proof uses character orthogonality, Jacobi sums, and induction on the sum of the base- p digits.

A.1. The coefficient field and notation. Let $k = \mathbf{F}_q$, $q = p^f$, and choose an unramified extension $K_0/\mathbf{Q}_p$ with residue field k . To construct it, lift a monic irreducible polynomial of degree f over $\mathbf{F}_p$ to a monic polynomial over $\mathbf{Z}_p$, and take its quotient ring R . It is finite free and p -adically complete over $\mathbf{Z}_p$, with $R/pR = k$. Every element with nonzero residue is a unit: lift an inverse modulo p and invert the remaining element of $1 + pR$ by its convergent geometric series. Every nonzero element is p^n times such a unit, since R is finite free and separated over $\mathbf{Z}_p$. Thus R is a discrete valuation ring with uniformizer p , and its fraction field is the required K_0 .

Hensel lifting of $X^{q-1} - 1$ gives a unique multiplicative lift $x \mapsto [x]$ from $k^\times$ into $R^\times$: its derivative is a unit at every nonzero residue root. The lifts multiply because their products are again roots with the prescribed residue. Choose a primitive p th root ζ and put

$$\varpi = \zeta - 1, \quad K_1 = K_0(\zeta), \quad \mathcal{O}_1 = \mathcal{O}_{K_1}.$$

For $p > 2$, the polynomial

$$\frac{(1+X)^p - 1}{X} = p + \binom{p}{2}X + \cdots + X^{p-1}$$

is Eisenstein; hence ϖ is a uniformizer of K_1 and its residue field is k . For $p = 2$, $K_1 = K_0$ and $\varpi = -2$ is already a uniformizer. All root-of-unity values below are viewed in K_1 ; the identities are algebraic identities among these values.

Let $\mathrm{tr}_k = \mathrm{Tr}_{k/\mathbf{F}_p}$ and $\psi(x) = \zeta^{\mathrm{tr}_k(x)}$. For $0 < a < q - 1$, write

$$a = \sum_{i=0}^{f-1} a_i p^i, \quad s_p(a) = \sum_i a_i, \quad a!_p = \prod_i a_i!.$$

Define the plus-sign Gauss sum

$$G_a = \sum_{x \in k^\times} [x]^{-a} \psi(x).$$

The convention of Section 5 is $\tau_k(\omega^a) = -G_a$, where $\omega(x) = [x]$.

A.2. A first-order calculation and a Jacobi sum.

Lemma A.1. *If $q > 2$, then $G_1 \equiv -\varpi \pmod{\varpi^2 \mathcal{O}_1}$. Moreover $G_{p^j} = G_1$ whenever $0 \leq j < f$ and $p^j < q - 1$.*

Proof. For each x , let $t_x \in \{0, \dots, p-1\}$ represent $\text{tr}_k(x)$. Then $\psi(x) = (1 + \varpi)^{t_x} \equiv 1 + t_x \varpi \pmod{\varpi^2}$. The exact identity $\sum_{x \neq 0} [x]^{-1} = 0$ follows by summing a nontrivial character of the cyclic group $k^\times$. Therefore

$$G_1 \equiv \varpi \sum_{x \neq 0} [x]^{-1} t_x \pmod{\varpi^2}.$$

The residue of the last sum is

$$\sum_{x \neq 0} x^{-1} \text{tr}_k(x) = \sum_{i=0}^{f-1} \sum_{x \neq 0} x^{p^i-1} = -1 \quad \text{in } k.$$

Indeed the $i = 0$ inner sum is $q - 1 = -1$, while every other exponent is strictly between 0 and $q - 1$ and its power sum is zero. A power sum vanishes by multiplying its variable by an element on which the corresponding character is nontrivial. Finally, Frobenius permutes $k^\times$ and preserves its absolute trace. Reindexing by the inverse Frobenius shows that $G_{pa} = G_a$, with indices reduced modulo $q - 1$. Iteration gives the last assertion. $\square$

Lemma A.2 (Jacobi sums when the digits add without carries). *Suppose $a, b > 0$, $a + b < q - 1$, and $a_i + b_i < p$ for every digit. Put*

$$J(a, b) = \sum_{x \in k \setminus \{0, 1\}} [x]^{-a} [1 - x]^{-b}.$$

Then

$$(307) \quad G_a G_b = J(a, b) G_{a+b},$$

and $J(a, b)$ is a unit with

$$(308) \quad J(a, b) \equiv - \prod_{i=0}^{f-1} \binom{a_i + b_i}{a_i} \pmod{\varpi \mathcal{O}_1}.$$

Proof. Expand the product of the Gauss sums and separate $t = x + y$. When $t = 0$ the contribution is zero because the character $[x]^{-(a+b)}$ is nontrivial. When $t \neq 0$, write $x = tu$, $y = t(1 - u)$. The two sums separate, giving (307).

For the reduction of the Jacobi sum, both exponents below are positive, so the terms at 0 and 1 may be included. In k the reduction is

$$\sum_{x \in k} x^{q-1-a} (1-x)^{q-1-b} = \sum_{j=0}^{q-1-b} (-1)^j \binom{q-1-b}{j} \sum_{x \in k} x^{q-1-a+j}.$$

The exponent is positive and smaller than $2(q-1)$. Its only possible positive multiple of $q-1$ occurs at $j = a$. This index occurs because $a + b < q - 1$. The reduction is consequently $-(-1)^a \binom{q-1-b}{a}$.

To evaluate this binomial coefficient, use in $\mathbf{F}_p[X]$ the identity

$$(1 + X)^{q-1-b} = \prod_{i=0}^{f-1} (1 + X^{p^i})^{p-1-b_i}.$$

Comparison of coefficients gives $\binom{q-1-b}{a} = \prod_i \binom{p-1-b_i}{a_i}$ modulo p ; each exponent on the right has a unique digit expansion. Since $a_i + b_i < p$, ordinary factorials give

$$\binom{p-1-b}{a_i} \equiv (-1)^{a_i} \binom{a_i+b_i}{a_i} \pmod{p}.$$

For odd p , $a \equiv \sum_i a_i \pmod{2}$, so the signs cancel. For $p = 2$ the same sign equality holds in the residue field. This proves (308). None of its factorials is zero modulo p , proving that $J(a, b)$ is a unit. $\square$

A.3. The leading coefficient.

Theorem A.3 (Leading Gauss congruence). *For every prime p , every $q = p^f$, and $0 < a < q - 1$,*

$$(309) \quad G_a \equiv -\frac{\varpi^{s_p(a)}}{a!_p} \pmod{\varpi^{s_p(a)+1}\mathcal{O}_1}.$$

In particular G_a has ϖ -valuation exactly $s_p(a)$, and

$$(310) \quad \tau_k(\omega^a) \equiv_{\times} \frac{\varpi^{s_p(a)}}{\prod_i a_i!}.$$

Here $X \equiv_{\times} Y$ means $X/Y \in 1 + \varpi\mathcal{O}_1$.

Proof. If $q = 2$ there is no index in the specified interval. Otherwise use induction on $s_p(a)$. For digit sum one, $a = p^j$, and Lemma A.1 proves the assertion. For larger digit sum choose j with $a_j > 0$ and put $c = a - p^j$. Then $c > 0$, and $c + p^j = a < q - 1$ is an addition without carries. Lemma A.2 gives $J(c, p^j) \equiv -a_j \pmod{\varpi}$. It is a unit, so the exact product formula and the induction hypothesis give

$$G_a = \frac{G_c G_{p^j}}{J(c, p^j)} \equiv -\frac{\varpi^{s_p(a)}}{a_j(a_j - 1)! \prod_{i \neq j} a_i!} \pmod{\varpi^{s_p(a)+1}}.$$

Division by the unit does not lower the error precision. This proves (309), including its nonzero leading coefficient. Changing from G_a to $-G_a$ proves (310) with the positive sign required in (20). The argument applies unchanged when $p = 2$ and $\varpi = -2$. $\square$

Equation (310) gives (20) after grouping the base- p digits into base- q digits.

APPENDIX B. FORMAL LOCAL RESIDUES AND THEIR TRACE COMPATIBILITY

This appendix proves the local identity used in the characteristic-two argument. A differential on $k((t))$ means a *formal differential* $h(t) dt$, with termwise differentiation of Laurent series, and

$$\text{Res}_t \left(\sum_n h_n t^n dt \right) = h_{-1}.$$

We prove substitution and trace compatibility, including for wildly ramified extensions.

B.1. Substitution and a finite free expansion.

Lemma B.1 (Formal substitution). *Let k be a field and let $u(t) \in t^e k[[t]]$ have exact order $e \geq 1$. For every $H(s) \in k((s))$,*

$$(311) \quad \text{Res}_t(H(u(t))u'(t) dt) = e \text{Res}_s(H(s) ds).$$

In particular the residue is unchanged by a change of uniformizer.

Proof. It suffices to treat $H(s) = s^n$: only finitely many of its Laurent monomials can contribute to the residue after substitution. Write $u = t^e v$ with $v \in k[[t]]^\times$. If $n = -1$, then $u'/u = e/t + v'/v$ and $v'/v \in k[[t]]$, proving the assertion. For $n \neq -1$ one must not divide by $n+1$ in characteristic dividing $n+1$. Instead, perform the calculation first over the torsion-free ring

$$R = \mathbf{Z}[c_0, c_0^{-1}, c_1, c_2, \dots], \quad v = c_0 + c_1 t + c_2 t^2 + \dots$$

The coefficient giving the residue uses only finitely many c_i and lies in this ring, also for negative n . The residue of a derivative is zero, so differentiating u^{n+1} gives $(n+1) \operatorname{Res}_t(u^n u' dt) = 0$. Torsion-freeness gives residue zero in R . Specializing the coefficients to any field proves the desired identity without division in that field. Linearity now proves (311); for a change of uniformizer, $e = 1$. $\square$

Lemma B.2 (A basis compatible with coefficients). *Let R be a commutative ring and $u(t) = t^e(c_0 + c_1 t + \cdots)$, where $c_0 \in R^\times$. Under $s \mapsto u(t)$, the module $R[[t]]$ is free over $R[[s]]$ with basis $1, t, \dots, t^{e-1}$. After inverting s it is $R((t))$, free over $R((s))$ with the same basis. These expansions and the resulting multiplication matrices commute with specialization of the coefficients.*

Proof. For $n = ej + i$, $0 \leq i < e$, the series $s^j t^i$ has leading term $c_0^j t^n$. Given a power series, cancel its lowest coefficient using the corresponding multiple of $s^j t^i$ and continue successively. This constructs an expansion $\sum_{i=0}^{e-1} H_i(s) t^i$, with $H_i \in R[[s]]$. It is unique: the first nonzero terms of the nonzero $H_i(s) t^i$ have different degrees modulo e , and hence cannot cancel. The algorithm uses only ring operations and c_0^{-1} . Every output coefficient is determined after finitely many operations, which proves compatibility with specialization. Since $s = t^e v(t)$ with $v(t)$ a unit, inverting s is equivalent to inverting t . This proves the assertions for Laurent series as well. In particular, the algebraic trace of multiplication by an element of $R((t))$ is an element of $R((s))$, obtained by the trace of its finite matrix, and it commutes with specialization. $\square$

B.2. Trace compatibility.

Lemma B.3 (Totally ramified substitution). *For a field k and $s = u(t)$ of positive order e ,*

$$(312) \quad \operatorname{Res}_s \left(\operatorname{Tr}_{k((t))/k((s))} (h(t)) ds \right) = \operatorname{Res}_t (h(t) u'(t) dt) \quad (h \in k((t))).$$

The trace here is the algebraic trace on the finite free extension. The identity remains valid even when $u' = 0$.

Proof. First suppose that k has characteristic zero. After a finite extension of the coefficient field, we can take an e th root of the leading coefficient of u and include all e th roots of unity. This scalar extension preserves the trace matrix in Lemma B.2, so an identity proved there descends. The binomial series then gives a formal parameter $z = t(c_0 + c_1 t + \cdots)^{1/e}$ with $s = z^e$. Write $h = \sum_n b_n z^n$. The extension $k((z))/k((s))$ is now Galois with automorphisms $z \mapsto \xi z$, $\xi^e = 1$. Thus

$$\operatorname{Tr}(h) = e \sum_{j \in \mathbf{Z}} b_{ej} s^j, \quad h ds = e \sum_n b_n z^{n+e-1} dz.$$

Both residues equal eb_{-e} . The change from t to z preserves residue by Lemma B.1. The trace is continuous for the Laurent-series topology, either from the displayed automorphisms or from its finite multiplication matrix, so these calculations apply to infinite Laurent series, not just Laurent polynomials.

Now take a field of arbitrary characteristic. Lift every coefficient of $u(t)$ and $h(t)$ to an independent indeterminate in a polynomial ring over $\mathbf{Z}$, inverting the nonzero leading coefficient of u . This ring is a characteristic-zero integral domain. The two residues in (312) belong to this ring: Lemma B.2 gives this for the trace, and the right side is an ordinary coefficient of a product and derivative. Each residue uses only finitely many of the indeterminates. Over the fraction field of this ring the characteristic-zero proof gives equality. It is therefore an equality in the ring itself. Specialize the coefficients to their values in k , using Lemma B.2 to specialize the trace. This proves (312) in every characteristic. This coefficient argument does not invert the possibly zero integer e in k and does not assume a tame extension. $\square$

Theorem B.4 (Local trace–residue identity). *Let E/F be a finite separable extension of equal-characteristic local fields with finite residue fields. For every formal differential η on E ,*

$$(313) \quad \operatorname{Tr}_{k_E/k_F} \operatorname{Res}_E(\eta) = \operatorname{Res}_F(\operatorname{Tr}_{E/F} \eta).$$

Here if s is a uniformizer of F and $\eta = x ds$, the trace of the differential is $\operatorname{Tr}_{E/F}(x) ds$.

Proof. We explain the Laurent-series presentations used in the proof. In an equal-characteristic complete discrete valuation field with finite residue field k of size q , each residue has a unique lift solving $X^q - X = 0$, by Hensel's lemma with derivative -1 . In characteristic p these lifts form a field: sums and products are again roots with the expected residues. Choosing a uniformizer t and successively subtracting residue coefficients gives a unique convergent Laurent expansion. Thus the field is $k((t))$. In a finite extension, these coefficient fields are compatible, because the lifts of elements of the smaller residue field are the same unique roots.

Write $F = k_F((s))$ and $E = k_E((t))$. The intermediate field $F' = k_E((s))$ is unramified over F ; its trace is the coefficientwise finite-field trace. In E , $s = u(t)$ has positive order e , and Lemma B.2 gives $[E : F'] = e$. Separability implies $u'(t) \neq 0$: otherwise $u \in k_E((t^p))$, and the intermediate extension $k_E((t))/k_E((t^p))$ would be purely inseparable of degree p , contradicting separability of E/F' . Therefore $ds \neq 0$ and every formal differential on E can be written as $x ds$.

Apply Lemma B.3 to E/F' . It gives $\text{Res}_{F'}(\text{Tr}_{E/F'}(x) ds) = \text{Res}_E(x ds)$ in k_E . Apply Tr_{k_E/k_F} and commute it with extraction of the s^{-1} coefficient. Transitivity of matrix traces proves (313). Residues do not depend on the chosen uniformizers by Lemma B.1. $\square$

Lemma B.5 (Logarithmic derivative of a norm). *For $v \in E^\times$ in the same separable extension,*

$$(314) \quad \frac{dN_{E/F}(v)}{N_{E/F}(v)} = \text{Tr}_{E/F} \left(\frac{dv}{v} \right).$$

Proof. Let D_F be differentiation with respect to a Laurent parameter of F , and let D_E be its extension to E ; separability gives this extension. Choose an F -basis $e_1, \dots, e_n$ of E . Let $M = M(v)$ be the matrix of multiplication by v , and let B be the matrix whose j th column is the coordinate vector of $D_E(e_j)$. Differentiating the coordinate identity for ve_j gives

$$D_F(M) + BM = M(D_E v) + MB,$$

where $D_F(M)$ is obtained by differentiating each entry and $M(D_E v)$ is the matrix of multiplication by $D_E v$.

Since $v \neq 0$, the matrix M is invertible. Multiply the displayed identity on the left by M^{-1} and take matrix traces. Cyclic invariance of the matrix trace gives

$$\text{tr}(M^{-1}D_F(M)) = \text{tr}(M^{-1}M(D_E v)) = \text{Tr}_{E/F}(D_E v/v).$$

Jacobi's determinant identity gives

$$\frac{D_F(\det M)}{\det M} = \text{tr}(M^{-1}D_F(M)).$$

Since $\det M = N_{E/F}(v)$, multiplication by the base differential gives (314). $\square$

Theorem B.4 and Lemma B.5 are used in the proof of Lemma 11.2.

APPENDIX C. ELEMENTARY DESCENT AND RAMIFICATION ARGUMENTS

The following lemmas are used to extend characters and to compute ramification. The results for extensions with Galois group C_ℓ^2 are proved in Appendix D.

Lemma C.1 (Cyclic Hilbert 90, in both forms). *Let E/F be cyclic of degree n , with generator σ . Then*

$$\ker N_{E/F} = \{y/\sigma(y) : y \in E^\times\}, \quad \ker \text{Tr}_{E/F} = \{\sigma(y) - y : y \in E\}.$$

Proof. Distinct field embeddings are linearly independent as maps. Indeed, take a nonzero relation of shortest length. Evaluating at tx and subtracting one embedding's value at t times the relation at x removes that embedding. Choose t on which another embedding differs from it. The resulting shorter nonzero relation is a contradiction.

If $a \in E^\times$ has norm one, put $b_0 = 1$ and $b_i = \prod_{j=0}^{i-1} \sigma^j(a)$ for $1 \leq i \leq n$. By the independence just proved the map $x \mapsto \sum_{i=0}^{n-1} b_i \sigma^i(x)$ is not identically zero. Choose x for which its value y is nonzero. Since $b_n = 1$ and $a\sigma(b_i) = b_{i+1}$, one has $a\sigma(y) = y$. This gives $a = y/\sigma(y)$. The reverse inclusion telescopes under the norm.

For the additive statement, the trace map is nonzero by the same independence, and hence is a surjective F -linear map to F . Its kernel has dimension $n - 1$. The map $\sigma - 1$ has kernel F , so its image also has dimension $n - 1$. Its image lies in the trace kernel by telescoping, proving equality. This proof applies when the characteristic divides n . $\square$

Lemma C.2 (Extension of characters). *Let H be a subgroup of $E^\times$ containing U_E^m for some $m \geq 1$, and let $\chi : H \rightarrow \mathbf{C}^\times$ be trivial on U_E^m . Then χ extends to a continuous character of $E^\times$. The same assertion holds if χ is first prescribed on several subgroups, agrees on their intersections, and defines a character on the subgroup they generate containing such a U_E^m .*

Proof. The group $E^\times/U_E^m$ is a product of an infinite cyclic group and the finite group $\mathcal{O}_E^\times/U_E^m$. Extend a character of a subgroup one generator at a time. If the first relation of a new generator g with the existing subgroup is $g^n = h$, choose a nonzero complex n th root of the prescribed value of h . If there is no such relation, choose any nonzero complex value. This defines an extension at each step, and finitely many steps suffice. Pullback to $E^\times$ is continuous because U_E^m is open and remains in the kernel. Apply this argument to the generated subgroup for the last assertion. $\square$

Lemma C.3 (The first wild ramification term). *Let K/F be a finite Galois extension of local fields, let σ be an inertia automorphism, and let π be a uniformizer of K . Write $v = v_K$ and suppose*

$$\sigma(\pi) = \pi(1 + c\pi^u), \quad v(c) = 0, \quad u \geq 1.$$

For $x \in K^\times$ with $v(x) = m$, one has $v(\sigma x - x) \geq m + u$. If m is prime to the residue characteristic, then equality holds.

Proof. Expand $x = \sum_{j \geq m} a_j \pi^j$ using the multiplicative lifts of residue representatives. Inertia fixes those lifts: they are the unique lifts of their residues among roots of unity of order prime to the residue characteristic (and zero). The series converges in the local field, with only finitely many negative powers. For any integer j , the binomial expansion at the principal unit gives

$$(1 + c\pi^u)^j - 1 \equiv jc\pi^u \pmod{\mathfrak{p}^{u+1}}.$$

This is also valid for negative j , by the convergent inverse series. Consequently the j th summand changes by an element of valuation at least $j + u$. When m is prime to the residue characteristic, the first summand changes with exact valuation $m + u$, and all following summands have strictly larger valuation. This proves the assertions. Without the condition $u \geq 1$, the first coefficient is instead $(\sigma\pi/\pi)^m - 1$, which can vanish even when m is prime to the residue characteristic. $\square$

Lemma C.4 (Transitivity of the different). *For a tower of finite separable local extensions $L/E/F$,*

$$\mathfrak{D}_{L/F} = \mathfrak{D}_{L/E} \mathfrak{D}_{E/F} \mathcal{O}_L.$$

Proof. Write $A = \mathcal{O}_F$, $B = \mathcal{O}_E$ and $C = \mathcal{O}_L$. These are finite free modules along the tower. Via the nondegenerate trace pairings their dual modules identify with the inverse differentials. The elementary adjunction

$$\mathrm{Hom}_A(C, A) \simeq \mathrm{Hom}_B(C, \mathrm{Hom}_A(B, A))$$

sends a functional ℓ to $c \mapsto (b \mapsto \ell(bc))$; its inverse evaluates at $b = 1$. Write the principal fractional B -ideal $\mathfrak{D}_{E/F}^{-1} = dB$. Under the trace identifications, the right side is $d \mathrm{Hom}_B(C, B) = d\mathfrak{D}_{L/E}^{-1}$. Transitivity of field trace shows that these identifications send $x \in L$ to the same functional $c \mapsto \mathrm{Tr}_{L/F}(xc)$, with no scalar discrepancy. Thus $\mathfrak{D}_{L/F}^{-1} = d\mathfrak{D}_{L/E}^{-1}$ as fractional C -ideals. Inversion gives the assertion. $\square$

Lemma C.5 (The discriminant of an integral order). *Let E/F be finite separable and let $B \subset \mathcal{O}_E$ be an $\mathcal{O}_F$ -order of full rank. Then its discriminant ideal is the field discriminant ideal multiplied by the square of its index ideal. In particular the field discriminant divides the order discriminant. For a totally ramified extension, the different exponent is at most the F -valuation of the order discriminant.*

Proof. Choose $\mathcal{O}_F$ -bases of B and $\mathcal{O}_E$, and let M be the inclusion matrix. The two trace Gram matrices satisfy $G_B = M^\top G_{\mathcal{O}_E} M$, so $\det G_B = (\det M)^2 \det G_{\mathcal{O}_E}$. This proves the first assertion and divisibility. The valuation of the field discriminant is the length over $\mathcal{O}_F$ of the trace-dual lattice modulo $\mathcal{O}_E$: this follows by diagonalizing the trace matrix over the discrete valuation ring. The trace dual is $\mathfrak{D}_{E/F}^{-1}$ by the definition of the inverse different. Its cyclic-prime specialization underlies the trace-ideal formula of [7, Theorem 3.7]. If the different exponent is d , this quotient has $\mathcal{O}_F$ -length $[k_E : k_F]d$, since each successive $\mathcal{O}_E$ -lattice quotient is k_E . In the totally ramified case $[k_E : k_F] = 1$, proving the stated bound. $\square$

APPENDIX D. NORM SUBGROUPS AND RAMIFICATION IN DEGREE ℓ^2

We prove the facts about C_ℓ^2 extensions used in the proof of Theorem 1.1. We use the cyclic-prime results of [7, Theorem 3.7, Corollary 3.9, and Proposition 3.3] and the standard local-field results recalled below. For their general forms, see [4, Chapters III–V, XIII–XIV].

For a finite separable extension E/F , write $e(E/F)$ and $f(E/F)$ for its ramification index and residue degree. We use $[E : F] = e(E/F)f(E/F)$ and the monogenic different formula

$$\mathfrak{D}_{E/F} = (f'(\pi)) \quad \text{if } \mathcal{O}_E = \mathcal{O}_F[\pi],$$

where f is the minimal polynomial of π over F . For a totally ramified extension of degree n , any uniformizer π of E satisfies this hypothesis. Indeed, the degree formula gives $F(\pi) = E$. In the basis $1, \pi, \dots, \pi^{n-1}$ the nonzero terms $a_i \pi^i$ have valuations $nv_F(a_i) + i$, which are distinct modulo n . If their sum is integral, each a_i is integral.

We also use the following choices of generators. In characteristic p , additive Hilbert 90 applied to 1 in a cyclic degree- p extension gives $\sigma z - z = 1$. Then $z^p - z \in F$ and $E = F(z)$. For a quadratic extension in characteristic different from two, choose u with $u \neq \sigma u$ and put $w = u - \sigma u$. Then $\sigma w = -w$, so $w^2 \in F$ and $E = F(w)$.

Lemma D.1 (Newton approximation). *Let R be a complete discrete valuation ring with valuation v , let $f \in R[X]$, and let $a \in R$. Suppose $s = v(f'(a)) < \infty$ and $v(f(a)) > 2s$. Then there is a unique root b in the ball $v(b - a) > s$. Moreover $v(b - a) \geq v(f(a)) - s$.*

Proof. The case $f(a) = 0$ is immediate, including uniqueness by the argument below. Otherwise iterate $a_{n+1} = a_n - f(a_n)/f'(a_n)$. Taylor's polynomial identity with integral coefficients shows inductively that $v(f'(a_n)) = s$, and, writing $r_n = v(f(a_n)) - 2s > 0$, that

$$v(a_{n+1} - a_n) = s + r_n, \quad r_{n+1} \geq 2r_n.$$

If an iterate is a root, stop. The derivative differences have valuation at least $s + r_n > s$, so they cannot change s ; the linear Taylor term cancels and the remaining terms have valuation at least $2(s + r_n)$. Thus the iterates are integral and converge to a root with the asserted error bound. For b, c in the stated ball, divided differences give $f(b) - f(c) = (b - c)(f'(a) + \varepsilon)$ with $v(\varepsilon) > s$. The second factor is nonzero; hence two roots there are equal. Rescaling the variable and the polynomial gives the same conclusion for the fractional ideals used in the preceding sections. $\square$

Lemma D.2 (Inertia, tame action, and subgroup numbering). *For a finite Galois extension K/F of local fields, let $G = \text{Gal}(K/F)$, let I be the kernel of its residue action, and put $q = [k_F]$. The residue action identifies G/I with the cyclic group $\text{Gal}(k_K/k_F)$. Its fixed field K^I/F is unramified. The quotient I/G_1 is cyclic of order prime to the residue characteristic p , and G_1 is a p -group. A lift of residue Frobenius conjugates I/G_1 by the q th power map. For $H \subset G$, the lower ramification groups on K/K^H are $H_j = H \cap G_j$ for every integer $j \geq 0$.*

Proof. Choose a generator $\bar{a}$ of k_K/k_F , lift its monic irreducible polynomial to $\mathcal{O}_F[X]$, and lift $\bar{a}$ to a root $a \in \mathcal{O}_K$ by Hensel's lemma. The field $F(a)$ has degree at most $f = [k_K : k_F]$ and residue degree at least f , hence degree exactly f and ramification index one. All residue conjugates of $\bar{a}$ lift uniquely inside $F(a)$, so $F(a)/F$ is Galois and reduction identifies its group with $\text{Gal}(k_K/k_F)$. Normality of K/F extends its automorphisms to K . An element of G acts trivially on the residue field precisely when it fixes a , by uniqueness of the simple-root lifts. This proves the assertions about I and G/I .

Choose a uniformizer π of K . Inertia fixes the multiplicative lifts of every residue element. Expanding integers in powers of π therefore tests lower ramification on π alone. In particular,

$$I \longrightarrow k_K^\times, \quad g \longmapsto \overline{g\pi/\pi}$$

is a homomorphism with kernel G_1 . For $j \geq 1$ the map

$$G_j \longrightarrow k_K, \quad g \longmapsto \overline{(g\pi/\pi - 1)/\pi^j}$$

is additive with kernel G_{j+1} . These statements follow by multiplying the two uniformizer ratios and reducing the first nonzero coefficient. The finite subgroup of $k_K^\times$ is cyclic and has order prime to p ; each additive image is a p -group. The filtration eventually equals 1, since every nonidentity inertia automorphism moves π . This proves the assertion about G_1 .

The tame residue is unchanged on replacing π by $u\pi$ for a unit u , since inertia fixes its residue. For a Frobenius lift ϕ , compute using $\pi' = \phi^{-1}\pi$:

$$\overline{(\phi g \phi^{-1})(\pi)/\pi} = \overline{\phi(g(\pi')/\pi')} = \overline{g\pi/\pi}^q.$$

Finally, membership in a lower ramification group is the condition $v_K(gx - x) \geq j + 1$ for all $x \in \mathcal{O}_K$. For an element of H this condition is identical over F and over K^H : both use the same normalized valuation on K . This proves the subgroup assertion. $\square$

Lemma D.3 (Ramification breaks in degree p^2). *Let K/F be totally ramified with $G = C_p^2$, where p is the residue characteristic. There are integers $1 \leq t \leq b$ such that the filtration is G through t , possibly one line H_0 of order p through b , and then 1. Choose $L_0 = K^{H_0}$ if $b > t$, and choose any degree- p field if $b = t$. The break of L_0/F is t ; the break of K/L_0 is b . Every other degree- p intermediate field L satisfies: L/F has break*

$$s = t + (b - t)/p,$$

and K/L has break t . In particular s is an integer, and $b = t + p(s - t)$. Inertia of order p , rather than p^2 , gives instead one unramified degree- p intermediate field and ramified degree- p intermediate fields whose breaks are unchanged by the unramified base extension.

Proof. Lemma D.2 gives $G_1 = G$. A decreasing filtration of C_p^2 can contain at most one proper nontrivial subgroup. The subgroup assertion gives the breaks of K/L_0 and K/L . For a uniformizer π of K , the monogenic different formula gives

$$d_{K/F} = v_K \prod_{g \neq 1} (\pi - g\pi) = \sum_{j \geq 0} (|G_j| - 1) = (p^2 - 1)(t + 1) + (p - 1)(b - t).$$

Here each nonidentity g is counted exactly $v_K(\pi - g\pi)$ times. Lemma C.4 and the cyclic-prime different formula in Section 3 give

$$d_{K/F} = (p - 1)(b + 1) + p d_{L_0/F} = (p - 1)(t + 1) + p d_{L/F}.$$

Solving these equalities gives the breaks of L_0/F and L/F , including the integrality of s .

If inertia has order p , its fixed field U is the unique unramified degree- p intermediate field. Every other degree- p intermediate field E is totally ramified over F . In $K = EU$, a uniformizer of E is still a uniformizer of K . The generator on K/U restricts to the generator on E/F , and $v_K|_E = v_E$. Thus its uniformizer difference has the same valuation in E and K . $\square$

Proposition D.4 (Distinct norm subgroups). *Let K/F have Galois group C_ℓ^2 . The norm subgroups $N_{L_0/F} L_0^\times$ and $N_{L/F} L^\times$ are distinct if L_0/F is the unramified degree- ℓ subextension, or, when K/F is totally ramified, L_0 is the field selected in Lemma D.3 and $L \neq L_0$. In the totally ramified one-break case all lower norm subgroups are pairwise distinct.*

Proof. Every subgroup $N_{L/F} L^\times$ has index ℓ by [7, Corollary 3.9]. If L_0/F is unramified, its nontrivial norm characters have conductor zero, whereas those of every ramified lower field have positive conductor. Equal norm subgroups would give equal character groups, contrary to their conductors. In the totally ramified two-break case, Lemma D.3 gives the distinct lower conductors $t + 1 < s + 1$, proving the assertion in the same way.

It remains to distinguish these subgroups when every extension K/L and L/F has break t . Here $\ell = p$. Fix $\pi \in K$ of valuation one and put

$$c_g = \overline{(g\pi/\pi - 1)/\pi^t} \in k, \quad k = k_F = k_K.$$

Lemma D.2 makes $g \mapsto c_g$ an injective additive map, with two-dimensional $\mathbf{F}_p$ -image V . For a line $H = \langle \sigma \rangle$, set $L_H = K^H$, $c = c_\sigma$, $\Pi_H = N_{K/L_H}\pi$, and $\varpi = N_{K/F}\pi$. For $\tau \notin H$ write $e = c_\tau$. The critical norm polynomial ([7, Proposition 3.3]) gives

$$P_H(X) = X^p - c^{p-1}X, \quad \overline{(\tau\Pi_H/\Pi_H - 1)/\Pi_H^t} = e^p - c^{p-1}e =: d_H \neq 0.$$

The second equality is the first applied to $N_{K/L_H}(\tau\pi/\pi) = \tau\Pi_H/\Pi_H$. Apply the same critical norm theorem now to L_H/F with the compatible uniformizers Π_H, ϖ . Its critical norm image is

$$d_H^p \ker \mathrm{Tr}_{k/\mathbf{F}_p},$$

whose annihilator is the line $d_H^{-p}\mathbf{F}_p$ in the residue trace pairing. This line is precisely the restriction of the norm-character group to U_F^t/U_F^{t+1} , by the norm-filtration theorem and the exact norm-character conductor.

Choose a basis c_1, c_2 of V and put $D = c_1c_2^p - c_1^pc_2 \neq 0$. For any ordered basis c, e of V ,

$$ce^p - c^pe = aD \quad \text{for some } a \in \mathbf{F}_p^\times, \quad d_H = aD/c.$$

Thus the annihilator line is $(c^p/D^p)\mathbf{F}_p$. Distinct lines H give distinct lines $c\mathbf{F}_p$, and Frobenius is bijective, so these annihilators are distinct. The norm-character groups, and hence the subgroups $N_{L_H/F}L_H^\times$, are distinct. $\square$

Lemma D.5 (Norm pullback of norm characters). *Let K/F have Galois group C_ℓ^2 , and let L_i, L_j be distinct degree- ℓ intermediate fields. Suppose $N_{L_i/F}L_i^\times \neq N_{L_j/F}L_j^\times$. Then*

$$S(L_j/F) \longrightarrow S(K/L_i), \quad \omega \longmapsto \omega \circ N_{L_i/F}$$

is an isomorphism. In particular it applies to the pairs (L_0, L) in Proposition D.4.

Proof. Norm transitivity shows that every displayed pullback is trivial on $N_{K/L_i}K^\times$. A nontrivial ω has kernel $N_{L_j/F}L_j^\times$, since that quotient has prime order ℓ . Its pullback cannot be trivial: otherwise one index- ℓ lower norm subgroup would be contained in the other and hence equal to it. Both character groups have order ℓ by FML, so the displayed homomorphism is an isomorphism. $\square$

Lemma D.6 (Quadratic norm characters). *Let K/F be biquadratic with quadratic intermediate fields L_1, L_2, L_3 , and let ω_i be the nontrivial character in $S(L_i/F)$. The characters ω_i are pairwise distinct and satisfy $\omega_3 = \omega_1\omega_2$. Consequently Lemma D.5 applies to every pair $L_i \neq L_j$.*

Proof. If $\mathrm{char} F \neq 2$, associate to a square class a the norm character of $F(\sqrt{a})/F$, and associate the trivial character to the square class of 1. Denote its value at b by (a, b) . The existence of a nonzero solution of

$$X^2 = aY^2 + bZ^2$$

is equivalent to $(a, b) = 1$. If a is nonsquare, any such solution has $Z \neq 0$ and gives $b = N(X/Z + (Y/Z)\sqrt{a})$; the split case is immediate. The conic is symmetric in a, b , so $(a, b) = (b, a)$. Multiplicativity in the second variable is the norm-character property; symmetry gives it in the first variable. A nonsquare class has nontrivial character by [7, Corollary 3.9]. Thus the map from square classes to quadratic characters is an injective homomorphism. The three fields have classes a, b, ab , proving the claim.

If $\mathrm{char} F = 2$, use the formula of Lemma 11.2:

$$\omega_f(u) = (-1)^{\mathrm{Tr}_{k/\mathbf{F}_2} \mathrm{Res}_F(f du/u)}.$$

The formula is additive in the Artin-Schreier class f . Lemma 11.2 also shows that the character is nontrivial for every nonzero class, whether ramified or unramified. The classes of the three fields are $f, g, f + g$, giving pairwise distinctness and the product identity. $\square$

Remark D.7. For odd ℓ , Proposition D.4 is applied to the field L_0 in Lemma D.3 and every other degree- ℓ intermediate field. Theorem 9.35 then gives equality of $\mathcal{A}_L$ for any pair by transitivity. For $\ell = 2$, Lemma D.6 gives distinct norm subgroups for every pair and the relation $\omega_3 = \omega_1\omega_2$.

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